

Sensation

CT

Troubleshooting Guide

Gantry

Troubleshooting Guide Gantry

This document is valid for:
SOMATOM Sensation 10/16/Cardiac
SOMATOM Sensation 40/ 64/ Cardiac 64/ Open

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Safety information

Only qualified and system trained service staff is allowed to perform CT system installation, service, maintenance, and quality assurance. Ensure that the most recent version of the service documentation is available.

To avoid any risk of injury to persons and/or damage to the system read and observe the **General Safety Notes** (TD00-000.860.01.xx.xx). Especially read and observe the **Product-specific safety notes** (CT02-023.860.01.xx.xx) ([CT02-023.860.01 / Product-specific safety notes](#)) which include very important safety-related information as well as information about handling screws and nuts, use of Loctite and instructions for torque wrench. You will find the following information in the “Product-specific safety notes”:

- Safety information related to the method of risk management ([CT02-023.860.01 / Safety warnings](#)).
- General safety information such as the handling of service documentation, CT system training requests, X-ray protection, electrical protection, service tools..([CT02-023.860.01 / General safety information](#)).
- General safety information regarding working in the gantry with the power off/on ([CT02-023.860.01 / Gantry](#)).
- Use of patient table as lifting device for replacing gantry parts([CT02-023.860.01 / Usage of patient table as lifting device](#)).
- Laser products ([CT02-023.860.01 / Laser Products](#)).
- Attachment of screws and nuts, use of Loctite, adjustment instructions for torque wrench([CT02-023.860.01 / Screw Connections](#)).

Information about the possibilities of switching off the gantry power

- Switching off the gantry power by switching the system to **COMP/ON** at the control box.
 - **Be aware that power is still supplied to the rotating part of the gantry (e.g DMS) after switching to COMP/ON! Switch off this power with circuit breaker F9 in the PDC.**
- Switching off the gantry power by using the **SYSTEM/OFF** switch at the control box.
 - **Be aware that power is still supplied to the rotating part of the gantry (e.g DMS) after activating the System/OFF button! Switch off this power via circuit breaker F9 in the PDC**
- Switching off gantry power by using the **SYSTEM/OFF** switch at the control box. Additionally, use the on-site **ON/OFF** switch to completely switch off system power.
 - **In this case, the complete system power is switched off .**

NOTE

Read and observe the safety information in the “Product-specific safety notes” before performing service work in the gantry. If the gantry power is switched off for any service work, always secure against accidental switch on.

Tools

Normally a standard tool kit is required for performing service as described in this document. If special tools are needed, they are described in the relevant replacement description in the chapter on “Tools and auxiliary equipment” .

An RMS meter is required for some measurements.

Only for Sensation Open with RTP-Option

The lateral lasers on the gantry front cover must be checked every time after the gantry front cover was opened.

Overview

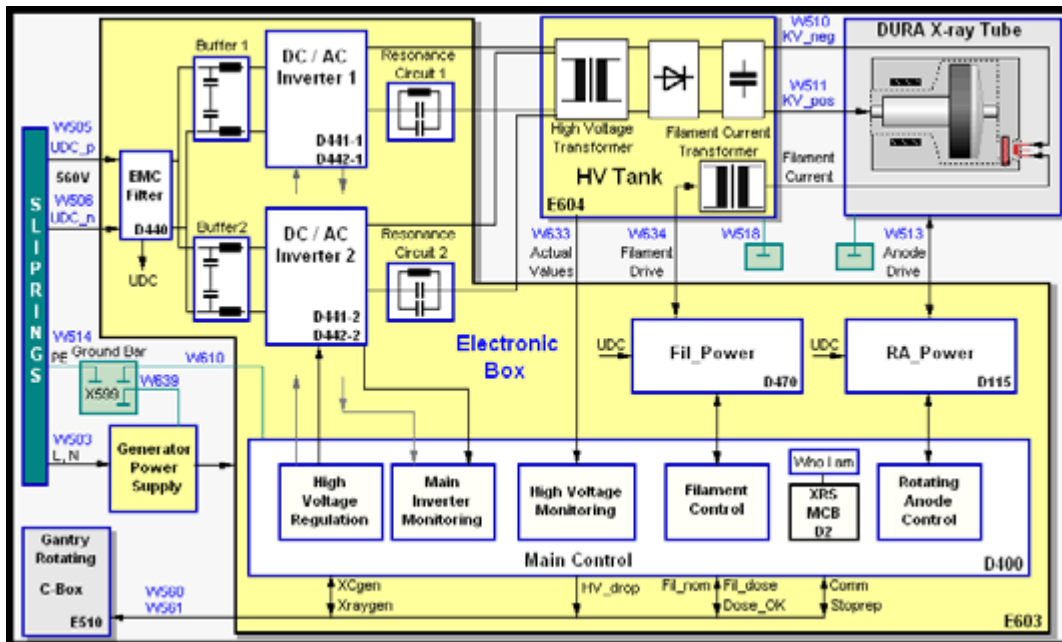


Fig. 1: Block diagram of XRS

The X-ray system main FRUs are:

- X-ray tube
- Electronic box - E603
- Fil/RAC - D470+D115
- Generator Power Supply - E606
- Main Control - D400
- High Voltage Tank - E604
- MCB
- D451

Problem isolation

This section contains information about isolating problems. **HV_Drops** explains the HV_Drop and its purpose; **HV testing** helps you decide which test to run for HV problems; **UDC information** is included because most XRS functions require correct voltage information. The **generator oscillator** procedure is provided as an option in case of problems due to adaptation, e.g. particular current settings not working. Other sections highlight problems that have been shown by experience to cause XRS problems.

HV_Drops

The actual high tension (P_UTact and N_UTact) is monitored by the D400. If either positive, negative, or both, are arcing (i.e. high tension breakdown greater than 30 kV), the D400 sets the HV_drop signal. In this case, the main inverter is stopped and restarted within 2 msec, to prevent high tension errors from occurring.

Notes regarding the HV_drop:

The purpose of the HV_drop is to eliminate the possibility of high tension being applied to only one side of the HT circuit. For instance, if during a 120 kV exposure, the 60 kV negative side "drops", the positive HT will not compensate and try to raise to 120 kV. If the positive side were allowed to rise to 120 kV, an error would occur when the HT reaches 82 kV.

The HV_drop is recorded in the tube history. HV_drop alone CANNOT be used to evaluate tube performance.

The firmware compares the number of HV_drops within 300 msec to a predefined value in the firmware. If this value is exceeded, an error is sent (typically XRS_47).

The HV_drop is transmitted to the ACU. The ACU will mark readings the HV_drop was active to allow the IRS to interpolate these readings.

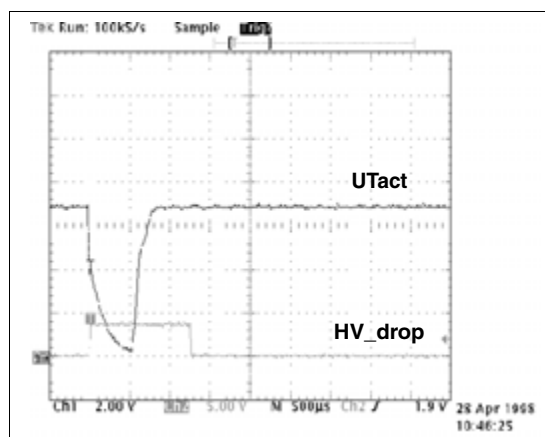


Fig. 2: HV_drop

(Fig. 2 / p. 13) shows an oscillogram of U_Tact compared to HV_drop. The key point is while the HV_drop is active for 1.25 msec, the HV collapses and is rebuilt, allowing the scan to continue without error.

Channel 1 = U_Tact

Channel 2 = HV_drop

HV testing

Troubleshooting the XRS is based on the error description. Whenever no specific component is identified (as in the case of high voltage failures), the following diagram may be used as a guideline.

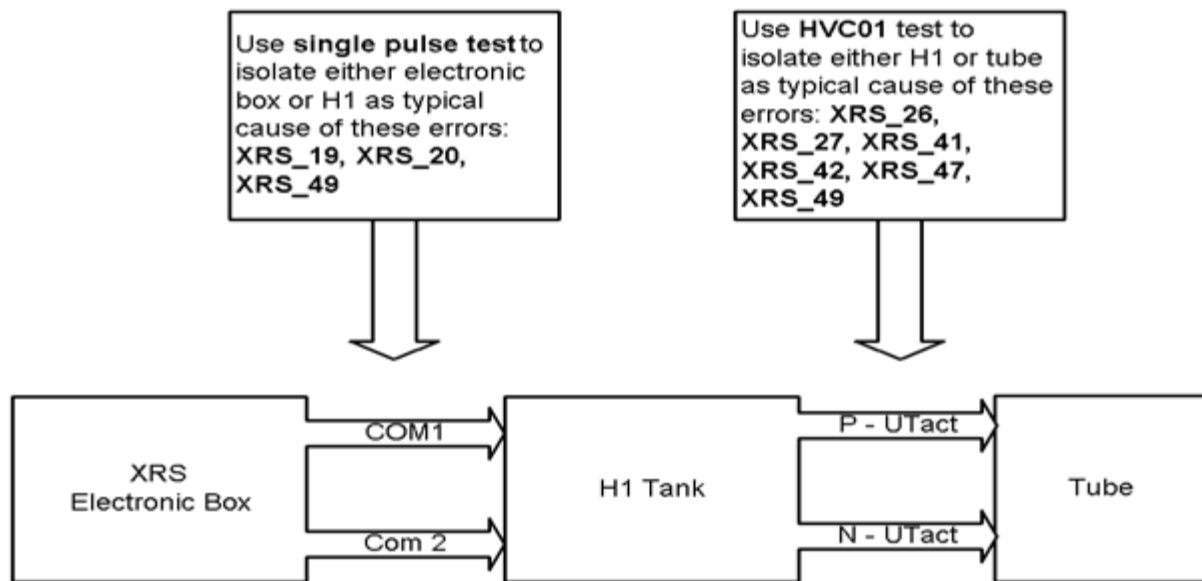


Fig. 3: Diagram

UDC voltage information

- The 560VDC UDC_P and UDC_N should be measured prior to starting the diagnostics. Many errors are related to improper voltage because the system uses this initial power for:
 - The ignition circuit of inverters (E-Box).
 - The filament power
 - The RAC power (FIL/RAC).
- The UDC voltage may vary because it depends on the line voltage. If a transformer is used (i.e. the line voltage is NOT 400V), the UDC voltage must be 560 V +/-10%. During a scan, the voltage drop of the UDC should never be more than 80 V below the actual UDC voltage. If the UDC voltage drops by more than 80 V during an exposure, a phase may be missing in the PDC.
 - ⇒ **In this case check the three F100 fuses.**
If one F100 is defective, replace all three. If the fuses are OK, check whether the line impedance meets specifications.
- UDC voltage is transferred as follows:
 - From D-450 X7-X10 in the PDC via the W3 cable to the X303 connection in the right leg of the gantry
 - From the X303 connection in the right leg of the gantry via the X315 cable to the slip ring connections X382 and X383.
 - From slip ring connections X382 and X383 via the W505/W506 cable to the E-box.

Generator oscillator procedure

Generator oscillator adaptation (Gen-Osc) is an automatic program. The purpose of the program is to adapt the inverter frequencies for maximum performance along the required frequencies. The program performs very short scans in different frequencies, and records the results in the main control D400.

- The procedure must be performed each time one of the following parts is replaced:
 - Generator electronic box
 - HV tank
 - D400

The frequency adaptation procedure must conclude successfully. If not, the current frequency parameters will be lost, making it impossible to scan. Failure of the procedure could be due to X-ray faults.

⇒ **If GenOsc fails, use dummy plugs (i.e. test plugs for HV tank) and perform GenOsc again with dummy plugs installed.**

Air in the HV tank

You can determine whether air is in the HV tank by simply turning the gantry by hand and listening closely to the HV tank. If you hear air bubbles (splashing noise), you have to change the HV tank.

1. Lower the patient table in order to open the gantry front door.
2. Switch **OFF** the system.
3. **Switch OFF power using the on-site ON-OFF switch!!**
4. Open the gantry front door and disable rotation with switch S 301.
5. Turn the gantry by hand (at least 360 degrees) and listen close to the HV tank for the sound of air bubbles.

⇒ **If there are any -> change the HV tank!**

Securing the stator cable

Improper fastening of anode stator cable

We know from field experience that an improperly secured W-513 anode stator cable causes errors. These are due to gravitational forces applied to the cable during rotation. (Fig. 4 / p. 16) illustrates proper cable attachment.

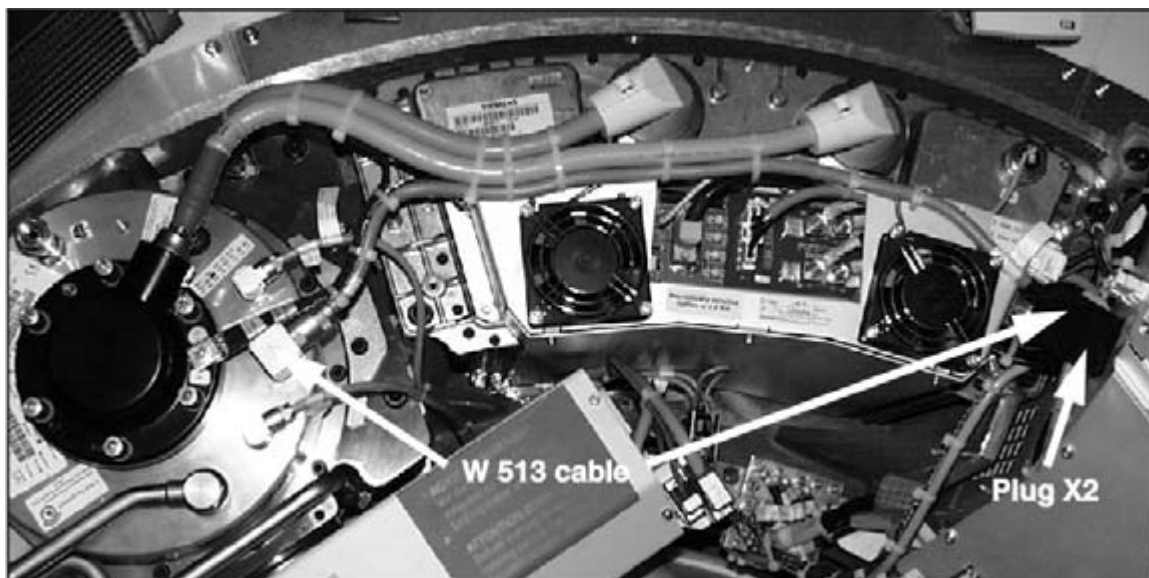


Fig. 4: Example of cable W_513 for rotation anode

Testing

Software and Hardware tests for the XRS system. Oscillograms of working and non working systems

HVC01 Dummy plug test

For high tension arcing or HT breakdown, the primary test utility **HVC01** is available in the test tools platform. The **HVC01** test lets you apply high tension to the HT circuit without tube current or gantry rotation. This allows the HT circuit to be systematically reduced using test plugs to isolate a high tension fault to either the tube or the HV tank.

Requirements:

- Dummy Plugs -> part number 16 21 791.

Features of the HVC01 test:

- kV can be varied from 70 kV to 145 kV.
- Anode Rotation.
- No filament current

NOTE

The HVC01 test is only allowed with test plugs installed. Tests with installed tube can result in misdiagnosis of failures.

Performing the HVC01 test

1. Lower the patient table in order to open the gantry front door.
2. Position the tube at 12 o'clock.
3. Switch off the gantry using the **COMP/ON** switch at the control box. Secure against accidental switch-on.
4. Open the gantry front door and disable rotation with switch S 301.
5. Disconnect HV cables from HV_Tank; install the "**Dummy Plugs**" instead.

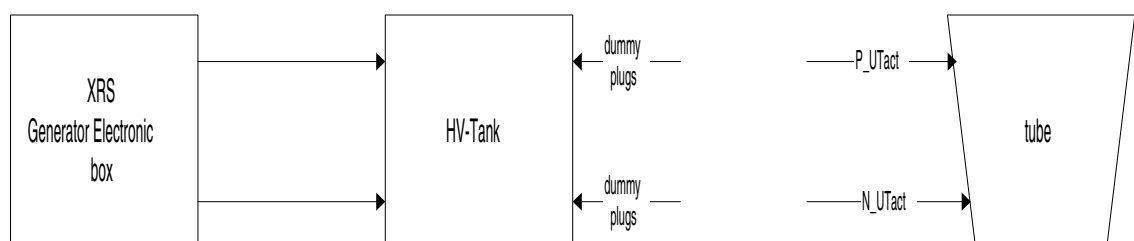


Fig. 5: Dummy plugs

6. Switch on rotation with switch S301 (necessary to reach Standby) then switch on the system (**SYS/ON**)
7. Select **Local Service -> Test Tools -> Controller -> XRS ->** select the **HVC01** test.

8. Select **Tube Voltage** and click **GO**.
9. Press the **Start** button -> vary the kV throughout the range from 70 kV to 145 kV.
10. After completing the measurement, evaluate the Event Log.
 - In case of an (XRS) error, the HV_Tank or Generator Electronic is at fault.
 - If no error occurs, the tube is defective.

⇒ **Before replacing the tube, try gettering to restore stable operation!**

HVC02 HV with rotation

For high tension arcing or HT breakdown, the test utility **HVC02** is available in the test tools platform. The **HVC02** test performs a scan with rotation but without tube current. Because the tube voltage can be increased up to 150 kV, this test verifies high voltage isolation of tube/H1 with rotation. **The HVC01 test is recommended over the HVC02 test, because it allows the user to choose between tube and HV tank.**

NOTE

Do not use this test with installed HV test plugs! Rotating the gantry with test plugs installed will damage parts in the gantry!

Performing the HVC02 test

1. Select **Local Service -> Test Tools -> Controller -> XRS ->** and the **HVC02** test.
2. Select **Tube Voltage** and **Rotation time**.
3. Click **GO**.
4. Press the **Start** button when prompted at the dialog bar.
 - ⇒ Follow the instructions in the dialog. The results are displayed based on the UT_act (kV) and IT_act (0 mA) readout from table T5E. The test will be terminated automatically after the output of T5E.

HVC03 Single pulse test

The Single Pulse test lets you isolate a fault between the Inverter (Electronic Box E603) and the HV_Tank by triggering the inverters for a single pulse. Please use a memory oscilloscope to measure the output of the single pulse test.

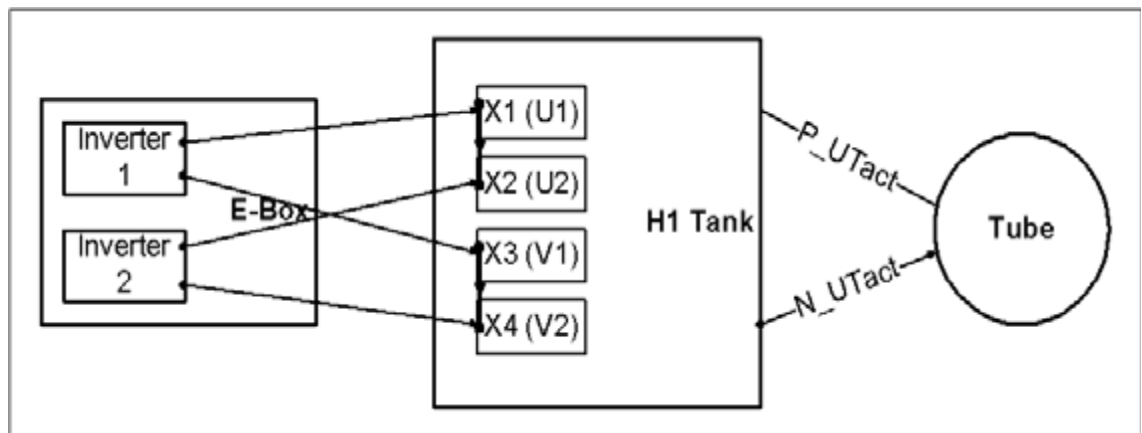


Fig. 6: Example of single pulse test diagram

Prerequisites

- Memory oscilloscope

Performing the HVC03 test

1. Lower the patient table in order to open the gantry front door.
2. Switch off the gantry using the **COMP/ON** switch at the control box. Secure against accidental switch-on.
3. Open the gantry front door and disable rotation with switch S 301.
4. Remove the generator electronic box cover.
5. Secure the D400 with 3 cover holding screws (board must be fixed).
6. Switch on rotation with switch S301 (necessary to reach Standby).
7. Switch the system **ON** at the control box (**SYS/ON**).
8. Select **Local Service -> Test Tools -> Controller -> XRS ->** and select the **HVC03** test.
9. Click **GO**
10. On **D-400**, connect a **MEMORY SCOPE** with two channels to:

Test point **X230- I_load (1V=50A)** Channel 1.

Test point **X260- Start_inv (Trigger)** Channel 2.

Each test point has a 0V point for reference.

Channel 2 is the trigger (Trigger neg.slope, single).

Adjust:

Channel 1 to 1V/division

Channel 2 to 5V/division

Time base 2 ms/division

11. Press the **Start** button when prompted in the dialog.

The test is terminated automatically. The single pulse test allows you to isolate a fault between the inverter (electronic box E603) and the HV_Tank by triggering the inverters for a single pulse. Test results are based on the evaluation of the I_{load} signal according to the following parameters (refer to (Fig. 7 / p. 20) and (Fig. 8 / p. 20)):

- The time for a single pulse must be 9 microseconds +/- 20%.
- The current should have a peak value of 150 A +/- 20%.
- Both a negative and positive pulse should be evaluated.

The following examples represent a nominal no-fault operational system.

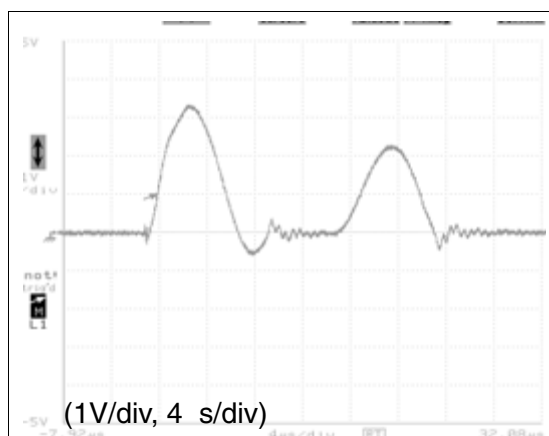


Fig. 7: Single pulse test oscillogram, peak 150A (positive)

The diagram (Fig. 7 / p. 20) represents the positive waveform of a single pulse. As noted in the cursors, the time between the rise and return to zero is measured. In this case, it is 8 μ s. The peak of the waveform can be evaluated as approaching 150 A.

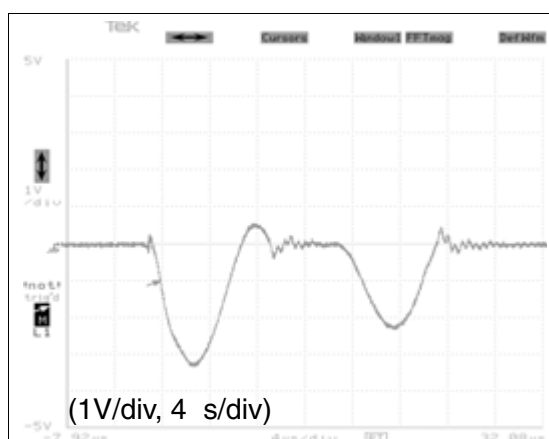


Fig. 8: Single pulse test oscillogram, peak 140A (negative)

As described in the positive waveform (Fig. 8 / p. 20), the period for this single pulse is 8 μ s. The current is approximately 140 A.

The following are examples of a faulty HV tank

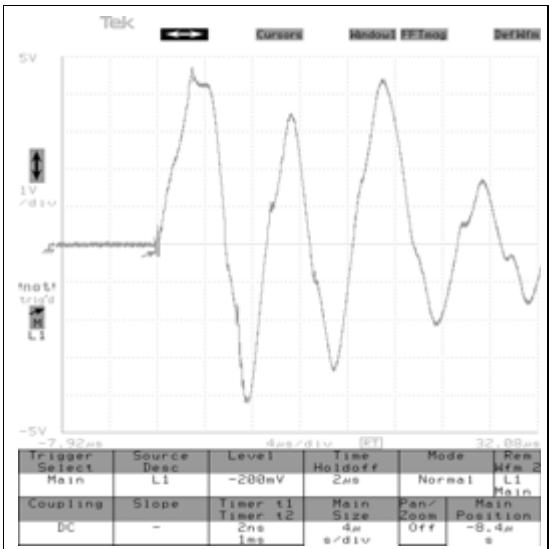


Fig. 9: I_{Load} -signal positive side (1V/div, μ s/div).

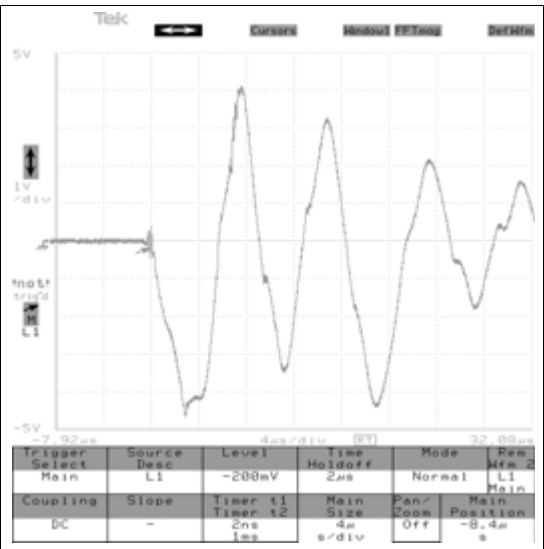


Fig. 10: I_{Load} -signal negative side (1V/div, μ s/div).

(Fig. 9 / p. 21) and (Fig. 10 / p. 21) represent a problem caused by a short circuit between x3, x1 and the wire to x4 (short circuit across the current sensor). The current flowing in the current sensors oscillates in this short circuit after switching off the inverter. The first pulse is switched off by the UCE- monitoring of the inverter. The duration of the first pulse is 6 msec, the maximum amplitude is 250 A. This short circuit has the same effects as a short circuit of one high voltage transformer inside the HV tank. A shorted transformer leads to a short circuit across x1 and x2, or x3 and x4, respectively (see (Fig. 6 / p. 19)).

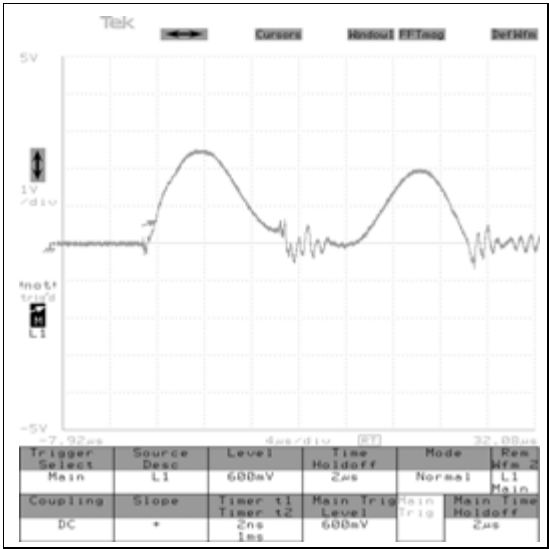


Fig. 11: I_{Load} -signal positive side (1V/div, μ s/div).

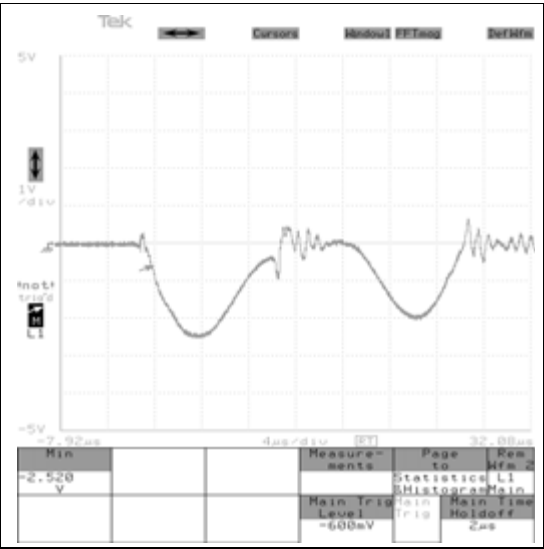


Fig. 12: I_{Load} -signal negative side (1V/div, μ s/div).

(Fig. 11 / p. 21) and (Fig. 12 / p. 21) are examples of an open circuit in the input, the high voltage transformer. x4 is not connected (no wire and no connection plate to x2). As a result, the stray inductance of the HV tank has twice the value as in normal operation since

one high voltage transformer is not connected. This causes such a large decrease in the resonance frequency that the first pulse is switched off by the pulse limiter. There is a very small follow-up pulse with inverse polarity. Duration of the first pulse: 12 μ s, max. amplitude 125 A.

RAC01 Rotating anode

The RAC 01 test starts up the rotating anode. Two different frequency steps may be selected (50 and 100 Hz). The system has to be in standby mode.

Performing the RAC01 test

1. Select **Local Service -> Test Tools -> Controller -> XRS -> RAC01** test.
2. Select a **Frequency** using the drop-down list.
3. Click **GO** and follow the instructions in the dialog. The results are displayed based on the fRAC last acceleration, fRAC actual, IRAC last acceleration, and IRAC actual read-out from the table T50. The test is terminated automatically after the output of T50.
 - ⇒ If test fails, measure anode output with an oscilloscope ([RAC01 Rotating anode / p. 22](#)) and measure anode stator resistance ([Anode stator resistance / p. 23](#)).

Scoping the anode stator

The anode stator can be checked with an oscilloscope. This test can help determine if the remanence signal is too low, signal noise is high or if the FIL/RAC is faulty.

Performing the measurement

1. Heat up the tube to full operating temperature, e.g. run extended warm-up.
2. Start a scan and directly after it finishes (i.e. before the anode is braked) turn the gantry power **OFF** via the COMP/ON switch at the control panel. The anode should still be rotating.
3. Remove X4 at the Tube and scope across the pins:-
 - 1-3
 - 1-4
 - 3-4
4. Wait until the rotation frequency is down to 40 Hz and the signal should be a clean sine wave with amplitude of 100mV p/p.
5. Low amplitude or noisy signal indicate tube problems. If the signal is within specifications, then the FIL/RAC is most likely the cause of problems.

Anode stator resistance

The anode stator resistance can be measured directly via the connector on the X-ray tube and via the cable connected to the E-Box. Improper connection and cable fastening are frequent causes for anode rotating errors.

- The following errors can be associated with the anode stator and cable connections.
 - XRS_50

The errors indicate problems with the anode stator rotation speed test.

Resistance measurement

NOTE

The measurement must be performed using a TRUE RMS meter (Effektiv Voltmeter).

Make sure that the anode starter has stopped completely prior to measurement. If it is still running, the measurement results will be incorrect.

1. Lower the patient table in order to open the gantry front door.
2. Switch **OFF** the system at the control box.
3. **Switch OFF power using the on-site ON-OFF switch!!**
4. Open the gantry front door and disable rotation with switch S 301.

5. To measure the anode resistance (including cable W513):

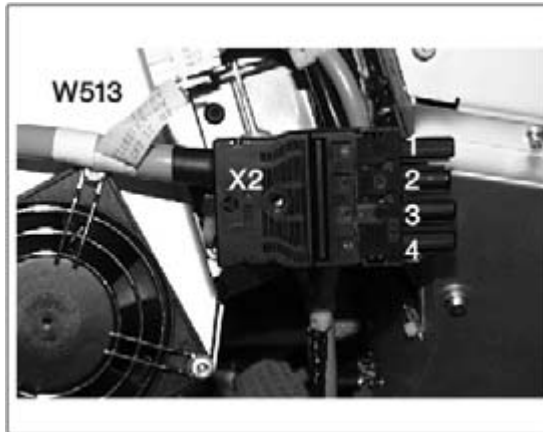


Fig. 13: Connector X2 for cable W513

- Disconnect connector x2 for cable W513 from the E-box side.
 - a.1 - 3
 - b.1 - 4
 - c. 3 - 4
- Measure approx. **3.6 ohms** between the following pins (including cable W513):
 - ⇒ **More important than the total value is that all 3 values are about equal.**

⇒ If any of the filaments are open, the following error is issued: XRS_48

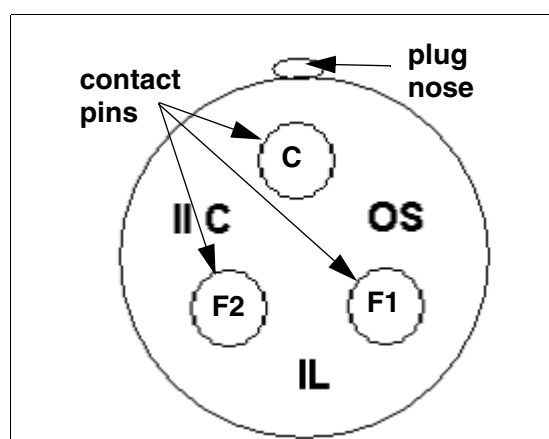
Filament resistance measurement

NOTE

The measurement must be performed using a **TRUE RMS meter (Effektiv Voltmeter)**.

Make sure that the anode starter has stopped completely prior to measurement. If it is still running, the measurement results will be incorrect.

1. Lower the patient table in order to open the gantry front door.
2. Switch **OFF** the system at the control box.
3. **Switch OFF power using the on-site ON-OFF switch!!**
4. Open the gantry front door and disable rotation with switch S 301.
5. Disconnect the cathode side of the HV cable from the HV tank.
6. Using a TRUE RMS meter (Effektiv Voltmeter), measure the filament resistance on the cable end.



Between **C**-(common) and **F2**-(large filament) -> **less than 2 ohms**; between **C**-(common) and **F1**-(small filament) -> **less than 2 ohms**.

If resistance not OK -> change tube

Fig. 15: Example of HV plug, front side

X-ray Timeout test

The **timeout test** is used to check the automatic monitoring of the total radiation duration in the customer modes TOPO, SEQUENCE, and SPIRAL. The test attempts a longer radiation duration than specified by the rated value of the loaded mode. The scan has to be cancelled after 108% of the rated value and the scanner has to switch to the ABORT state.

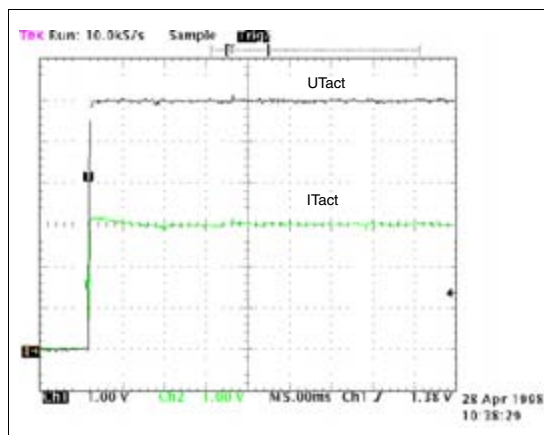
For the TOPO and SEQUENCE modes, all selectable scan times can be checked. For the mode SPIRAL, a rotation time and three different lengths of spiral ("min", "medium" and "max") can be selected.

Performing the X-ray timeout test

1. **Select Local Service -> Test Tools -> Controller -> X-ray Timeout.**
2. Selections:
 - Topo:** Select the length of the Topogram.
 - Sequence:** Select the scan time.
 - Spiral:** Select a rotation time and a spiral length (possible values for spiral length are: min., max., and medium). If "min." or "max." is selected, the spiral length is calculated based on the minimum/maximum possible speeds and the selected rotation time. If "medium" is selected, a mean speed is used.
3. Click **Go**.
4. Press the **START** button when prompted and follow the instructions in the dialog bar. Expect an abort, if radiation duration monitoring is active. The test is terminated automatically after the results are output in the dialog.
5. Perform a System Continue after the timeout test.
 - ➡ If test fails -> change D400

Oscillograms

The following oscillograms are from key test points in the XRS. They contain a description of the test point along with suggestions for evaluating the waveforms. Where applicable, the scan parameters are noted.



The oscillogram (Fig. 16 / p. 27) compares UTact and ITact for a 120 kV 300 mA scan.

Channel 1 = UTact

Channel 2 = ITact

Fig. 16: UTact and ITact

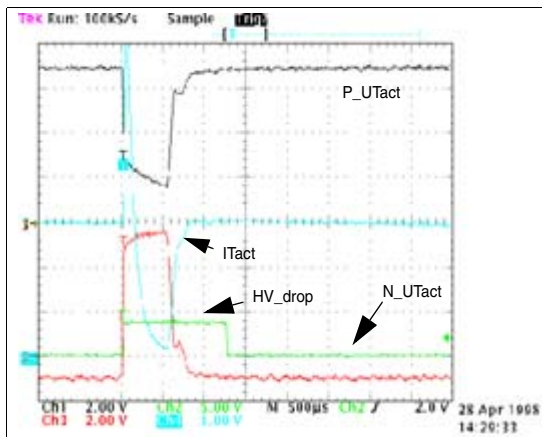


Fig. 17: Effects of HV_drop

The oscillogram (Fig. 17 / p. 28) describes the effects of HV_drop by showing the HT delay triggering HV_drop, as well as the corresponding rise in tube current. After approximately 500 msec, the inverter is restarted and HT is restored.

Channel 1 = P_UTact

Channel 2 = HV_drop

Channel 3 = N_UTact

Channel 4 = ITact

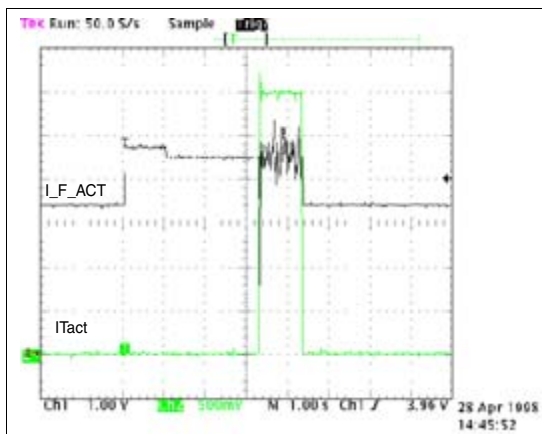


Fig. 18: F_ACT and ITact

The oscillogram (Fig. 18 / p. 28) shows the I_F_ACT signal and ITact. This oscillogram represents the changes in the filament during a period from standby to preparation through the completion of a 1 sec exposure (corresponding to the rise of the ITact waveform) and return to standby. The trace demonstrates the 3.33 V at standby corresponding to 5 A filament standby current. The regulation noted on the filament waveform is due to tube current regulation.

Channel 1 = I_F_ACT

Channel 2 = ITact

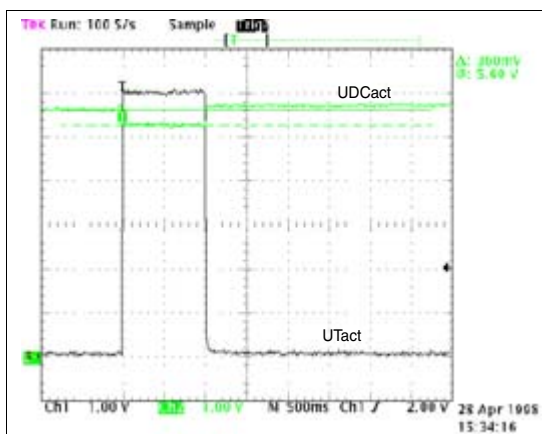


Fig. 19: UTact and UDC

The oscillogram (Fig. 19 / p. 28) shows **UTact** compared to **UDCact**. The UDC depends on the line voltage and may vary. Note the 36 volt drop in UDCact during the exposure. This loss should remain less than 80V from the actual UDC.

Channel 1 = UTact

Channel 2 = UDCact

NOTE

If a transformer is present (Installations with line voltage other than 400 V), the UDC should not vary from 560 V +/- 10%.

If UDC is dropping by more than 80 V during an exposure, a possible cause is a missing phase in the PDC. Check fuses F100. If one of the three fuses is blown, all fuses must be replaced.

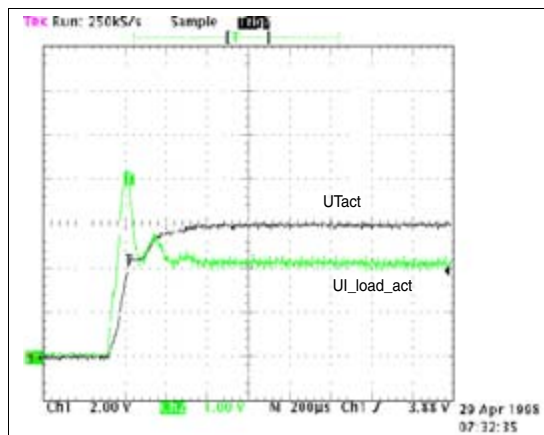


Fig. 20: UAct and UI_load_act

The oscillogram (Fig. 20 / p. 29) shows the inverter current UI_load_act compared to UAct. This exposure is 60 kW.

Channel 1 = UAct

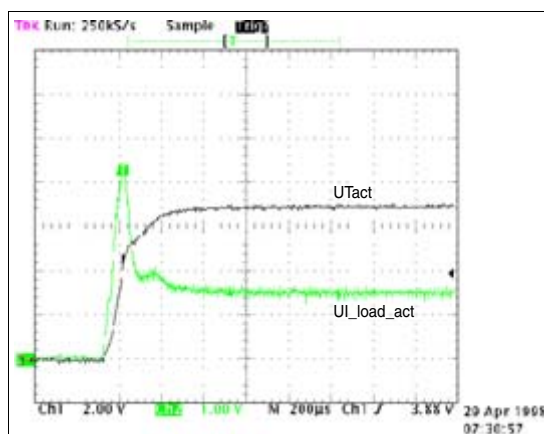


Fig. 21: UAct and UI_load_act

The oscillogram (Fig. 21 / p. 29) shows the inverter current UI_load_act compared to UAct. This exposure is 42 kW.

Channel 1 = UAct

Channel 2 = UI_load_act

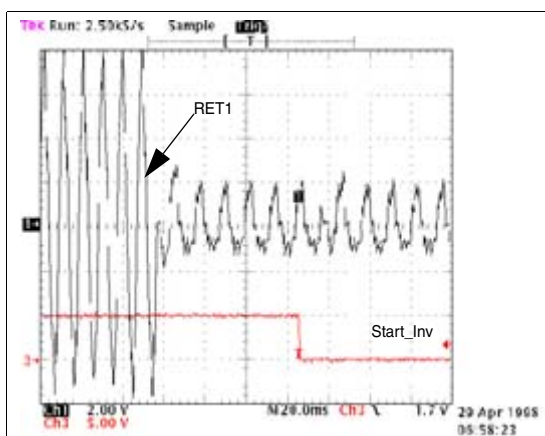


Fig. 22: RET1 and Start_inv

The oscillogram (Fig. 22 / p. 30) shows the RET1 signal in relationship to the Start_inv. Note the additional current for accelerating the anode compared to the running current.

Channel 1 = RET1

Channel 2 = Start_inv

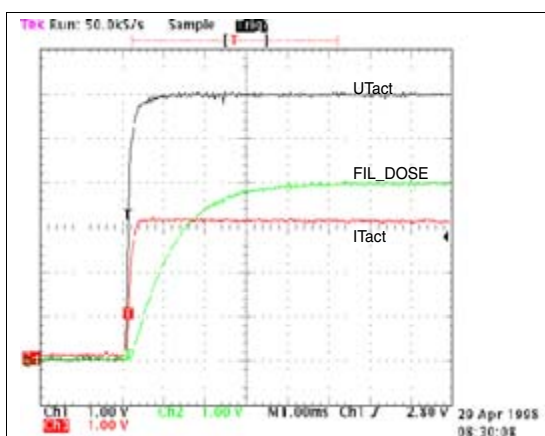


Fig. 23: Dose

The oscillogram (Fig. 23 / p. 30) includes the UAct and ITact for a 120 kV/300 mA exposure. The third signal is the FIL_DOSE signal from the DLP.

Channel 1 = UAct

Channel 2 = FIL_DOSE

Channel 3 = ITact

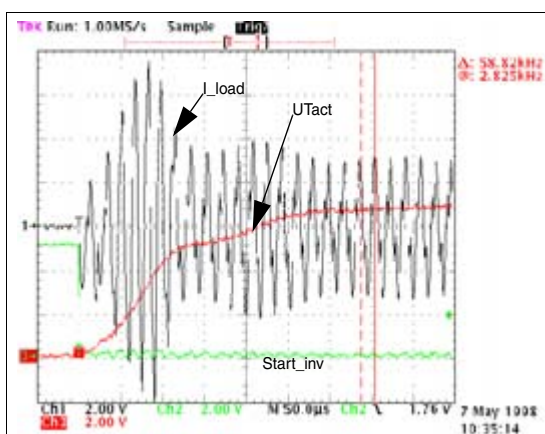


Fig. 24: Inverter function

The oscillogram (Fig. 24 / p. 30) shows the I_load compared to the UAct. Note the higher current as HT increases. Once 140 KV is reached, the I_load stabilizes. Scan parameters are 140 kV at 300 mA. Compare to (Fig. 22 / p. 30).

Channel 1 = I_load

Channel 3 = UT_act

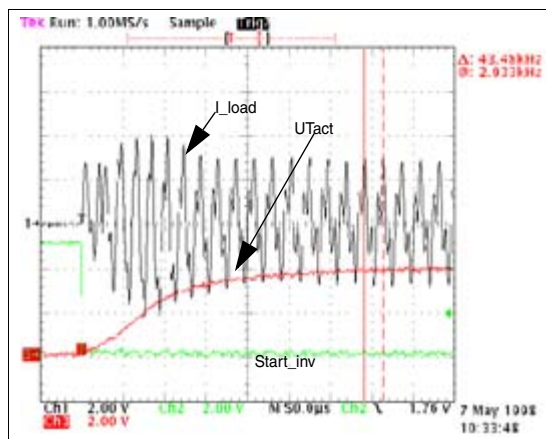


Fig. 25: Inverter function

The oscillogram (Fig. 25 / p. 31) shows the same signal relation as (Fig. 21 / p. 29). The scan parameters here are 80 kV at 260 mA.

Channel 1 = I_{load}

Channel 2 = $Start_{inv}$

Channel 3 = $U_{T_{act}}$

Power supplies

The XRS power supply is supplied via **Mains2** (230 V permanent). As a result, it is always under power even if the gantry is switched OFF. When the gantry is switched OFF, the DPU signal **Gen-ena** is low and inhibits the output voltages at slots 1-3 via the **INHIP** signals; in this case, only 48 V are available. When the gantry is turned ON, the DPU removes the **INHIP** signals and all supplies are available.

Tab. 1 E606 and UDC Power Supply

Power Supply	Voltage	Tolerance	Measuring Point	E606 Location	Notes
E606	5V	+/- 5%	At E606	Slot 1	INHIP + 5 V inhibits output
E606	15V	+/- 5%	At E606	Slot 2	INHIP15 V inhibits output
E606	-15V	+/- 5%	At E606	Slot 3	INHIP+15 inhibits output
E606	15V	+/- 5%	At E606	Slot 3	INHIP-15 inhibits output
E606	24V	+/- 5%	At E606	Slot 4	Slots 4 and 5 are connected in series to make 48VP
E606	24V	+/- 5%	At E606	Slot 5	
UDC	560V Nominal	Depends on line voltage	D400 X130 UDCact	n.a	

Slot1	Slot2		Slot3	Slot4	Slot5
*INHIP +5v	*INHIP 15v	*INHIP-15	S3a_15v	Not used	Not used
S1_5v	S2b_15v		S3a_0v	Link Wire	E606_48v+
Not used	S2a_15v		No connector	Not used	Not used
S1_0v	S2b_0v	*INHIP+15	S3b_15v	Not used	Not used
Not Used	S2a_0v		S3b_0v	E606_48v-	Link Wire

Fig. 26: XRS power supply connections

Notes:

- Each of the INHIP connectors has two (2) wires at one end (* indicates the end with the wires).
- INHIP+5 and INHIP15 should be secured with a small amount of silicone sealant.

Fuses, Measuring Point and LED's

The following table lists the field replaceable fuses, measuring points and LED's which are part of the XRS.

Measuring Points

The following is a listing of the most important measuring points available at the D400. Each Test Field (X110, X130 Etc.) has a 0V point for reference.

Tab. 2 Common test points

Test Point	Test Field	Parameters	Description	Status Table
FIL_DOSE	X110	0...10V	Analog dose signal for dose regulation from the DLP	T5E
FIL_NOM	X110	0...10V	Analog signal from DOM to regulate filament in Servo-Mode	N/A
UTact	X210	1V=20kV	Actual high tension voltage	T5E
I_F_ACT	X110	1V=1.5A	RMS value from I_load_FIL (standby 5A = 3.33V)	T5E
UDCact	X130	1V=100V	Actual UDC value	T5B
RET1	X140	1V=4.65A	Analog current to one phase of anode motor windings	N/A
F_ROT	X140	high=12V Low=5V	Remanent signal from anode rotation	N/A
UI_load_act	X200	1V=50A	Mean value of actual Inverter current	N/A
I_load	X230	1V=50A	Actual output current of main inverter	N/A
N_UTact	X210	1V=10kV	Actual value of negative component of high tension	N/A
P_UTact	X210	1V=10kV	Actual value of high tension	N/A
ITact_100mA	X230	1V=100mA	Actual tube current	T5E
Start_Inv	X260	digital	Low active signal which starts the Inverters	N/A
HV_drop	X270	digital	High active signal which indicates either one side or both sides have had a high tension disturbance (see note)	T5E

LED's

Table shows signals, which are available via LED on the D400 to help troubleshooting.

Tab. 3 List of LED's

Signal	Colour	Description
Start_inv	Green	Start of Inverters
X-ray_on	Green	Generated when HT is greater than 20 kV
P_UTmax	Red	The positive component of high tension has reached 82 kV
N_UTmax	Red	The negative component of high tension has reached -82 kV
Supply_Volt	Red	One of the D400 power supplies are out of tolerance
Start_fil_inv	Green	Start signal for the filament inverter
FIL_Inv_Fault	Red	An error has occurred with the filament circuit
RAC_Inv_Fault	Red	An error has occurred with the RAC circuit
Inv_Fault	Red	An error has occurred in the main inverter
XC	Green	Start signal for the high tension circuit
Temp_Error	Red	The XRS temperature is out of tolerance
FLEX_Fault	Red	The FLEX chip did not initialize properly

Fuses

Tab. 4 List of field replaceable fuses

Fuse	Value	Load
F1	6.3A	E606
F2	2A	XRS Fans M1.....M4
F3	2A	RAC/FIL and HV_TANK fans
F4,F5	30A	RAC/FIL
F100	80A	Generator

Overview

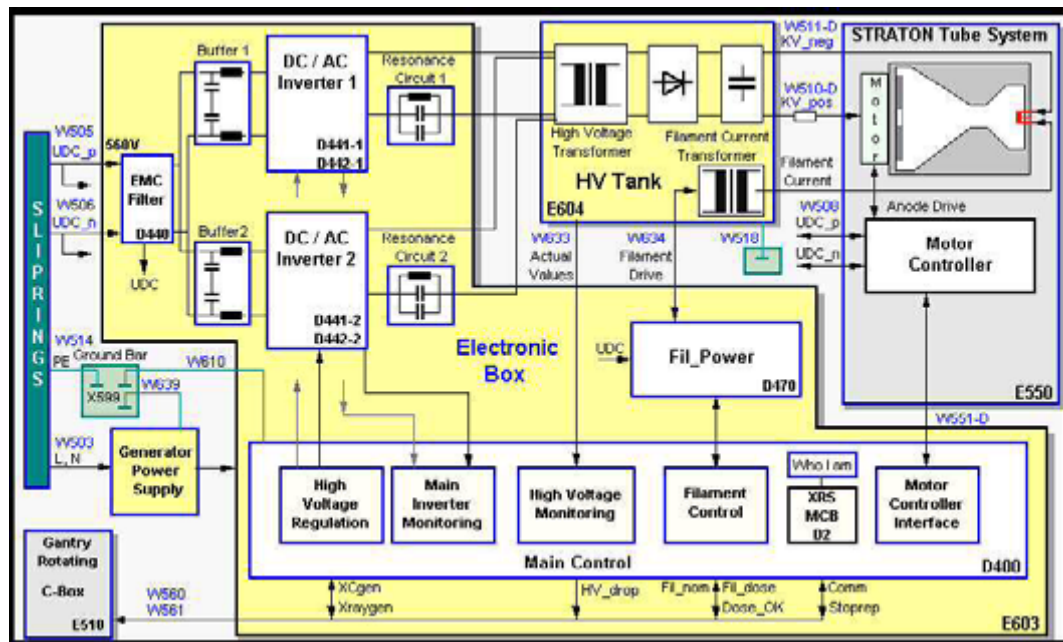


Fig. 27: XRS (STRATON) block diagram

⇒ Block diagrams are for illustrative purposes only. Refer to the functional description for current and type-specific information.

The main FRU's of the X-ray system are:

- X-ray tube
- Electronic box - E603
- Fil power- D470
- Generator power supply - E606
- Main control - D400
- High voltage tank - E604
- MCB
- D451 (stationary part)
- HV cables

Problem isolation

This section contains information about isolating problems. **HV_Drops** explains the HV_Drop and its purpose; **HV testing** helps you decide which test to run for HV problems; **UDC information** is included because most XRS functions require correct voltage information. **Rotate controller flash codes** describes the error codes that can be read from the flashing red LED on the motor controller for the tube. The **generator oscillator** procedure is provided as an option in case of problems due to adaptation, e.g. particular current settings not working. Other sections highlight problems that have been shown by experience to cause XRS problems.

HV_Drops

The actual high tension (P_UTact and N_UTact) is monitored by the D400. If either positive, negative, or both, are arcing (i.e. high tension breakdown greater than 30 kV), the D400 sets the HV_drop signal. In this case, the main inverter is stopped and restarted within 2 msec, to prevent high tension errors from occurring.

Notes regarding the HV_drop:

The purpose of the HV_drop is to eliminate the possibility of high tension being applied to only one side of the HT circuit. For instance, if during a 120 kV exposure, the 60 kV negative side "drops," the positive HT will not compensate and try to increase to 120 kV. Were the positive side allowed to rise to 120 kV, an error would occur when the HT reached 82 kV.

The HV_drop is recorded in the tube history. HV_drop alone CANNOT be used to evaluate tube performance.

The firmware compares the number of HV_drops within 300 msec to a value set in the firmware. If this value is exceeded, an error is sent (typically XRS_47).

The HV_drop is transmitted to the ACU. The ACU marks readings when the HV_drop was active to allow the IRS to interpolate these readings.

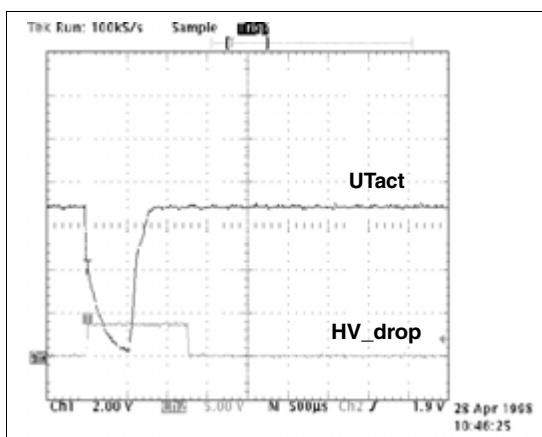


Fig. 28: HV_drop

(Fig. 28 / p. 36) shows an oscillogram of U_Tact compared to HV_drop. The key point is the 1.25 msec when the HV_drop is active; during this period, the HV collapses and is rebuilt, allowing the scan to continue without error.

Channel 1 = U_Tact

Channel 2 = HV_drop

HV testing

Troubleshooting the XRS is based on the error description. Whenever no specific component is identified (as in the case of high voltage failures), the following diagram may be used as a guideline.

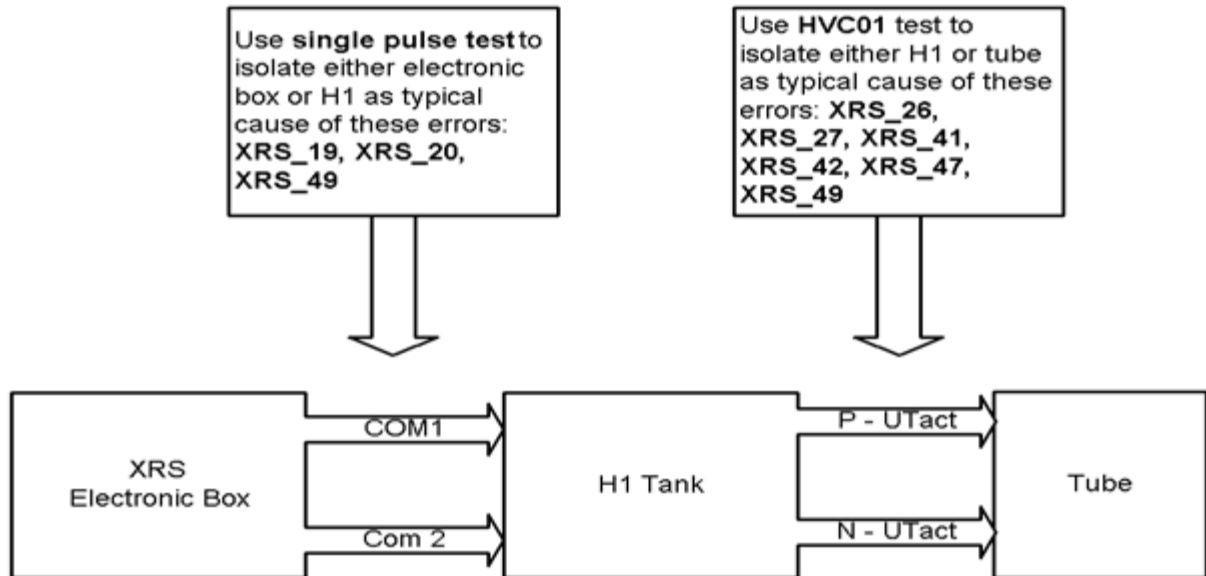


Fig. 29: Diagram

UDC voltage information

- The 560 V DC UDC_P and UDC_N should be measured prior to starting the diagnostics. Many errors are related to improper voltage because the system uses this initial power for:
 - The ignition circuit of inverters (E-box)
 - The filament power
 - The motor controller
- The UDC voltage may vary because it depends on the line voltage. If a transformer is used (i.e. the line voltage is NOT 400V), the UDC voltage has to be 560 V +/-10%. During a scan, the voltage drop of the UDC should never be more than 80 V below the actual UDC voltage. If the UDC voltage drops by more than 80V during an exposure, a phase may be missing in the PDC.
 - ➡ **In this case, check the three F100 fuses.**
If one F100 is defective, replace all three. If the fuses are OK, check whether the line impedance meets specifications.
- UDC voltage is transferred as follows:
 - From D-450 X7-X10 in the PDC via the W3 cable to the X303 connection in the right leg of the gantry.
 - From the X303 connection in the right leg of the gantry via the X315 cable to the slip ring connections X382 and X383.
 - From slip ring connections X382 and X383 via the W505/W506 cable to the E-box.

- From slip ring connections X22c and X23c via the W508 cable to X22 of the Straton tube assembly and subsequently through two fuses to X1 of the tube motor controller.

Generator oscillator procedure

Generator oscillator adaptation (Gen-Osc) is an automatic program. The purpose of the program is to adapt the inverter frequencies for maximum performance along the required frequencies. The program performs very short scans in different frequencies, and records the results in the main control D400.

- The procedure has to be performed each time one of the following parts is replaced:
 - Generator electronic box
 - HV tank
 - D400

The frequency adaptation procedure has to conclude successfully. If not, the current frequency parameters will be lost, making it impossible to scan. Failure of the procedure could be due to X-ray faults.

⇒ **If GenOsc fails, use dummy plugs (test plugs for HV tank) and perform GenOsc again with dummy plugs installed.**

Rotate controller flash codes

You can determine the error code of the rotate motor controller by reading the flash sequence of the red LED located near X21 of the controller. Should an error be detected, the LED will continually flash a code with a number of short flashes followed by a long flash. For example:

<LONG><short><short><short><LONG><short><short><short><LONG>

would indicate error code 3 (over current error).

The error codes are:

1. Speed error (under or over)
2. Under voltage error
3. Over current error
4. Over voltage error
5. Motor protection activated
6. Controller protection activated
7. Motor temperature high
8. Controller or heatsink temperature high

The controller cannot be exchanged. If defective, the tube assembly has to be replaced.

Air in the HV tank

You can determine whether air is in the HV tank by simply turning the gantry by hand and listening closely to the HV tank. If you hear air bubbles (splashing noise), you have to change the HV tank.



1. Lower the patient table to open the gantry front door.
2. Switch **OFF** the system.
3. **Switch OFF power using the on-site ON/OFF switch!**
4. Open the gantry front door and disable rotation with switch S 301.
5. Turn the gantry by hand (at least 360 degrees) and listen close to the HV tank for the sound of air bubbles.

⇒ **If there are any -> change the HV tank!**

Testing

Software and hardware tests for the XRS system. Oscillograms of working and non-working systems

HVC01 dummy plug test

For high tension arcing or HT breakdown, the primary test utility **HVC01** is available in the test tools platform. The **HVC01** test lets you apply high tension to the HT circuit without tube current or gantry rotation. This allows the HT circuit to be systematically reduced using test plugs to isolate a high tension fault to either the tube or the HV tank.

Requirements:

- Dummy plugs -> part no. 16 21 791.

Features of the HVC01 test:

- kV can be varied from 70 kV to 150 kV
- Anode rotation
- No filament current

NOTE

The HVC01 test is only allowed with test plugs installed. Tests with installed tube can result in misdiagnosis of failures.

Performing the HVC01 test



1. Lower the patient table in order to open the front door of the gantry.
2. Position the tube at 12 o'clock.
3. Switch off the gantry using the **COMP/ON** switch at the control box. Secure against accidental switch on.
4. Open the front door of the gantry and disable rotation via switch S 301.
5. Disconnect HV cables from HV_Tank; loosen the two Allen screws and unscrew the holding ring; then install the "**Dummy Plugs**".

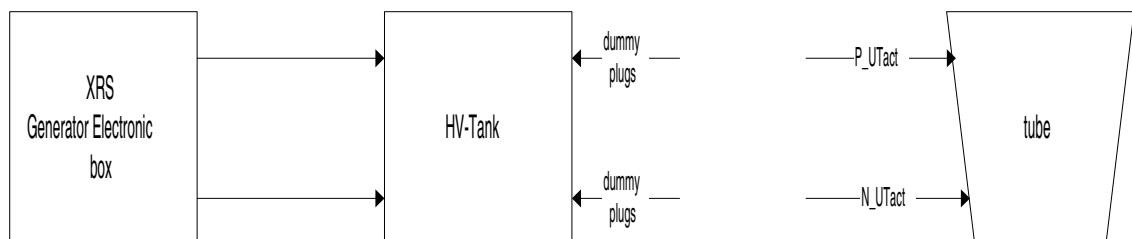


Fig. 30: Dummy plugs

6. Switch on rotation via switch S301 (necessary to reach standby) and then switch on the system (**SYS/ON**).

7. Select **Local Service -> Test Tools -> Controller -> XRS ->** and select the **HVC01** test.
8. Select **Tube Voltage** and click **GO**.
9. Press the **Start** button -> vary the kV throughout the range from 70 kV to 150 kV. The test runs for 80 seconds.
 - ⇒ Follow the instructions in the dialog. The results are displayed based on the UT_act (kV) and IT_act (0 mA) readouts from table T5E. The test is terminated automatically after the output of T5E.
10. After completing the measurement, evaluate the Event Log.
 - In case of an (XRS) error, either the HV_Tank or generator electronics is at fault. Next, run the HVC3 Test.
 - If no error occurs, the tube or HV cables are defective.
 - ⇒ **Before replacing the tube, try gettering to restore stable operation!**

HVC02 HV with rotation

For high tension arcing or HT breakdown, the test utility **HVC02** is available in the test tools platform. The **HVC02** test performs a scan with rotation but without tube current. Because the tube voltage can be increased up to 150KV, this test verifies high voltage isolation of tube/H1 with rotation.

NOTE

Do not perform this test with HV cables installed! Remove the HV cables completely before running this test. The HV cables are different. Make sure you reinstall them in the same position.

Performing the HVC02 test



1. Lower the patient table to open the front door of the gantry.
2. Position the tube at 12 o'clock.
3. Switch off the gantry using the **COMP/ON** switch at the control box. Secure against accidental switch on.
4. Open the gantry front door and disable rotation via switch S 301.
5. **Remove** HV cables from gantry; **Identify and mark the HV cables**. Loosen the 2 Allen screws, and unscrew the holding rings on H1 and the STRATON tube assembly. Cut tie wraps as needed and remove the HV cables from the gantry.
 - ⇒ **It is very important to reinstall the HV cables in their original positions.** Refer to (Fig. 38 / p. 46) for information on how to identify the cables.
6. Install the "dummy plugs" in the HV tank. **Only use the "short" type of dummy plugs** (part no. 16 21 791). Other types will hit the gantry frame during rotation. Ensure that all other cables are secure; use tie wraps as needed
7. Rotate the gantry 360 degrees by hand, and ensure that all cables are safe and secure. Make sure the dummy plugs do not contact the gantry frame.

8. Switch on rotation via switch S301 and switch on the system (**SYS/ON**).
9. Select **Local Service -> Test Tools -> Controller -> XRS ->**, and select the **HVC02** test and rotation speed, e.g. 1 second normally.
10. Select **Tube Voltage** and click **GO**.
11. Press the **Start** button -> The test runs for approx. 1 second regardless of rotation speed. Vary the kV throughout the range from 70 kV to 150 kV.
 - ⇒ Follow the instructions in the dialog. The results are displayed based on the UT_act (kV) and IT_act (0 mA) readout from table T5E. The test is terminated automatically after the output of T5E.
12. After completing the measurement, evaluate the Event Log.
 - In case of an (XRS) error, either the HV_Tank or generator electronics is at fault. Next, run the HVC3 Test.
 - If no error occurs, the tube or HV cables are defective.
 - ⇒ **Before replacing the tube, try getting to restore stable operation!**

HVC03 Single pulse test

The single pulse test allows you to isolate a fault between the inverter (electronic box E603) and the HV_Tank by triggering the inverters for a single pulse. Please use a memory oscilloscope to measure the output of the single pulse test.

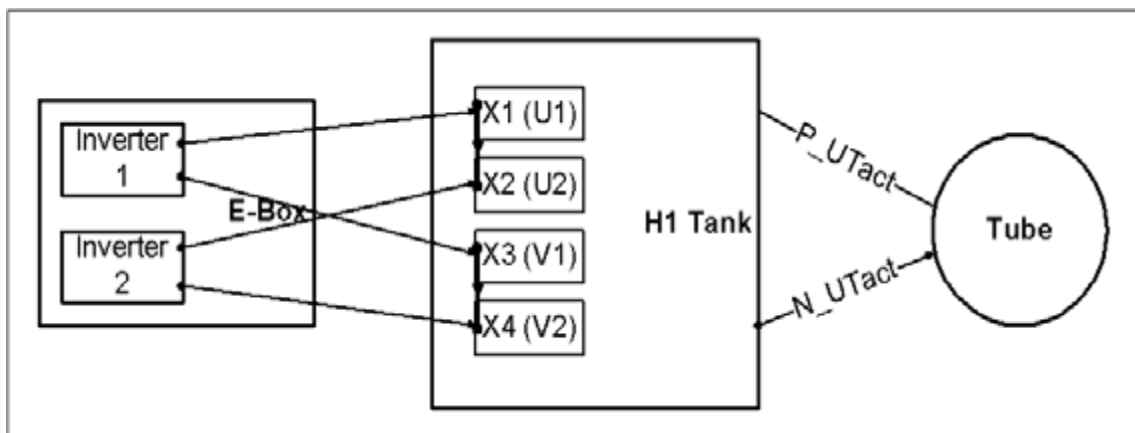


Fig. 31: Example of single pulse test diagram

- ⇒ The Sensation Open has one inverter only which is connected to X2, X3.

Prerequisites

- Memory oscilloscope

Performing the HVC03 test



1. Lower the patient table to open the front door of the gantry.
2. Switch off the gantry using the **COMP/ON** switch at the control box. Secure against accidental switch on.
3. Open the gantry front door and disable rotation via switch S 301.

4. Remove the generator electronic box cover.
5. Secure the D400 with three holding screws for the cover (except Somatom Open).
6. Switch on rotation via switch S301 (necessary to reach standby).
7. Switch the system **ON** at the control box (**SYS/ON**).
8. Select **Local Service** -> **Test Tools** -> **Controller** -> **XRS** -> and select the **HVC03** test.
9. Click **GO**.
10. On **D-400**, connect a **MEMORY SCOPE** with two channels to:

Test point **X230- I_load (1V=50A)** Channel 1.

Test point **X260- Start_inv (Trigger)** Channel 2.

Each test point has a 0 V point for reference.

Channel 2 is the trigger (Trigger neg. slope, single).

Adjust:

Channel 1 to 1 V/division

Channel 2 to 5 V/division

Time base 2 ms/division

11. Press the **Start** button when prompted in the dialog.

The test is terminated automatically. The single pulse test allows you to isolate a fault between the inverter (electronic box E603) and the HV_Tank by triggering the inverters for a single pulse. Test results are based on the evaluation of the I_load signal according to the following parameters (refer to [\(Fig. 32 / p. 43\)](#) and [\(Fig. 33 / p. 44\)](#)):

- The time for a single pulse has to be 9 microseconds +/- 20%.
- The current should have a peak value of 150 A +/- 20%.
- Both a negative and positive pulse should be evaluated.

The following examples represent a nominal no-fault operational system.

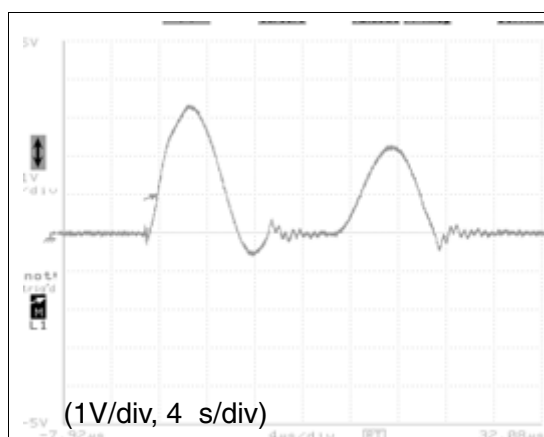


Fig. 32: Single pulse test oscillogram, peak 150A (positive)

The diagram [\(Fig. 32 / p. 43\)](#) represents the positive waveform of a single pulse. As noted in the cursors, the time between the rise and return to zero is measured. In this case, it is 8 μ s. The peak of the waveform can be evaluated as approaching 150 A.

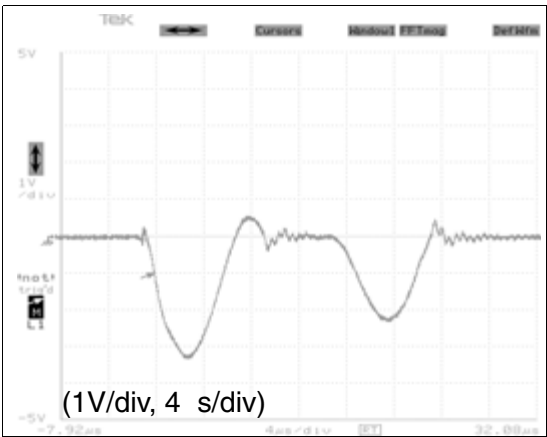


Fig. 33: Single pulse test oscillogram, peak 140A (negative)

As described in the positive waveform (Fig. 33 / p. 44), the period for this single pulse is 8 μ s. The current is approximately 140 A.

The following are examples of a faulty HV tank

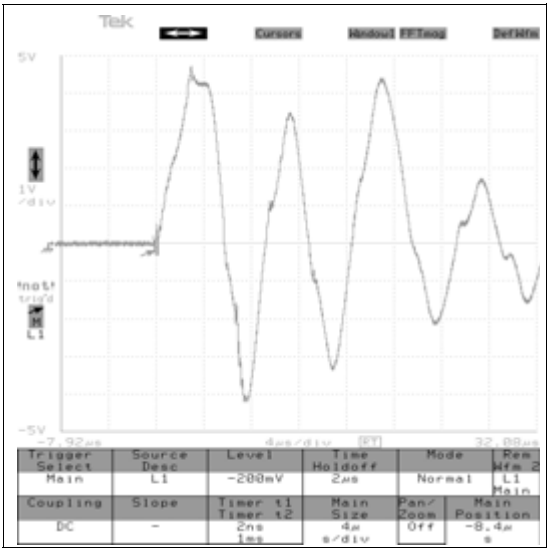


Fig. 34: I_Load-signal positive side (1V/div, μ s/div).

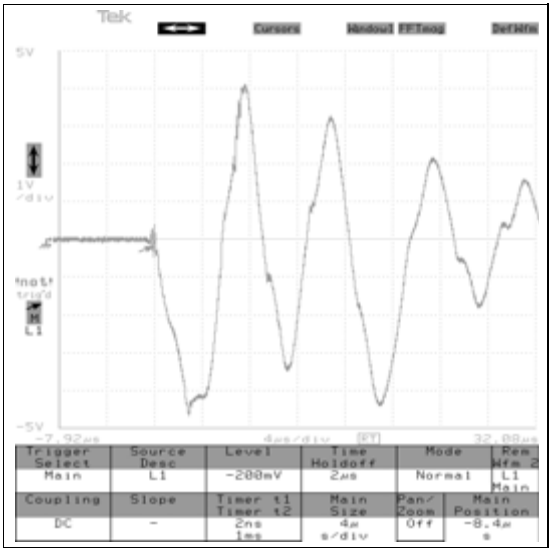


Fig. 35: I_Load-signal negative side (1V/div, μ s/div).

(Fig. 34 / p. 44) and (Fig. 35 / p. 44) represent a problem caused by a short circuit between x3, x1, and the wire to x4 (i.e. short circuit across the current sensor). The current flowing in the current sensors oscillates in this short-circuit after switching off the inverter. The first pulse is switched off by the inverter’s UCE monitoring. The duration of the first pulse is 6 msec, the maximum amplitude is 250 A. This short-circuit has the same effect as a short-circuit of one high voltage transformer inside the HV tank. A shorted transformer leads to a short-circuit across x1 and x2, or x3 and x4, respectively (see (Fig. 31 / p. 42)).

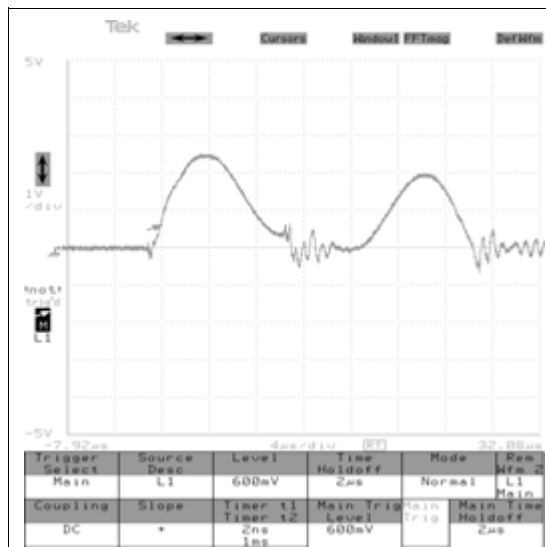


Fig. 36: I_{Load} -signal positive side (1V/div, μ s/div).

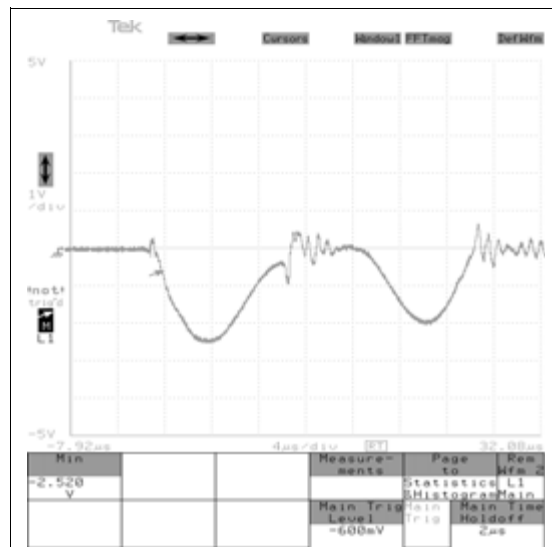


Fig. 37: I_{Load} -signal negative side (1V/div, μ s/div).

(Fig. 36 / p. 45) and (Fig. 37 / p. 45) are examples of an open circuit in the input, the high voltage transformer. x4 is not connected (no wire and no connection plate to x2). As a result, the stray inductance of the HV tank has twice the value as in normal operation since one high voltage transformer is not connected. This causes such a large decrease in the resonance frequency that the first pulse is switched off by the pulse limiter. There is a very small follow-up pulse with inverse polarity. Duration of the first pulse: 12 μ s, max. amplitude 125 A.

RAC01 rotating anode

The RAC 01 test starts up the rotation motor. The system has to be in standby mode.

Performing the RAC01 test

1. Select **Local Service -> Test Tools -> Controller -> XRS -> RAC01** test.
2. Select a **Frequency** using the drop-down list.
3. Click **GO** and follow the instructions in the dialog. The results are displayed based on the fRAC last acceleration, fRAC actual, IRAC last acceleration, and IRAC actual read-out from the table T50. The test is terminated automatically after the output of T50.
 - ➡ If test fails, check for error codes on the motor controller ([Rotate controller flash codes / p. 38](#)) and replace tube or D400 board.

FIL01 filament test

This test checks the function of the filament current regulator. **The system has to be in standby mode.** No radiation is released during this test procedure.

Performing the FIL01 test

1. Select **Local Service -> Test Tools -> Controller -> XRS -> FIL01** test.
2. Select **Tube current** and **Static time**.
3. Click **GO** and follow the instructions in the dialog. The resulting IF_act (mA) is displayed based on the readout from table T50. The test is terminated automatically after the output of T5E.
 - ⇒ If the above test fails, check the HV cables, replace the filament assembly, and replace the STRATON tube. The filament current is transferred to the tube via an inductive coupling. As a result, it is not possible to measure the filament directly.

STRATON HV cables

Identifying and testing Straton HV cables

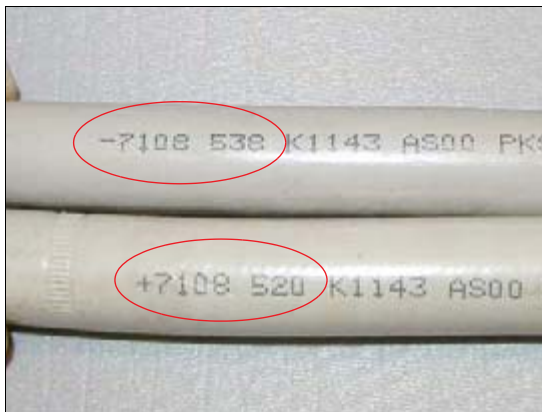


Fig. 38: Identifying HV cable

- Each cable has a part number and indicates polarity as shown in the illustration (old type shown)
- The cathode (-ve) cable (part no. 7108538 old type, 7110294 new type with shrinkable tubing at the end) has approx. zero ohms when measured from one plug to the other with all 3 pins isolated from each other.
- The anode (+ve) cable (part no. 7108520 old type, 7110286 new type with shrinkable tubing at the end) has approx. 160 ohms when measured from one plug to the other with all 3 pins shorted together.

X-ray timeout test

The **timeout test** is used to check the automatic monitoring of the total radiation duration in the customer modes TOPO, SEQUENCE, and SPIRAL. The test attempts a longer radiation duration than specified by the rated value of the loaded mode. The scan has to be cancelled after 108% of the rated value, and the scanner has to switch to the ABORT state.

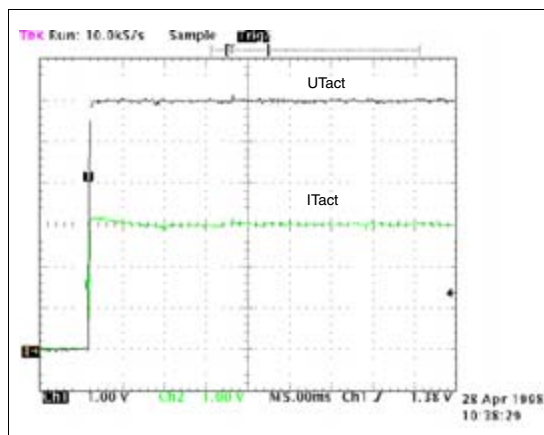
For the TOPO and SEQUENCE modes, all selectable scan times can be checked. For the SPIRAL mode, a rotation time and three different lengths of spiral ("min.", "medium," and "max.") can be selected.

Performing the X-ray timeout test

1. **Select Local Service -> Test Tools -> Controller -> X-ray Timeout.**
2. Selections:
 - Topo:** Select the length of the topogram.
 - Sequence:** Select the scan time.
 - Spiral:** Select a rotation time and a spiral length (possible values for spiral length are: min., max., and medium). If "min." or "max." is selected, the spiral length is calculated based on the minimum/maximum possible speeds and the selected rotation time. If "medium" is selected, a mean speed is used.
3. Click **Go**.
4. Press the **START** button when prompted and follow the instructions in the dialog bar. Expect an abort, if radiation duration monitoring is active. The test is terminated automatically after the results are output in the dialog.
5. Perform a System Continue after the timeout test.
 - ➡ If test fails -> change D400.

Oscillograms

The following oscillograms are from key test points in the XRS. They contain a description of the test point along with suggestions for evaluating the waveforms. Where applicable, the scan parameters are noted.



The oscillogram (Fig. 39 / p. 47) compares UTact and ITact for a 120 kV 300 mA scan.

Channel 1 = UTact

Channel 2 = ITact

Fig. 39: UTact and ITact

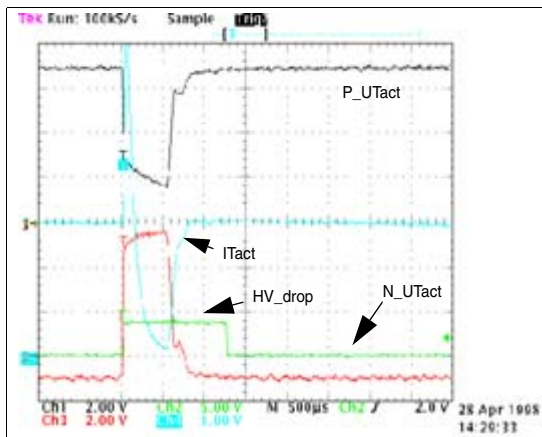


Fig. 40: Effects of HV_drop

The oscillogram (Fig. 40 / p. 48) describes the effects of HV_drop by showing the HT delay triggering HV_drop, as well as the corresponding rise in tube current. After approx. 500 msec, the inverter is restarted and HT is restored.

Channel 1 = P_UTact

Channel 2 = HV_drop

Channel 3 = N_UTact

Channel 4 = ITact

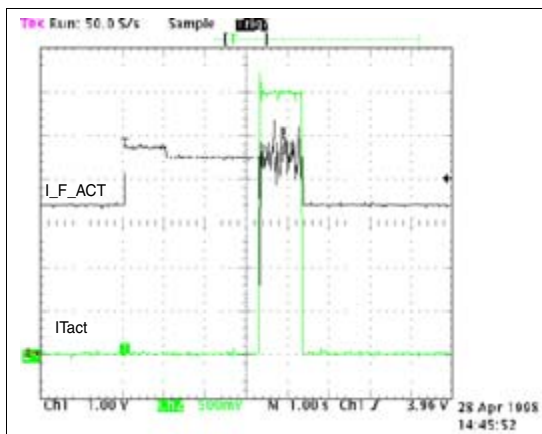


Fig. 41: F_ACT and ITact

The oscillogram (Fig. 41 / p. 48) shows the I_F_ACT signal and ITact. This oscillogram represents the changes in the filament during a period from standby to preparation through the completion of a 1 sec exposure (corresponding to the rise of the ITact waveform) and return to standby. The trace demonstrates the 3.33 V at standby corresponding to 5 A filament standby current. The regulation noted on the filament waveform is due to tube current regulation.

Channel 1 = I_F_ACT

Channel 2 = ITact

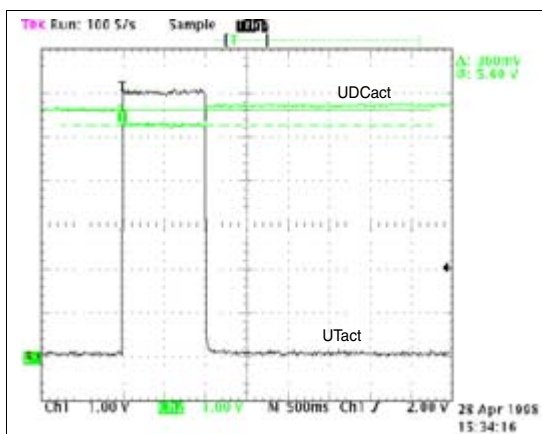


Fig. 42: UTact and UDC

The oscillogram (Fig. 42 / p. 48) shows **UTact** compared to **UDCact**. The UDC depends on the line voltage and may vary. Note the 36 volt drop in UDCact during the exposure. This loss should remain less than 80 V from the actual UDC.

Channel 1 = UTact

Channel 2 = UDCact

NOTE

If a transformer is present (e.g. installations with line voltage other than 400 V), the UDC should not vary from 560 V $\pm$ 10%.

If UDC is dropping by more than 80 V during an exposure, a possible cause is a missing phase in the PDC. Check fuses F100. If one of the three fuses is blown, all fuses have to be replaced.

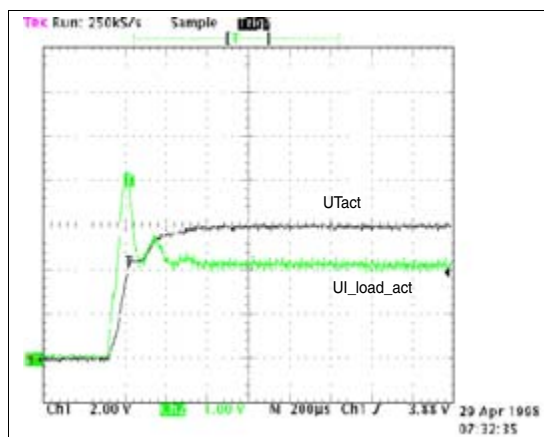


Fig. 43: U_{Tact} and UI_{load_act}

The oscillogram (Fig. 43 / p. 49) shows the inverter current UI_{load_act} compared to U_{Tact} . This exposure is 60 kW (this setting is not possible on Sensation Open!).

Channel 1 = U_{Tact}

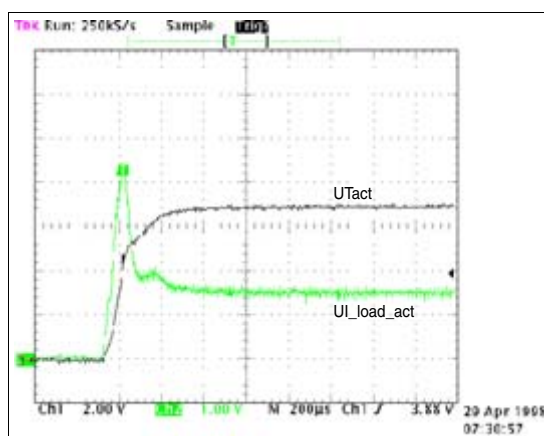


Fig. 44: U_{Tact} and UI_{load_act}

The oscillogram (Fig. 44 / p. 49) shows the inverter current UI_{load_act} compared to U_{Tact} . This exposure is 42 kW.

Channel 1 = U_{Tact}

Channel 2 = UI_{load_act}

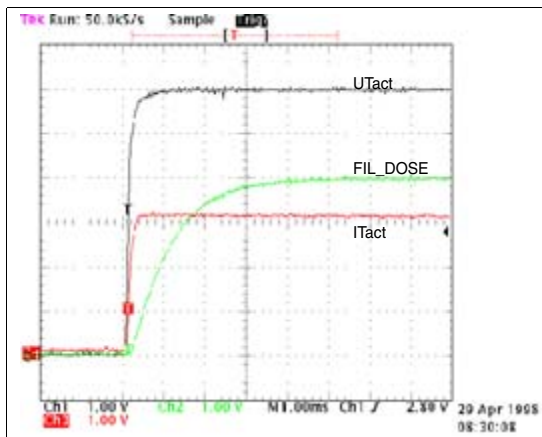


Fig. 45: Dose

The oscillogram (Fig. 45 / p. 50) includes the U_{Tact} and I_{Tact} for a 120 kV/300 mA exposure. The third signal is the FIL_DOSE signal from the DLP.

Channel 1 = U_{Tact}

Channel 2 = FIL_DOSE

Channel 3 = I_{Tact}

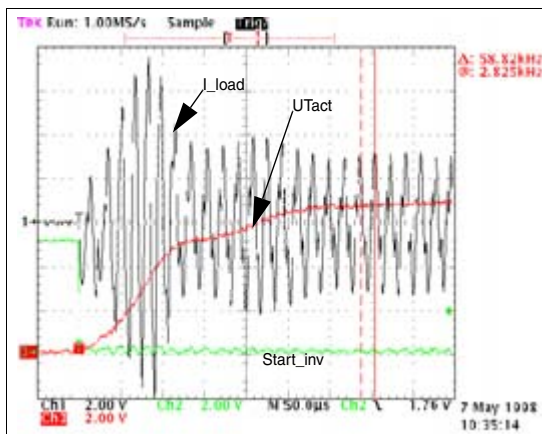


Fig. 46: Inverter function

The oscillogram (Fig. 46 / p. 50) shows the I_{load} compared to the U_{Tact}. Note the higher current as HT is increasing. Once 140 kV is reached, the I_{load} stabilizes. Scan parameters are 140 kV at 300 mA.

Channel 1 = I_{load}

Channel 3 = U_{Tact}

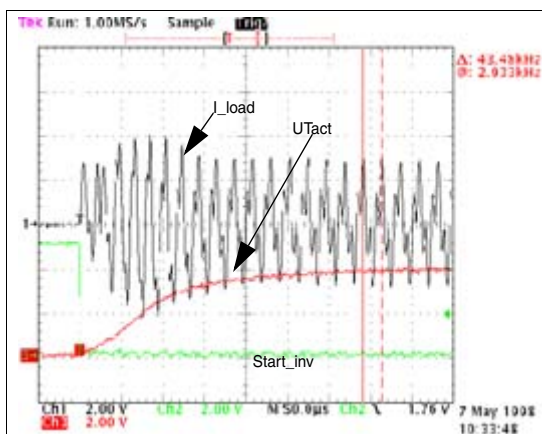


Fig. 47: Inverter function

The oscillogram (Fig. 47 / p. 50) shows the same signal relation as (Fig. 44 / p. 49). The scan parameters here are 80 kV at 260 mA.

Channel 1 = I_{load}

Channel 2 = Start_inv

Channel 3 = U_{Tact}

Error tracing Lust Drive Controller.

Preface

Exchanging the Lust Drive Controller (called hereafter MC) to solve Anode Drive related problems as described below may only be carried out on Straton System with the following revisions:

STRATON	7115301	Revision 3 and higher
STRATON	7737807	Revision 8 and higher
STRATON	7738110	Revision 4 and higher

When exchanging the MC it is mandatory to replace also its main fuses and vice versa

Under certain circumstances the MC drive communicates a detailed error message after a MC related failure. This detailed error message is called "DKS info" and may be found in the eventlog under the error code 3055 CT_XRS E 50 02 37. In order to generate this message the intermediate circuit must not be interrupted after an error but the system has to run through reset, and the MC has to be still alive. For interpreting the DKS info see details below.

Error tracing description

There are two error situations which may be solved by exchanging the Lust Drive Controller on STRATON X-ray tube assemblies:

1. Permanent failures of the Drive Controller

3040 CT_XRS control error XRS (E 50 00 28 00 00 12) operating LEDs at the MC are dark (even though intermediate circuit is alive)

⇒ Replace MC

3040 CT_XRS control error XRS (E 50 00 28 00 00 0e)

⇒ Check for additional DKS info

If no DKS info is present

⇒ Replace MC

When DKS info E 50 02 37 00 00 01 - 07 occurs

⇒ Replace MC

When DKS info E 50 02 37 00 00 81, 83, 85, 86, 87, 88 occurs

⇒ Replace MC

When DKS info E 50 02 37 00 00 82 occurs and intermediate circuit voltage at the MC fuses is > 425V

⇒ Replace MC

Else

⇒ Replace Straton XTA

When DKS info E 50 02 37 00 00 84 occurs and intermediate circuit voltage at the MC fuses is < 625V

⇒ Replace MC

Else

⇒ Replace Straton XTA

2. Sporadic XRS 3050 errors

These errors reflect to anode drive frequency measurements done by the XRS. When these errors occur sporadically and are not triggered by simultaneous tube arcing (which may be the case when the connection XRS - MC is disturbed), most probably the interface group of the MC is broken

⇒ Replace MC

Power supplies Sensation 40/64/Cardiac64/Open.

The XRS is supplied by **Mains1** (230 V switched) which is powered off when the system is switched to COMP/ON at the control box. The type of fans has been changed to 24V DC. The fans are supplied from Slot 4, Slot 5 providing the 48V supply to the DPU. The inhibit signal from DPU is no longer used.

Tab. 5 E606 and UDC Power Supply

Power Supply	Voltage	Tolerance	Measuring Point	E606 Location	Notes
E606	15 V	+/- 5%	At E606	Slot 2	
E606	-15 V	+/- 5%	At E606	Slot 3	
E606	15 V	+/- 5%	At E606	Slot 3	
E606	5 V	+/- 5%	At E606	Slot 4	
E606	24V	+/- 5%	At E606	Slot 4	Used for XRS fans
E606	48 V	+/- 5%	At E606	Slot 5	48V for DPU board
UDC	560 V Nominal	+/- 10%	D400 X130 UDCact	n.a	

Tab. 6 XRS power supply connections

Slot 1	Slot 2	Slot 3	Slot 4	Slot 5
Not used	Not used	S3a_15V	S4a_24V	No connector
Not used	S2b_15V	S3a_0V	S4a_0V	E606_48V+
Not used	S2a_15V	No connector	No connector	E606_48V+
Not used	S2b_0V	S3b_15V	S4b_5V	E606_48V-
Not used	S2a_0V	S3b_0V	S4b_0V	E606_48V-

- INHIP connectors are not used on these models.

Fuses, measuring points, and LED's

The following table lists the field replaceable fuses, measuring points, and LED's that are part of the XRS.

Measuring points

The following is a listing of the most important measuring points available at the D400. Each Test Field (X110, X130 Etc.) has a 0 V point for reference.

Tab. 7 Common test points

Test Point	Test Field	Parameters	Description	Status Table
FIL_DOSE	X110	0 to 10 V	Analog dose signal for dose regulation from the DLP	T5E
FIL_NOM	X110	0 to 10 V	Analog signal from DOM to regulate filament in Servo-Mode	N/A
UTact	X210	1V=20kV	Actual high tension voltage	T5E
I_F_ACT	X110	1V=1.5A	RMS value from I_load_FIL (standby 5A = 3.33V)	T5E
UDCact	X130	1V=100V	Actual UDC value	T5B
RET1	X140	-	Not used with STRATON tube	N/A
F_ROT	X140	-	Not used with STRATON tube	N/A
UI_load_act	X200	1V=50A	Mean value of actual inverter current	N/A
I_load	X230	1V=50A	Actual output current of main inverter	N/A
N_UTact	X210	1V=10kV	Actual value of negative component of high tension	N/A
P_UTact	X210	1V=10kV	Actual value of high tension	N/A
ITact_100mA	X230	1V=100mA	Actual tube current	T5E
Start_Inv	X260	digital	Low active signal which starts the inverters	N/A
HV_drop	X270	digital	High active signal which indicates either one side or both sides have had a high tension disturbance (see note)	T5E

LED's

Table shows signals available via LED on the D400 to help troubleshooting.

Tab. 8 List of LED's

Signal	Color	Description
Start_inv	Green	Start of inverters
X-ray_on	Green	Generated when HT is greater than 20 kV
P_UTmax	Red	The positive component of high tension has reached 82 kV
N_UTmax	Red	The negative component of high tension has reached -82 kV
Supply_Volt	Red	One of the D400 power supplies is out of tolerance
Start_fil_inv	Green	Start signal for the filament inverter
FIL_Inv_Fault	Red	An error has occurred with the filament circuit
RAC_Inv_Fault	Red	Not used with STRATON tube
Inv_Fault	Red	An error has occurred in the main inverter
XC	Green	Start signal for the high tension circuit
Temp_Error	Red	The XRS temperature is out of tolerance
FLEX_Fault	Red	The FLEX chip did not initialize properly

Fuses

Tab. 9 List of field replaceable fuses

Fuse	Value	Load
F1	6.3 A	E606
F2	2 A	XRS fans ¹
F2	3.15 A	XRS fans ²
F3	2 A	RAC/FIL and HV_TANK fans
F4, F5	30 A	RAC/FIL
F100	80 A	Generator

1. On Sensation 10 / 16 / Cardiac, 230V AC fans are used.
2. With Sensation 40 / 64 / Cardiac 64 / Open, 24V DC fans are used.

Overview

Block diagram

The block diagram shows all field replaceable parts and their connections.

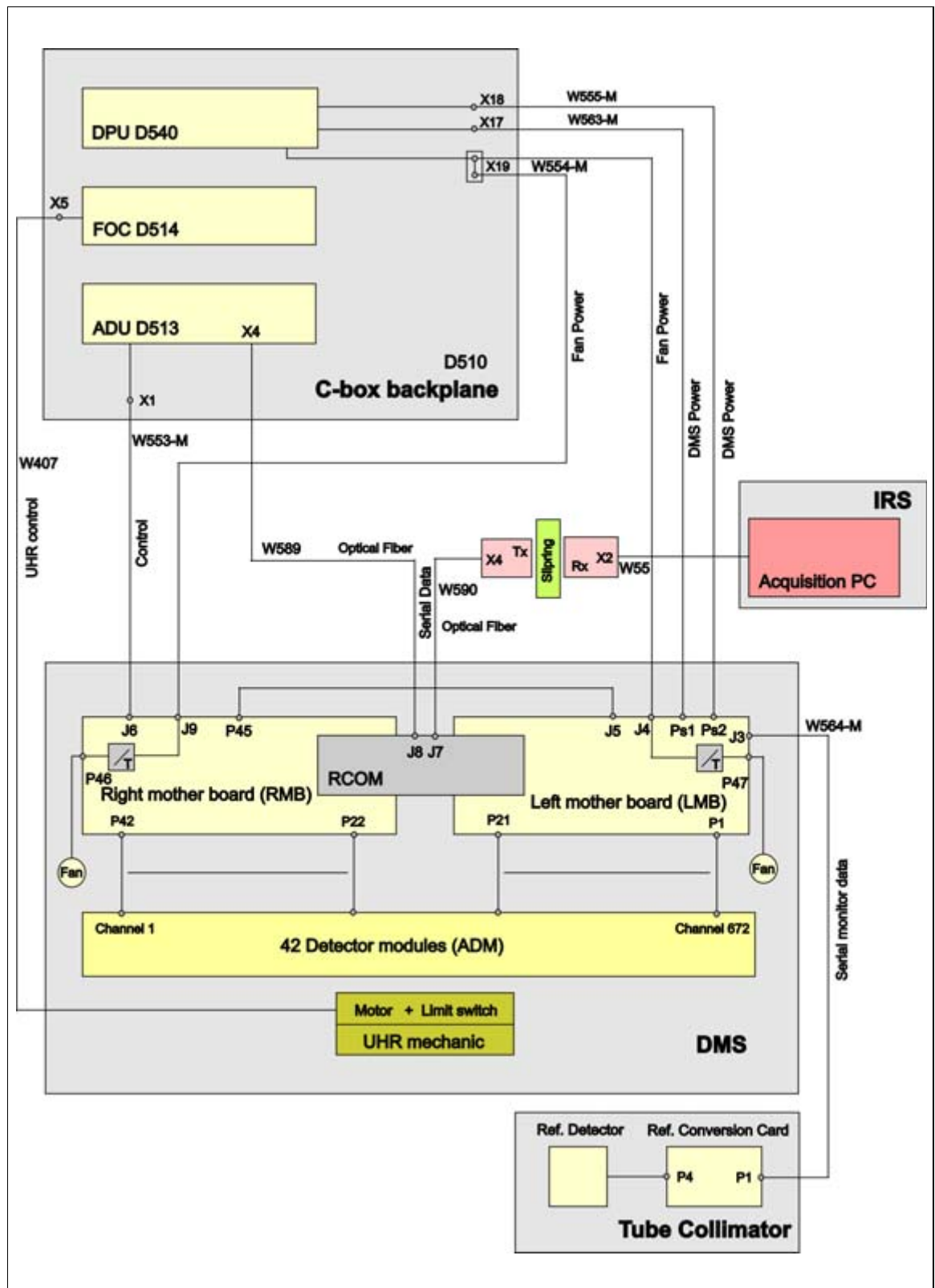


Fig. 48: Block diagram of DMS

General hints

Here you will find general hints you should always keep in mind when working on DMS problems.

NOTE

Follow the C-MOS guidelines when handling the DMS.

- Always start troubleshooting with the ROI/Ring([ROI/Ring / p. 65](#)) and use the position found as a reference for the expected position of the defect (two positions possible).
Normally, you start with the image the customer is complaining about.
- To determine a defective slot, always use tests that output a result across all channels.
- Check whether artifact is also visible in topogram. If yes, position a marker and perform a topogram to define the affected side of the detector.
- To distinguish between module (ADM), backplane (right or left motherboard) and RCOM, it's possible to cross the cable connection between module and backplane.
⇒ If location of error changes when cable is crossed: ADM is defective.
- Before starting extensive troubleshooting of weak rings, be sure environmental conditions and tune-up is ok([Environmental Condition / p. 88](#)).
- Depending on the software installed, only two slice combinations are useful for troubleshooting ([Tab. 10 / p. 58](#)):
- Before starting extensive troubleshooting, be sure you troubleshoot the right side.
To distinguish between DMS and IRS, use the Disc Test ([Disc test / p. 83](#)).
- Don't forget to also use test modes with rotation when troubleshooting.
- Test results are stored in two different ways:
 - as HTML files or
 - as TXT files

Normally HTML files are used to store test results.

However, the plots of the DMS tests, in particular, are stored in a text file (via the Extract to File button). Please note: the file name has to be changed manually. Otherwise, the file will be overwritten each time the soft button is used.

The file may be displayed using Excel (e.g. after manual remote transfer).

Test differences between Sensation 16 / Sensation 10

Tab. 10 Differences between Sensation 16 and Sensation 10

Sensation 16 with VA50 installed	Sensation 16 with VA60 installed	Sensation 10 with VA60 installed
Test mode/slices	Test mode/slices	Test mode/slices
16*1.5mm	16*1.5mm	16*1.5mm
16*0.75mm	16*0.75mm	10*0.75mm

Ring, band, and streak artifacts

This chapter provides a basic overview of different types of image quality problems.

General

Possible reasons for ring and band (mostly 16 channel) artifacts are ([Fig. 48 / p. 57](#)):

- Defective detector element (ADM)
- Defective detector cable
- Defective backplane (LMB or RMB)
- Defective RCOM
- Bad calibration
- Bad tune-up tables
- Problem with balancing
- Bad environmental condition

Partial ring occurs due to weighting during image reconstruction in spiral mode.

Strong rings

In most cases we expect a defect in a detector module (ADM).

Start troubleshooting with the following DMS tests:

1. Signal value ([Signal value / p. 67](#))
2. Offset value ([Offset value / p. 67](#))
3. Sinugram ([Sinugram / p. 79](#))

Strong bands (8 or 16 channel) may also result from a defective backplane or RCOM.

Weak rings

A ring is visible in homogeneous areas such as the cerebrum, if the difference compared to the neighboring channel is about 1 ppm.

In most cases we expect the stability of the element affected (ADM) to be insufficient. Before starting extensive troubleshooting, ensure:

- That all environmental conditions are OK ([Environmental Condition / p. 88](#))
- That tune-up was performed correctly
- That the DMS is powered on over night.

Start troubleshooting using DMS tests:

1. BasCal stability ([Basecal stability / p. 71](#))
2. Signal linearity ([Signal linearity / p. 74](#))
3. Sinugram ([Sinugram / p. 79](#))

Evaluate the physicist's line in the CT image text (lower left corner of image) in order to use the information as a basis for generating raw data sets of the parameter set identical to the one the customer is complaining about.

Streak artifacts

Possible reasons:

- Defective detector module:

If the problem is intermittent, a streak artifact tangential to a circle (position of the defective channel) occurs.

⇒ Use the popcorn noise to isolate the defective channel ([Popcorn noise / p. 66](#))

- Patched channels:

Streak artifact tangential to a circle (position of the patched channel) occurs.

⇒ Check if there are too many (adjacent) patched channels ([Using Reports to view details of the Defective Channels Correction / p. 112](#))

- Alignment adjustment:

Fine streak artifacts not correlated tangentially to any circle.

⇒ Check whether MTF measurement (Quality) is within tolerance.

Other artifacts

Some other artifacts can be caused by:

- The tube
- The tube side collimator
- The tube cooling system
- The DPU

Pattern over the whole image

If you have patterns (e.g. grid pattern) in your image:

- Perform tests or test images with and without X-ray, to find out if the problem is related to radiation.

If problem is related to radiation:

- ⇒ - Check the reference channel ([Checking the reference channel / p. 62](#))
 - Check (replace) tube
 - Check (replace) tube collimator

If problem is not related to radiation:

- ⇒ - Check the DPU ([Power supply / p. 87](#))
 - Check the ADM's at slot 1 and 42 (slice normalization)

Inhomogeneous areas

To identify the source of the problem, evaluate the Basecal values (contact CS HSC 22).

The reason for inhomogeneous areas in the image (with radiation) are mainly:

- Air (hydrogen) in the tube cooling system:

The artifact is not stable but changing from image to image.

- ⇒ The Dura Akron and the cooling system can be vented

- Defective BaseCal tables (visible mainly in the cerebrum):

If calibration was performed with an absorber (contrast media) in the scan field:

The artifact is stable (in the same image area) in all images.

- ⇒ Check the report for entries.
(**Options > Local Service > Reports**)
- ⇒ Clean the plexi-strip and check for other absorbers in the scan field.
Repeat calibration.

Inhomogeneous slices

If the mean value (of the same image area) differs from slice to slice :

- ⇒ Check the ADM's of slots 1 and 42 (air slice normalization)

Hint:

The task of the air slice normalisation is to equalize all 16 slices.

For this purpose some of the channels of the ADM's of slot 1 and slot 42 are used to calculate a mean value for each slice. All slices are now compared and equalized.

Checking the reference channel

There is no special test for the reference channel, but some tests also display the values for the reference channel:

- Offset value ([Offset value / p. 67](#))
- Signal value ([Signal value / p. 67](#))

If the test results for both tests are in tolerance, the reference channel works fine.

- Analyze ([Analyze / p. 77](#))

Use the **Dose Monitor** ([Fig. 56 / p. 78](#)) function to check the reference channel.

The advantage of this test is that you see also the transient response of the dose.

- Tests which provides also information about the reference channel:
 - Offset value ([Offset value / p. 67](#))
 - Signal value ([Signal value / p. 67](#))
 - Spectral linearity ([Spectral linearity / p. 73](#))
 - Signal linearity ([Signal linearity / p. 74](#))
 - Analyze ([Analyze / p. 77](#))

Tools for troubleshooting

Here you will find a list of all the software tools available for handling DMS problems.

General

- Common for all test results providing a **plot** :

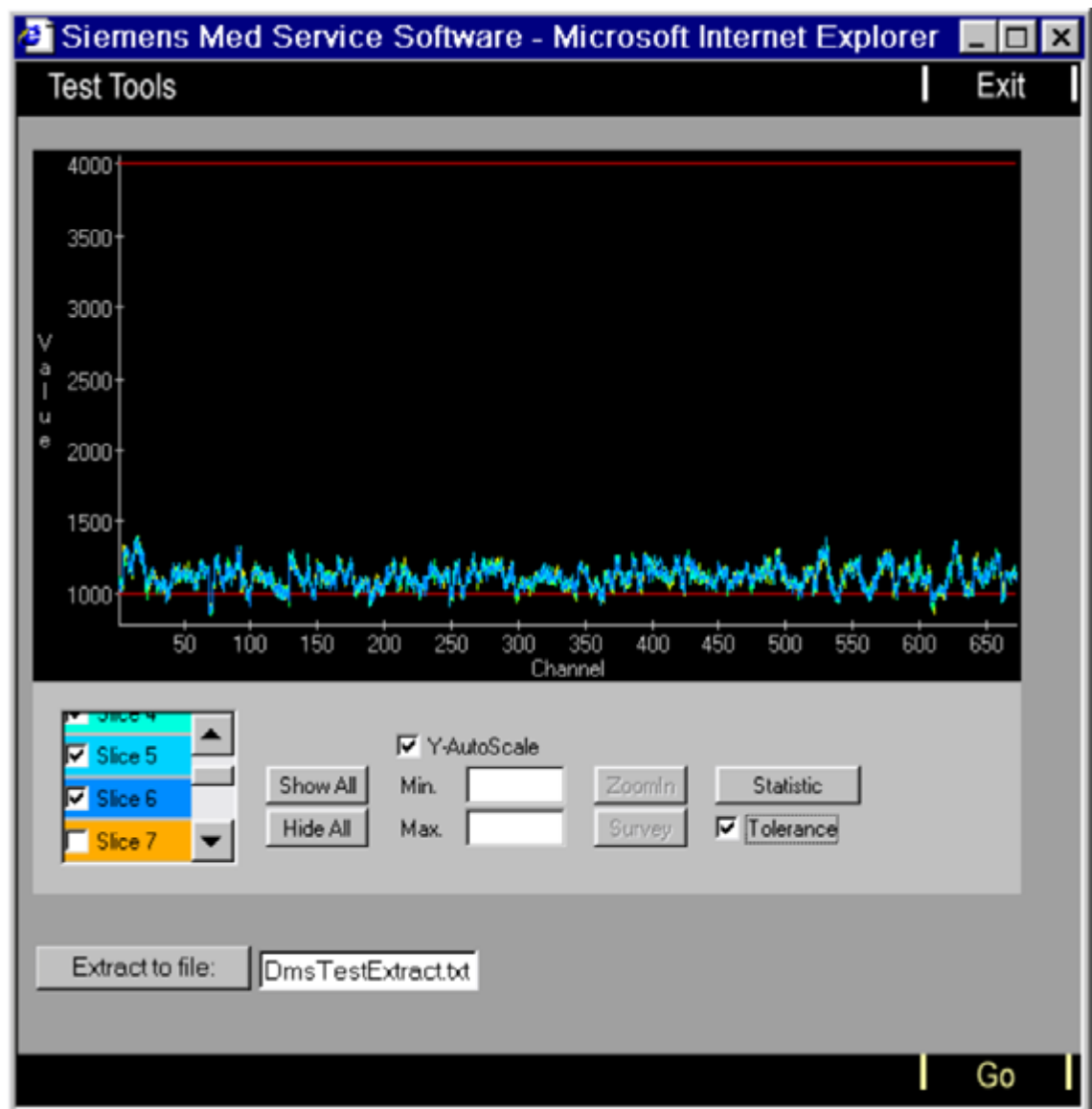


Fig. 49: Example for show plot

- The button **Show Plot** shows the graphical output of the table.
It's possible to select all slices via **Show All**, none via **Hide All**, or a single slice via a list box. It's also possible to zoom in (or out via **Survey**) by using the left mouse button. Mark the area of interest, while holding down the left button. Then press **Zoom In**.
- Position the cursor at the desired location and wait for the display of:
 - channel
 - slice
 - value at that position
 The button **Statistic** displays the following for all slices: Minimum, Maximum, Mean and StdDev.
The min. and max values are shown only if the check box **Y-Auto Scale** is not marked.
- The button **Extract to file** is used to store the result of the test to
c:\Somaris\service\extract\DmsTestExtract.txt
Hint: If you want to store more than 1 file, you have to change the file name manually.
- The display of the tolerances in the plot may be switched on and off using the checkbox for **Tolerance**
- Tolerances are available for all test results.
As a result, test result pages display the following at the top of each page: **OK** for tests passed or **Not OK** for tests failed.
- In case of **values out of tolerance**, slice and channel are output.
Click on the channel number and read out: Channel xxx, Slot xx and Plug Pxx.
The Plug xx corresponds with the Phillips sorting order (labeling at DMS).

Band assessment tool

Scope

- This test performs automatic test scans and runs an algorithm to find ring structures in water phantom images. The disadvantage is that only the range of the phantom is checked (20 or 30 cm).

The prerequisite is a phantom consisting of 3 parts:

- 20 cm body
- 20 cm head
- 30 cm body

The phantom will **not** be delivered with the system

- Normally this test is used only in the factory.

How to find

- **Options > Local Service > Test Tools > Band Assessment**

How to use

1. Position phantom so it lines up with the engraved mark.
The table position is checked.
2. Select the desired modes.
Possible choices: Select all, Deselect all and Select box
3. Press **Go**
 - ⇒ If there are results out of tolerance continue troubleshooting ring artifacts.

ROI/Ring**Scope**

- The **ROI** function (region of interest) is used to calculate the mean value and the standard deviation of a user-defined area.
- The **Ring** function shows the correlation of a position in an image with the 2 possible channels in the DMS that have been involved in creating data at that position.

How to find

- **Options > Local Service > Test Tools > ROI/Ring**

How to use

1. Select the desired order for sorting the images in the list box by activating the corresponding radio button. (You may have to wait.)
2. Mark the image of your choice in the list box.
3. Press the button **Load selected image**
 - ⇒ The image will be displayed
4. Select the radio button **Ring**
5. Use the left mouse button to move ring to desired position.
Use center mouse button for windowing.
 - ⇒ Read out the 2 possible channels (channel1, channel2) causing the problem.
 - ⇒ Use this information as basis for further troubleshooting.

Offset image**Scope**

- The offset image is calculated from an offset corrected raw image added to a synthetic disc. The raw image is acquired without radiation.
As a result, the artifacts visible in the offset image are not related to the X-ray or the UFC of the detector but are instead related mainly to the electronics of the integrator section.

How to find

- **Options > Local Service > Test Tools > DMS > Offset Image**

How to use

1. Select the radio button for rotation on or off.
Start testing with rotation.
2. Select a scan time from 5 possible times (0.420 - 1.500 secs).
Start with the scan time the customer is complaining about.
3. Press **Go**
4. Wait for the Press **START** key to appear and then press the start button.
 - ⇒ 16 offset images are calculated.
Images have to be without rings, bands and streaks.
Check images with window 550 and center 0.
In case of problems:
 - ⇒ Continue with offset value.
5. Select radio button image is correct (or not correct)
6. Interpretation of the results:
The images must show a homogeneous noise structure and must be without:
 - Rings or partial rings
 - Streaks
 - Any visible structure.

Hint:

The offset images will be stored under Service Patient.

The offset measurement will be aborted in case of an active patient.

Popcorn noise**Scope**

- The principle of the popcorn noise test (spike test) is to monitor a number of offset scans, in order to find unstable channels. As a result, ppm thresholds are defined for the values. If these thresholds are exceeded several times (e.g. max. number of spikes), the test identifies the corresponding channels.
- It is difficult to find suitable tolerances for this test.
This test should mainly be used to find sporadic problems.
(For instance, to find the worst channel.)

How to find

- **Options > Local Service > Test Tools > DMS > Popcorn noise**

How to use

1. Select max. numbers of spikes
2. Select lower threshold
3. Select upper threshold
4. Select number of measurements
5. Select test with or without rotation
Start troubleshooting with rotation
6. Press **Go**

⇒ Check results to find channels outside the threshold in use.

Hint: The value for “number of spikes” must be smaller than the “number of measurements”. Otherwise, a result will not be output, even though the threshold has been exceeded.

Offset value**Scope**

- This test is used to check the mean values and noise without X-ray.

How to find

- **Options > Local Service > Test Tools > DMS-Test > Offset**

How to use

1. Select the desired mode -> (Tab. 10 / p. 58)
2. Press **Go**
3. Wait for the Press **START** key to appear and then press the start button.
⇒ Replace modules that exceed the threshold
4. Description of the **result page**

The result of the test is shown in two tables:

- **Offset value**
- **Offset noise**

Both tables show Slice, Minimum, Maximum and Test Result.

Signal value**Scope**

- This test is used to check the mean values and noise with X-ray.
A mode with 16 x 1.50 mm is used.

How to find

- **Options > Local Service > Test Tools > DMS-Test > Signal Value**

How to use

1. Press **Go** to start the test
 - ⇒ If there are channels out of tolerance, exchange the corresponding module.
2. Description of the **result page**

The result of the test is shown in three tables:

- **Relative Signal Value**

The result shows the average for the neighbourhood of +/- 5 channels.

This table shows Slice, Minimum, Maximum and Test Result.

- **Relative Standard Deviation**

This table shows Slice, Minimum, Maximum and Test Result.

In addition, the monitor value is shown.

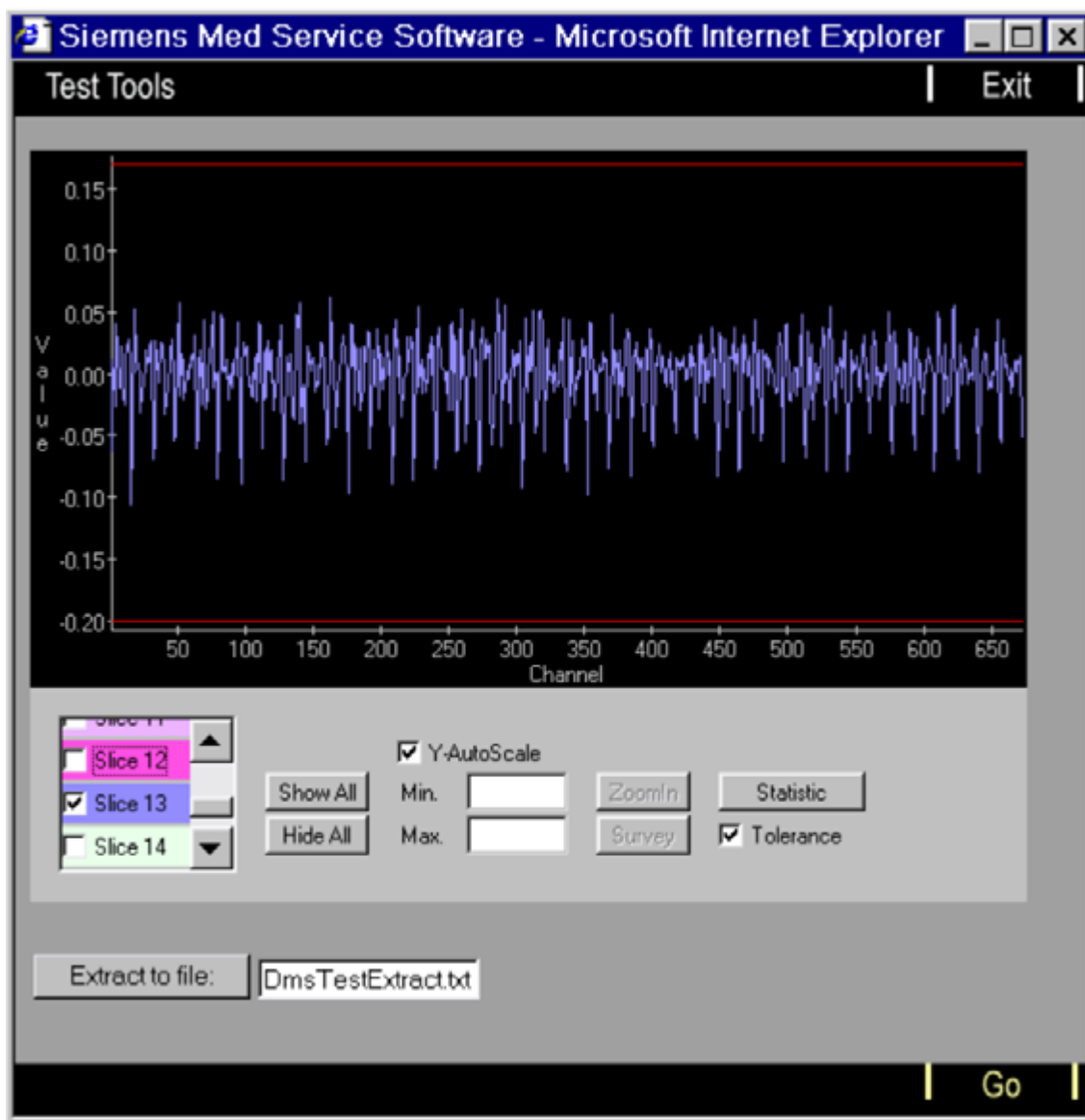


Fig. 50: Example for relative standard deviation

- **Absolute Signal Value**

This table shows Slice, Minimum, Maximum and Test Result.

In addition, the monitor value is shown.

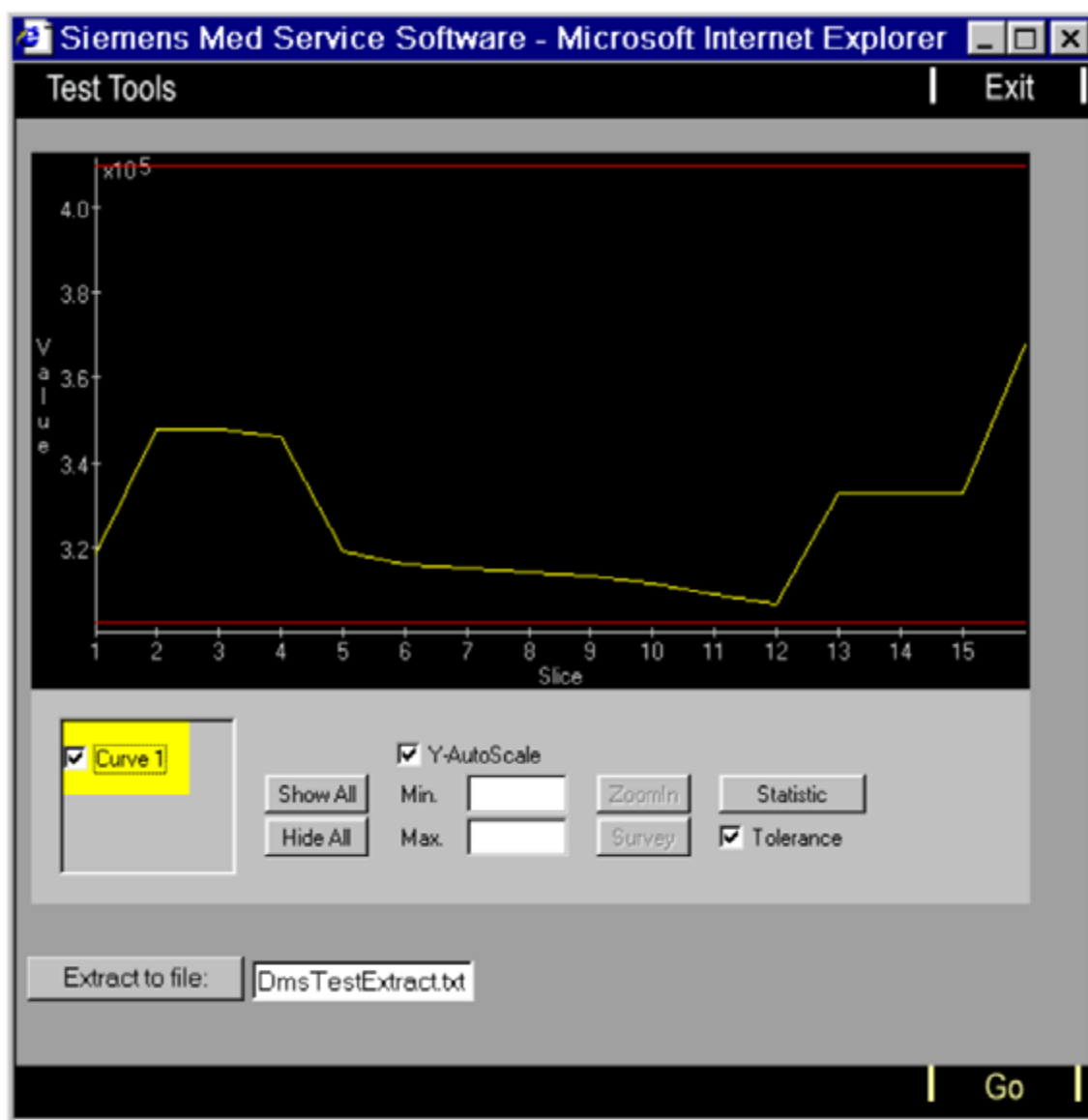


Fig. 51: Example of absolute signal value

Common for all tables is:

- The button **Show Plot** graphically displays the table.

It's possible to select all slices via **Show All**, none via **Hide All**, or a single slice via a checkbox. It's also possible to **Zoom In** (or out via **Survey**) by using the left mouse button. Mark the area of interest, while holding the left button.

The button **Statistic** displays the following for all slices: Minimum, Maximum, Mean and StdDev.

The check box **Y-AutoScale** allows for automatic scaling of the y-axis

- For values out of tolerance, slice and channel are output.
Click on the channel number and read out: Channel xxx, Slot xx and Plug Pxx.
The Plug xx corresponds to the Phillips sorting order (labeling at DMS).
- The button **Extract to file** is used to store the result of the test to disk.

Basecal stability

Scope

- The main function of the basecal stability test is to subtract the last two basecals of one given mode in order to find unstable channels.
The structure of the basecal names is: basecal_slice_focus_algorithmn_time_organ.
They are stored in c:\Somaris\service\datastore\icstables

How to find

- **Options > Local Service > Test Tools > DMS-Test > BasCal Stability**

How to use

1. Select the desired base calibration table from the list box.
Use the basecal corresponding to the mode the customer is complaining about.
2. Press **Go**
 - ⇒ The result page is shown.
 - ⇒ Before exchanging modules, compare the result with the results of ROI/Ring to verify whether the customer image problem is found.

3. Activate **Show Plot**

⇒ Evaluate the plot

In addition to the standard functions you can define the Segments which should be shown.

Hint: A basecal is referred to as a 360 degree rotation and is divided into 10 angular segments.

It's normal for BaseCal plots of UHR modes to contain 2 spikes at the border of the UHR mechanics (channels 174 and 501).

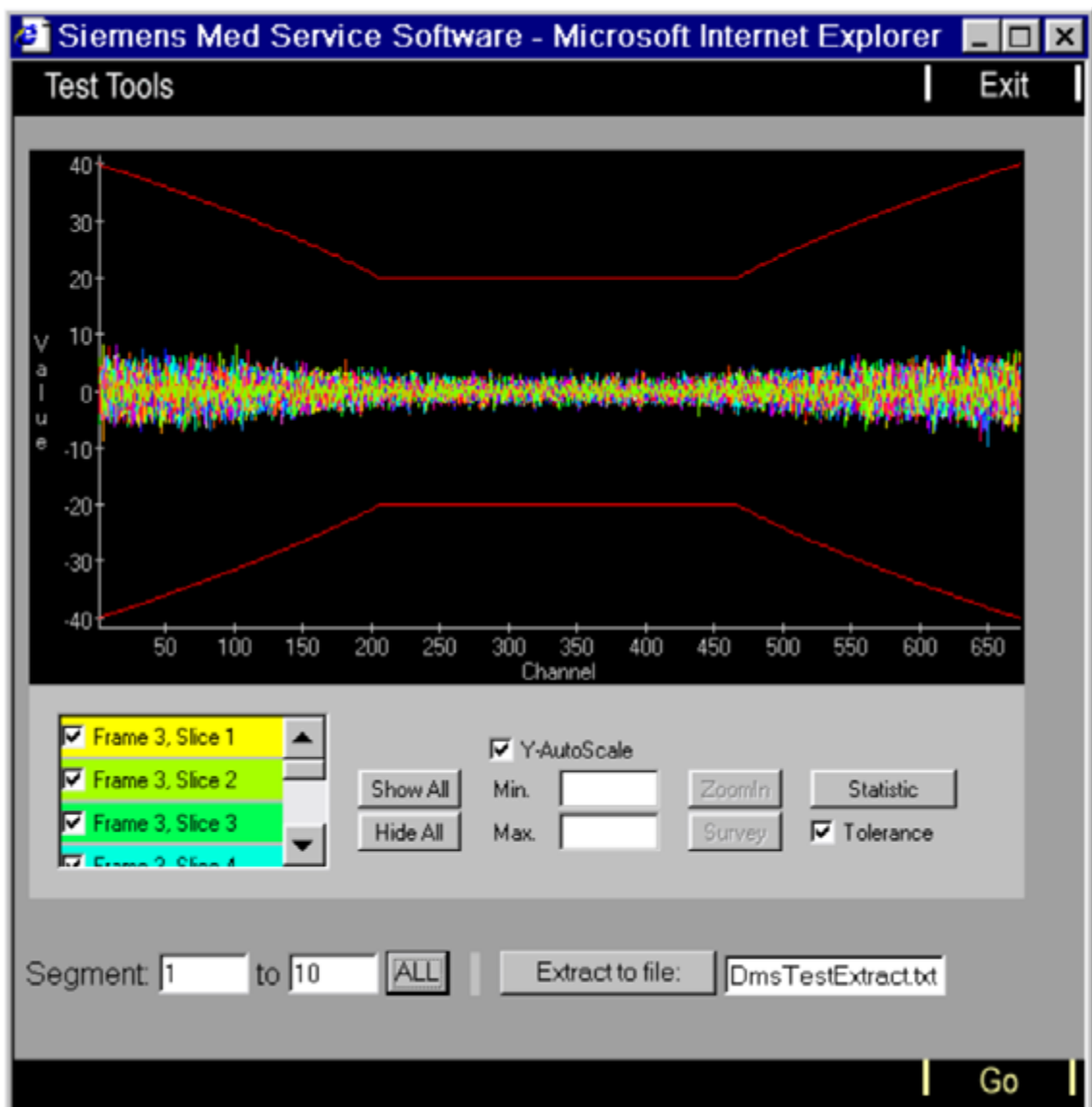


Fig. 52: Example of basecal stability

Spectral linearity

Scope

- This test performs two scans with different tube voltages (120 kV and 140 kV).
Subtracting the mean values of these scans shows the deviation caused by the spectral non-linearity.
The result of this test should be used **only in addition to other tests** to confirm a channel as satisfactory or unsatisfactory.
- This test is normally used only in the factory.

How to find

- **Options > Local Service > Test Tools > DMS-Test > Spectral Lin**

How to use

1. Wait for the Press **START key** to appear and then press the Start button.
 - ⇒ The test result page is displayed.
 - ⇒ If there are channels out of tolerance call CS HSC 22.

2. Press **Show Plot**

⇒ The plot is displayed.

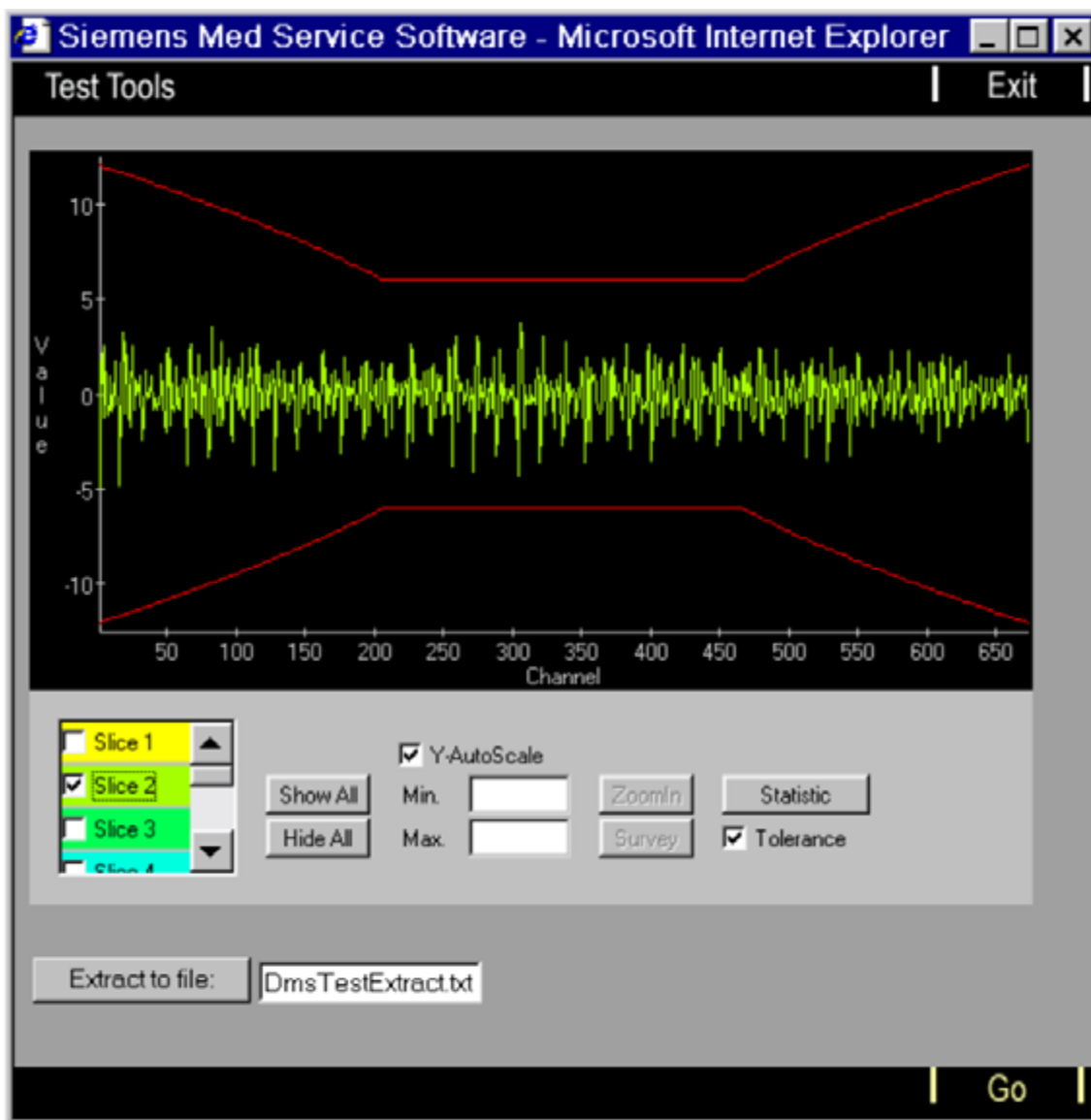


Fig. 53: Example of spectral linearity

Signal linearity

Scope

- This test performs two scans with different tube currents (100mA and 300 mA).

Subtracting the mean values of these scans shows the deviation caused by the signal non-linearity.

How to find

- Options > Local Service > Test Tools > DMS-Test > Signal Lin**

How to use

1. Wait for the Press START key to appear and press the Start button
 - ⇒ The test result page is displayed.
 - ⇒ If there are channels out of tolerance, change the corresponding ADM
2. Press **Show Plot**
 - ⇒ The plot is displayed

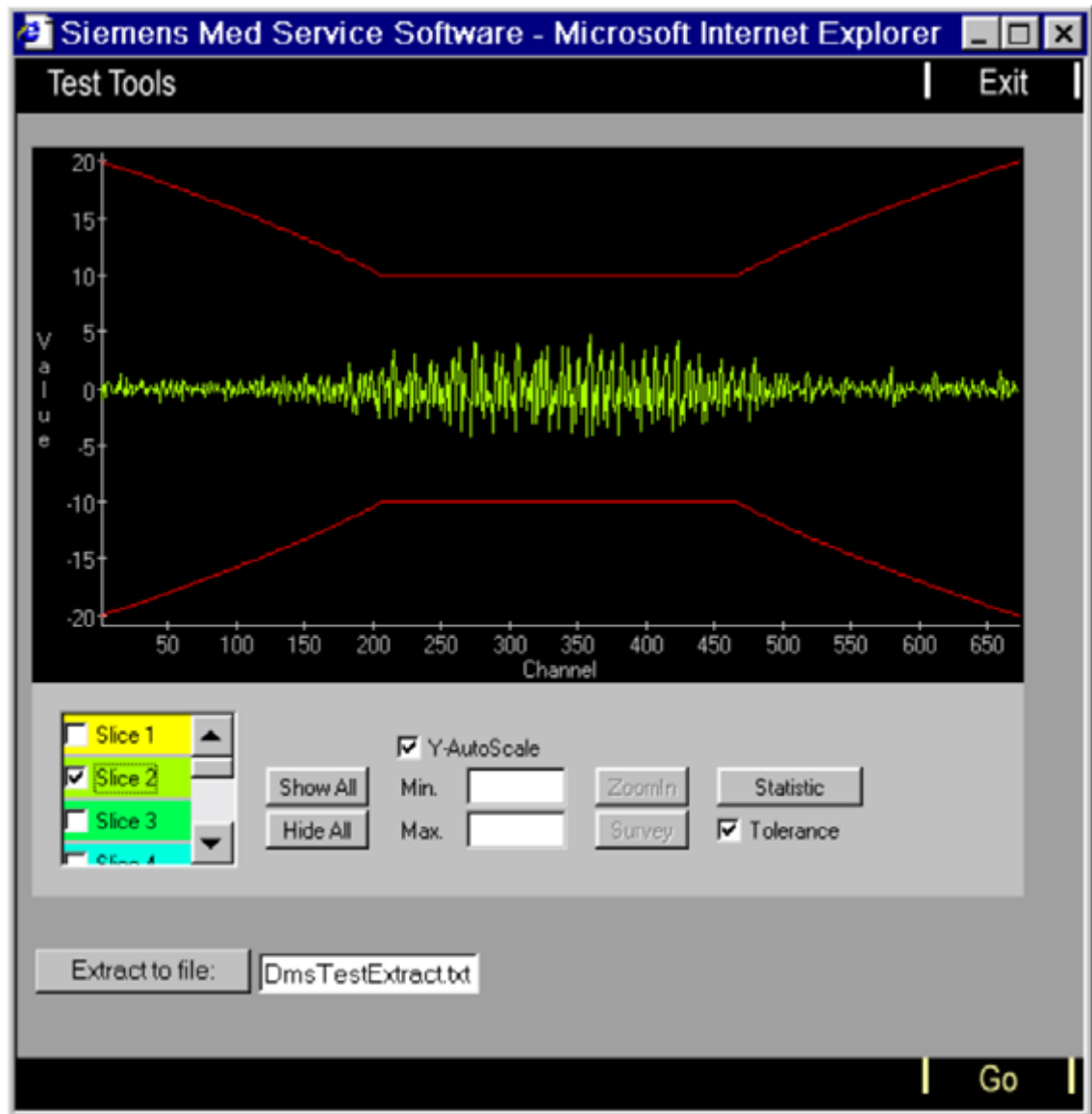


Fig. 54: Example of signal linearity

Assignment test**Scope**

- The assignment test is an addressing test between the RCOM and the 42 detector modules (ADM's)

How to find

- **Options > Local Service > Test Tools > Assignment**

How to use

1. Wait for the Press **START** key to appear and then press the Start button
 - ⇒ Wait for the result page
 2. Activate **Show Plot**
 - ⇒ A continuous staircase from top left to lower right should be visible for all 16 slices.
- Hint: In addition to the standard function, you can define:
- **Segment** 1 to 1160 (frames)
 - and
 - **FFS** 1, 2 or all

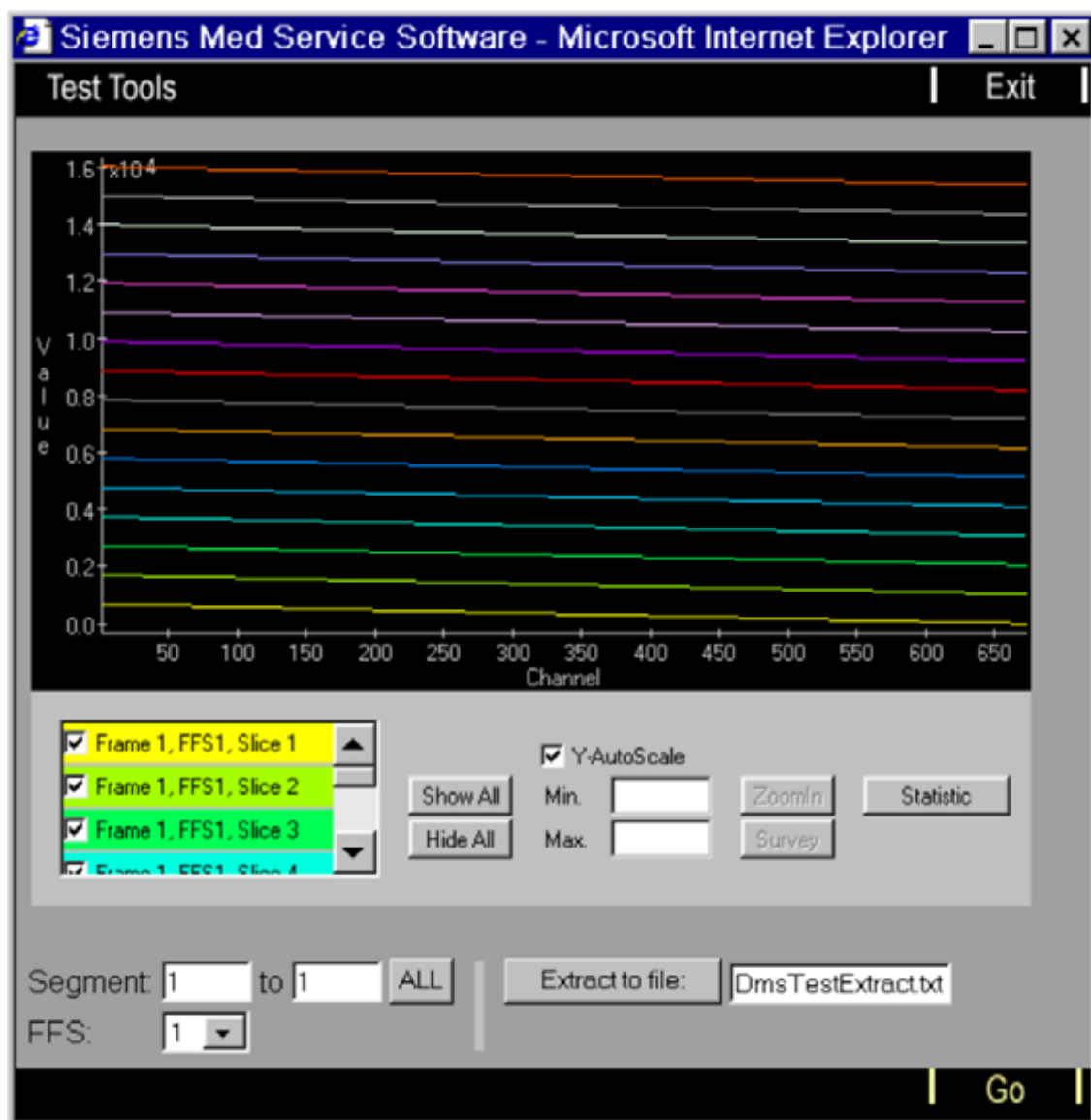


Fig. 55: Example of assignment test

3. Interpretation of the result

Possible defective part may be:

- defective detector module (ADM)
- defective backplane (motherboard)
- defective RCOM

Analyze

Scope

- Analyze is used to check and plot header data of a raw data set.

Useful header data are, e.g.

- Integration time
- Dose monitor
- DMS Temperatures

How to find

- **Options > Local Service > Test Tools > Analyze**

How to use

1. Select the radio button **ICS-RawData**
2. Mark the desired raw data set in the list box
3. Press **Go**
 - ➡ View is automatically selected and the plot appears.
4. Select **Special**
5. Mark the desired checkbox
 - ReadingNo
 - TubeAngle
 - MaximumValue
 - ErrorInfo
 - **IntegrationTime**

- Dose monitor

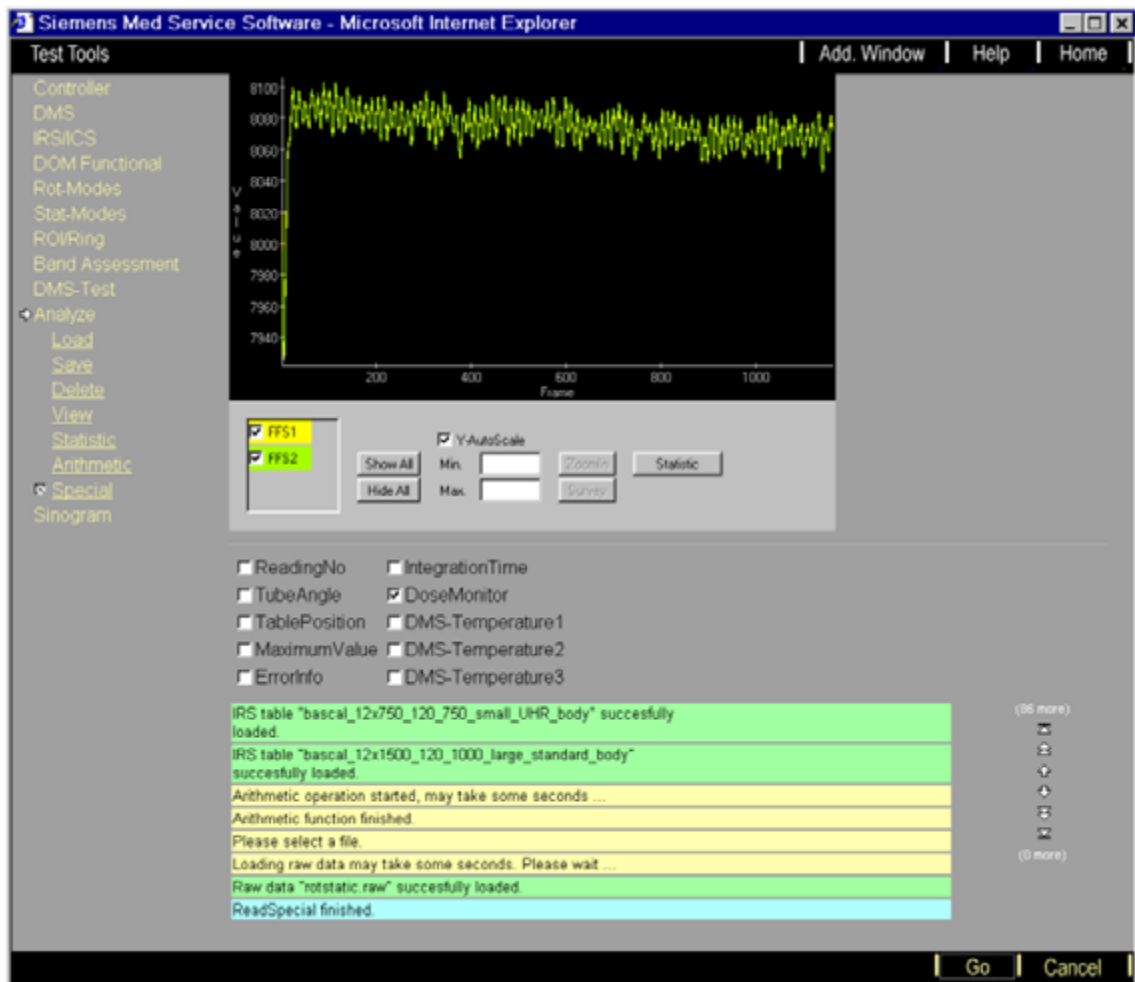


Fig. 56: Example of dose monitor

- DMS-Temperature 1

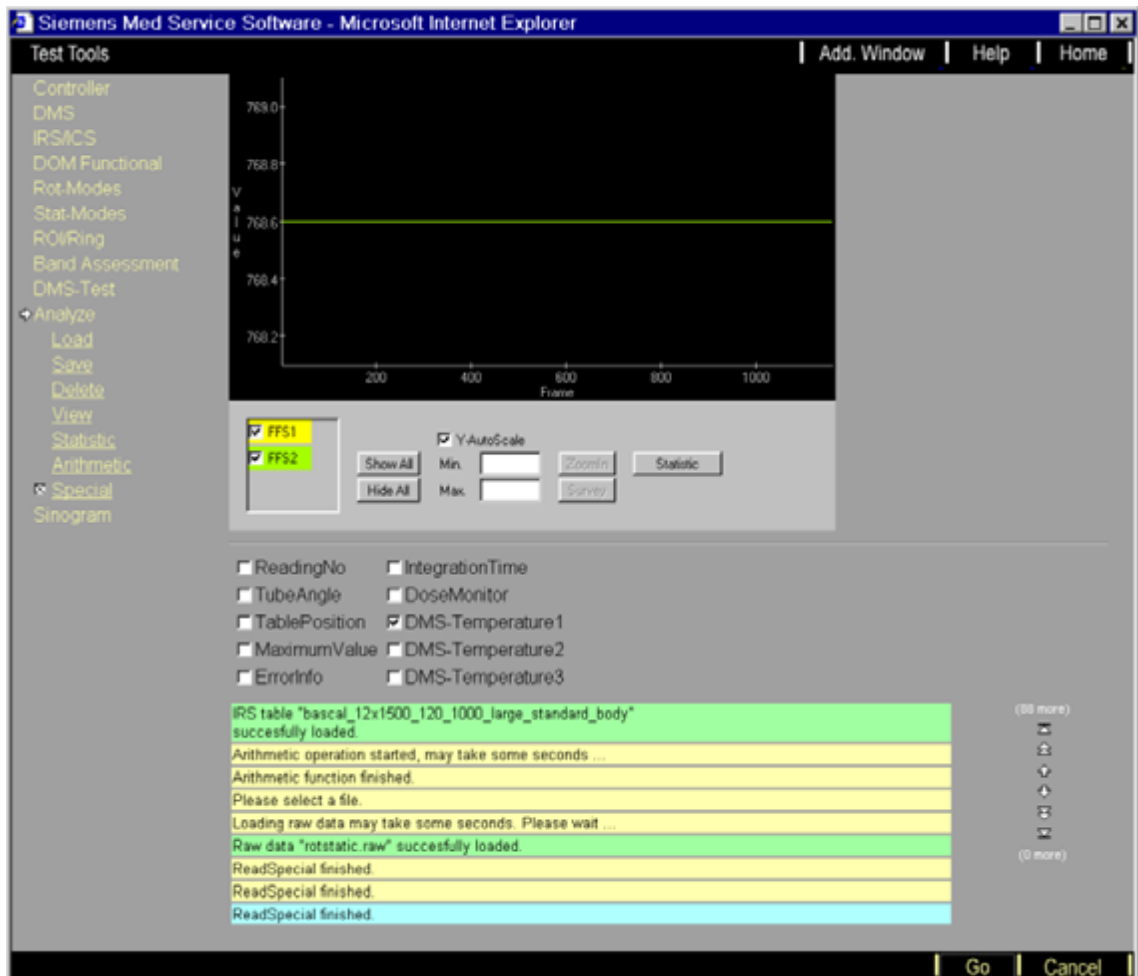


Fig. 57: Example of DMS temperature 1

- DMS-Temperature 2

- (DMS -Temperature 3 is **not used**)

and press **Go**

⇒ The corresponding curve is displayed.

⇒ The detector temperature is shown (in a strange way)

Hint: To translate the result to degrees celsius proceed as follows:

- Convert the number shown from decimal to hexadecimal

- Divide the low byte by 8

- Add high byte and low byte

(For example, 680 corresponds to 31 degrees Celsius)

Sinugram

Scope

- The output of the sinugram is an image showing all channels at the x-axis as well as all frames (projections) at the y-axis.

- Raw data sets of air scans and of scans made with an homogeneous absorber can be evaluated.
- The sinugram is a useful tool to find weak rings

How to find

- **Options > Local Service > Test Tools > Sinugram**

How to use

1. Mark a file in the list box Raw Data Files.
2. Mark or unmark the checkboxes for **Convolve with Kernel** and **Median filtering** .
Start with both checkboxes unmarked.
3. Press **Go**
 - ⇒ The plot is displayed and you can select the slice via the select box.
 - ⇒ Using the left mouse, mark a position to display the corresponding frame and channel.
 - ⇒ Mean value and standard deviation of a marked area is shown, if you use the left mouse button to mark the desired area.

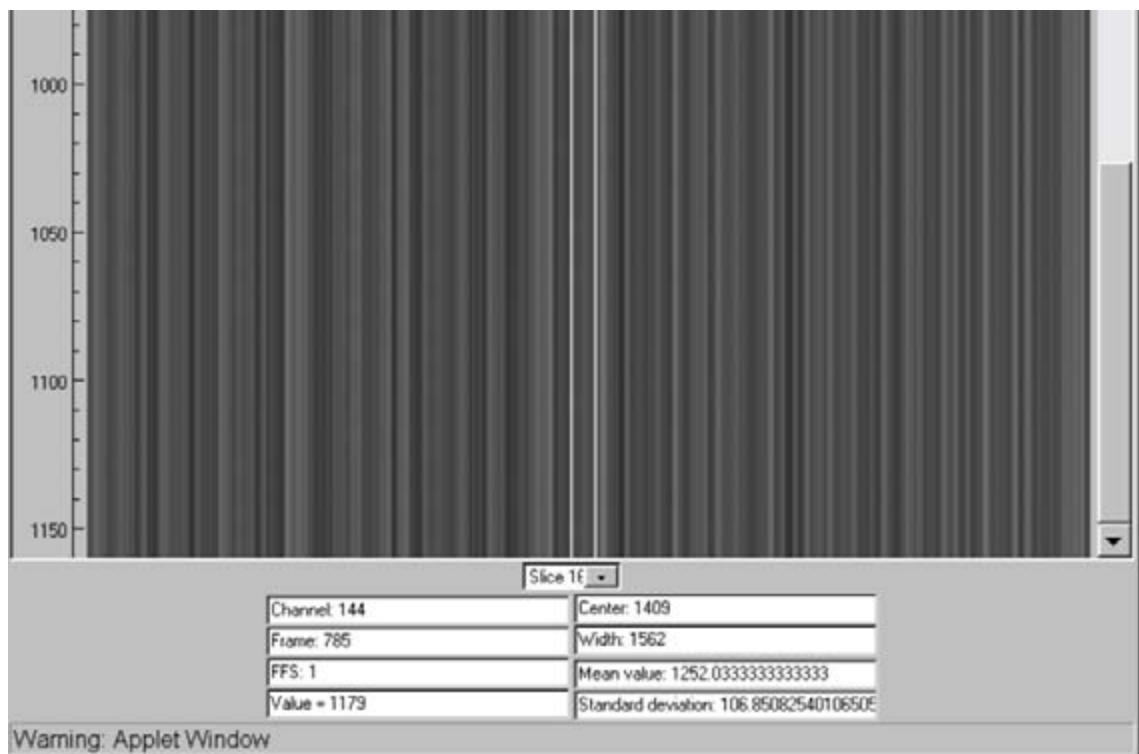


Fig. 58: Sinugram

4. Interpretation of results:

- A vertical line shows up in all frames if a channel is defective.
- A vertical dotted line shows up if a channel is defective but not in all frames.
- A horizontal line shows up if a frame is defective (projection).
- A dot shows up if a channel is defective in one frame.
- A sine wave shows up if a stationary absorber is in the scan field.
(Absorber is not moving with DMS)

Data link

Scope

- This test checks the physical link level of the data link.

For this purpose, test data packets are sent from the RCOM to the receiver (F-REC, D23) of the IRS.

The received packets are checked by performing a CRC (cycling redundancy check).

The test includes the following parts:

RCOM (partial) -- W590 -- Tx -- Slip ring -- Rx -- W55 -- F-REC (partial)

Various **predefined modes** (Auto test modes) allow the user to identify relations between the error and:

- the X-ray (various tube positions, different power)
- Rotation (various rotation times)
- In addition, the following **freely selectable modes** are available.

Before using these modes, define in

Options > Local Service > Test Tools > Stat-Modes / Rot-Modes
the desired parameter.

- Static 20.dat
- Static 20.dat, X-ray
- Rot20.dat
- Rot20.dat, X-ray

How to find

- **Options > Local Service > Test Tools > DMS-Test > Data Link**

How to use

1. Select the desired mode from the list box
Standard test parameters should be Rot 0.50s and X-ray off
2. Press **Go**

3. Wait for the Press **START** key to appear and then press the Start button.
 - ⇒ Several scans are performed
 - ⇒ The results page is shown
 - Important: No CRC Errors are allowed (CRC error sum:0)
 - ⇒ In case of errors, troubleshoot the data path

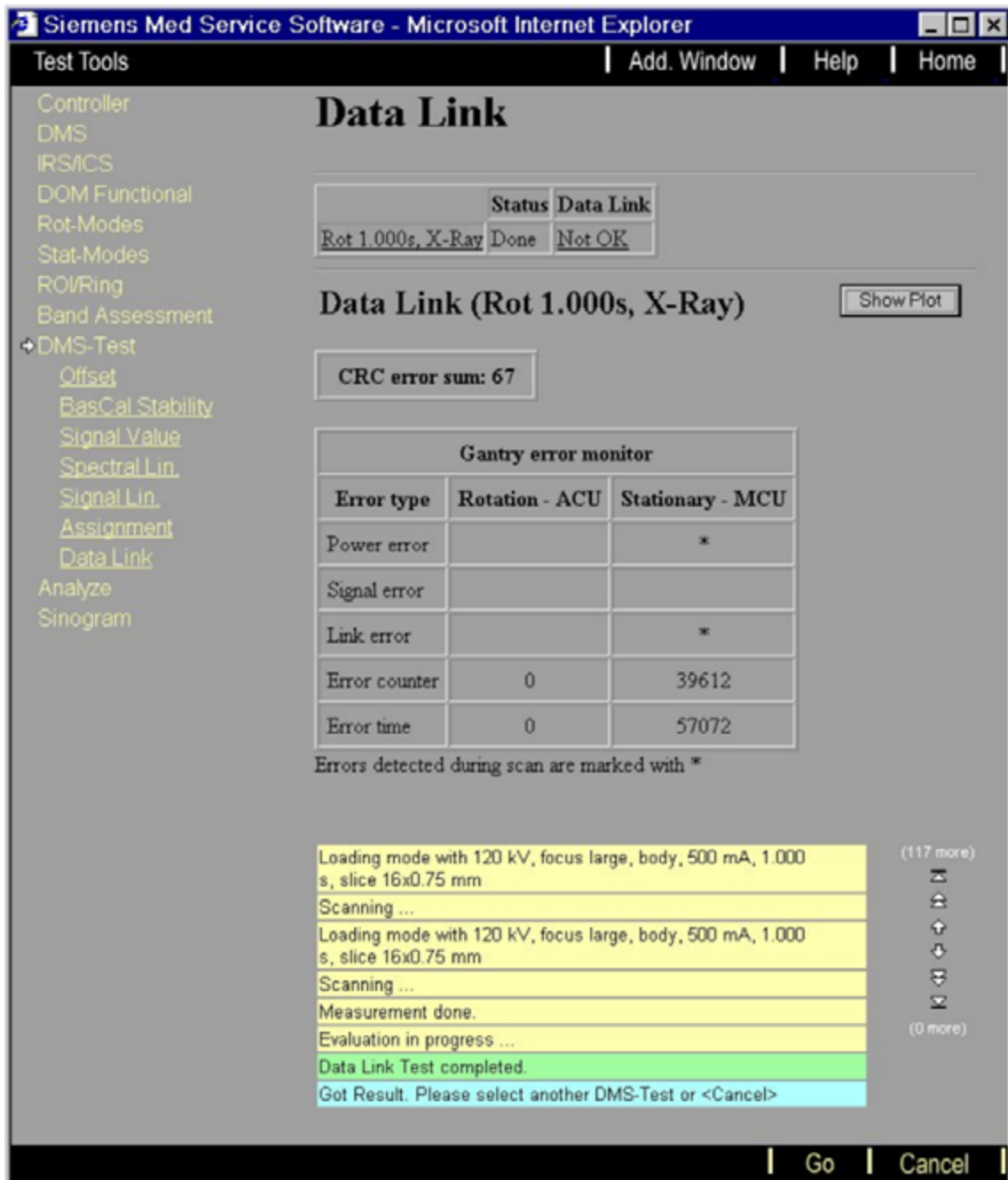


Fig. 59: Example of data link test with error

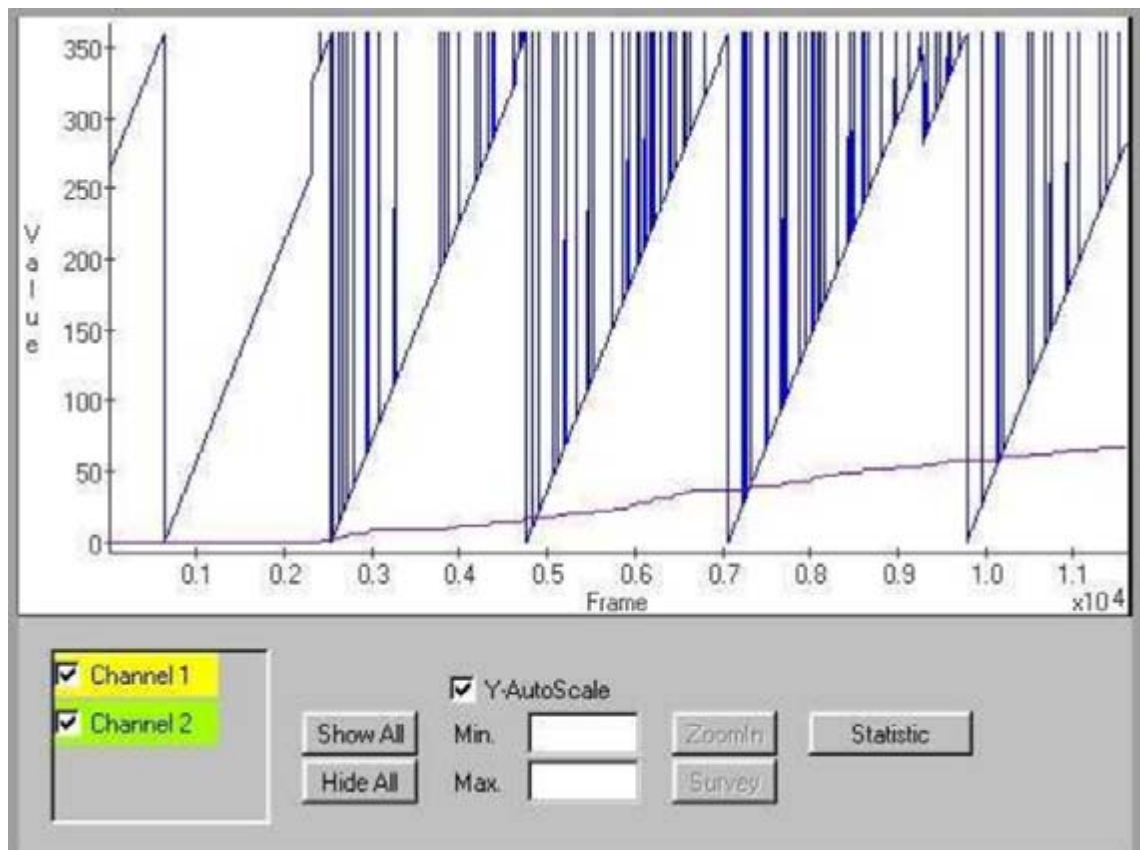


Fig. 60: Example of show plot at data link test

Channel 1 = Tube angle

Channel 2 = CRC error

Disc test

Scope

- The Disc Test uses a predefined test pattern, stored in RCOM. This test pattern is transmitted from RCOM to F-REC and, in addition, all reconstruction steps are performed on that test data. The IRS checks via CRC the reconstructed data.

It is also possible to use a test pattern from the receiver (IRS).

In case of error, this test may be used to distinguish between the DMS and the IRS (including data transmission path).

How to find

- Options > Local Service > Test Tools > IRS/ICS > IRS Disc Test**

How to use

1. Select the source of test data by activating the corresponding radio button:
 - **Transmitter with rotation** (data source is RCOM)
 - **Transmitter without rotation** (data source is RCOM)
 - **Receiver** (data source is IRS)
2. Press **Go**
3. Wait for the Press **START** key to appear and then press the Start button
 - ⇒ The results page is shown
 - After test has finished, IRS Disc Test successful is printed out
 - A disc image must be visible (ignore the rings)
 - Check for error messages (in the Somaris log as well)
 - ⇒ In case of error, continue testing the data path.

DMS mechanics

If there are problems with the UHR mechanics, follow the advice provided in the error message.

In addition, you may perform the following tests:

Mechanics

- Check whether mechanics are operating smoothly, by moving UHR in and out by hand.
(The prerequisite is to switch on the service switch S1 at D514.)([1/ Fig. 62 / p. 91](#))
 - ⇒ If mechanics are not moving smoothly:
Check for parts blocking the UHR.

Light barrier

- If the problem is related to the light barrier:
 - Check its functionality using the service switch S1 at D514 and move the UHR in and out by pressing S3 (UHR out) or S4 (UHR in) ([1/ Fig. 62 / p. 91](#)).
Ensure UHR LED is activated at both mechanical end stops.
 - ⇒ If LED is not activated at end stop on one side:
Remove the protective cover, position the UHR by hand so that the flag is out of the light barrier.
Use, e.g., a feeler gauge to check whether the light barrier is properly working.
If light barrier is OK (electrically)
→ adjust the flag for the UHR light barrier.
 - If light barrier is not ok:
 1. Replace D 514
 2. Replace the light barrier (UHR limit switch card)
 - For adjustment of light barrier, refer to the document Replacement of Parts: Gantry.

UHR motor

- If the problem is related to the UHR Motor
 - Check its functionality using the service switch S1 at D514 and move UHR in and out by pressing S3 (UHR out) or S4 (UHR in) ([1/ Fig. 62 / p. 91](#)).
Observe whether the UHR is moving.
 - ⇒ If UHR is not moving:
Disconnect plug X5 at backplane of C-box D510 and measure at pin 2 and 3 the resistance of the UHR motor. 10 Ohms is standard.
Measure at various motor positions (after moving the UHR by hand):
 1. If resistance is out of tolerance, replace UHR motor.
 2. If resistance is in tolerance, replace D 514

- To check the output for the UHR motor at D 514, connect a DVM at X5 pins 2 and 3, and check if the output is activated when S3 or S4 are pressed (with service switch S1 on).

Because the output is modulated pulse wide, this is only a core test.

A precise measurement is only possible with an oscilloscope and differential inputs, or by using an isolation transformer. The pulse wide voltage should be approx. 20 V

⇒ If there is no output at D514:

Replace D514

- Check the time necessary to move from the outer endstop to the inner endstop:
4 - 6 seconds are expected.

⇒ If the time is significantly higher:

1. Check mechanics
2. Replace D514
3. Replace UHR motor

Adjustment

- To ensure the UHR is well adjusted in the phi-direction, perform a MTF measurement (Quality). If the results are in tolerance, the adjustment is OK.

If the results are not in tolerance, perform the COMB/ALI adjustment.

Other sources of problems

To troubleshoot image quality related problems, especially weak rings, be sure to check the event log as well for environmental problems and error messages concerning the following parts.

Power supply

The power supply for the DMS is the DPU. To check the supply, perform a measurement at the front side of the DPU or at the left motherboard by the DMS. (Fig. 48 / p. 57)

- To check the DPU manually, evaluate the clearly marked test points at the DPU: (2/ Fig. 62 / p. 91)

DPU lable	3P3_C	5P75_C	5N75_c	24V_Hfan
Voltage (V)	3.4	5.75	- 5.75	24.0
Tolerance	+/- 5%	+/- 4%	+/- 4%	+/- 5%
Noise (mV)	< 20	< 20	< 20	< 240

- To check the voltages at the **left DMS backplane** (LMB), evaluate the following test points:

Test point	Ground	Description
TP 3	TP 8 (DGND)	+ 2.5 V output from VR1
TP 4	TP 8 (DGND)	+ 2.5 V output from VR2
TP 5	TP 8 (DGND)	+ 2.5 V output from VR3
TP 10	TP 9 (AGND)	- 5.75 V after filter and ripple attenuator
TP 11	TP 9 (AGND)	+ 5.75 V after filter and ripple attenuator
TP 12	TP 8 (DGND)	+ 3.3 V for RCOM, after filter
TP 13 and TP 2	TP 8 (DGND)	+3.3 V for MB and ADMs, after filter
TP 14	TP 15 (24V_Return)	24 V for left temperature control, after filter
TP 6	TP 8 (DGND)	5 V output from RCOM

- To check the voltages at the **right DMS backplane** (RMB), evaluate the following test points:

Test point	Ground	Description
TP 10	TP 6 (DGND)	+ 2.5 V output from VR1
TP 11	TP 6 (DGND)	+ 2.5 V output from VR2
TP 12	TP 6 (DGND)	+ 2.5 V output from VR3

Test point	Ground	Description
TP 5	TP 7 (AGND)	- 5.75 V after ripple attenuator
TP 4	TP 7 (AGND)	+ 5.75 V after ripple attenuator
TP 9	TP 6 (DGND)	+ 3.3 V for MB and ADMs after filter
TP 2	TP 3 (24V_Return)	+ 24 V for right temperature control, after filter

ADU

The **ADU** (D513) consists of two controllers:([Fig. 61 / p. 90](#))

- **DOM** which performs the dose modulation
- **ACU** which controls data acquisition

Checking the DOM section

1. **Options > Local Service > Test Tools > Controller > DOM**
2. Select the radio button
Selftest or
Dose signal test
3. Press **Go**
 - ⇒ If the test was successful, **Test passed** is output
 - ⇒ Otherwise, check Somaris log for errors

Checking the ACU section

1. **Options > Local Service > Test Tools > Controller > ACU**
2. Select the radio button **DAS Communication signals**
3. Press **Go**
 - ⇒ If the test was successful, **Test passed** is output
 - ⇒ Otherwise, check Somaris log for errors

Environmental Condition

Environmental conditions are a possible reason only in the case of weak ring artifacts.

Check the Somaris log for errors and warnings .

Gantry

Inside the gantry, the temperature and the humidity is regulated.

To read out the actual values for temperature and humidity, use the control window.

The output of the table may be selected for decimal or hexadecimal.

- ⇒ To read out the actual **temperature** of the gantry, perform the following:
 1. **Options > Local Service > Control > Table load/modify**
 2. Select radio button **CONTROL SYSTEM** under TABLE SOURCE
 3. In the list box TABLE NAMES select **T00**
 4. Press **Go**

The parameter set for the table T00 is shown.
 Use the slider to move to the desired information
 - gantry temperature [grad C]

- ⇒ To read out the actual **humidity** of the gantry, perform the following:
 1. **Options > Local Service > Control > Table load/modify**
 2. Select radio button **CONTROL SYSTEM** under TABLE SOURCE
 3. In the list box TABLE NAMES select **T00**
 4. Press **Go**

The parameter set for the table T00 is shown.
 Use the slider to move to the desired information
 - gantry humidity [%]

DMS

- Environmental conditions for the DMS are:
 - **Temperature :**
 - ⇒ Check the Somaris log for errors and warnings concerning temperature problems.
 - ⇒ Ensure that the DMS is powered on over night.
 - ⇒ If raw data is available for the images the customer is complaining about, use Analyze to read out the temperature from the header data.
 This indicates the temperature at the time of data acquisition.
 - ⇒ To read out the actual temperatures for the DMS, perform the following:
 1. **Options > Local Service > Control > Table load/modify**
 2. Select radio button **CONTROL SYSTEM** under TABLE SOURCE
 3. In the list box TABLE NAMES select **TCC**
 4. Press **Go**

The parameter set for the table TCC is shown.
 Use the slider to move to the desired information
 - left DMS temp (grad)
 - right DMS temp (grad)
 Hint: ensure that the radio button Dec is activated for decimal display.
 A difference between left and right temperatures is normal.
 - **Humidity :**
 - ⇒ Check the Somaris log for errors and warnings concerning humidity problems.

Troubleshooting the data path

General

NOTE

Follow the C-MOS guidelines when handling the DMS or the data path.

- It is recommended a spare be on hand for W55 (Fig. 61 / p. 90) (30 m optical fiber).
- Always start troubleshooting by analyzing the error log.

Tab. 11 Overview of the gantry angle in relation to the tube position

Gantry angle (deg)	0	30	60	90	120	150	180	210	240	270
Tube position (x o'clock) from the front	3	4	5	6	7	8	9	10	11	12
Tube position from the front	east (right)			south (bottom)			west (left)			north (up)

Block diagram

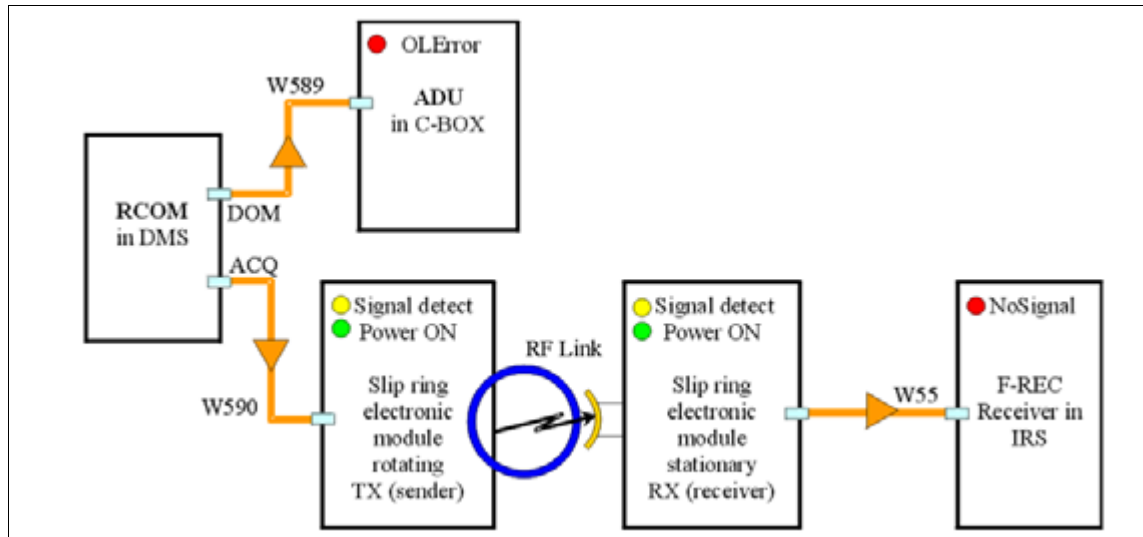


Fig. 61: Block diagram of data path

Description of parts

- **RCOM**

- ADU

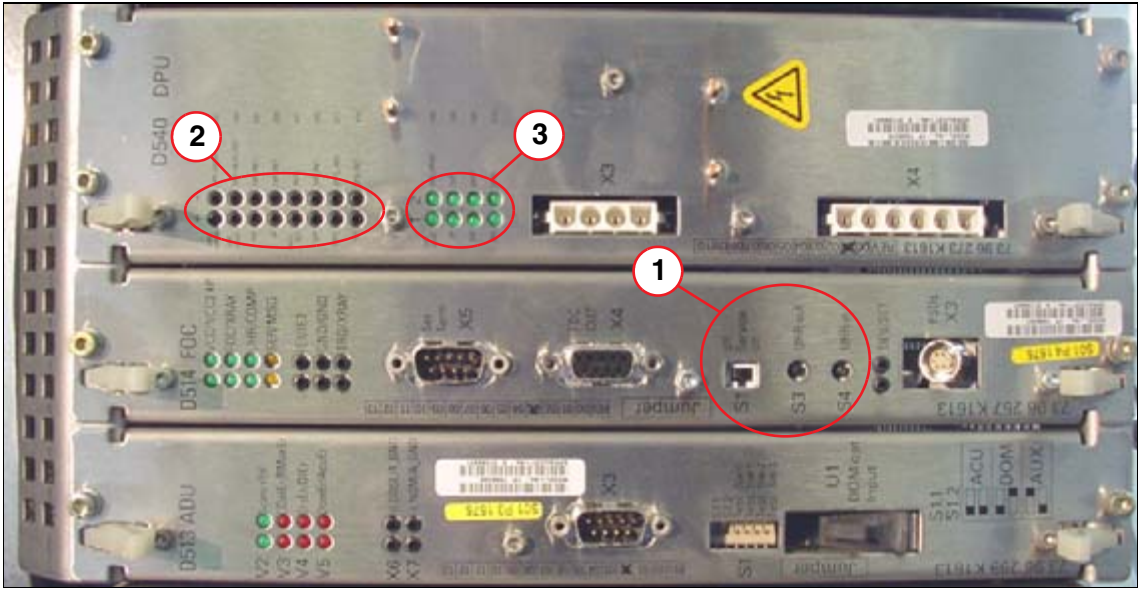


Fig. 62: C-box

Pos. 1 Service switches for the UHR

Pos. 2 Test points for the DPU output voltages

Pos. 3 LED's for the DPU output voltages

V2	□ □	Scan / +5V
V3	□ □	CDatEr / RMuxEr
V4	□ □	SrsEr / OLEr
V5	□ □	DomEr / AcuEr

Fig. 63: LED's of the ADU

- Transmitter

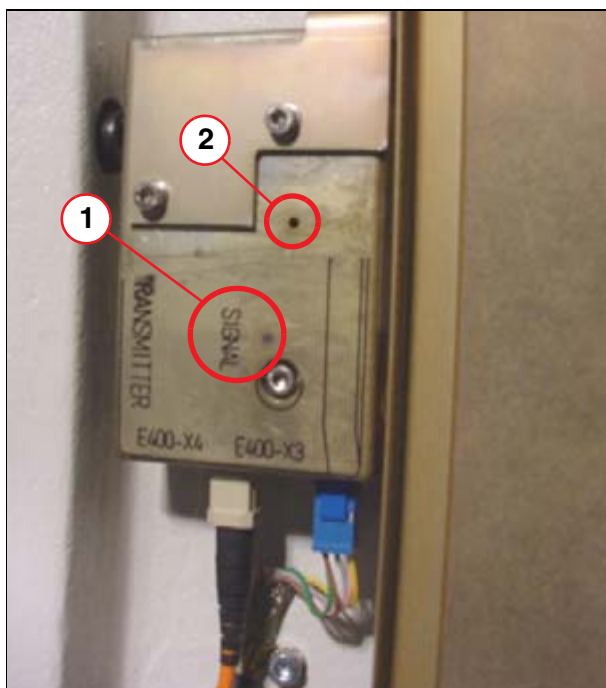


Fig. 64: Slip ring transmitter

Pos. 1 LED SIGNAL

Pos. 2 LED POWER

- **Receiver (slip ring)**

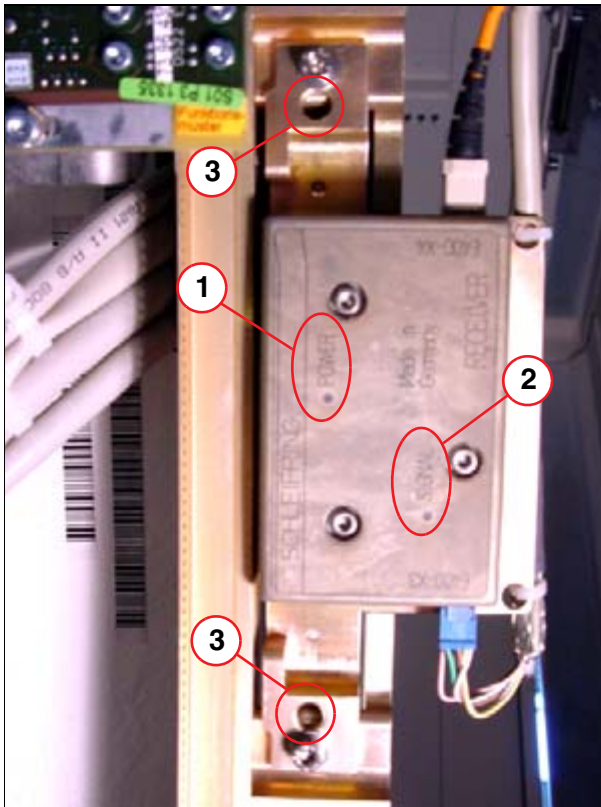


Fig. 65: Slip ring receiver

- Pos. 1 LED POWER
Pos. 2 LED SIGNAL
Pos. 3 Marker for Z-adjustment of slip ring receiver

- **F-REC (IRS)**

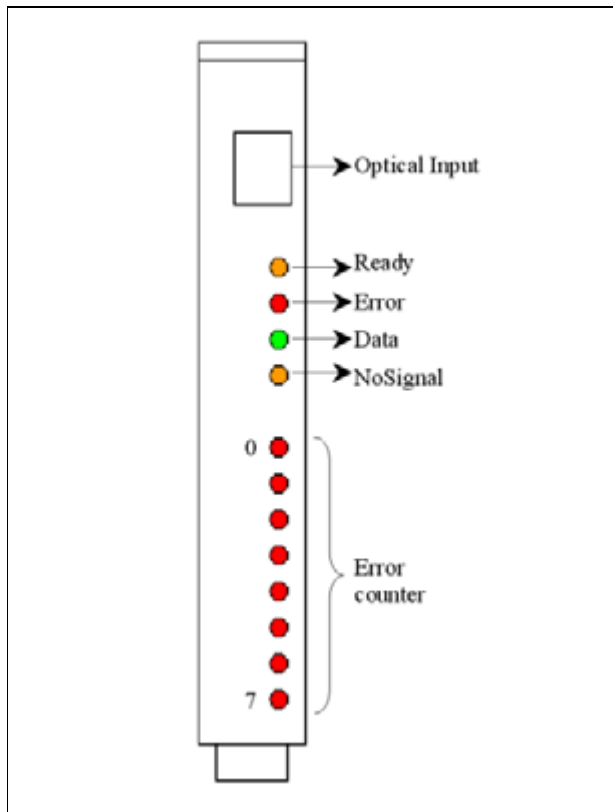


Fig. 66: F-REC (D23) in IRS acquisition PC

Tools for handling problems

There are many ways to check for transitory errors in the data path.

Log book

At end of scan, the IRS software counts the data errors and issues a warning in the error log.

Also, the signal detections at ADU, transmitter, receiver and F-REC are monitored at ADU and MCU. A malfunction will generate a warning in the error log.

Error counter at F-REC

At the F-REC of the IRS, a CRC check is performed. The synchronisation of the data link is also monitored at the F-REC.

At the F-REC (D23 in acquisition PC) you will find 8 red LED's labeled ERR0 through ERR7. (Fig. 66 / p. 94)

This counter is only active during a scan and will be reset at the next mode load.

In order to use this counter as an error detector:

- Load a mode and perform a scan
- Manipulate the data path (e.g. replace hardware or swap cable)

- Load a mode again and perform another scan
 - ⇒ Compare the error counts before and after
 - ⇒ Under normal circumstances, the error counter doesn't count at all. (LED's ERR0 - ERR7 are off)

Data link test

This test checks the physical link level of the data link. [\(Data link / p. 81\)](#)

A detailed description is located in Tools for troubleshooting.

Checking the data path

If the problem is permanent start troubleshooting here.

After the quick check, a detailed description of how to identify the source of a link problem follows.

Quick check

A quick check may be performed by checking the following status LED's: [\(Fig. 61 / p. 90\)](#)

- Check the **OLError** LED at the ADU (red LED) [\(Fig. 63 / p. 91\)](#)
 - If LED is off
 - ⇒ Link RCOM → ADU is working
- Check the **Signal detect** LED at the slip ring transmitter module (yellow LED) [\(1/ Fig. 64 / p. 92\)](#)
 - If LED is on
 - ⇒ Link RCOM → transmitter is working
- Check the **Signal detect** LED at the slip ring receiver module (yellow LED) [\(2/ Fig. 65 / p. 93\)](#)
 - If LED is on
 - ⇒ Link transmitter → receiver is working (slip ring system)
- Check the **No Signal** LED at the F-REC in the IRS (red LED) [\(Fig. 66 / p. 94\)](#)
 - If LED is off
 - ⇒ Link receiver → F-REC is working

RCOM - ADU

- Check the OLError LED at the ADU front plate. [\(Fig. 63 / p. 91\)](#)
 - In case of error, LED will be on or blinking:
 - ⇒ Possible causes:
 - RCOM's DOM output is defective
 - Optical fiber cable W589 is defective
 - ADU is defective
 - DOM FW on ACU not loaded

- Checking the DOM output of RCOM (Fig. 61 / p. 90)
 - Use the ACQ output instead of the DOM output.
If the OLError LED turns off using the ACQ output
 - ⇒ RCOM's DOM output is defective.
 - A second option for checking the RCOM's DOM output involves connecting a spare optical fiber (known to be good) from RCOM's DOM output to the slip ring's transmitter module.
 - ⇒ If the signal detect LED (2/Fig. 65 / p. 93) switches on at the slip ring receiver, RCOM's DOM output is OK.
 - A third option for checking the DOM's output is to connect a spare optical fiber (known to be good) from RCOM's DOM output to the F-REC in the IRS.
 - ⇒ If the NoSignal LED (Fig. 66 / p. 94) switches off when connecting, RCOM's DOM output is OK.
 - If all three tests fail:
 - ⇒ Replace RCOM
- Checking W589 (Fig. 48 / p. 57)

Replace W589 with an optical fiber known to be good

 - If the OLError LED at the ADU (Fig. 63 / p. 91) switches off using the spare cable
 - ⇒ Replace W589
 - If the OLError LED at the ADU (Fig. 63 / p. 91) stays on using the spare cable (and the RCOM's DOM output was checked):
 - ⇒ 1. Reload DOM firmware
2. Replace ADU D513

RCOM - F-REC

- Check the NoSignal LED at the F-REC in the IRS (Fig. 66 / p. 94)

In case of error LED is on or blinking:

- ⇒ Possible causes (Fig. 48 / p. 57)
 - RCOM ACQ output defective
 - W590 defective
 - Slip ring transmitter defective
 - Slip ring system defective (mechanical reasons)
 - Slip ring receiver defective
 - W55 defective
 - F-REC in IRS defective

As quick check, connect the DOM output of RCOM directly to the F-REC input, using a spare optical cable.

If the NoSignal LED at the F-REC is still on

- ⇒ Problem is at the RCOM or F-REC

If NoSignal LED at the F-REC turns off

- ⇒ Problem is located at (Fig. 61 / p. 90)
W590, transmitter, slip ring system, receiver and W55.
See the subsequent chapters to identify the source of the problem.

RCOM - Transmitter

- Check the Signal Detect LED at the transmitter module (1/Fig. 64 / p. 92).

In case of error, LED is off: (Fig. 48 / p. 57)

- ⇒ Possible causes
 - RCOM's ACQ output is defective
 - W590 is defective
 - Transmitter has no power
 - Transmitter is defective

- Checking the RCOM's ACQ output (Fig. 61 / p. 90)

- Use the RCOM's DOM output instead of ACQ output.

The prerequisite for this test is that the RCOM's DOM output is working fine.

If the Signal Detect LED at the transmitter (1/Fig. 64 / p. 92) switches on now:

- ⇒ RCOM's ACQ output is defective.

- A second option for checking the DOM's ACQ output is to use a spare cable (known to be good) and make a connection from RCOM's ACQ output to the ADU.

- ⇒ If the OLEError LED (Fig. 63 / p. 91) switches off, the RCOM's ACQ output is OK.

- A third option for checking the RCOM's ACQ output is to use a spare cable (known to be good) and make a connection from RCOM's ACQ output to the F-REC in the IRS.

- ⇒ If the NoSignal LED at the F_REC (Fig. 66 / p. 94) switches off, then RCOM's ACQ output is OK.

- If all three tests fail:

- ⇒ Replace RCOM

- Checking W590 (Fig. 61 / p. 90)

If the RCOM's ACQ output was checked and the LED Signal Detect at transmitter module is not activated at all, use a spare cable instead of W590.

If the Signal Detect LED at the transmitter module (1/Fig. 64 / p. 92) now switches on:

- ⇒ Replace W590

- Check the PowerON LED at the transmitter (2/Fig. 64 / p. 92)

If LED is off

- ⇒ Possible causes
 - DPU
 - C-box backplane
 - Power cable W591

- Checking the transmitter (Fig. 61 / p. 90)

There is no explicit test for the transmitter.

If the RCOM's ACQ output, cable W590 and the power for the transmitter was checked and all tests were passed:

- ⇒ Replace transmitter

Transmitter - Receiver

- Check the Signal Detect LED at the slip ring receiver module ([2/Fig. 65 / p. 93](#))

In case of error LED is off

- ⇒ Possible causes:
 - Transmitter (RF output) defective
 - slip ring system defective
 - Receiver has no power
 - Receiver defective

- Checking the transmitter

There is no test to check the RF output.

- To check the mechanics of the slip ring system, refer to ([Mechanical slip ring system / p. 104](#))
- Check the PowerOn LED at the receiver module ([1/Fig. 65 / p. 93](#))

In case power LED is off

- ⇒ Possible causes:
 - MCU is defective
 - Power cable is defective W375
 - Receiver is defective

Receiver - F-REC

- Check the NoSignal LED at the F-REC in the IRS ([Fig. 66 / p. 94](#))

In case of error, LED will be on

- ⇒ Possible cause:
 - Power for slip ring receiver is missing
 - Receiver defective
 - W55 defective
 - F-REC in IRS defective

- Check the PowerOn LED at the slip ring receiver ([1/Fig. 65 / p. 93](#))

In case power LED is off

- ⇒ Possible causes:
 - MCU is defective
 - Power cable is defective
 - Receiver module at slip ring is defective

- Checking receiver output ([Fig. 61 / p. 90](#))

The prerequisite is that both LED's at the receiver module are on. ([1,2/Fig. 65 / p. 93](#))

Remove W589 from the ADU (observe whether the OLError LED switches on when removing)

and connect a known for good spare cable from the output of the receiver module on slip ring to the input of ADU.

- ⇒ If the OLError LED at the ADU is turned off, output of receiver on the slip ring is ok.

- Checking W55 ([Fig. 61 / p. 90](#))

The prerequisite is that both LED's at receiver module are on. ([1,2/ Fig. 65 / p. 93](#))

Use a spare optical cable to replace W55.

⇒ If the NoSignal LED at the F-REC ([Fig. 66 / p. 94](#)) is still on, install a new W55.

- Checking the F-REC ([Fig. 66 / p. 94](#))

If all tests above have been performed unsuccessfully

⇒ Replace the F-REC in the IRS

Problem isolation

In case of sporadic problems in the data path, it is necessary to distinguish between different possible sources of problems.

If the data transmission is operating with transitory errors, scan will be aborted when errors accumulate above an acceptable threshold.

- The main tools for isolating the component causing the problem include:
 - Error counter at the F-REC ([Fig. 66 / p. 94](#))
 - Data link test ([Data link / p. 81](#))
 - In addition, always check the error log

Optical cable

The optical fiber between the RCOM and slip ring or the slip ring and F-REC may be partially broken. One reason for malfunction may be contamination by carbon dust (connections inside the gantry).

So the first step in isolating the optical cable is to clean the optical connectors.

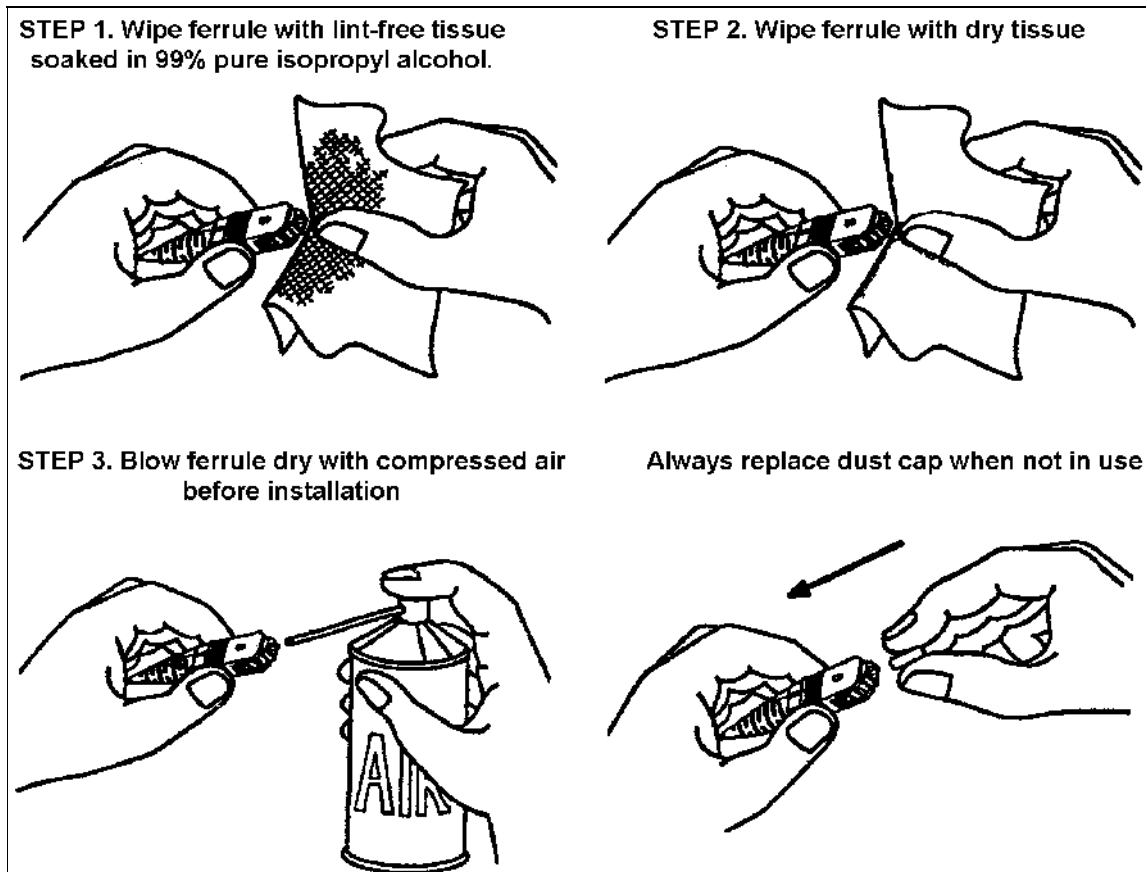


Fig. 67: Cleaning optical connectors

1. Perform a test scan and make a note of the error counter at the F-REC.
 2. Use 99% pure isopropyl alcohol and a lint free tissue to clean the coupling.
 3. Wipe coupling with dry tissue.
 4. Use compressed air to dry coupling before installation.
 5. Perform a test scan and compare the error counter at F-REC with step1.
- ⇒ In case errors are not reduced significantly, replace optical fiber.

Rotation

The frequency inverter that drives the rotation motor generates electrical noise between rotational and static grounds.

This noise is injected in the RF link between transmitter and receiver electronics.

To check if the problem is related to rotation, use the data link test: [\(Data link / p. 81\)](#)

1. Perform a test scan using the predefined **Rot20.dat** mode (Auto test mode) in the data link test, and note the error counter of the result page of this test.
2. Perform a second test scan using the predefined **Static20.dat** mode (Auto test mode) and compare the error counter of the result page with step 1.

⇒ In case errors are reduced significantly, replace frequency inverter.

Hint: The frequency inverter for the rotation also generates interference in the static mode. Therefore, only the data link test is useful for testing.

X-ray

The E-BOX generates HF equalization currents through the PE brush connection which connects the rotational with the static ground. These equalization currents inject noise in the RF link between the transmitter and receiver electronics.

Before checking the relation to the X-ray, check that there is no relation to rotation!

To check whether the problem is related to the X-ray, proceed as follows:

1. Perform a test scan in Rot-Mode and note the error counter at the F-REC.
2. Perform the same test scan without X-ray and compare the error counter at the F-REC with step 1.

If there is a significantly higher error count with the X-ray on:

⇒ Check:

1. **UDC line** for an unsatisfactory connection

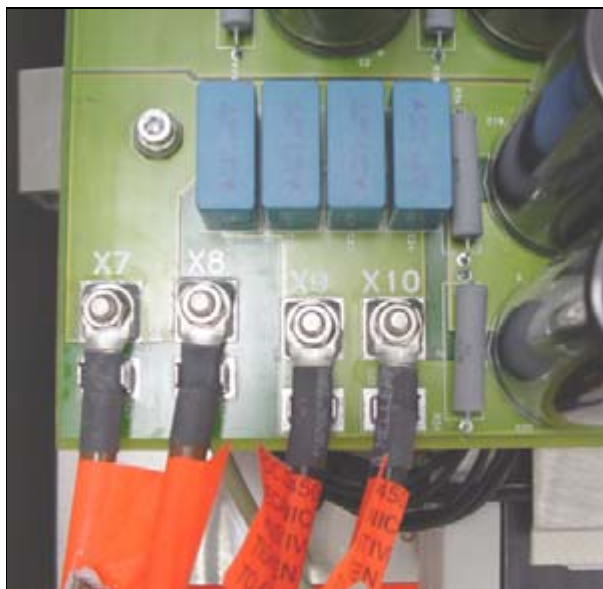


Fig. 68: UDC connection at PDC

- In PDC at board D450: X7, X8, X9, X10 tighten with a torque of 4.8 Nm.
- Installation sequence is: connection block, cable, washer, nut.

- In the right gantry stand: terminal block X303 (all 8 screws).

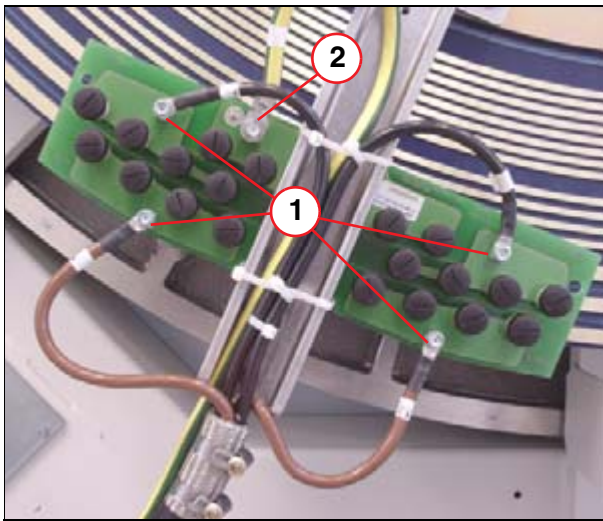


Fig. 69: UDC and ground brush assembly

Pos. 1 UDC 6 Nm

Pos. 2 Ground connection 8 Nm

- At back side of gantry: X382 brush holder. Use a torque of 6 Nm.



Fig. 70: UDC connection at generator electronics

- At E-Box D440 X1 and X2. Correct sequence of assembling is:
- connection block, cable, washer, nut. Use a torque of 4.8 Nm.

2. Power line from **E-Box to H1** . Correct sequence of assembly is:

- connection block, cable, washer, nut. Use a torque of 4.8 Nm.

3. All **ground connections**

If 1-3 was without success

⇒ Replace E-Box

The data link test and its predefined modes may also be used. ([Data link / p. 81](#))

Ground connection

The slip ring is extremely sensitive to HF common mode noise between the rotational and static grounds.

- ⇒ Check ground connection between gantry and PDC
- ⇒ Check ground connection between stator and rotor.
 - Screw on the brush holder. Use a torque of 8 Nm.
 - Carbon brushes (length ok and freely moving)
 - Slip ring surface. Inspect the surface visually for mechanical damage.
- ⇒ Ground connection at E-Box X603. Check with a torque of 4.8 Nm



Fig. 71: Ground connections at rotor

Termination of transmitter's antenna

The output of the transmitter module is connected to two semicircular micro-strip structures. Each micro-strip structure consists of two 180 degree segments connected in parallel to the transmitter's output. Each segment is terminated with a 36 ohm resistor.

If a resistor is broken due to mechanical stress, we expect a malfunction when the receiving antenna is positioned in the range of the 180 degrees with broken resistor.

NOTE

Follow the C-MOS guidelines when handling transmitter or receiver.

- Measurement of the resistor:
 - Remove the fiber and W591 from the transmitter module (Fig. 61 / p. 90)
 - Remove the transmitter module to access the measurement pins (Fig. 72 / p. 104)
 - Perform measurement in accordance with the table

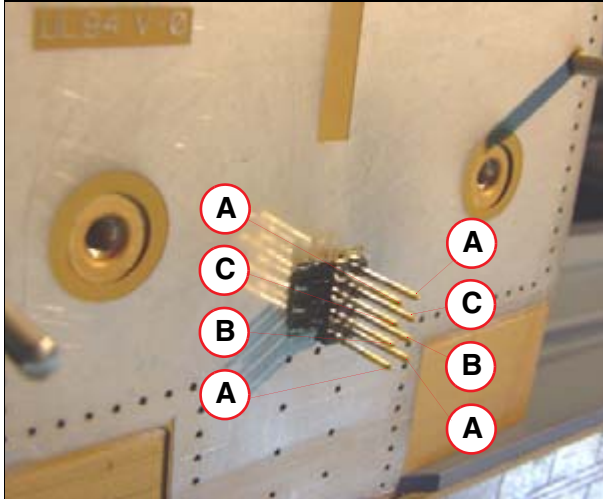


Fig. 72: Measurement of termination resistors

Measurement point	Expected value	Tolerance
B to A	18 Ohm	1 Ohm
B to C	36 Ohm	2 Ohm
C to A	18 Ohm	1 Ohm
1, 4, 5 or 8 to GND	0 Ohm	1 Ohm
2, 3, 6 or 7 to GND	18 Ohm	1 Ohm
2 or 3 to 6 or 7	36 Ohm	2 Ohm

Mechanical slip ring system

A visual inspection of the structure of the sending antenna is only possible after removing the receiver module and the receiving antenna. This provides access to an area the size of the receiver module. To access the whole antenna structure, rotate the gantry by hand. There should be no connection between the pads.

Mainly, there are three mechanical problems possible

- Radial

To check radial misalignment, use a feeler gauge and check the gap between the light barrier at DLP D522 and the ring. Move gantry by hand to check the gap for all 360 degrees.

A deviation of maximal +/- 0.3 mm is allowed in the radial direction.

- ⇒ If the result is out of tolerance, check whether slip ring is mounted centrally by inserting 8 mm locating pins. Check all four holes and adjust, if necessary, by loosening and fixing the corresponding screws.
(In case of adjustment, 4 pins are needed.)

- Axial

To check axial misalignment, use a feeler gauge and check the gap between the light barrier at DLP D522 and the slotted ring. Move gantry by hand to check the gap for all 360 degrees.

Another possibility is to observe the adjustment marks at the receiving antenna while rotating the gantry by hand.

There is no adjustment possible.

- Z- adjustment of receiver antenna.

The receiver antenna is adjustable in the z-direction and in parallel.

(For adjustment, refer to Replacement of Parts: Stationary Gantry.)

Jitter at data link

If the jitter, generated in RCOM or along the data path exceeds the jitter tolerance at F-REC or the jitter tolerance of F-REC is not within specification, transitory errors may occur. To check the jitter, special measurement equipment is required.

- ⇒ Otherwise (if no eye diagram analyzer is available) replace:
 - RCOM
 - F-REC

Handling of defective channels

Scope

The software automatically detects bad channels.

In order to find a bad channel, the Defective Channels procedure performs 4 scans without and with radiation.

From the result of these 4 measurements, the channels are determined to be good or bad.

Defective channels are handled in 3 different ways:

- During each CheckUp the defective channel procedure is performed.
- Manual start of defective channel procedure from tune-up platform.
- Manual entry of a defective channel in the table.

Manual entry in the tune-up table is only necessary in case of a very sporadic problem.

The inner center of the detector and the first and last slot (air slice normalization) have to be without defective channels: slot 1, slot 21, slot 22 and slot 42.

General Information

Information for software VA50 installed

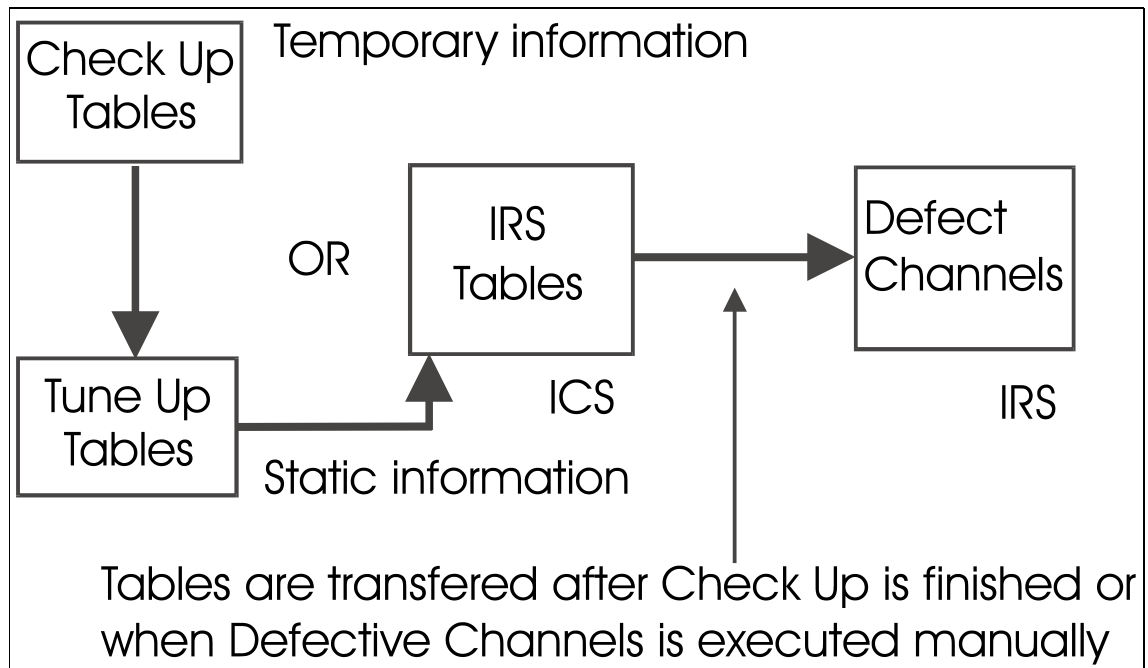


Fig. 73: Overview (VA50)

1. Automatic defective channel detection **does not** delete previously recognized defective channels from the defective channel table while creating and saving the new defective channels IRS tables. For this reason, it is **not** necessary to create new tune-up tables after automatic channel correction has been performed.
2. You may patch a channel manually, if necessary, as well as remove the manually patched channel again. A partial tune-up is not necessary in this case.
3. After certain part replacements (e.g. module) all channels marked defective and related to these parts have to be removed manually (see guided tours). Subsequently, automatic channel correction has to be performed (to create current defective channel status). A partial tune-up as described in the guided tours follows.
4. Manually patched channels are not automatically deleted from the program. These channels have to be deleted manually.

Information for software VA60 installed

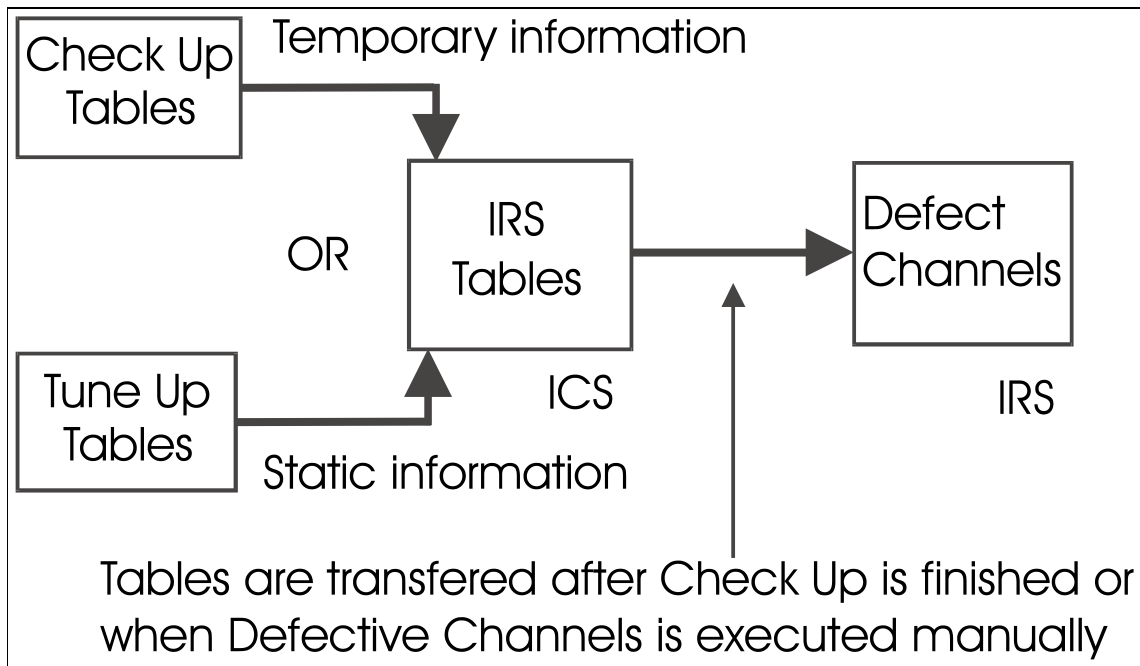


Fig. 74: Overview

1. Automatic defective channel detection does not delete previously recognized defective channels from the defective channel table while creating and saving the new defective channels IRS tables.

Manually entered defect entries remain in the tune-up tables.

For this reason, it is absolutely **necessary** to create new tune-up tables after automatic channel correction has been performed.

2. You may patch a channel manually if necessary as well as remove the manually patched channel again. A partial tune-up is not necessary in this case.

3. After certain part replacements (e.g. module) all channels marked defective and related to these parts have to be removed manually (see guided tours). Subsequently, automatic channel correction has to be performed (to create current defective channel status). A partial tune-up as described in the guided tours follows.

4. Manually patched channels are not automatically deleted from the tune-up tables. These channels have to be deleted manually.

Overview channel translation

2x600	12x750	16x750	16x1500	12x1500	2x5000	8x1500	2x1000
			1				
			2				
			3	1			
			4	2			
		1	5	3		1	
	1	3	6	4	1	2	
	2	4	7	5		3	
	3	5	8	6		4	1
1	4	6	9	7		5	2
2	5	7	10	8		6	
	6	8	11	9		7	
	7	9	12	10		8	
	8	10	13	11			
	9	11	14	12			
	10	12	15				
	11	13	16				
	12	14					
		15					
		16					

Description of channel number translation between the tables
Hint: mode 2x5000 is constructed from mode 8x1500

Fig. 75: Channel translation (Sensation 16)

2x600	8x750	10x750	16x1500 / 2x12000	12x1500 / 6x3000	10x1500	8x1500 / 2x5000	2x1000
			1	-	-	-	-
			2	-	-	-	-
			3	1	-	-	-
			4	2	1	-	-
-	-	-	5	3	2	1	-
-	-	-	6	4	3	2	-
-	1	2	7	5	4	3	1
-	2	3	8	6	5	4	2
-	3	4	9	7	6	5	-
-	4	5	10	8	7	6	-
-	5	6	11	9	8	7	-
-	6	7	12	10	9	8	-
-	7	8	13	11	10	-	-
-	8	9	14	12	-	-	-
-	9	10	15	-	-	-	-
-	10	-	16	-	-	-	-

Description of channel number translation between the tables
Hint:
Mode 2x12000 is constructed from mode 16x1500.
Mode 6x3000 is constructed from mode 12x1500.
Mode 2x5000 is constructed from mode 8x1500.

Fig. 76: Channel translation (Sensation 10)

Translation of slice number for modes

Slice number 'z' in mode 16x750	correspondents to slice number 'z' in mode 16x1500	Slice number 'z' in mode 10x750	correspondents to slice number 'z' in mode 16x1500 = 2x12000
-	1	-	1
-	2	-	2
-	3	-	3
-	4	-	4
1 and 2	5	-	5
3 and 4	6	1	6
5 and 6	7	2 and 3	7
7 and 8	8	4 and 5	8
9 and 10	9	6 and 7	9
11 and 12	10	8 and 9	10
13 and 14	11	10	11
15 and 16	12	-	12
-	13	-	13
-	14	-	14
-	15	-	15
-	16	-	16

Fig. 77: Translation of slice number for modes

Information about the 3 defective channel table groups

- **CheckUp** tables

- Generated automatically at every checkup.
- Table starts blank at every checkup (temporary table).
- Table contains information about the modes used (software dependant)
([Tab. 10 / p. 58](#)) .

- **TuneUp** tables

- Software VA50 tables maintain old entries (static tables).
Software VA60 tables maintain defect entries until next automatic defective channel correction is performed via tune-up platform.
- Table contains information about the modes used (software dependant)
([Tab. 10 / p. 58](#))
- Generated while performing a tune-up
- Only these tables may be edited.

- **IRS** tables

- Tables are derived by merging checkup and tune-up tables. During tune-up they are derived only from tune-up tables.
- Table contains information about the modes used (software dependant)
([Tab. 10 / p. 58](#)) .
- The tables are located on the ICS and are transferred at every checkup to the IRS.

Important:

- The information about the defects in the IRS table is derived as follows:
 - for **16 x 750 or 10*750** : all information is copied from the tune-up tables.
 - for **16 x 1500**: all information is copied from the tune-up tables.
 - for **all other slice combinations** the information is derived from these tables (Fig. 75 / p. 109) ff.

How to find

Options > Local Service > TuneUp > Table Generation > Defective Channels

How to use

The direction for counting the channels is in accordance with the Siemens convention (clockwise).

The direction of counting the slices is from head to feet.

Detector segment corresponds with detector slot.

After starting the program Defective Channels a window pops up, enabling:

Editing the correction tables

1. Click **Correction Tables** , select a table and wait:
 - **CheckUp** (read only)
 - **TuneUp**
 - **IRS** (read only)
2. Click the button for the desired detector segment (slots 1- 42)
Hint: A segment containing bad elements is marked in red.
3. After a while, the user interface will show the corresponding channel / slice matrix.
 - ⇒ Now select the desired channel and slice. Select the element with the left mouse, and insert a 1 to mark the element as defective.
 - ⇒ To remove a bad channel entry change the 1 to 0.

Starting automatic detection

Remove all absorbers!

1. Ensure that the checkbox **Automatic Detection after Editing** has been checked.
2. Click **Go**
 - ⇒ Measurements are performed automatically.
 - ⇒ To repeat an automatic detection in the case of unstable results, leave the upright checkbox checked and click Go.
3. Unmark the checkbox **Automatic Detection after Editing** .

4. Click **Go**
 - ⇒ You will be asked to store or reject the results.
5. Click **Go**
 - ⇒ The IRS tables are transferred to the IRS
 - ⇒ If you click **Cancel**, no modifications to the tune-up and IRS tables will be stored.

Viewing the detector statistics

Select **Detector Statistics**

- ⇒ The number of defective elements are shown for the various collimations (tune-up and checkup).

Using Reports to view details of the Defective Channels Correction

1. Open an additional window and choose **Reports** .
2. From the menu bar TuneUp/QA/Tests choose:
TuneUp > Table Generation > Def. Channels
3. Double click Report to view the results.
 - ⇒ **Detector Statistics** shows all currently defective channels.
 - Table Editing History** shows the history of the detection.
 - **Added by Automatic Detection** and
 - **Manually Added/Removed**
 - Table Conditions** show all affected slice combinations within the IRS table.
 - Affected Modules** shows the segment number (slot) and, in addition, whenever replacements are necessary (change = has to be replaced).
 - Comments** shows additional information, in case a comment was added by the customer service engineer.

Appendix

Channel list

Note that Siemens counts the slots clockwise (e.g. output of DMS tests), while they are marked counterclockwise on the cradle (Phillips slot count order).

The inner six slots (slots 19 - 24) have to be replaced simultaneously.

Slots 1 and 42 are used for air slice normalization and are not permitted to contain defective detector modules.

Channels Siemens	Slot count Siemens	Slot count Phillips	Channels Siemens	Slot count Siemens	Slot count Phillips
1 - 16	1	42	337 - 352	22	21
17 - 32	2	41	353 - 368	23	20
33 - 48	3	40	369 - 384	24	19
49 - 64	4	39	385 - 400	25	18
65 - 80	5	38	401 - 416	26	17
81 - 96	6	37	417 - 432	27	16
97 - 112	7	36	433 - 448	28	15
113 - 128	8	35	449 - 464	29	14
129 - 144	9	34	465 - 480	30	13
145 - 160	10	33	481 - 496	31	12
161 - 176	11	32	497 - 512	32	11
177 - 192	12	31	513 - 528	33	10
193 - 208	13	30	529 - 544	34	9
209 - 224	14	29	545 - 560	35	8
225 - 240	15	28	561 - 576	36	7
241 - 256	16	27	577 - 592	37	6
257 - 272	17	26	593 - 608	38	5
273 - 288	18	25	609 - 624	39	4
289 - 304	19	24	625 - 640	40	3
305 - 320	20	23	641 - 656	41	2
321 - 336	21	22	657 - 672	42	1

C-box block diagram

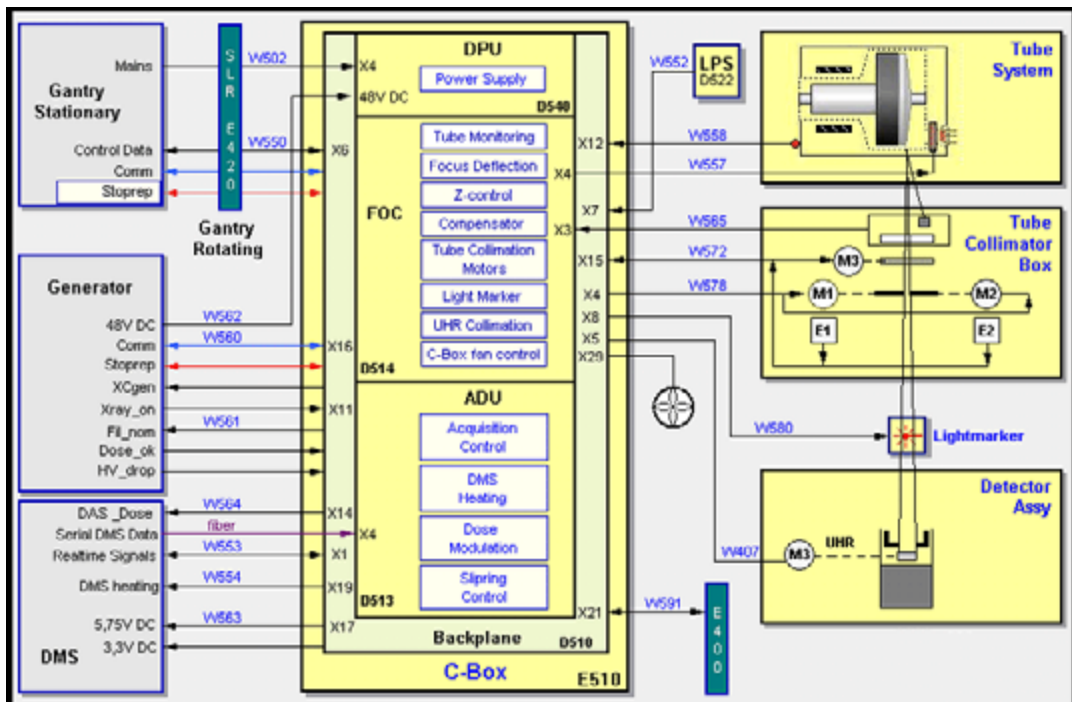


Fig. 78: Block diagram of C-box AKRON tube

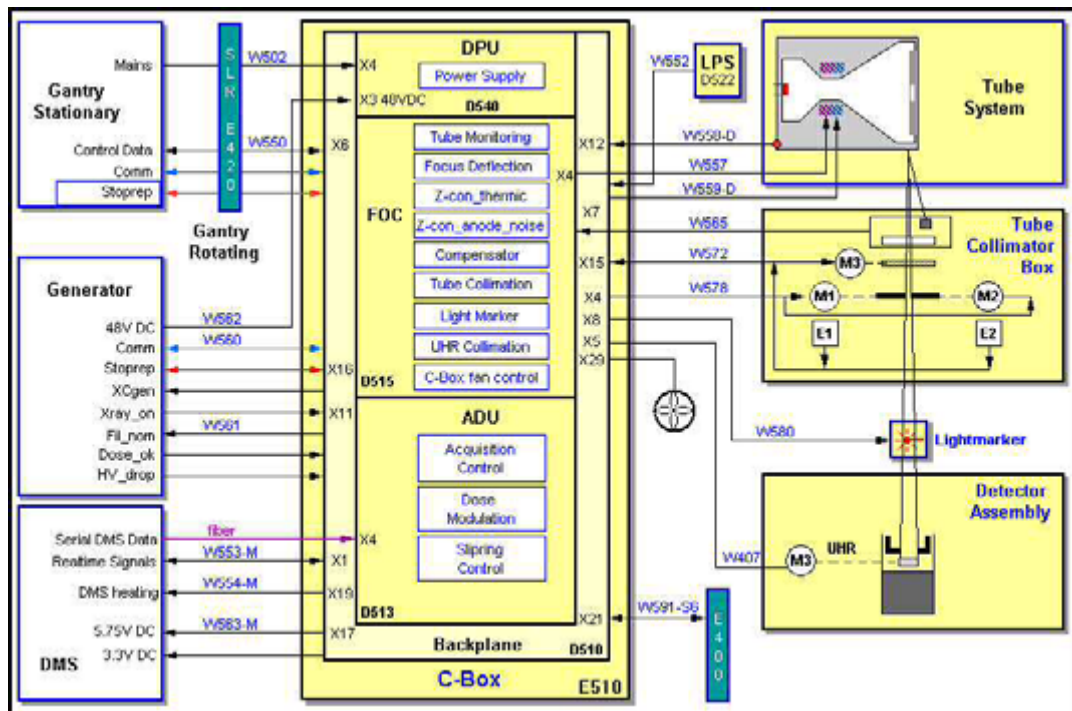


Fig. 79: C-Box (STRATON) block diagram

⇒ Block Diagrams are for illustrative purposes only. Refer to the functional description for current and type-specific information.

ADU D513

The ADU handles the following functions

- AP pulses and tube position.
- Dose modulation.
- Communication to MCU.
- Communication to DMS.
- Monitoring DMS data transmission over slip rings.

Refer to [\(Fig. 78 / p. 114\)](#) for functions.

Self tests

The following self tests are performed by the ADU firmware when power is switched on:

- Communication to MCU
- Communication to DMS

LED switches and test points

The table shows signals that can be evaluated on D513 ADU

Tab. 12 ADU LEDs

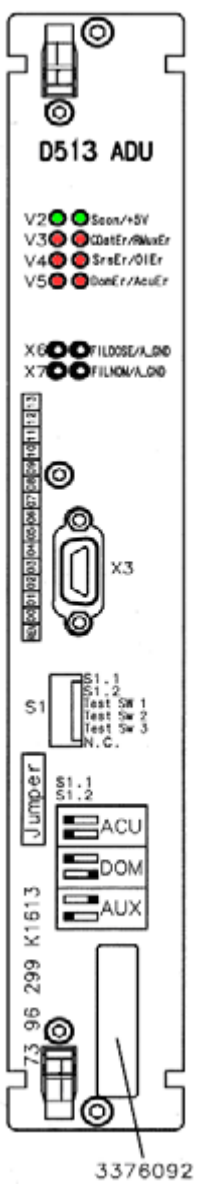
	LED	Color	Description
	V2	Green	+5V is available on the D513
	V2	Green	Scan Active
	V3	red	Control_dat Link error
	V3	red	Resi-mux Link error
	V4	red	SRS optical link error
	V4	red	DAS optical link error
	V5	red	DOM error
	V5	red	ACU error
	Test point	Label	Description
	X6	Fil_Dose	Actual dose from PSD
	X7	Nom_dose	Nominal dose
	X3	X3	Service terminal connection
	Service Switch		Description
	S1.1	Select output on X3 (service terminal connection) according to table on front plate of ADU board.	
	S1.2		
	Test Sw-1	Reserved	
	Test Sw-2	Reserved	
	Test Sw-3	Reserved	

Fig. 80: ADU board

FOC D514

The FOC board has the following functions:

- Beam deflection, Phi_Z_Mon amplifier
- Controlling tubeside collimator and compensator
- Controlling the UHR positioning (if used)
- Tube temperature sensors
- Controlling rotating light markers
- Controlling the C_Box fan (if used)

Refer to (Fig. 78 / p. 114) for functions.

Self tests

The following self tests are performed by the firmware when power is switched on:

- Voltage monitoring of the DPU power supply
- Mechanical movements of the tube collimator, UHR collimator, and compensator
- Checking the tube sensors (temperature, pressure)

LED service switches and test points

When **S1** is in the "On" position, mechanical movement is not possible during the INIT phase.

The tube collimator cannot be moved using service switches.

The air scan compensator cannot be moved using service switches.

Tab. 13 ZCON LED"s

LED	Color	Signal	Description
V1	Green	+5V	+5V supply applied to D514 FOC
V1	Green	+24V	+24V supply applied to D514 FOC
V2	Green	FOC	Deflection active
V2	Green	XRAY	X-ray detected
V3	Green	UHR	UHR collimator is in/out
V3	Green	COMP	Compensator is in/out
V4	Yellow	SER	Service Switch (S1 on D514) for UHR collimator is active
V4	Green	MSG	No function

Tab. 14 Service switches

Switch	Label	Function
S1	Service	Switches D514 to service mode
S2	UHR_OUT	Moves the UHR out of the scan field
S3	UHR_IN	Moves the UHR into the scan field

Tab. 15 Test points

Test point	Function
IE1	Index encoder 1
IE2	Index encoder 2
GND	Ground connection
GND	Ground connection
TRIG	Unused
XRAY	X-ray detected
ABV (dev)	Focus deviation
SOLL (set)	Focus set point

Phi_Z_Mon

Sensation16/Cardiac:-RZ1 scans (0.5 second scan time) use a static focus deflection. RZ2 scans (scan time greater than 0.5 seconds) use a dynamic focus deflection.

Sensation 40/64/Cardiac64/Open has up to 4 RPP modes, refer to the([DMS Sensation 40/64 / Cardiac64 / Open / p. 128](#)) troubleshooting guide for information.

The Phi_Z_Mon can be checked using an Ohm meter. Results can be affected if the Phi_Z_Mon is exposed to light. The expected result should be approx. 50 KOhm.



To check the Phi_Z_Mon

1. Switch **OFF** the gantry at the control panel
2. Disconnect X3 from the front panel of the FOC board
3. For Phi monitor, measure across Cable W565 X3 pins 1.3
4. For Z-monitor, measure across Cable W565 X3 pins 4.6

Note:

The connector X3 is a so-called “Lemosa” connector. Looking at the connector from the front, begin counting for pin 1 to the left and continue in a clockwise direction. Refer to the cable specification within the functionality description for more information.

AKRON Tube Monitoring

The tube cooling is monitored in the following ways:

- Tube temperature monitoring
- Tube oil pressure monitoring

Warnings and errors

The temperature of the oil in the X-ray tube is monitored by the temperature sensor S1 and the temperature switch S2 (80°C). The temperature thresholds for S1 are set by the firmware located on D514(FOC). D514(FOC) also monitors the temperature and pressure inside the tube. With the location of temperature sensor S1 near the anode (the hottest part of the X-ray tube), the temperature at S1 is always higher than at temperature switch S2, which is located near the stator. S1 will always send a warning message first, if the oil temperature becomes too hot. S3 monitors the oil pressure.

The software/firmware monitors the tube cooling

Temperature of oil near temperature sensor S1 is between 70°C and 80°C

The temperature gradient is > 15°C/min.

- ⇒ Reaction: Warning from the host at the time of the error and each time the system is started.

Temperature of oil near temperature sensor S1 is >80°C

- **Reaction:**
 - ⇒ Fatal error, the host will not permit a scan to be run. Reset is necessary.
- **Possible reasons:**
 - Pump or fan in the tube cooling system is defective.
 - Oil hose is bent.
 - Tube load calculator is malfunctioning.
 - Temperature sensor S1 is defective. (Measure the resistance between X6.1 and X6.2: approx. 10 kOhm at 25 °C.)

Temperature of oil near temperature switch S2 is > 80°C

- **Reaction:**
 - ⇒ Fatal error, the host will not permit a scan to be run. Reset is necessary.
- **Possible reasons:**
 - Pump or fan in the tube cooling system is defective.
 - Oil hose is bent.
 - Tube load calculator is malfunctioning.
 - Temperature switch S2 is defective. <pin Nos TO MEASURE S2>

Oil pressure in tube measured with pressure switch S3 is > 1.6 bar

- **Reaction:**

- ⇒ Interruption of power to the X-ray tube.
Fatal error, the host will not permit a scan to be run.

- **Possible causes of error:**

- Excess oil in the tube cooling system.
- Expansion membrane is defective or misadjusted.
- Pressure switch S3 is defective. (Measure between X6.3 and X6.4.)
- Temperature sensor S1 and/or temperature switch S2 is defective.
- Oil temperature is too high; refer to error situations 1 and 2.

STRATON tube monitoring

The tube cooling is monitored in the following ways:

- Tube temperature monitoring
- Tube oil pressure monitoring

Warnings and errors

The temperature of the oil in the X-ray tube is monitored by the temperature sensor S1 and the temperature switch S2 (115°C). S4 monitor the oil pressure. The temperature thresholds for S1 are set by the firmware located on D514(FOC). D514(FOC) also monitors the temperature and pressure inside the tube.

The software/firmware monitors the tube cooling

Temperature of oil near temperature sensor S1 is between 105°C and 115°C

- ⇒ Reaction: Warning from the host at the time of the error and each time the system is started.

Temperature of oil near temperature sensor S1 is >115°C (+/- 5°C)

- **Reaction:**
 - ⇒ Fatal error, the host will not permit a scan to be run. Reset is necessary.
- **Possible reasons:**
 - Pump in the tube cooling system is defective.
 - Oil hose is bent.
 - Tube load calculator is malfunctioning.
 - Temperature sensor S1 is defective. (Measure the resistance between X12.3 and X12.5: approx. 10 kOhm at 25 °C.)

Temperature of oil near temperature switch S2 is > 115°C (+/- 5°C)

- **Reaction:**
 - ⇒ Fatal error, the host will not permit a scan to be run. Reset is necessary.
 - ⇒ Reset of temperature switch at 100°C (+/- 4°C)
- **Possible reasons:**
 - Pump in the tube cooling system is defective.
 - Oil hose is bent.
 - Tube load calculator is malfunctioning.
 - Temperature switch S2 is defective.

Oil pressure in tube measured with pressure switch S3

- **S3 is not used anymore**
 - ⇒ S3 is jumpered out and only used in the factory (for old systems)
 - ⇒ Ex April 2006, the switch S3 is not applicable.

D522 LPS

Testing the D522 LPS:

- Rotate the gantry by hand, and observe the AP0 and AP90 LED"s on the D301 MCU. As the gantry rotates, these signals should toggle. Once per revolution, an index pulse should be noted at the index LED.
- The AP0 and AP90 signals are generated at the D522 LPS and transmitted from the D513 (ADU) to the MCU via the control data slip ring.

DPU (Sensation 10/16/Cardiac)

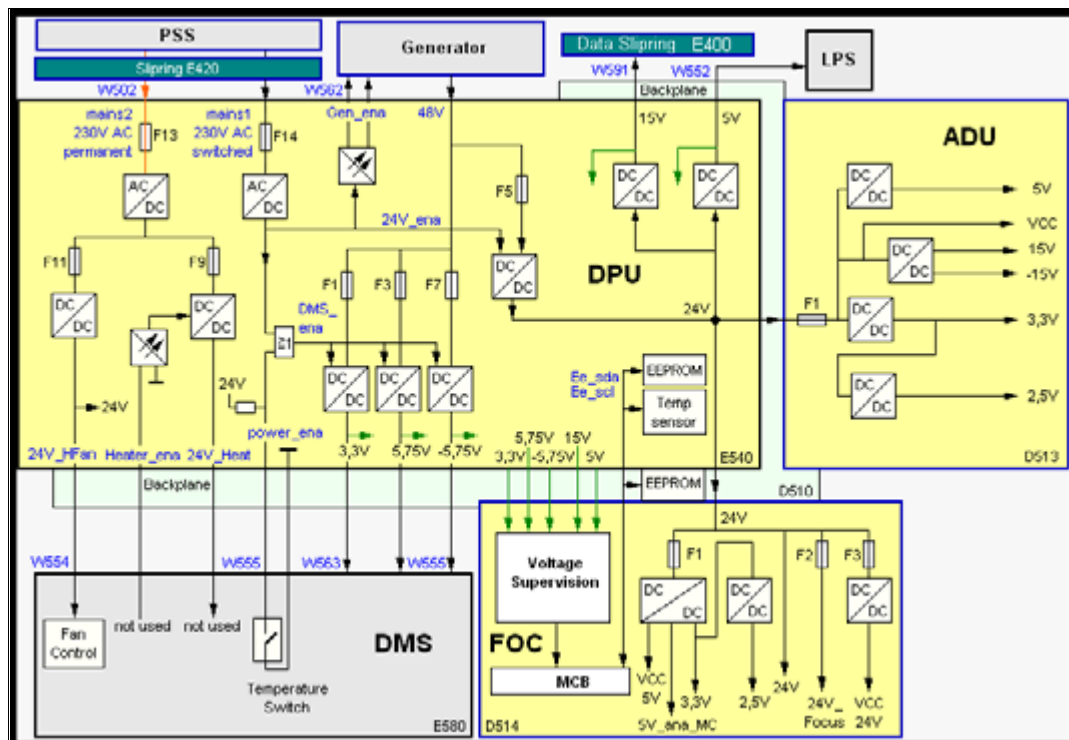


Fig. 81: Block diagram of DPU

The DPU uses two sources to generate the voltages. Mains2 (230 V permanent) and 48 V from the XRS (see table for details). Mains1 (230 V switched) is not used to generate voltages. Mains1 is only used to produce the **24v_ena** signal.

- When the gantry is **ON**:
 - Mains2 is present and 24V_Hfan and 24V_Heat are supplied to the DMS (note: 24V_Heat is not used).
 - XRS supply 48V is present.
 - Mains1 is on and the **24v_ena** signal enables the 24 V, 5 V and 15 V for the C-box. (It also enables the remaining XRS supplies via the **Gen_ena** signal.)
 - **24v_ena** is logic OR'd with the **power_ena** signal from DMS. This enables the 3.3V, 5.75V and -5.75V supplies for the DMS via the **Dms_ena** signal.
 - When the gantry is **OFF**:
 - Mains2, 24V_Hfan and 24V_Heat are still present
 - XRS supply 48V is still present
 - Mains1 is off, **24v_ena** is low and the output from 24V, 5V and 15V are inhibited.
 - Since **24v_ena** is low, 3.3 V, 5.75 V and -5.75 V (DMS supplies) are now controlled by the **power_ena** signal from the DMS.
- ⇒ In normal operation, the DMS supplies are always enabled if mains2 is present (gantry is on). If it is not, (gantry is off) then the **power_ena** signal from the DMS controls the DMS supplies. As a result, unless the DMS temperature rises above 42°C, the **DMS supplies will always be ON**.

Tab. 16 DPU_1 power supply

DPU	3P3_C	5P75_C	5N75_C	24P	5P	15P	24V _Hfan	24V _Heat
Voltage	3.4	5.75	-5.75	24.0	5.0	15.0	24.0	24.0
Tolerance	+/-5%	+/-4%	+/-4%	+/-4%	+/-4%	10.26	+/-5%	+/-5%
Current	20	20	13	6.6	0.2	1.33	7.0	1.25
Fuse	F1	F3	F7	F5	F5	F5	F13,11	F13,9
Source voltage	48V_P	48V_P	48V_P	48V_P	48V_P	48V_P	Mains 2	Mains 2
Noise (mv) (10 Hz to 100 kHz)	< 20	< 20	< 20	<240	na	na	< 240	< 240

Evaluate the DPU power supply according to the table above. Test points are clearly marked on the DPU board. No adjustment is possible.

Note:

The voltage measured on the test point may be higher than that in the table above. More details will be available in the subsequent version.

DPU (Sensation 40/64/Cardiac 64/Open)

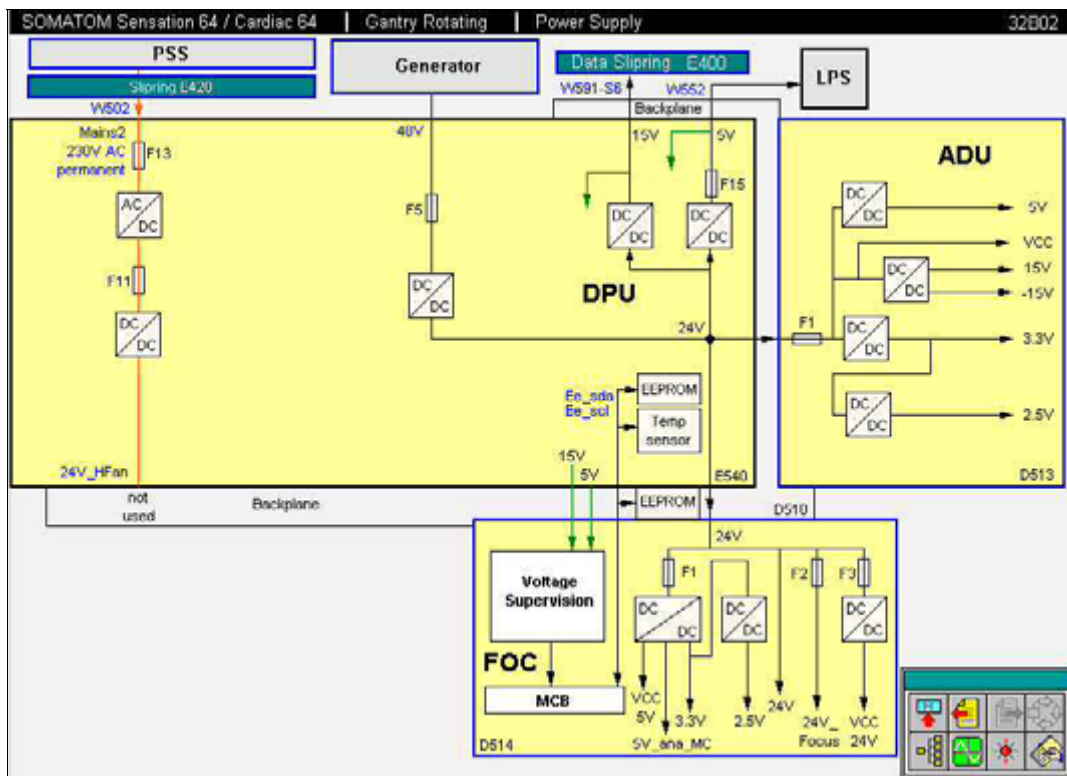


Fig. 82: Block diagram DPU (Sensation 40 / 64 / Cardiac 64 / Open)

The DPU uses only the 48V from the XRS to generate the supply voltages for ADU and FOC. Mains1 (230 V switched) is not used any more. The DPU wiring is essentially the same, however, since the XRS power supply is powered by mains1, the 48V supply is not present when the gantry is turned off. The Gen_ena signal is made redundant and is therefore no longer connected to the XRS supply. Also, the DMS has now a separate supply (E607). Mains 2 (230 V permanent) is present on the DPU board but is used internally only. Mains 2 voltage can be switched off with breaker F9 in the PDC.

- When the gantry is **ON**:
 - XRS supply 48V is present.
 - Mains 2 (230 V permanent) is switched on
- When the gantry is **OFF**:
 - XRS supply 48V is switched off
 - Mains 2 (230 V permanent) is switched on

Tab. 17 DPU_1 power supply

DPU	24P	5P	15P
Voltage	24.0	5.0	15.0
Tolerance	+/-4%	+/-4%	10V - 26V
Current	6.6	0.4	1.33
Fuse	F5	F5/15	F5
Source voltage	48V_ P	48V_ P	48V_ P
Noise (mv) (10 Hz to 100 kHz)	<240	n.a.	n.a.

The block diagram shows all field-replaceable parts and their connections.

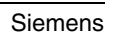


Fig. 83: Block diagram of DMS with UHR comb

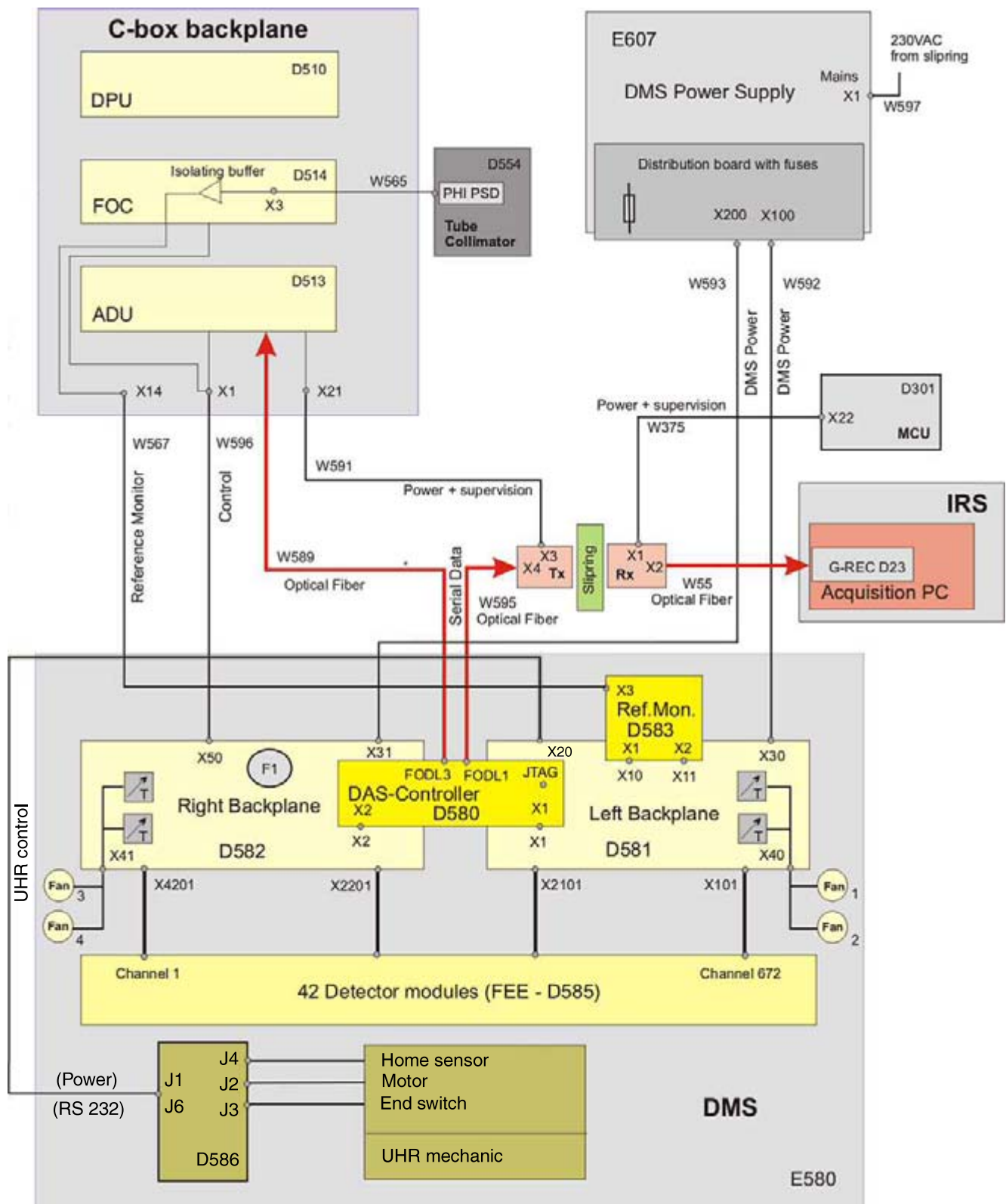


Fig. 84: Block diagram of DMS with advanced UHR comb

Data flow inside DMS

This figure shows an overview of the routing of the measurement data

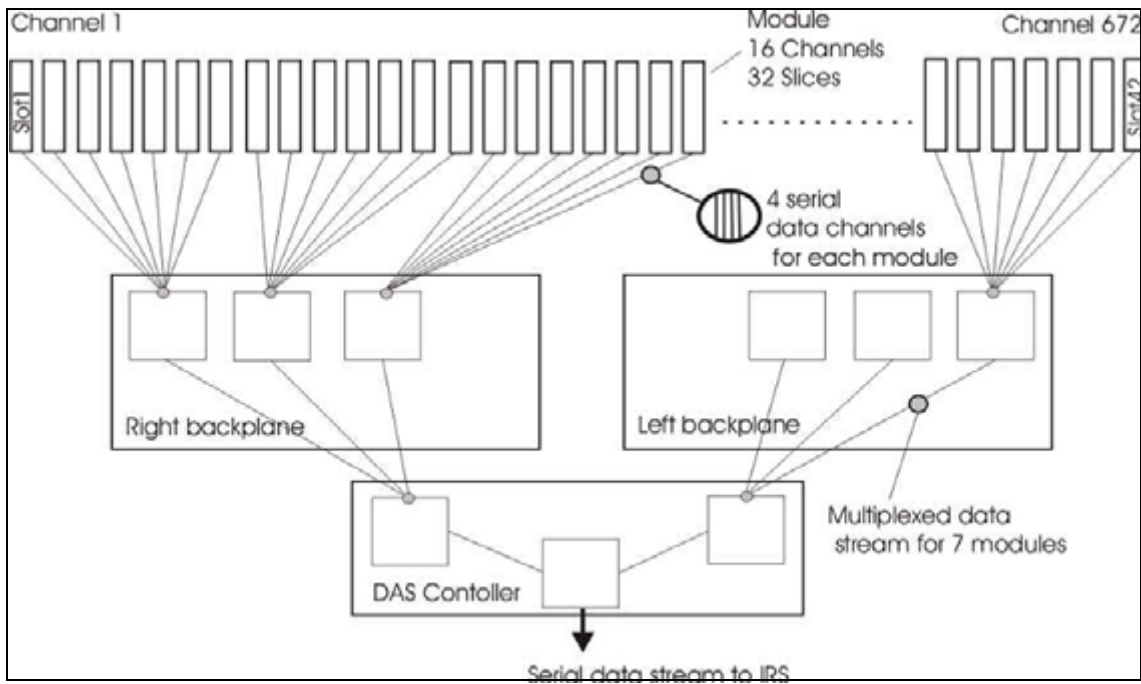


Fig. 85: Data flow of DMS

Replaceable parts inside the DMS

This image shows all replaceable parts inside the DMS

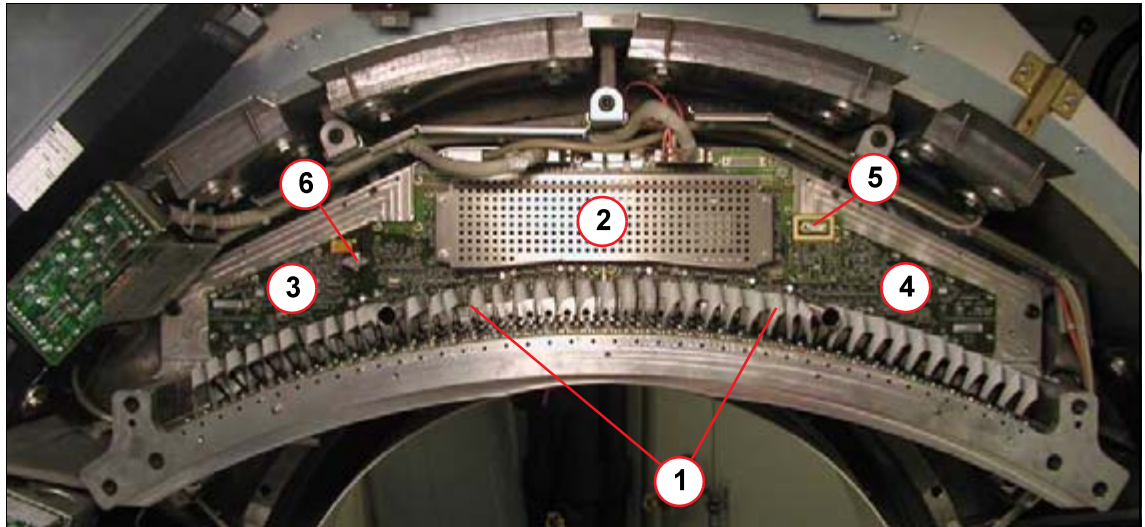


Fig. 86: DMS without cover

- Pos. 1 Modules
- Pos. 2 DAS controller
- Pos. 3 Right backplane
- Pos. 4 Left backplane
- Pos. 5 Reference monitor board
- Pos. 6 Temperature switch F1

Orientations

All orientations used are shown in the next figure:

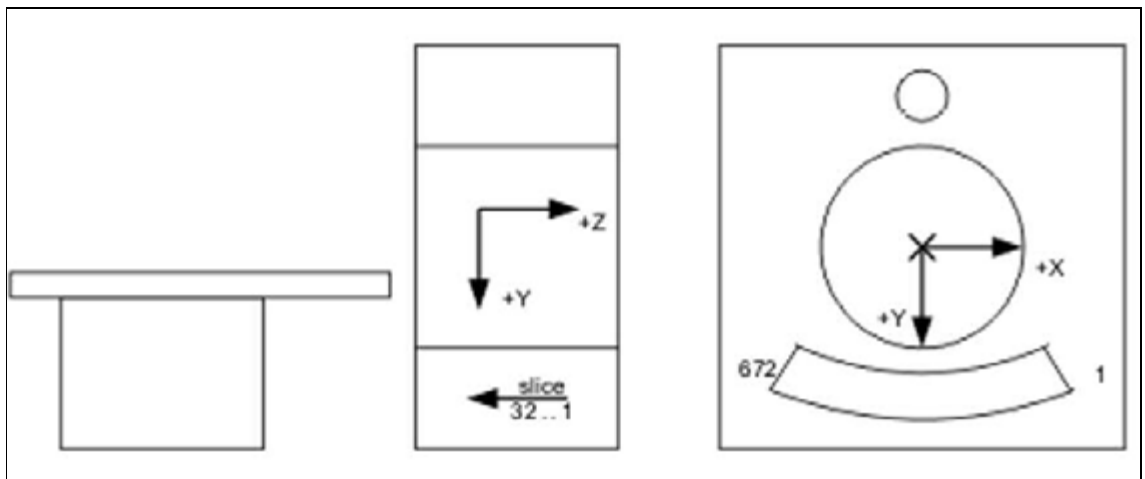


Fig. 87: Orientations

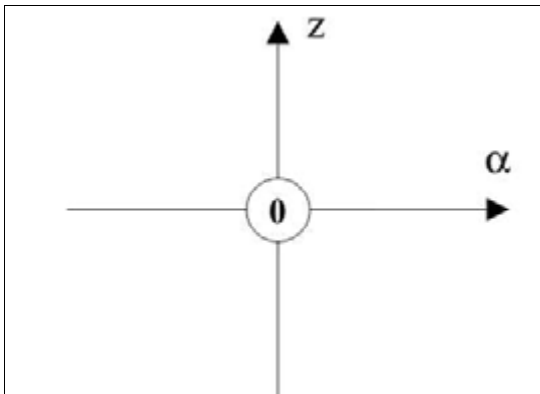
The list of channels has been appended. ([Channel list / p. 217](#))

Overview -Focus movement

Focal spot movement in the Z-direction is a feature with Sensation 40/64/Cardiac64/Open.

The next figures show an overview of all possible / used combinations.

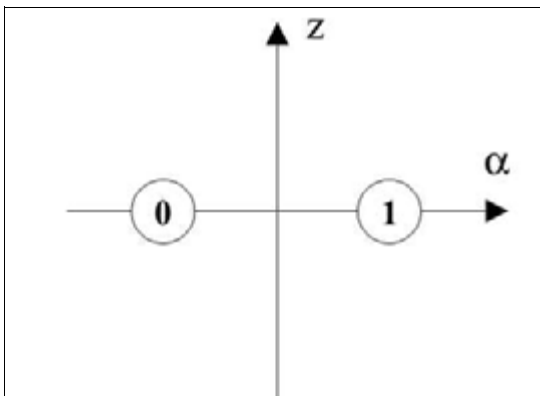
RPP = reading per projection. FFS = flying focal spot.

**RPP 1**

No FFS

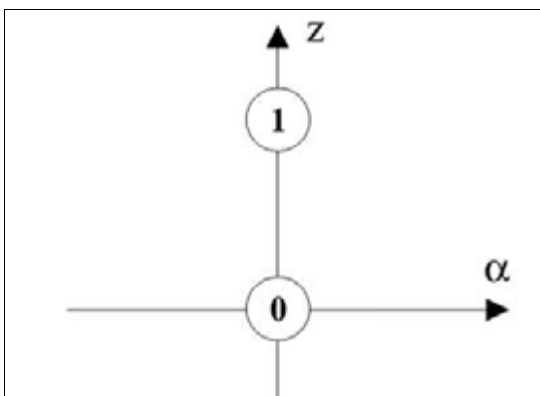
Used in

- Sequence
- 1.2 mm collimation

**RPP 2 x 1**

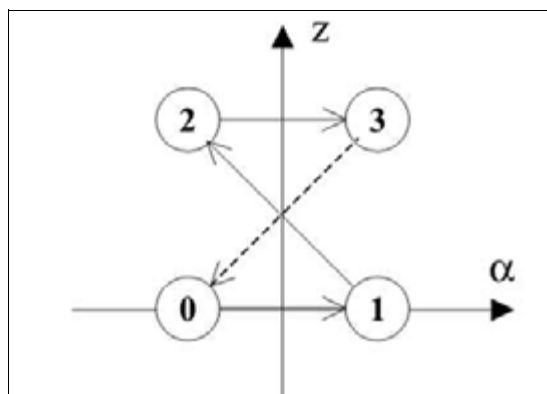
Traditional FFS

- Sequence
- 1.2 mm collimation

**RPP 1 x 2**

Pure z FFS

- 0.6 mm spiral with rotation time < 1 s

**RPP 2 x 2**

Combined FFS

- 0.6 mm spiral with rotation time 1 s

General helpful notes

In this section we are providing you with general helpful notes. Please keep these in mind when working on DMS problems.

NOTE

Please adhere to the C-MOS guidelines when handling the DMS.

NOTE

With software VB20, a selection of DMS tests are combined under item "Test Tools" in the local service platform. If you select these DMS test sequences called "Guided DMS-Test", the collective tests will run automatically in a row.

- Begin your troubleshooting with the ROI/Ring and note the positions located as a reference for the position expected to hold the defect (two positions are possible).

Normally, you start with the image the customer is complaining about.

- To determine a defective slot, always use tests that output results across all channels.
- Check whether the artifact is visible in the topogram as well. If yes, position a marker and take a topogram to define the affected detector side.
- Before starting the extensive troubleshooting of weak rings, be sure that the environmental conditions and the tune-up are ok.

To perform a **complete tune-up**, go to Tuneup Detector and start with ZAdjust.

Perform this step prior to escalating an image quality problem.

- Only two slice combinations are useful for troubleshooting :
 - 32 x 0.6 mm
 - 24 x 1.2 mm
 - Before starting extensive troubleshooting, be sure you troubleshoot the right area.
- To distinguish between DMS and IRS, use the Disk Test .

- To distinguish between module and backplane, it is possible to **swap** the **module cable** at the backplane.
If the problem moves when swapping the cable, the problem is related to the module.
- Use also test modes with rotation for troubleshooting.
- Test results are stored in two different ways:
 - As HTML files
Path:c:\Somaris\service\html\report\htmlreport
 - As TXT files
Path:C:\Somaris\service\extract
 - Additionally, it's sometimes helpful to use a screenshot to report the result plots:
 - Activate the window you intend to capture
 - Use Ctr / PrtSc to capture the window
 - Start > Programs > Accessories > Paint
 - Use Edit > paste
 - File > save as and use file type gif to reduce the size

Normally, HTML files are used to store test results.

However, the plots of the DMS tests are usually stored in a text file (via Extract to File button). Please note: the file name has to be changed manually. Otherwise, the file will be overwritten each time the softkey is being used.

The file may be displayed using Excel (e.g. after manual remote transfer).

Differences regarding Sensation 40, 64, Cardiac 64 and Open

Different with respect to:	Sensation 40	Sensation 64	Cardiac 64	Open
UHR Phi comb ¹	Yes	Yes	Yes	No
Z Focus Check	Yes	Yes	Yes	Yes
UHR Phi-z comb ²	No	Yes	Yes	No

1. Installed in Sensation 40 and 1. version of Sensation 64/Cardiac64

2. Installed in 2. version of Sensation 64/Cardiac64 (needs software >=VB20)

Ring, band, streak and other artifacts

This chapter provides a basic overview of different types of image quality problems. The most useful way for starting troubleshooting is explained.

General

Possible reasons for ring and band (mostly 4, 8 or 16 channels) artifacts are:

- Defective detector element
- Defective detector cable
- Defective backplane
- Defective DAS controller
- Bad DMS power supply
- Insufficient calibration (e.g. contrast media at plexi ring)
- Insufficient tune-up tables
- Problem with balancing
- Insufficient environmental conditions
- Defective IRS

Partial rings occur due to weighting during image reconstruction in spiral mode.

Strong rings

In most cases we suspect that the detector module is defective.

To permanently patch out the defect, perform a defect channel detection under TuneUp. ([Handling of defective channels / p. 212](#))

Check the entry in Affected Modules to determine if the defect can stay in or not.

In addition, start troubleshooting with the following DMS tests:

1. Signal value ([Signal value / p. 149](#))
2. Offset value ([Offset value / p. 147](#))
3. Sinogram ([Sinogram / p. 170](#))

Strong bands (4, 8 or 16 channels) may also result from a defective backplane or DAS Controller.

Weak rings

A ring is visible in homogeneous areas such as the cerebrum, if the difference compared to the neighboring channel is about 1 ppm.

In most cases we expect the stability of the element affected to be insufficient. Before starting extensive troubleshooting, ensure:

- That all environmental conditions are OK

- That tune-up was performed correctly
- That the DMS is powered on overnight.

Start troubleshooting using DMS tests:

1. BasCal stability ([BaseCal stability / p. 154](#))
2. Signal linearity ([Signal linearity / p. 158](#))
3. Sinogram ([Sinogram / p. 170](#))
4. Generate images with and without balancing ([How to switch off balancing / p. 217](#))
and compare them to find out if the balancing algorithm is causing the problem

Hint:

Evaluate the physicist's line in the CT image text (lower left corner of image) to use the information as a basis for generating raw data sets of the parameter set identical to the one the customer is complaining about. ([How to interpret the physicist's line / p. 218](#))

Rings in outer slices only

If the rings are visible in the outermost slices only, start troubleshooting with

1. ZCheck (TuneUp) ([Z check of tube collimator / p. 178](#))
 - ⇒ IF out of tolerance, repeat ZAdjust
2. Check the slice width to verify that it is within constancy
 - ⇒ If out of tolerance, check if FOC and tube collimator are working

Sporadic strong rings

To permanently patch out the defect, perform a defect channel detection under TuneUp. ([Handling of defective channels / p. 212](#))

In case of a very sporadic ring problem:

1. Ask the customer to perform a **checkup in case of problems**
2. Use the ROI/RING function to analyze the customer's image ([ROI/Ring / p. 143](#))
3. Analyze the history of the problem channel (use the Bad Channel History program for this purpose). Look for entries corresponding with one of the results found in ROI/RING
4. Manually patch out the problem channel ([Editing the correction tables / p. 215](#))

Sporadic weak rings

In case of sporadic weak rings, ensure that all steps for weak rings have been performed.

In addition, try and find the problem using BaseCal

1. Ask the customer to calibrate in case of problems.
2. Perform the BasCal stability test ([BaseCal stability / p. 154](#))
 - ⇒ The result (difference between a satisfactory and unsatisfactory BaseCal) will show the location of the problem.

Streak artifacts

Possible reasons:

- Defective detector module:
If the problem is intermittent, a streak artifact tangential to a circle (position of the defective channel) occurs.
 - ⇒ Use the popcorn noise to isolate the defective channel ([Popcorn noise / p. 145](#))
- Patched channels:
Streak artifact tangential to a circle (position of the patched channel) occurs.
 - ⇒ Check if there are too many (adjacent) patched channels
Evaluate the result of the unsatisfactory channel detection. ([Using Reports to view details of the Defective Channels Correction / p. 216](#))
Check if Affected Modules show slots that are marked →change
- Alignment adjustment:
Fine streak artifacts not correlated tangentially to any circle.
 - ⇒ Check whether the MTF measurement (Quality) is within tolerance.
- Tube arcing
 - ⇒ Check the tube history for this particular scan to see if tube arcing was detected.
Getter tube

Other artifacts

Some artifacts may be caused by:

- The tube
- The tube-side collimator
- The tube cooling system
- The DMS power supply

Pattern across the entire image

If you have patterns (e.g. grid pattern) in your image:

- Perform tests or test images with and without X-ray to find out if the problem is related to radiation.

If the problem is related to radiation:

- ⇒ - Check the reference channel ([Checking the reference monitor / p. 140](#))
- Check (replace) tube
- Check (replace) tube collimator

If the problem is not related to radiation:

- ⇒ - Check the DMS power supply ([Power supply / p. 186](#))
- Check the modules at slots 1 and 42 (slice normalization)

Inhomogeneous areas

To identify the source of the problem, evaluate the BaseCal values (contact CS HSC 22).

The reasons for inhomogeneous areas in the image (with radiation) are mainly:

- Defective BaseCal tables (visible mainly in the cerebrum sequence):

If calibration was performed with an absorber (contrast media) in the scan field:

The artifact is stable (in the same image area) in all images.

- ⇒ Check the report for entries.
(**Options > Service > Local Service > Reports**)
- ⇒ Clean the plexi strip and check for other absorbers in the scan field.
Repeat calibration.

Inhomogeneous slices

If the mean value (of the same image area) differs from slice to slice :

- ⇒ Check the modules of slots 1 and 42 (air slice normalization)

Hint:

The task of the air slice normalization is to equalize all slices.

For this purpose, some of the channels of the module of slot 1 and slot 42 are used to calculate a mean value for each slice. All slices are now compared and equalized.

Poor image quality due to wrong scan parameter or usage

Using a CT is getting to be more and more complex.

There are also non-technical reasons for certain types of image quality problems

- Windmill artifact

If you browse through a stack of images, the image contains an artifact that moves like a windmill. It is especially visible at the edges of bones.

Windmill artifacts occur due to undersampling or interpolation.

- ⇒ Compare the pitch used with the Siemens default values
- ⇒ Check if the reconstructed slice width is 1.25 times the collimated slice width.
- ⇒ Involve Application in troubleshooting

- Artifacts due to patient positioning or use of positioning aids. (mainly streaks)

Head:

- ⇒ Check to see if the correct head holder was being used.

Body:

- ⇒ When using the CT-slicker, bend the edges down and glue them to the table

Checking the reference monitor

The monitor signal is generated in a different way than at S16.

Here the PHI-Z-PSD is used not only for dose control, but also for generating the reference signal for the detector. The following are part of this chain:

PSD -- cable W565 -- board D554 -- FOC D514 -- Cbox backplane D510 -- cable W567 -- D583 -- D581

To track down a problem we can:

- **Check the monitor value in TA0**

⇒ To read out the monitor values, perform the following:

1. **Options > Service > Local Service > Control > Table load/modify**
2. Select radio button **CONTROL SYSTEM** under TABLE SOURCE
3. In the list box TABLE NAMES select **TA0**
4. Press **GO**

The parameter set for the table TA0 is shown

Search for **PHI monitor neg signal** and **PHI monitor pos signal**

The sum of both has to be in the range of 0 to 5000 [mV]. The table shows the value of the last scan.

→If values are plausible, D554, W565 and partially D514 are ok

- **Check the dose values in the tube history**

⇒ To read out the monitor values, go to:

Options > Service > Local Service > Report > Tube Current History
and evaluate the value for **dose actual**

→If values are plausible, D554, W565 and partially D514 are ok

- **Check Reference Monitor board**

⇒ Perform scans with and without W567, X3 connected at the reference monitor board.

→Ok, if the values change to offset the value when disconnected

- Use tests which provide information about the reference channel:

- Offset ([Offset value / p. 147](#))
- Signal value ([Signal value / p. 149](#))
- Spectral linearity ([Spectral linearity / p. 156](#))
- Signal linearity ([Signal linearity / p. 158](#))
- Analyze ([Analyze / p. 162](#))

If the offset and signal values are within tolerance, the reference monitor is operating satisfactorily.

- Use Analyze

Use the **Dose Monitor** function to check the reference monitor. ([Analyze / p. 162](#))

The advantage of this test is that you also see the transient response of the dose.

Tools for troubleshooting

Here you will find a list of all software tools available for handling DMS problems.

General

- Common for all test results providing a **plot** :

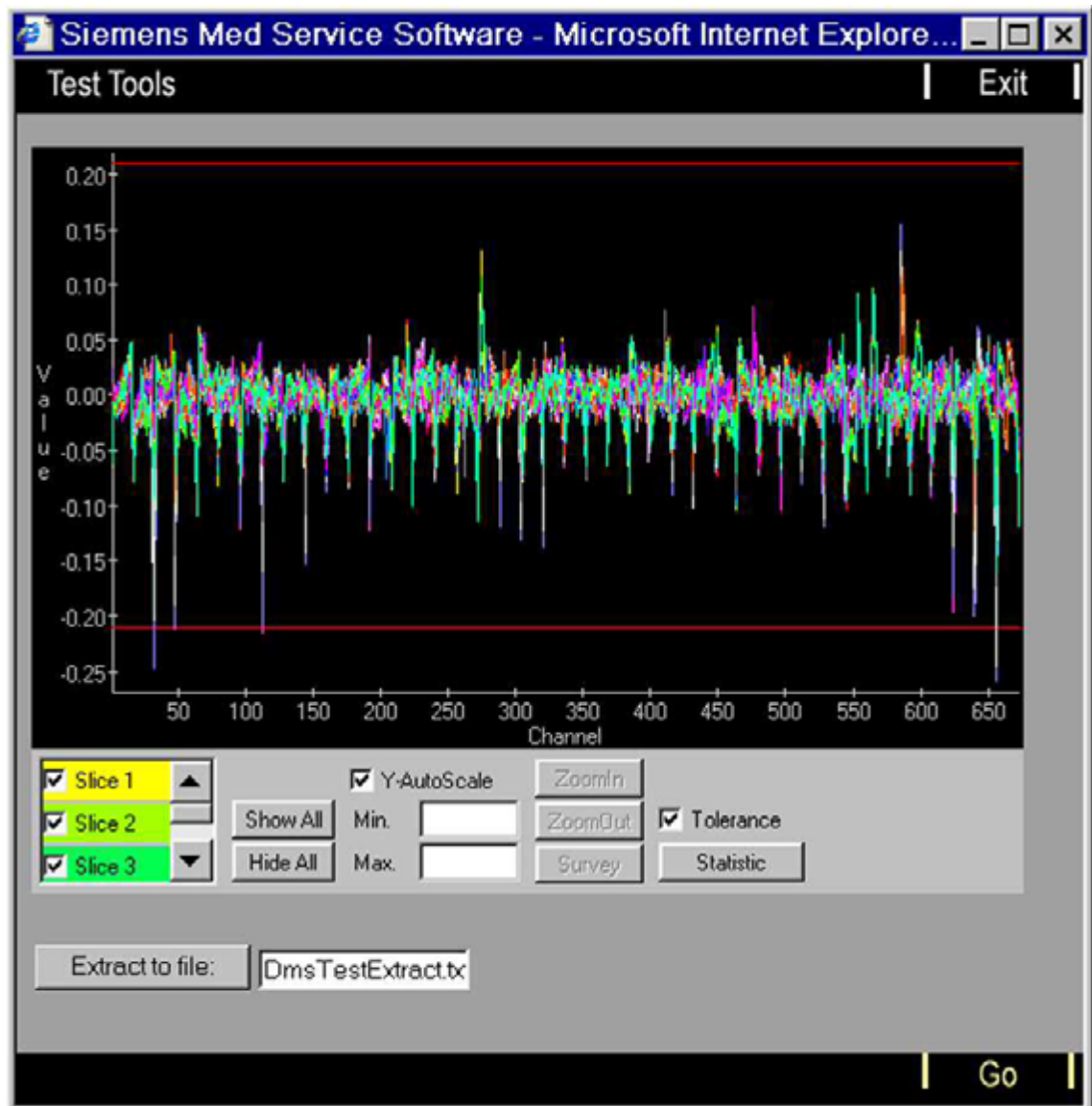


Fig. 88: Relative Signal Value

- The button **Show Plot** shows the graphical output of the table.
It's possible to select all slices via **Show All**, none via **Hide All**, or a single slice via a list box. It's also possible to zoom in (or out via **Survey**) by using the left mouse button. Mark the area of interest while holding down the left button. Then press **Zoom In**.
- Position the cursor at the desired location and wait for the display of:
 - channel
 - slice
 - value at that positionThe button **Statistic** displays the following for all slices: Minimum, Maximum, Mean, and StdDev.
The min. and max. values are shown only if the checkbox **Y-Auto Scale** is not marked.
- The button **Extract to file** is used to store the result of the test to
c:\Somaris\service\extract\DmsTestExtract.txt
Hint: If you want to store more than one file, you have to manually change the file name.
- The display of the tolerances in the plot may be switched on or off using the checkbox for **Tolerance**
- Tolerances are available for all test results.
As a result, test result pages display the following at the top of each page: **OK** for tests passed or **Not OK** for tests failed.
- In case of **values out of tolerance**, slice and channel are output.
Click on the channel number and read out: Channel xxx and Slot xx.

Band assessment tool

Scope

- This test performs automatic test scans and runs an algorithm to find ring structures in water phantom images. The disadvantage is that only the range of the phantom is checked (20 or 30 cm).

In the factory a special phantom consisting of 3 parts is used:

- 20 cm body
- 20 cm head
- 30 cm body

The phantom is not delivered with the system!

NOTE

With installed software $\geq$ VB20, the "Band assessment test" can be performed with the 20cm quality phantom as well. In this case the test does not cover the 30cm body section!

How to find

- **Options > Service > Local Service > Test Tools > Band Assessment**

How to use

1. Position phantom so it lines up with the engraved mark.
The table position is checked.
2. Select the desired modes.
Possible choices: Select all, Deselect all, and Select box
3. Press **Go**
 - ⇒ If results are out of tolerance, continue troubleshooting the ring artifacts.

ROI/Ring**Scope**

- The **ROI** function (region of interest) is used to calculate the mean value and the standard deviation of a user-defined area.
- The **Ring** function shows the correlation of a position in an image to 2 possible channels in the DMS that have been involved in creating data at that position.

How to find

- **Options > Service > Local Service > Test Tools > ROI/Ring**

How to use

1. Select the desired order for sorting the images in the list box by activating the corresponding radio button. (You may have to wait.)
2. Mark the image of your choice in the list box.
3. Press the button **Load selected image**
 - ⇒ The image will be displayed
4. Select the radio button **Ring**
5. Use the left mouse button to move the ring to the desired position.
Use the center mouse button for windowing.
 - ⇒ Read out the 2 possible channels (channel1, channel2) causing the problem.
 - ⇒ Use this information as a basis for additional troubleshooting.

Offset image

Scope

- The offset image is calculated from an offset-corrected raw image added to a synthetic disk. The raw image is acquired without radiation.

As a result, the artifacts visible in the offset image are not related to the X-ray or the UFC of the detector. Instead they are primarily related to the electronics of the integrator section.

NOTE

With the Sensation 40/64/Cardiac64/Open systems the offset images may have ring artefacts. Therefore the offset images can not be used for evaluation or decision to replace the DMS.

How to find

- **Options > Service > Local Service > Test Tools > DMS > Offset Image**

How to use

1. Select the radio button for rotation on or off.
Start testing with rotation.
2. Select a scan time from 4 possible times .
Start with the scan time the customer is complaining about.
3. Press **Go** and wait for the result
 - ⇒ The offset images are calculated.
Images have to be without rings, bands, or streaks.
Check images with window 550 and center 0.
In case of problems:
 - ⇒ Continue with the offset value.
4. Select "radio button image is correct" (or not correct)
5. Interpretation of the results:
The images must show a homogeneous noise structure and must be without:
 - Rings or partial rings
 - Streaks
 - Any visible structures.

Hint:

The offset images will be stored under Service Patient.

The offset measurement will be aborted in case of an active patient.

Popcorn noise

Scope

- The principle of the popcorn noise test (spike test) is to monitor a number of offset scans to find unstable channels. As a result, ppm thresholds are defined for the values. If these thresholds are exceeded several times (e.g. max. number of spikes), the test identifies the corresponding channels.
- It is difficult to find suitable tolerances for this test.
In general, this test should be used to find sporadic problems.
(For instance, to find the worst channel.)

How to find

- **Options > Service > Local Service > Test Tools > DMS > Popcorn noise**

How to use

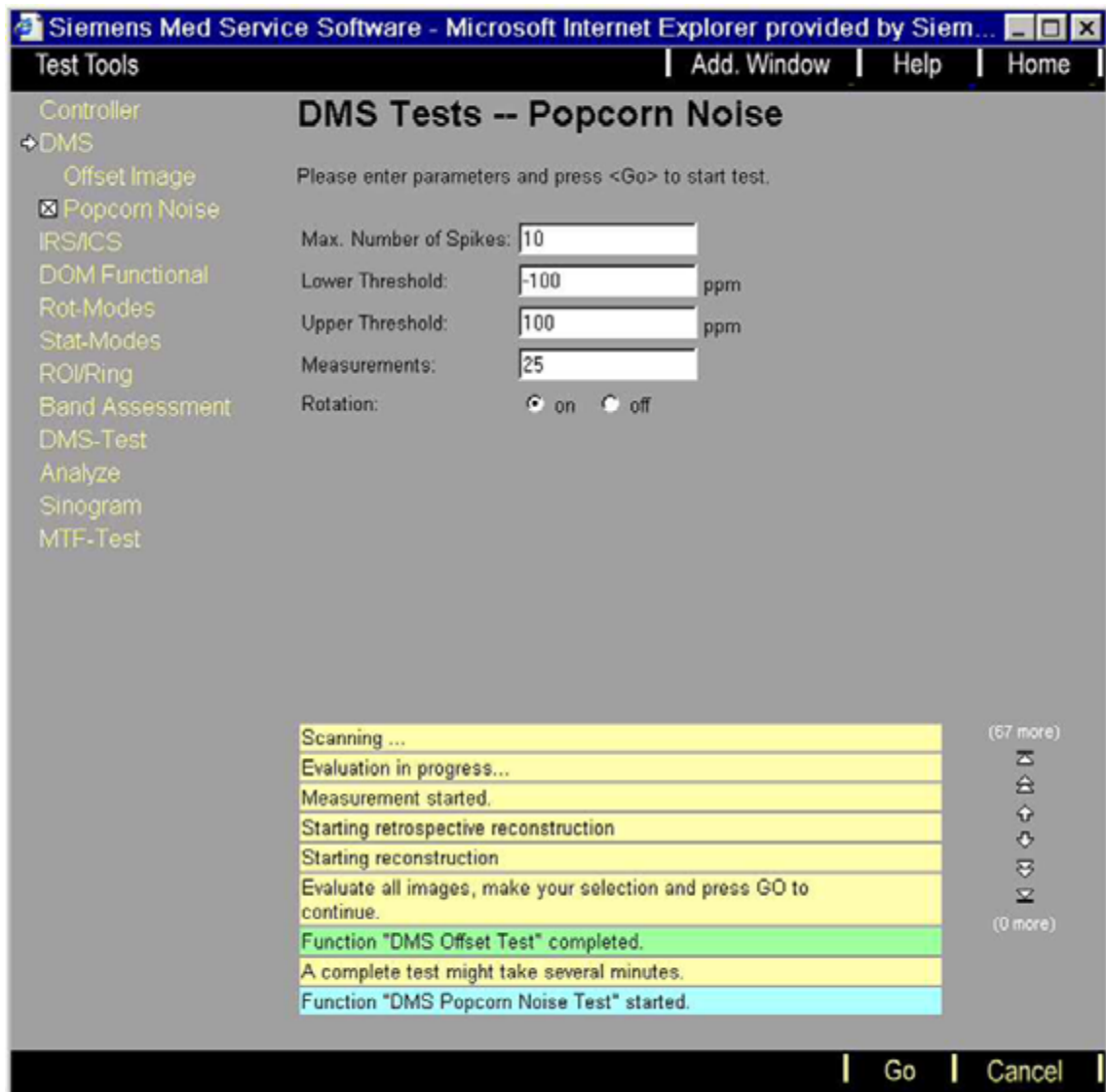


Fig. 89: Example of popcorn test

1. Select "max. number of spikes"
2. Select "lower threshold"
3. Select "upper threshold"
4. Select "number of measurements"
5. Select "test with or without rotation"
Start troubleshooting with rotation
6. Press **Go**
 - ⇒ Check results to find channels that lie outside the threshold.
 - ⇒ Lower the value for the upper threshold from 100 to 50, 30 and 20 to get the worst channel output

Offset value

Scope

- This test is used to check the mean values and noise without X-ray.

How to find

- **Options > Service > Local Service > Test Tools > DMS-Test > Offset**

How to use

1. Select the desired mode
2. Press **Go**
3. Wait for the result
 - ⇒ Replace modules that exceed the threshold
4. Description of the **result page**

The results of the test are shown in two tables:

 - **Offset value**
 - **Offset noise**

Both tables show Slice, Minimum, Maximum and Test Result.
Use the Show Plot button to display the plots:

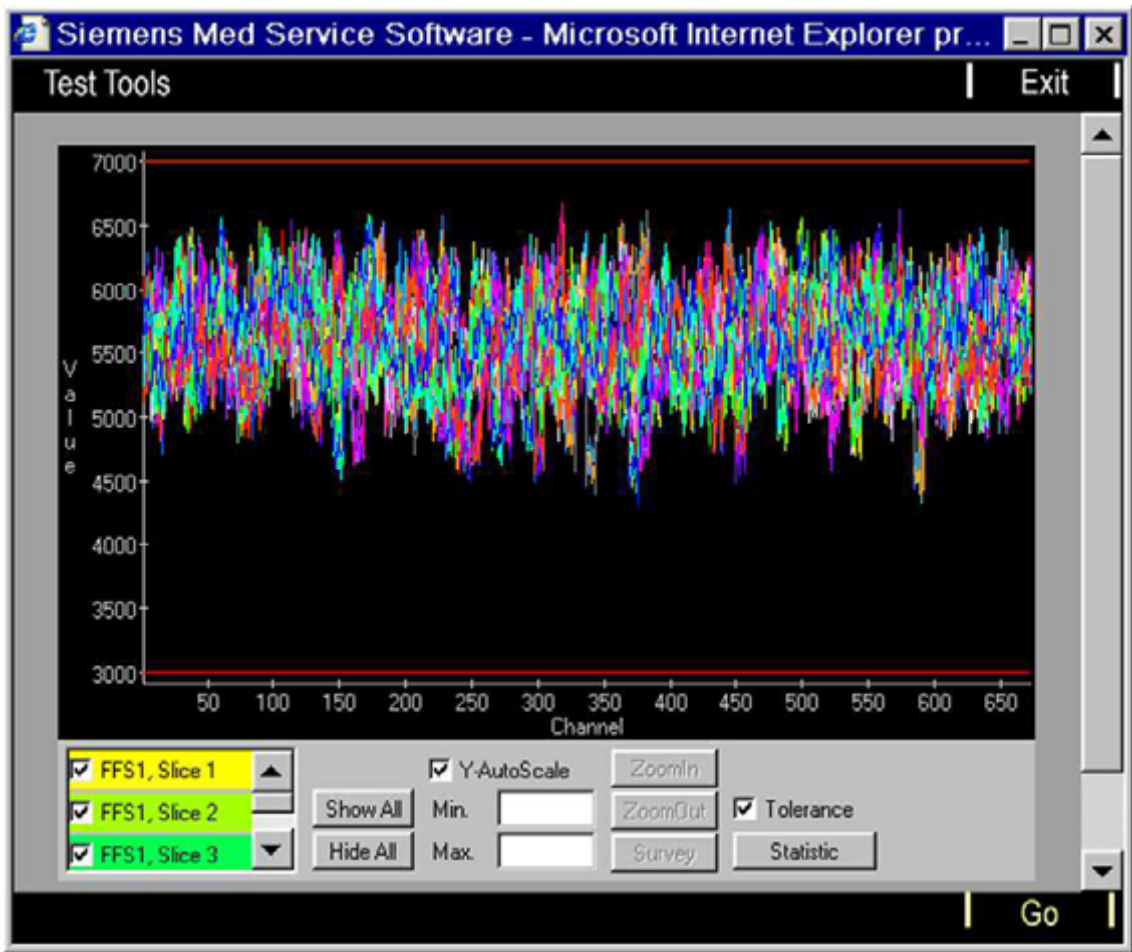


Fig. 90: Example of offset value result

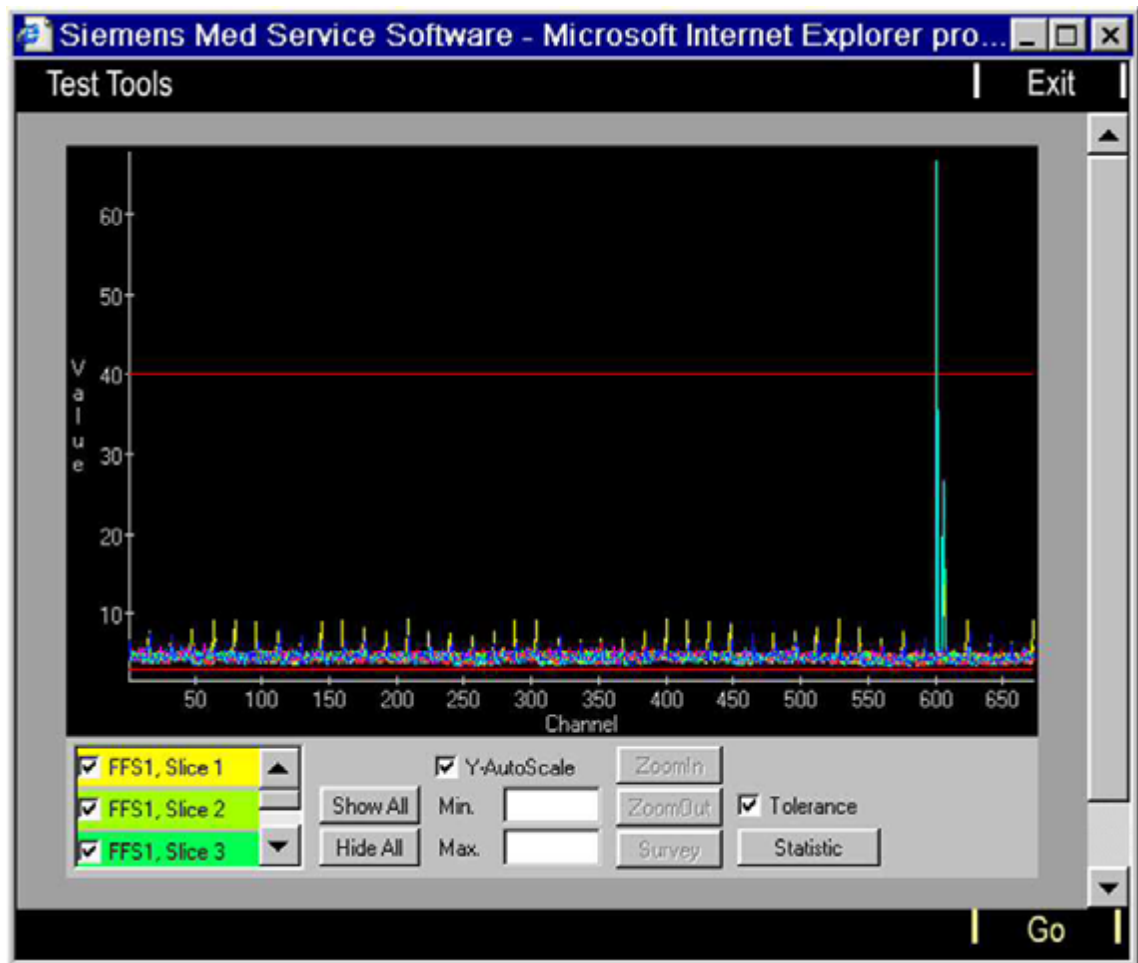


Fig. 91: Example of offset noise result with error

Signal value

Scope

- This test is used to check the mean values and noise with X-ray.
A mode with 32 x 0.6 mm or 24 x 1.2 mm is used.

How to find

- Options > Service > Local Service > Test Tools > DMS-Test > Signal Value

How to use

1. Wait for the Press **START** key to appear and then press the Start button
 ➞ If channels are out of tolerance, exchange the corresponding module.

2. Description of the result page

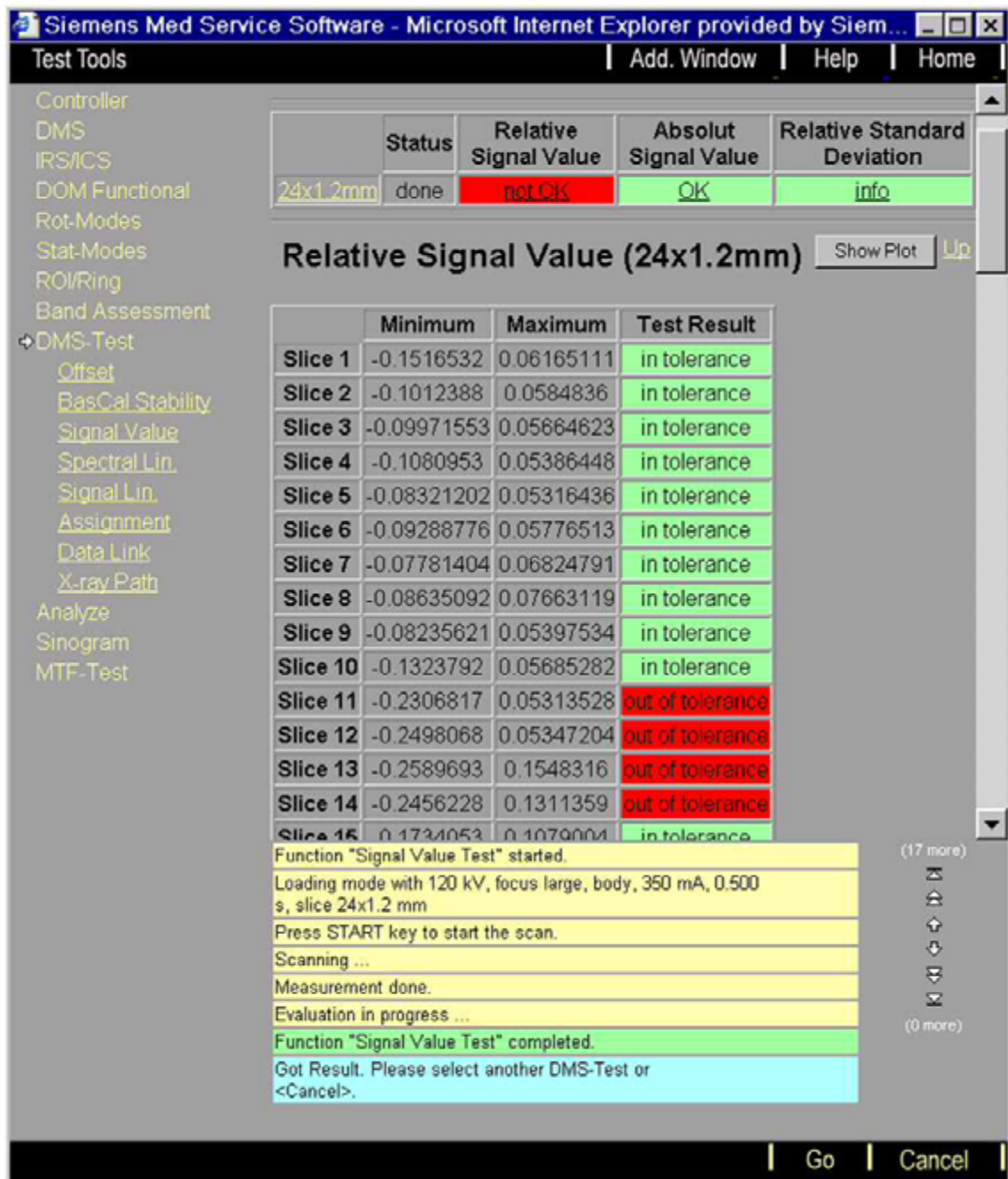


Fig. 92: Signal Value Test

The results of the test are shown in three tables:

- **Relative Signal Value**

The result shows the averaged signal of the channels in relation to the neighborhood.
This table shows Slice, Minimum, Maximum, and Test Result.

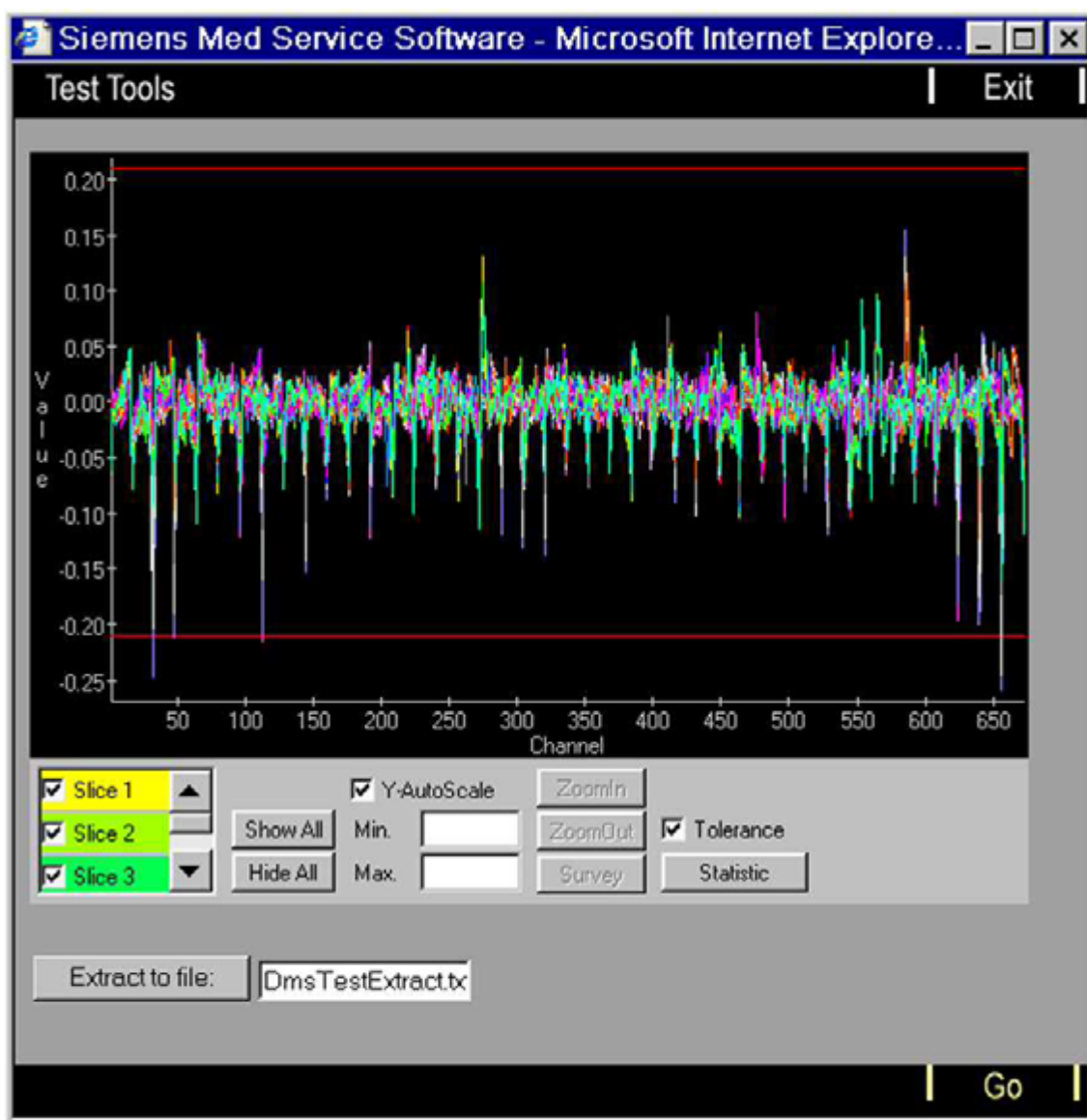


Fig. 93: Relative Signal Value

- Relative Standard Deviation

This table shows Slice, Minimum, Maximum, and Test Result.

In addition, the monitor value is shown.

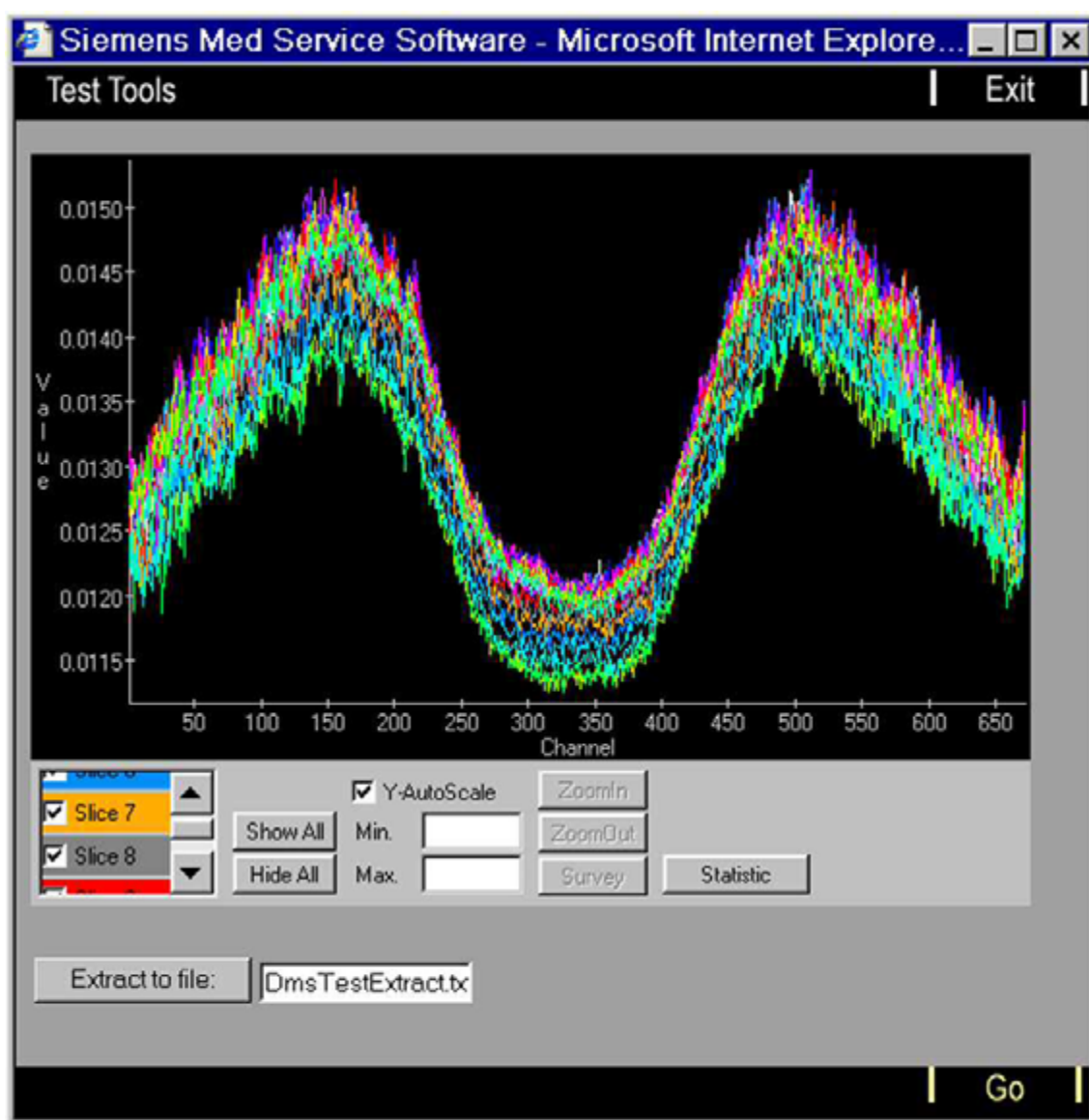


Fig. 94: Relative Standard Deviation

- **Absolute Signal Value**

This table shows Slice, Minimum, Maximum, and Test Result.
In addition, the monitor value is shown.

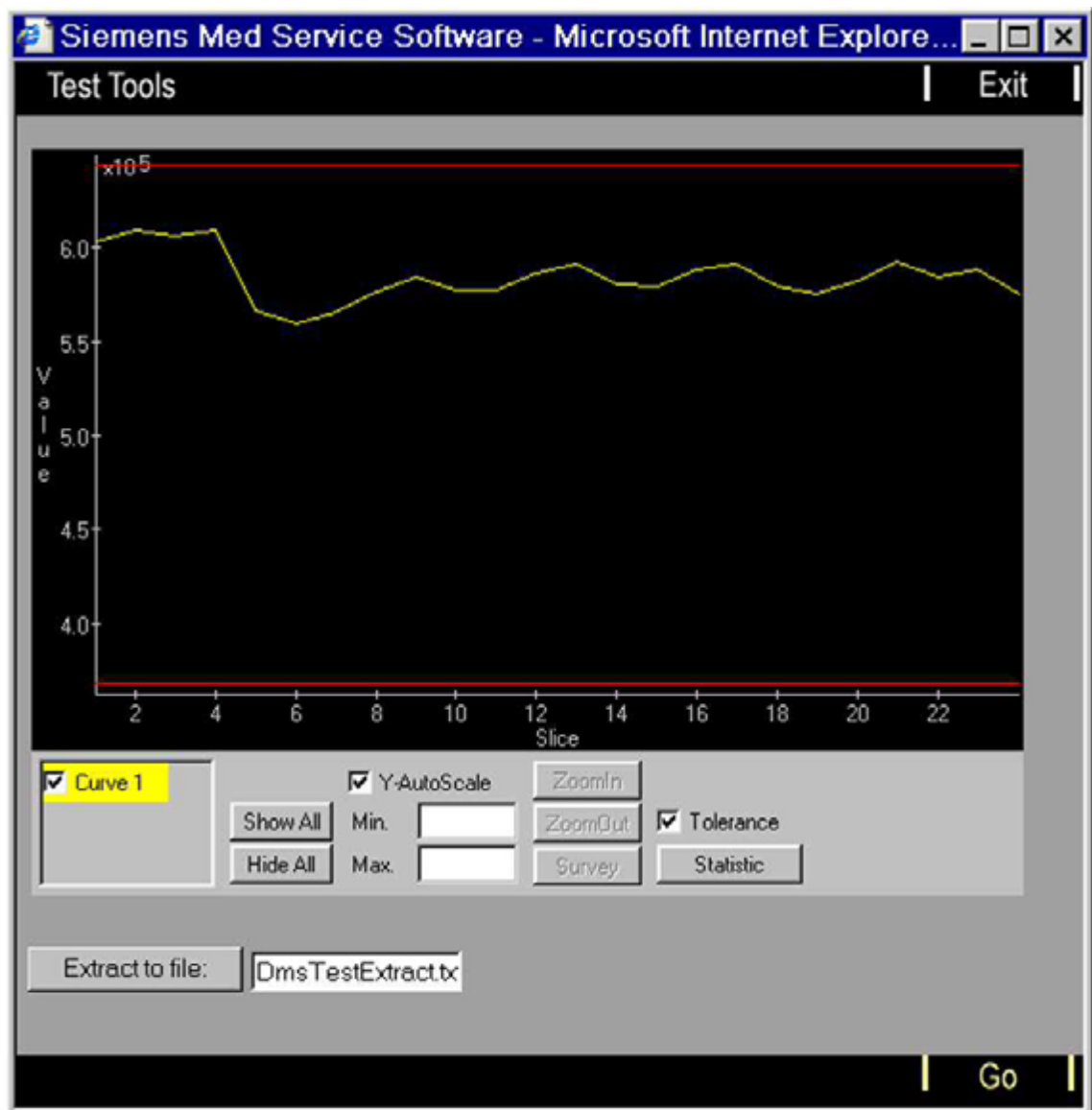


Fig. 95: Absolute Signal Value

Common for all tables is:

- The button **Show Plot** graphically displays the table.
It's possible to select all slices via **Show All**, none via **Hide All**, or a single slice via a checkbox. It's also possible to **Zoom In** (or out via **Survey**) by using the left mouse button. Mark the area of interest, while holding down the left button
The button **Statistic** shows the following for all slices: Minimum, Maximum, Mean, and StdDev.
The check box **Y-AutoScale** allows automatic scaling of the Y-axis
- For values out of tolerance, slice and channel are output.
Click the channel number and read out: Channel xxx and Slot xx.
- The button **Extract to file** is used to store the result of the test to disk.

BaseCal stability**Scope**

- The main function of the BaseCal stability test is to subtract the last two BaseCals of one given mode in order to find unstable channels.

The structure of the BaseCal names is:

basecal_slice_ tube voltage_ scantime_ focus size_ algorithmn_ organ_ focus type.

They are stored in c:\Somaris\service\datastore\icstables

How to find

- **Options > Service > Local Service > Test Tools > DMS-Test > BasCal Stability**

How to use

1. Select the desired base calibration table from the list box.
Use the BaseCal corresponding to the mode the customer is complaining about.

2. Press **Go**

- ⇒ The result page is shown.
- ⇒ Before exchanging modules, compare the result with the results of ROI/Ring to verify whether the customer's image problem has been located.

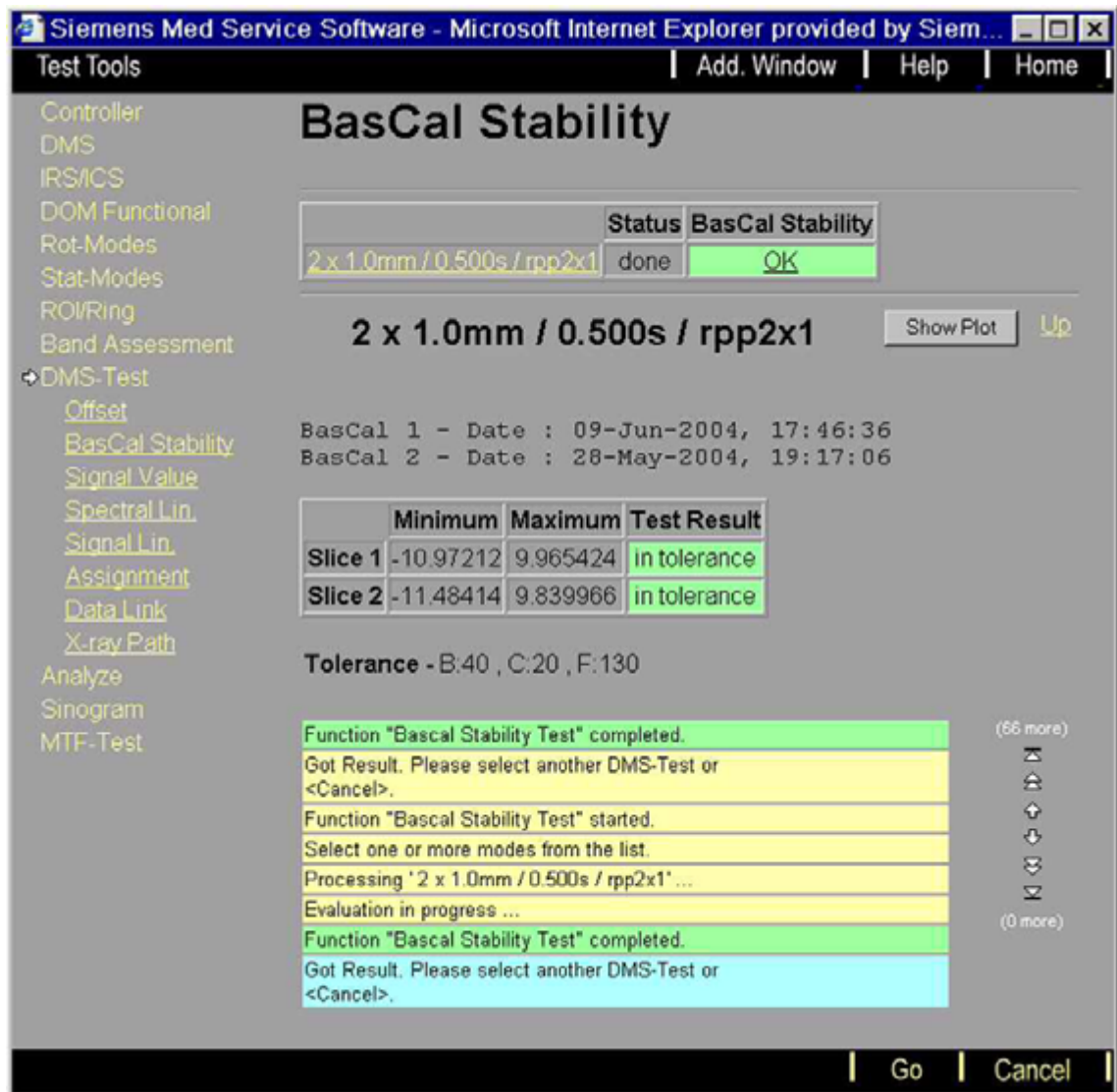


Fig. 96: Example of BaseCal stability test

3. Activate **Show Plot**

⇒ Evaluate the plot

In addition to the standard functions, you can define the Segments which should be shown.

Hint: A BaseCal is referred to as a 360 degree rotation and is divided into 10 angular segments.

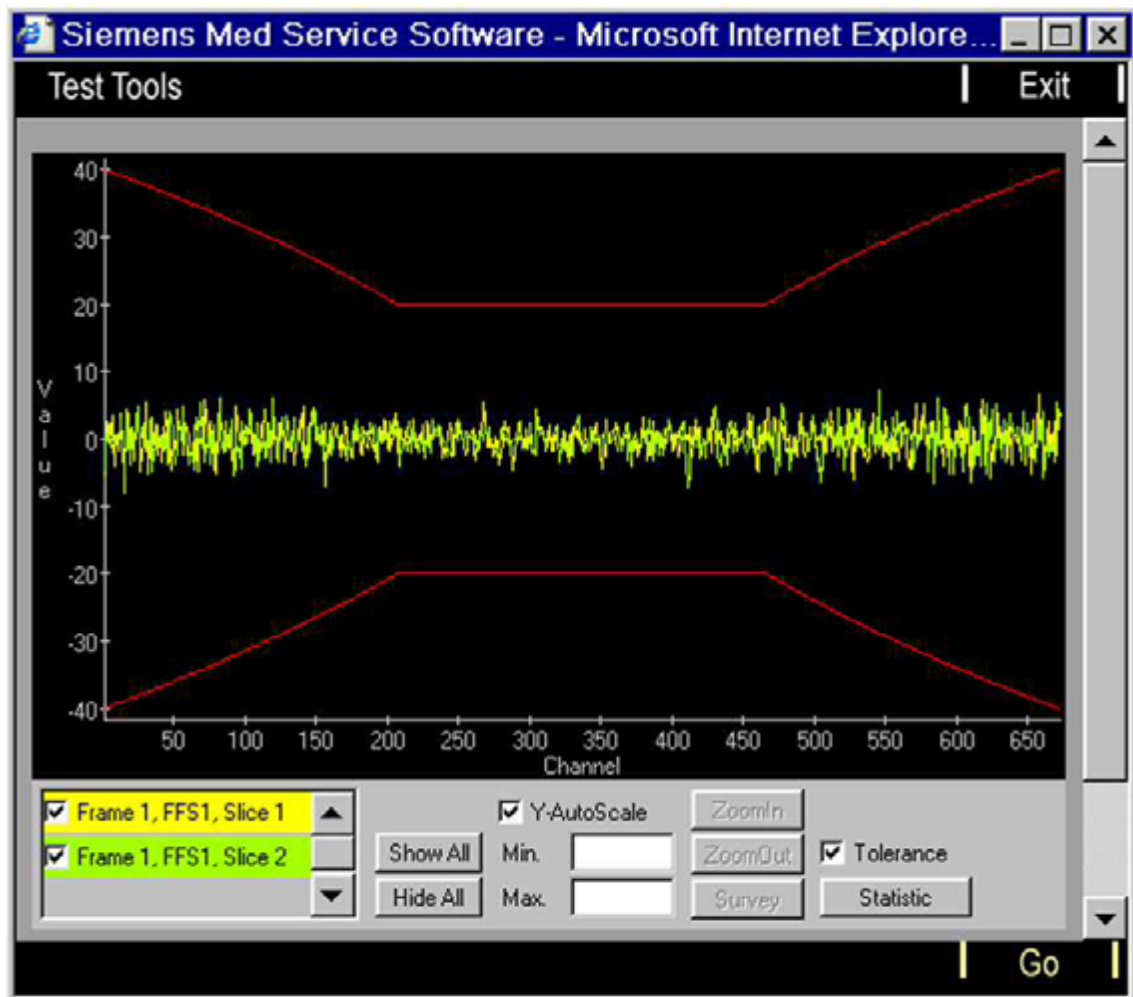


Fig. 97: Example of BaseCal stability result

Spectral linearity

Scope

- This test performs two scans with different tube voltages (120 kV and 140 kV).
Subtracting the mean values of these scans shows the deviation caused by the spectral non-linearity.
The result of this test should be used **only in addition to other tests** to confirm a channel as satisfactory or unsatisfactory.
- This test is normally used in the factory only.

How to find

- Options > Service > Local Service > Test Tools > DMS-Test > Spectral Lin

How to use

1. Wait for the Press **START key** to appear and then press the Start button.
 - ⇒ The test result page is displayed.
 - ⇒ If there are channels out of tolerance, call CS HSC 22.
2. Press **Show Plot**
 - ⇒ The plot is displayed.

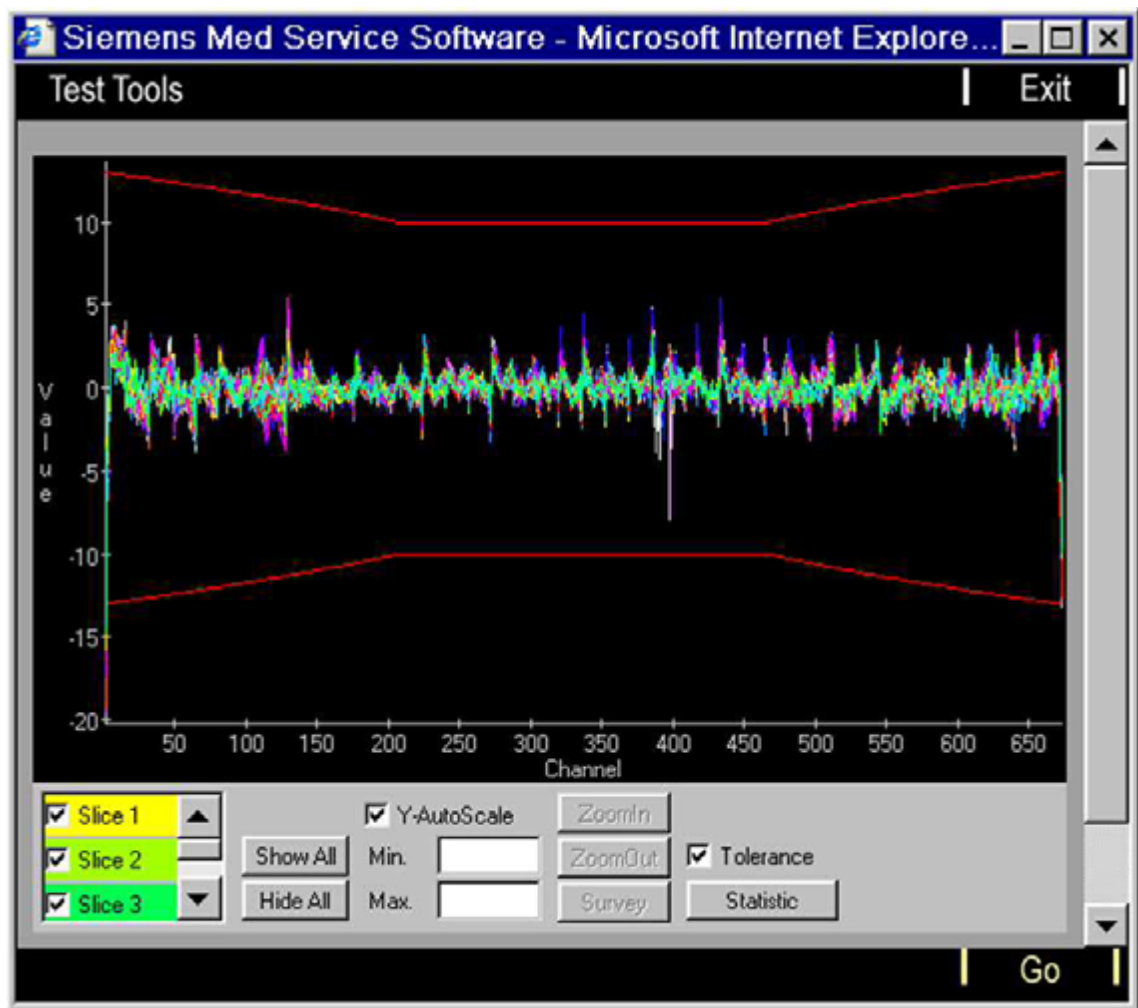


Fig. 98: Example of spectral linearity result

Signal linearity

Scope

- This test performs two scans with different tube currents (100mA and 300 mA).
Subtracting the mean values of these scans shows the deviation caused by the signal non-linearity.
In addition, you may use phantoms in the scan field to check the low signal linearity.

How to find

- **Options > Service > Local Service > Test Tools > DMS-Test > Signal Lin**

How to use

1. Wait for the Press START key to appear and press the Start button
 - ⇒ The test result page is displayed.
 - ⇒ In case channels are out of tolerance, change the corresponding module

2. Press **Show Plot**

⇒ The plot is displayed

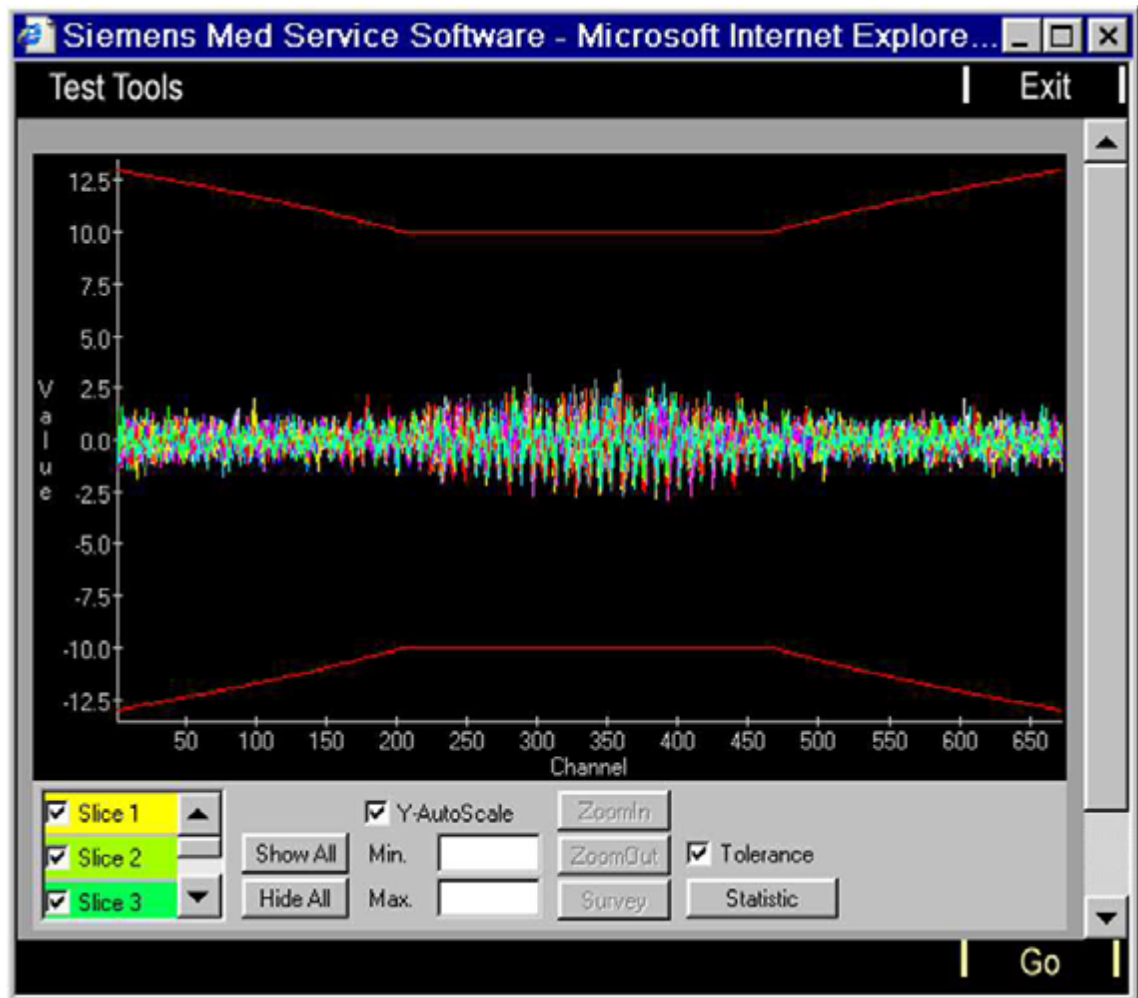


Fig. 99: Example of signal linearity result

Assignment test

Scope

- The assignment test is an addressing test between the DAS Controller and the 42 detector modules

How to find

- Options > Service > Local Service > Test Tools > Assignment**

How to use

- Wait for the Press **START** key to appear and then press the Start button

⇒ Wait for the result page

2. Activate **Show Plot**

⇒ A continuous staircase from upper left to lower right should be visible for all 16 slices.

Hint: In addition to the standard function, you can define:

- **Segment** 1 to 1160 (frames)

and

- **FFS** 1, 2 or all

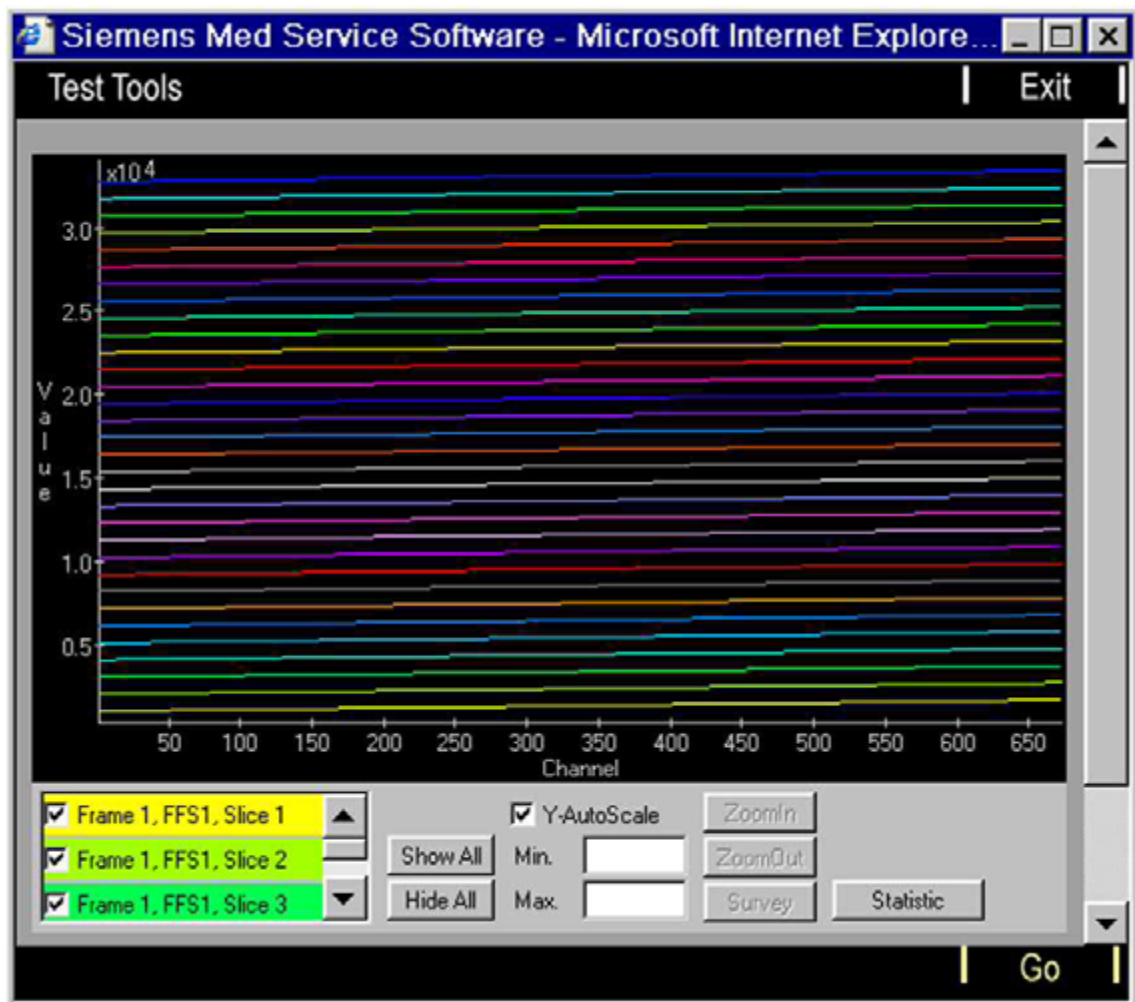


Fig. 100: Example of assignment test result

3. Interpretation of the result

The possibly defective part may be:

- defective detector module
- defective backplane
- defective DAS Controller

X-ray Path

Scope

- This test involves finding any “contamination” in the X-ray path.
A scan is performed and the 2 focal spots are subtracted.
Depending on the position of an object in the X-ray path, the plot looks different.
 - If the object is near the detector, the contamination covers only a few channels.
(Contamination through contrast media)
A peak at one channel is most likely a damaged detector collimator
 - If the object is near the tube (or tube collimator), the plot will cover a lot of channels and the deviation in the plot looks “weak”.
With software version VB19X, the result of this test is high-pass filtered, so an object close to tube-side is not visible.
The subtraction has to be performed manually, using analyze. ([How to manually perform a focus difference / p. 220](#))

How to find

- **Options > Service > Local Service > Test Tools > DMS-Test > X-ray Path**

How to use

1. When the test is selected, a mode with 120kV and 350mA is automatically loaded.
2. Wait for the Press **START key** to appear and then press the Start button.
 - ⇒ The test result page is displayed.
 - ⇒ If channels are out of tolerance, contact CS HSC 221 for further advice.
3. Press **Show Plot** to see the plot
 - ⇒ Capture the plot and supply it to CS HSC 221 for additional evaluation.

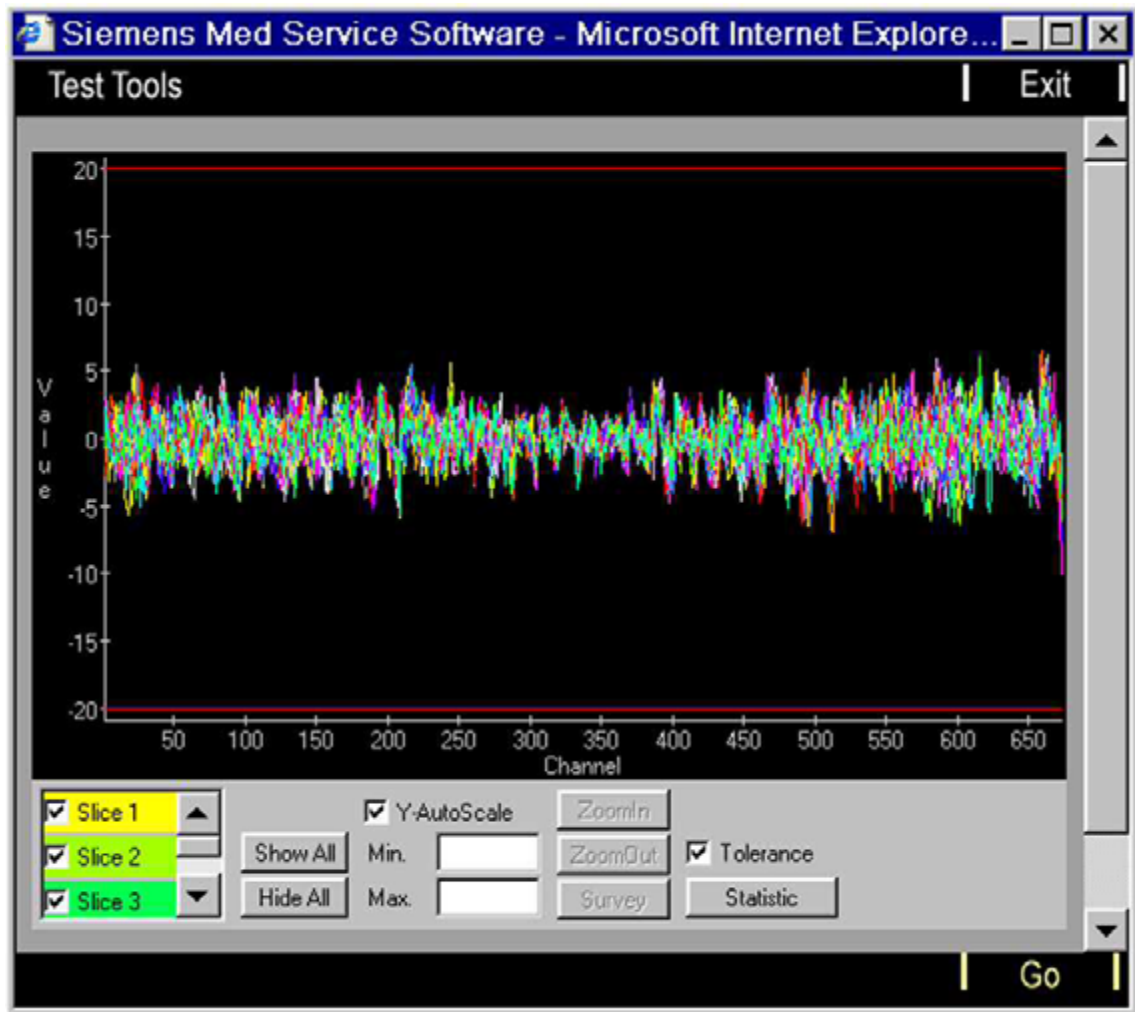


Fig. 101: Example of X-ray path result

Analyze

Scope

- Analyze is a versatile tool, used to evaluate raw data as well as header data
The following functions are useful for field services:
 - Integration time
 - Dose monitor
 - DMS Temperatures
 - Raw Data Frequency Analysis

How to find

- Options > Service > Local Service > Test Tools > Analyze**

How to use

- Using the macros under Special

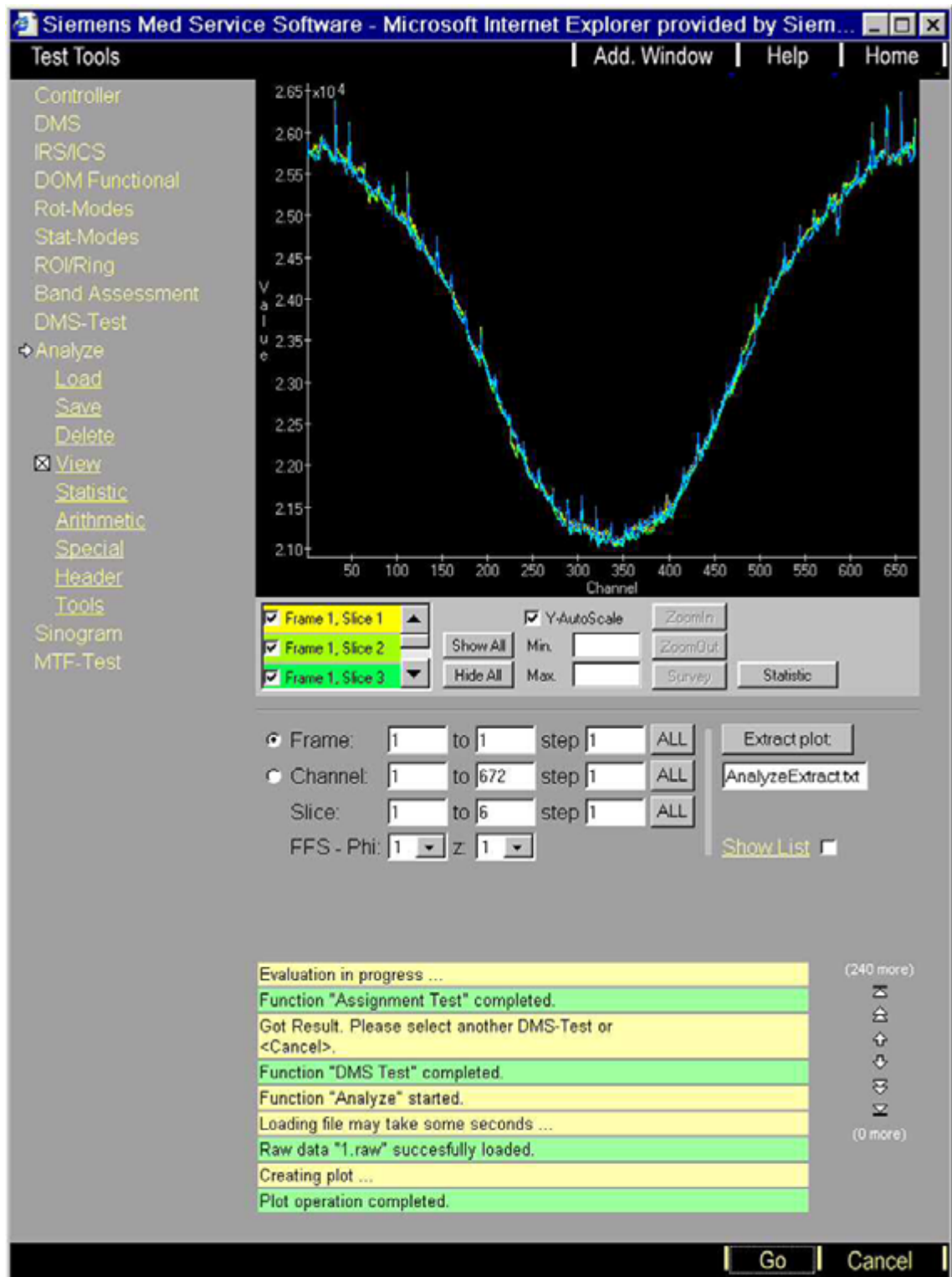


Fig. 102: Example of analyze window

1. Select the radio button **ICS-RawData**
2. Mark the desired raw data set in the list box

3. Press **Go**

⇒ View is automatically selected and the plot appears.

4. Select **Special**

5. Mark the desired checkbox

- ReadingNo
- TubeAngle
- MaximumValue
- ErrorInfo
- CRC Error
- **IntegrationTime**
- **Dose monitor**
- **DMS-Temperature 1**
- **DMS-Temperature 2**
- **DMS-Temperature 3**
- **DMS-Temperature 4**

and press **Go**

⇒ The corresponding curve is displayed.

- **Example for integration time**
Select Tube Angle and Integration Time

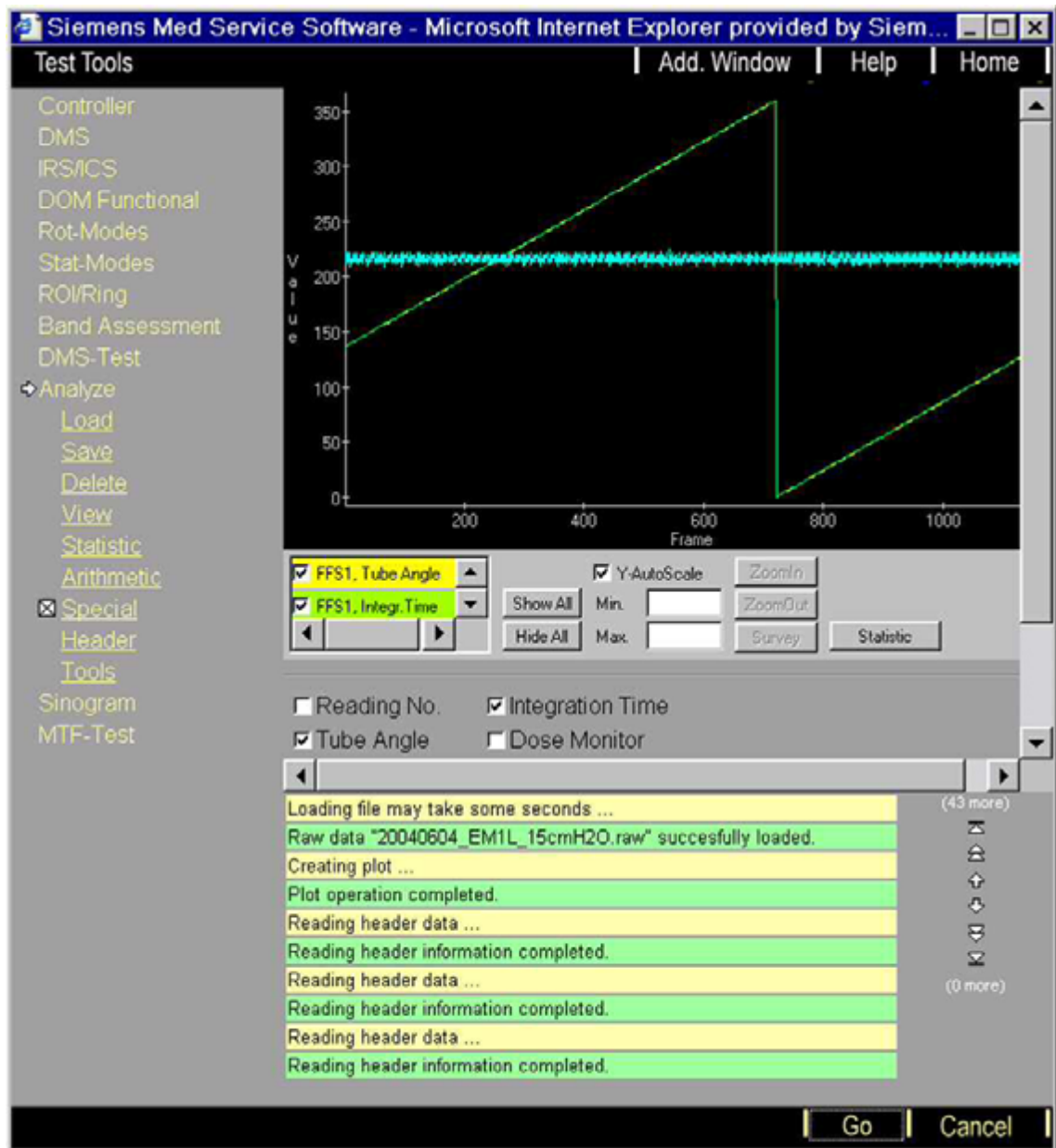


Fig. 103: Example of integration time

- ⇒ The Integration time and the tube angle are displayed.
Use the Zoom function to display details.
Use the left mouse button and point to the curves to display the values.

- Example for dose monitor

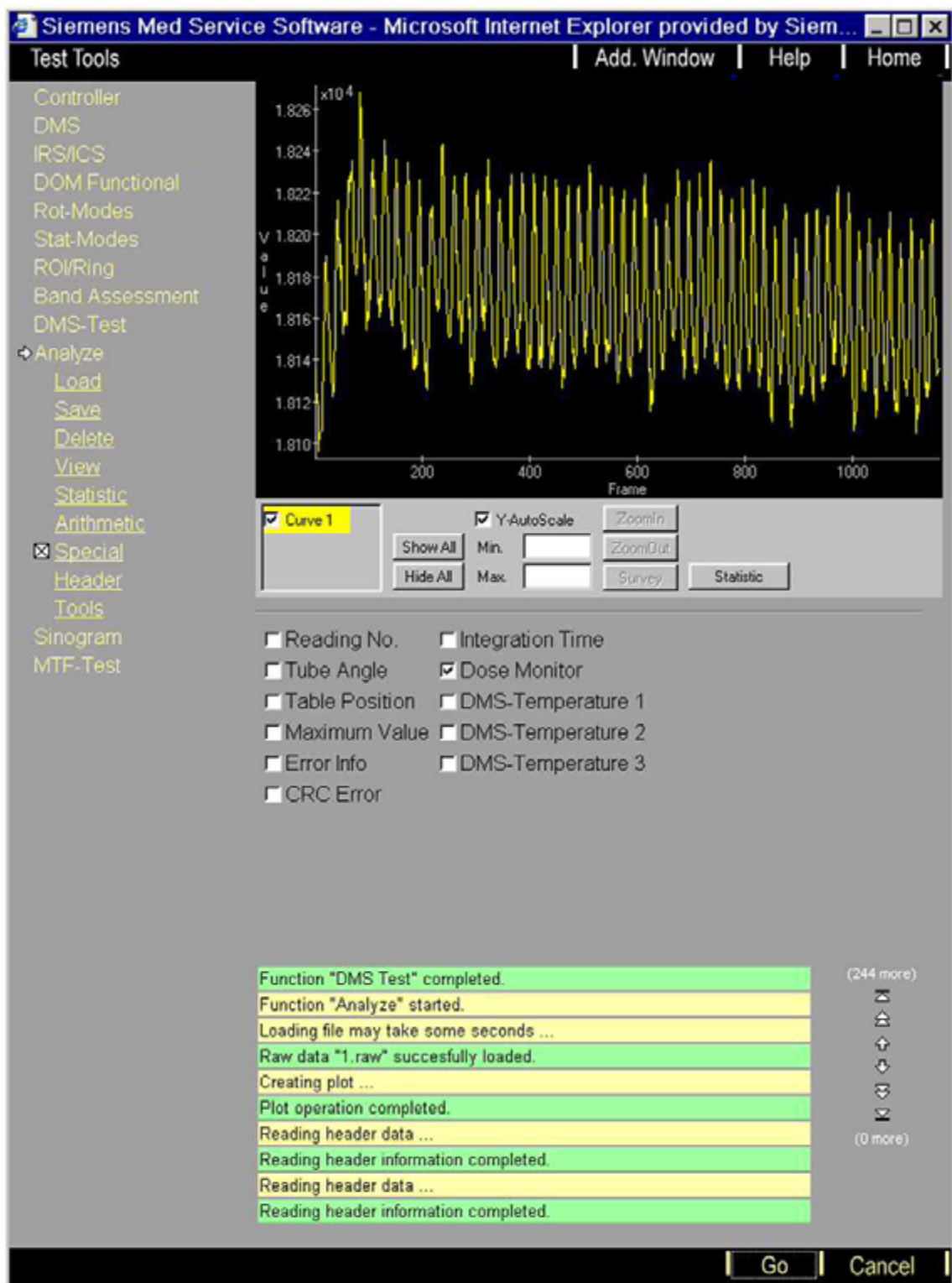


Fig. 104: Example of dose monitor

⇒ The transient response of the dose is displayed

- Example for DMS temperature



Fig. 105: Example of DMS temperature

⇒ The detector temperature is shown on a Centigrade scale.

- **Using the raw data frequency analysis under Analyze**
 - Load a static mode with **2 x 1.0 mm**
 - Perform 2 scans of that mode and use the second one

- Open an additional window
- Start **Options > Service > Local Service > TestTools > Analyze**
- Select **ICS-Rawdata** mark **rotstaticraw** and press **Go**
- Select **Tools**
- Select **Raw Data - Frequency Analysis** and press **Go**
- Select the channels
Start with 305 to 368
- Do not select the checkbox for fusing the slices
- Do not select Separate FFS
- Select all slices

As shown in the example, press Go and zoom the result, if necessary.
The x-axis shows the frequency, and the Y-axis shows the amplitude

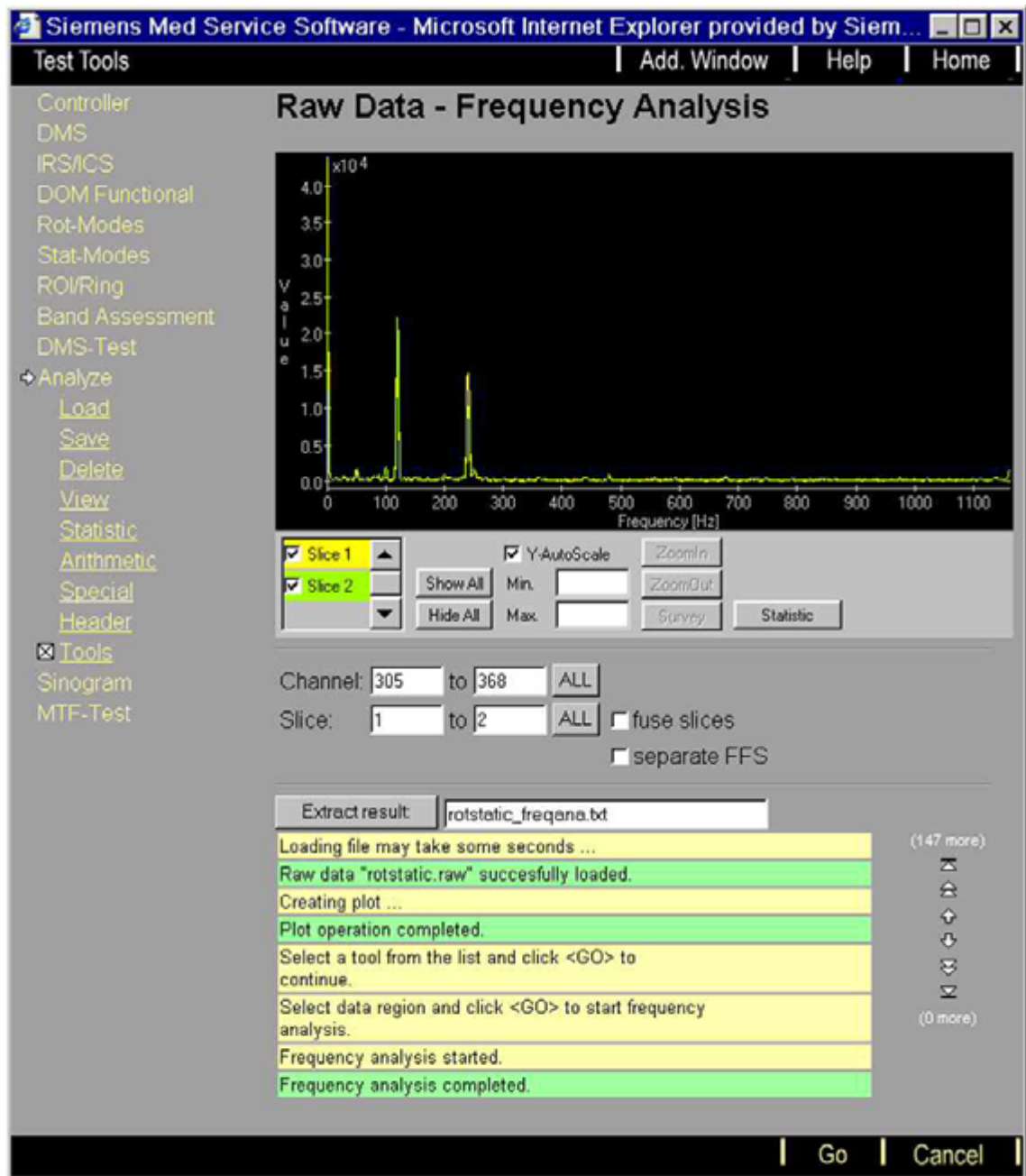


Fig. 106: Example of raw data frequency analysis

- ⇒ The amplitude at the first peak (here 120 Hz) shows the quality of anode rotation (wobbling anode)
- ⇒ The amplitude at the first harmonic shows the quality of the emitter (tube)
- ⇒ Because we currently do not have that much experience send the plot to CS HSC 221 for additional evaluation.

Sinogram

Scope

- The output of the sinogram is an image showing all channels at the X-axis as well as all frames (projections) at the Y-axis.
- Raw data sets of air scans and of scans made with a homogeneous absorber can be evaluated.
- The sinogram is a useful tool for locating weak rings

How to find

- **Options > Service > Local Service > Test Tools > Sinogram**

How to use

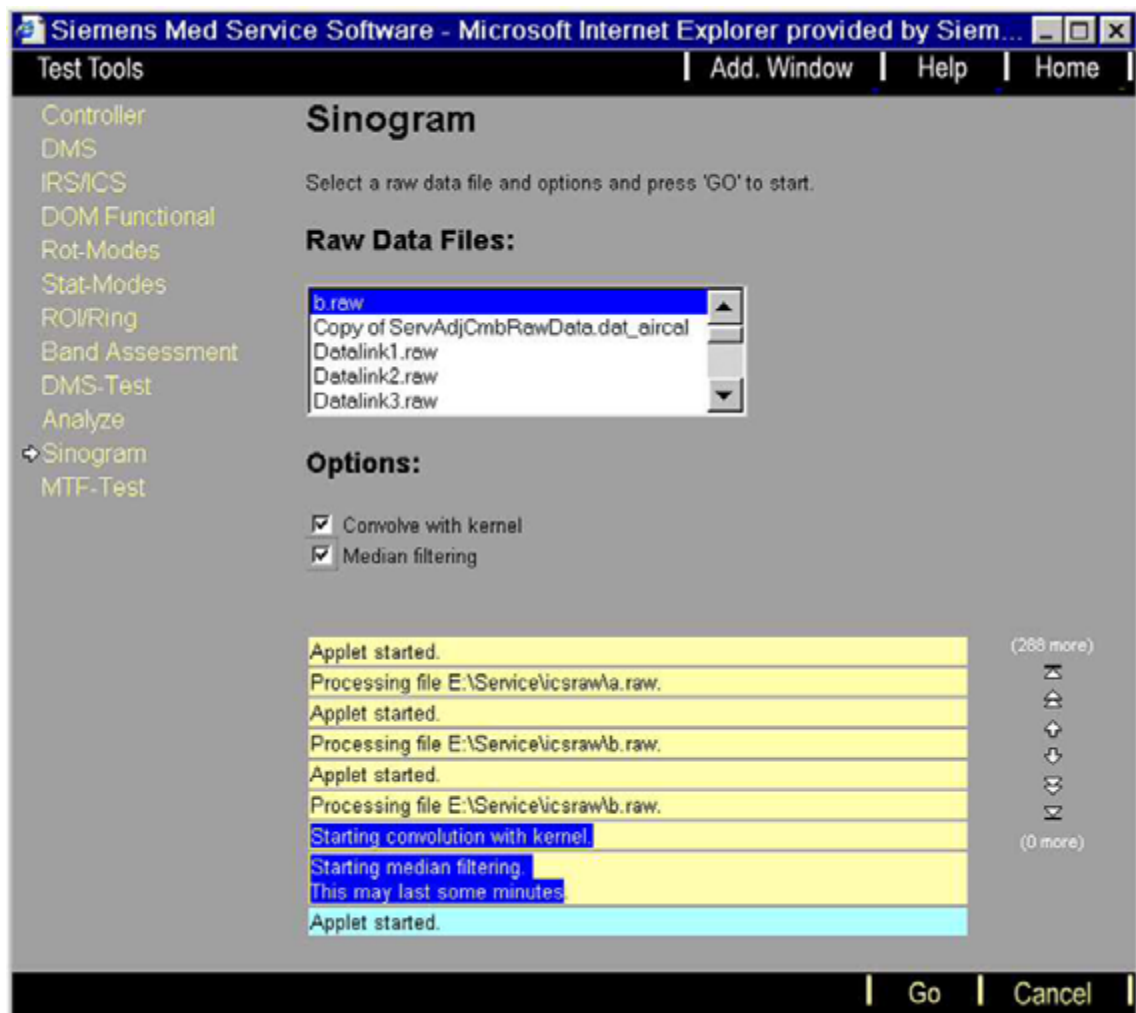


Fig. 107: Example of sinogram test

1. Mark a file in the list box Raw Data Files.
2. Mark or unmark the checkboxes for **Convolve with Kernel** and **Median filtering**.
Start with the marked checkbox for Convolve with Kernel.

3. Press **Go**

- ⇒ The plot is displayed and you can select the slice via the select box.
- ⇒ Using the left mouse, mark a position to display the corresponding frame and channel.
- ⇒ The mean value and standard deviation of a marked area are shown, if you use the left mouse button to mark the desired area.

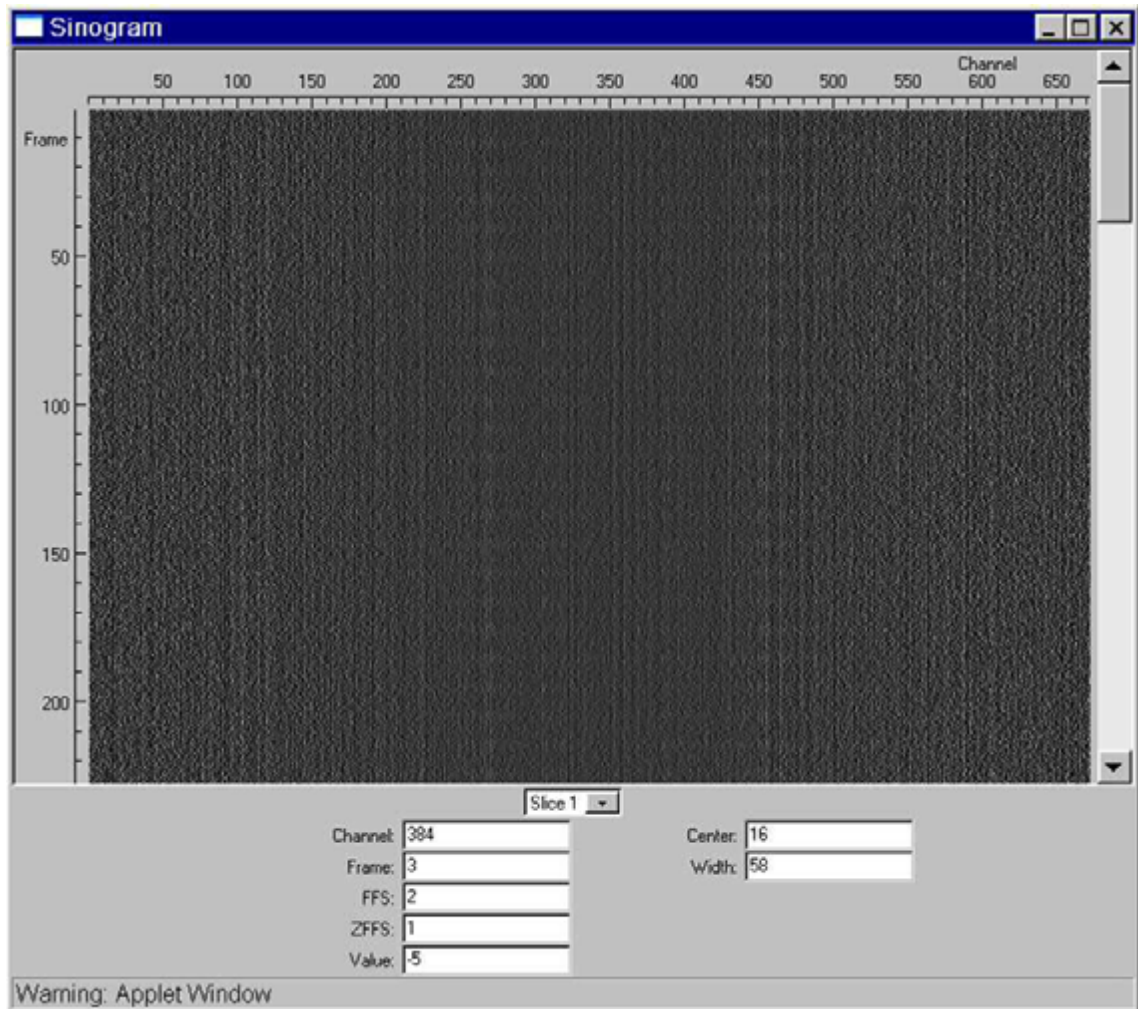


Fig. 108: Example of sinogram result

4. Interpretation of results:

- A vertical line shows up in all frames, if a channel is defective.
- A vertical dotted line shows up, if a channel is defective but not throughout all frames.
- A horizontal line shows up, if a frame is defective (projection).
- A dot shows up, if a channel is defective in one frame.
- A sine wave shows up, if a stationary absorber is located in the scan field.
(Absorber is not moving with DMS)

Data link

Scope

- This test checks the physical link level of the data link.

For this purpose, test data packets are sent from the DAS Controller to the receiver (G-REC, D24) of the IRS.

The received packets are checked by performing a CRC (cycling redundancy check).

The test includes the following parts:

DAS Controller (partial) -- W590 -- Tx -- Slip ring -- Rx - - W55 -- G-REC (partial)

Various **predefined modes** (Auto test modes) allow the user to identify relations between the error and:

- the X-ray (various tube positions, different power)
- Rotation (various rotation times)
- In addition, the following **freely selectable modes** are available.

Before using these modes, define in

Options > Service > Local Service > Test Tools > Stat-Modes / Rot-Modes

the desired parameter.

- Static 20.dat
- Static 20.dat, X-ray
- Rot20.dat
- Rot20.dat, X-ray

How to find

- **Options > Service > Local Service > Test Tools > DMS-Test > Data Link**

How to use

1. Select the desired mode from the list box
Standard test parameters should be Rot 0.50s and X-ray off
2. Press **Go**

3. Wait for the Press **START** key to appear and then press the Start button.
 - ⇒ Several scans are performed
 - ⇒ The results page is shown
 - Important: No CRC Errors and Data Errors are allowed (CRC error Sum:0 and Data Error Sum:0)
 - ⇒ In case of errors, troubleshoot the data path

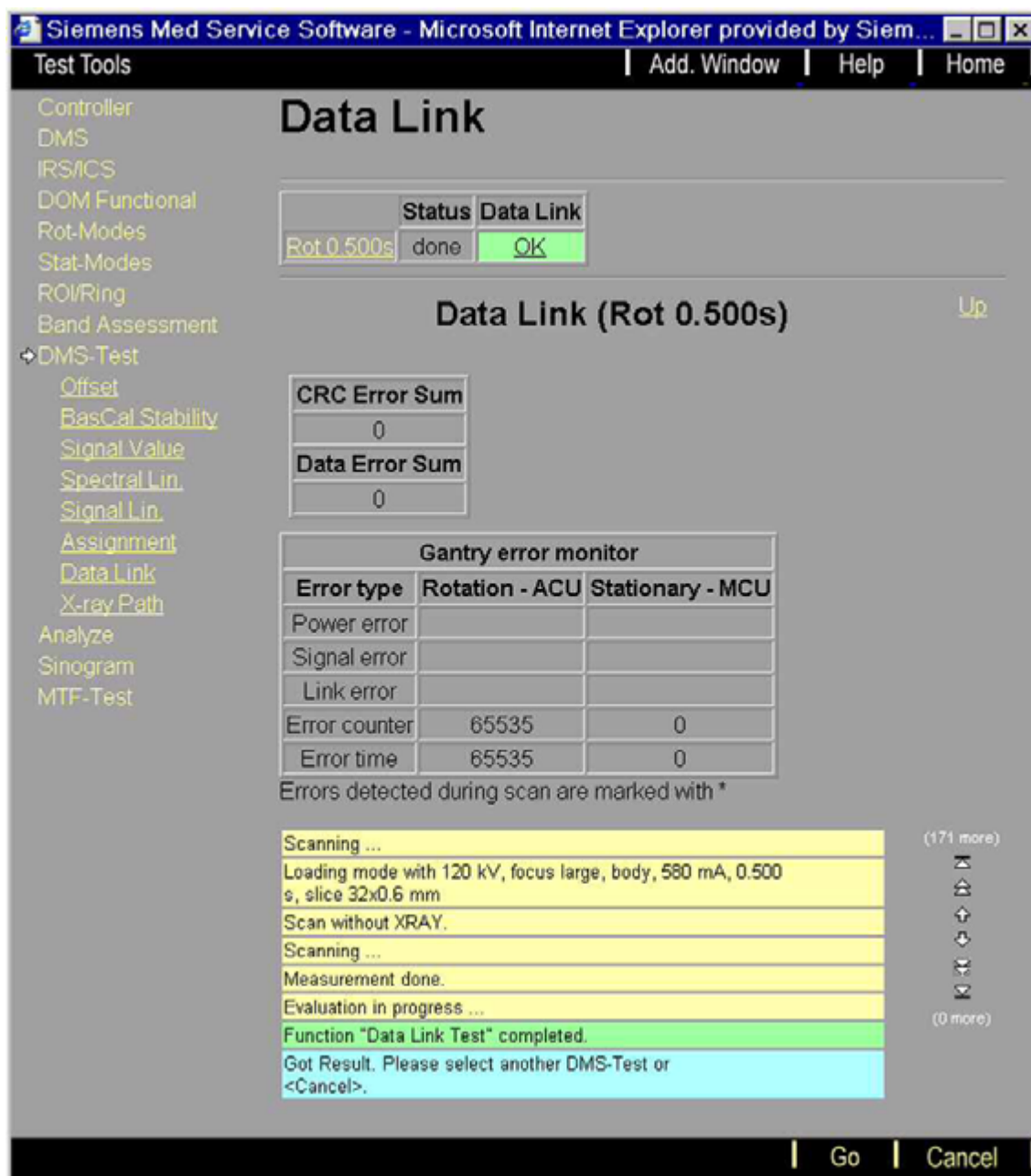


Fig. 109: Example of data link test

For details with respect to interpreting results, refer to: Troubleshooting the data path

Disk test

Scope

- The Disk Test uses a predefined test pattern stored in the DAS Controller. This test pattern is transmitted from the DAS Controller to G-REC and, in addition, all reconstruction steps are performed on that test data. The IRS checks the reconstructed data via CRC. It is also possible to use a test pattern from the receiver (IRS).
In case of error, this test may be used to distinguish between the DMS and the IRS (including data transmission path).

How to find

- **Options > Service > Local Service > Test Tools > IRS/ICS > IRS Disc Test**

How to use

1. Select the source of test data by activating the corresponding radio button:
 - **Transmitter with rotation** (data source is DAS Controller)
 - **Transmitter without rotation** (data source is DAS Controller)
 - **Receiver** (data source is IRS)
 - Select a mode
2. Press **Go**
3. Wait for the Press **START** key to appear and then press the Start button
 - ⇒ The results page is shown
 - After the test has finished, "IRS Disc Test successful" is printed out
 - A disk image must be visible (ignore the rings)
 - Check for error messages (in the Somaris log as well)
 - ⇒ In case of error, continue testing the data path.

MTF test

Scope

- The MTF (Modulation Transfer Function) shows the spatial resolution at high contrast. For this purpose, a 0.22mm tungsten wire phantom is used.

How to find

- **Options > Service > Local Service > Test Tools > MTF-Test**

How to use

- Follow the software

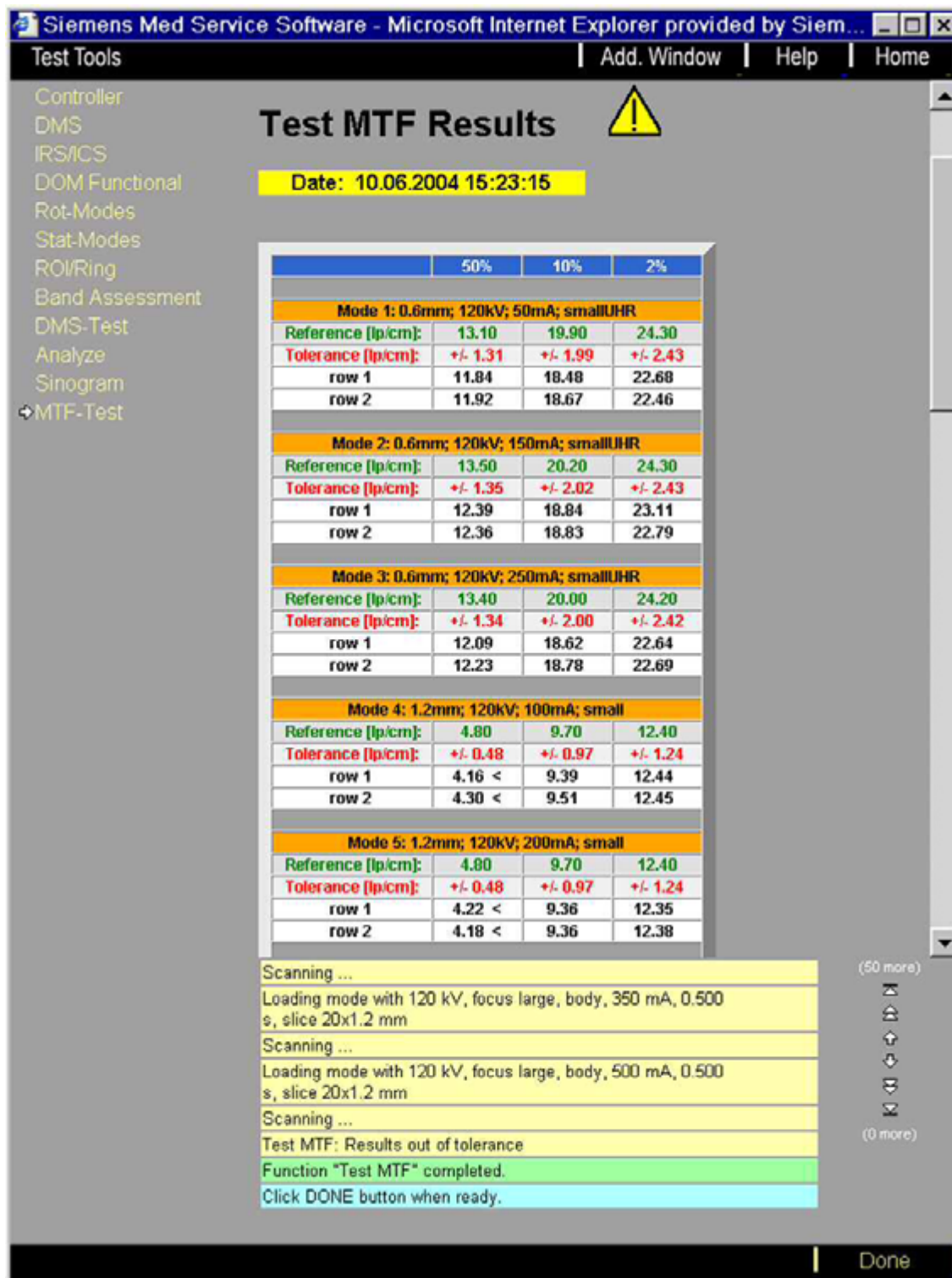


Fig. 110: Example of MTF test result

- ➡ If result is slightly out of tolerance
 - Perform FocAli and Z-Focus check
 - If FocAli solves the problem, continue with tune-up X-Ray Tube TuneUp

- ⇒ If the result is not plausible
 - Troubleshoot the FOC board and the tube collimator (PHI-PSD)
- ⇒ If only the UHR modes are out of tolerance
 - Perform CombAli alignment

Z position check of modules

Scope

- The detector modules are no longer fixed with guide pins, so it is possible to position the modules at a wrong position during replacement. With software $\geq$ VB20x, an automatic module position check is implemented in the guided tours. Follow the software instruction in this case.

For systems with installed software $\leq$ VB20, the following check has to be performed after replacing a detector module.

The Z alignment of the module is tested by evaluating 2 air scans with a different tube collimator position.

How to use

A workaround using analyze is listed below:

1. Remove all absorbers, use a static mode, and select the following parameters:
120 kV, 300 mA, 20 x 1.2 mm, 1 sec and all processing steps
(the checkbox for PHIRPP and Focus large is also selected)
It is important to select the checkbox "save raw data"
2. Press **Go** to load the selected mode
3. Wait for the Press **START key** to appear and then press the Start button
4. Load the same mode again by pressing **Go**
5. Open an **Additional Window** and select **Control** and then **General**
6. Select **Collimator** and change the opening from 9.8mm to 8.0mm and press **Go**
7. Press the **Start** button
8. Open **Analyze** and load the first raw data set
Identify the raw data set by evaluating the time stamp
(rotstaticraw"Date and time")

9. Select **Arithmetic, Subtract** and under **ICS-Rawdata** the raw data set from the second scan

In addition select the following for both data sets:

Frame 1 - 1160

Channel 1 - 672

Slice 1-20

FFS: Phi:1, Z:1

and press **Go**

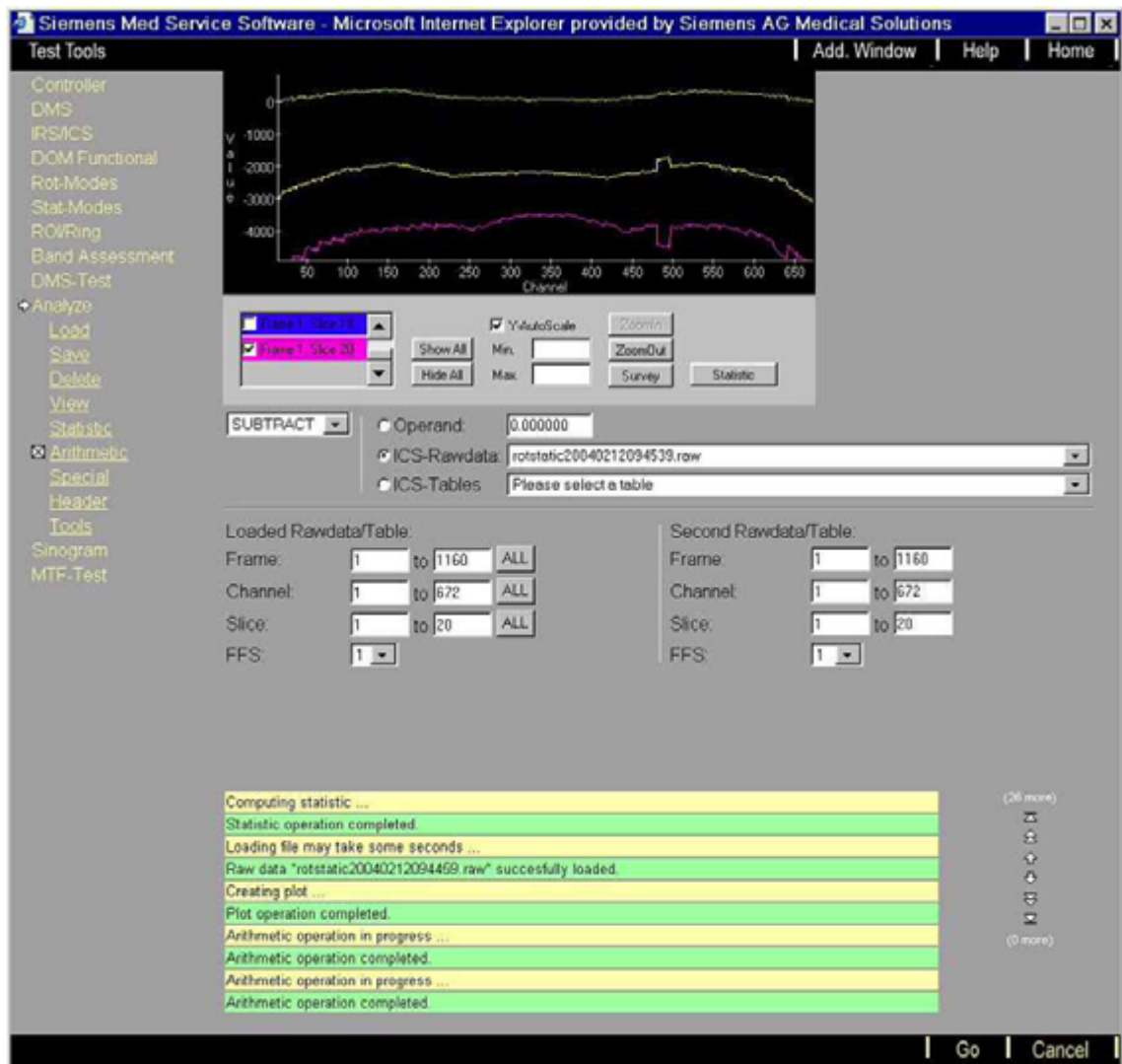


Fig. 111: Unsatisfactory Z position of module

- ⇒ In this example, two slots are not aligned properly
 You see the drop in the graph in the first and last slice.
 Hint: The two outer slices may also be positioned above the zero line.
 It depends which raw data set is selected first.

10. Check if there is a drop of 16 channels, especially at the position where you performed repairs.

If yes, loosen the module exchanged previously and press it toward the rotating plane while you are fixing the situation. Repeat the test.

Z check of tube collimator

Scope

- This test checks the Z monitor (PSD) and Z control

How to find

- **Options > Service > Local Service > Tuneup > Expert Mode > ZCheck**

How to use

- Follow the software
 - ⇒ If the result is slightly out of tolerance
 - Perform a tune-up. Follow the guided tour for Tube Collimator
 - ⇒ If the result is not plausible
 - Troubleshoot the tube collimator and the FOC board

Z Focus check

Scope

- This test checks the Z focus movement.
Also the control of PHI movement is checked.
The PHI closed loop control comprises the FOC board and the PHI PSD.
The Z-feed forward control is performed only by the tube system.

How to find

- **Options > Service > Local Service > Tuneup > Expert Mode > ZFocus Check**

How to use

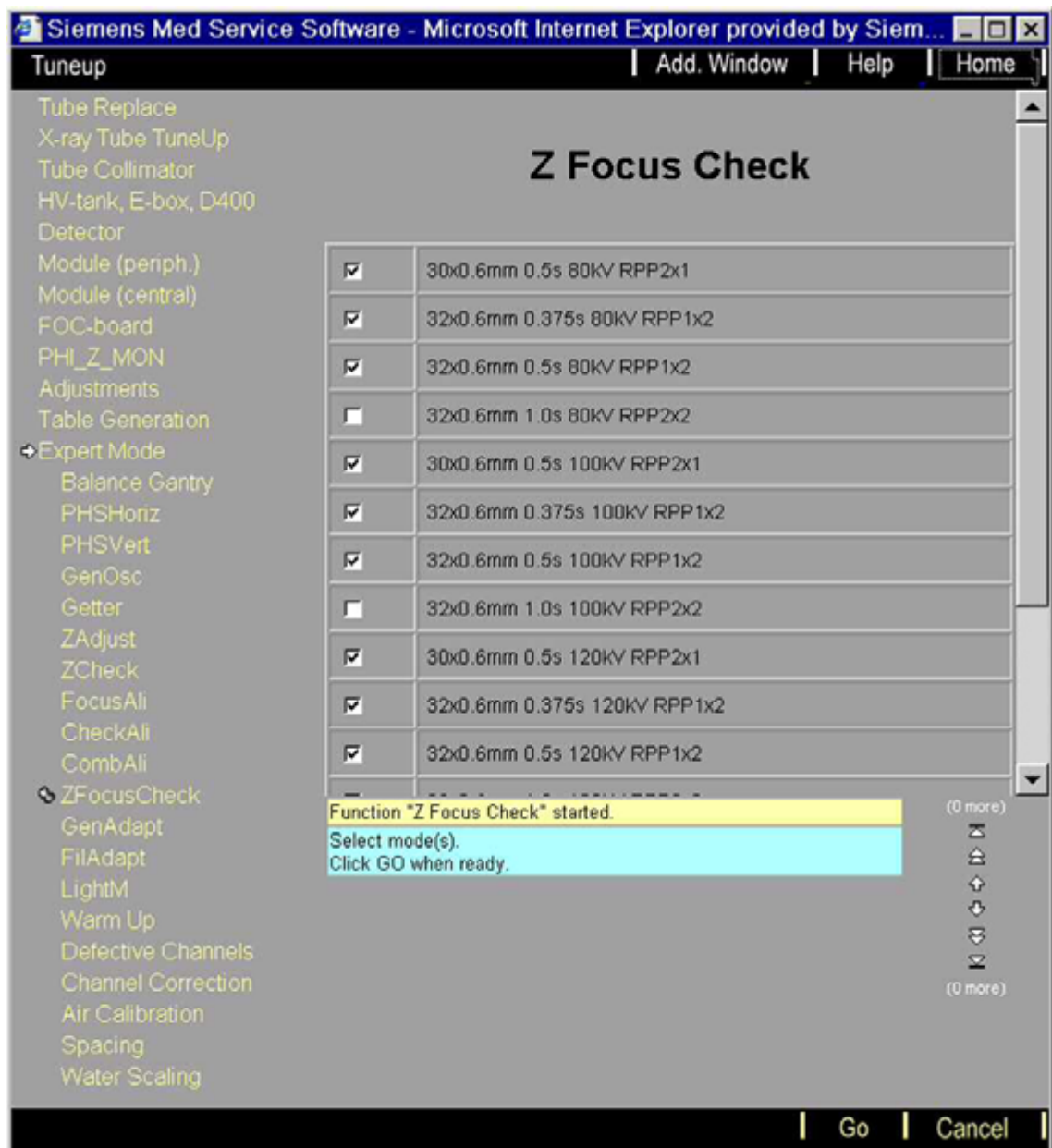


Fig. 112: Z Focus Check

- Follow the software

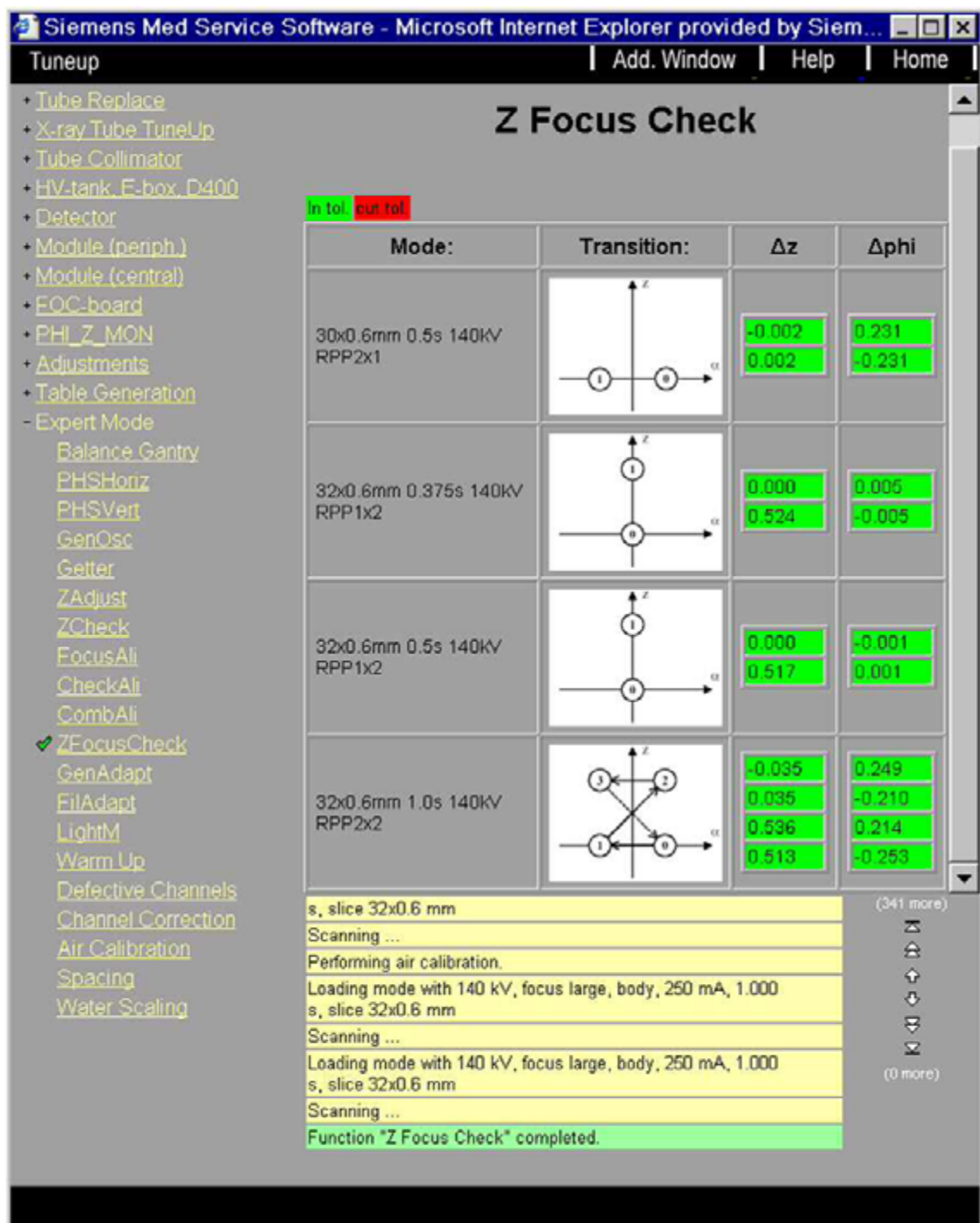


Fig. 113: Example of Z focus test result

- ⇒ If the result is out of tolerance
 - Repeat the test to find out if the result is failing constantly
 - Perform a ZAdjust and FocAli and repeat the test
 - Exchange X-Ray Tube
- Hint: **Contact CS HSC 221 before ordering the tube**

Check alignment

Scope

- This test checks if the focus deflection is working properly.
Phase offset and amplitude have to be in range.

How to find

- **Options > Service > Local Service > Tuneup > Expert Mode > CheckAli**

How to use

- Follow the software

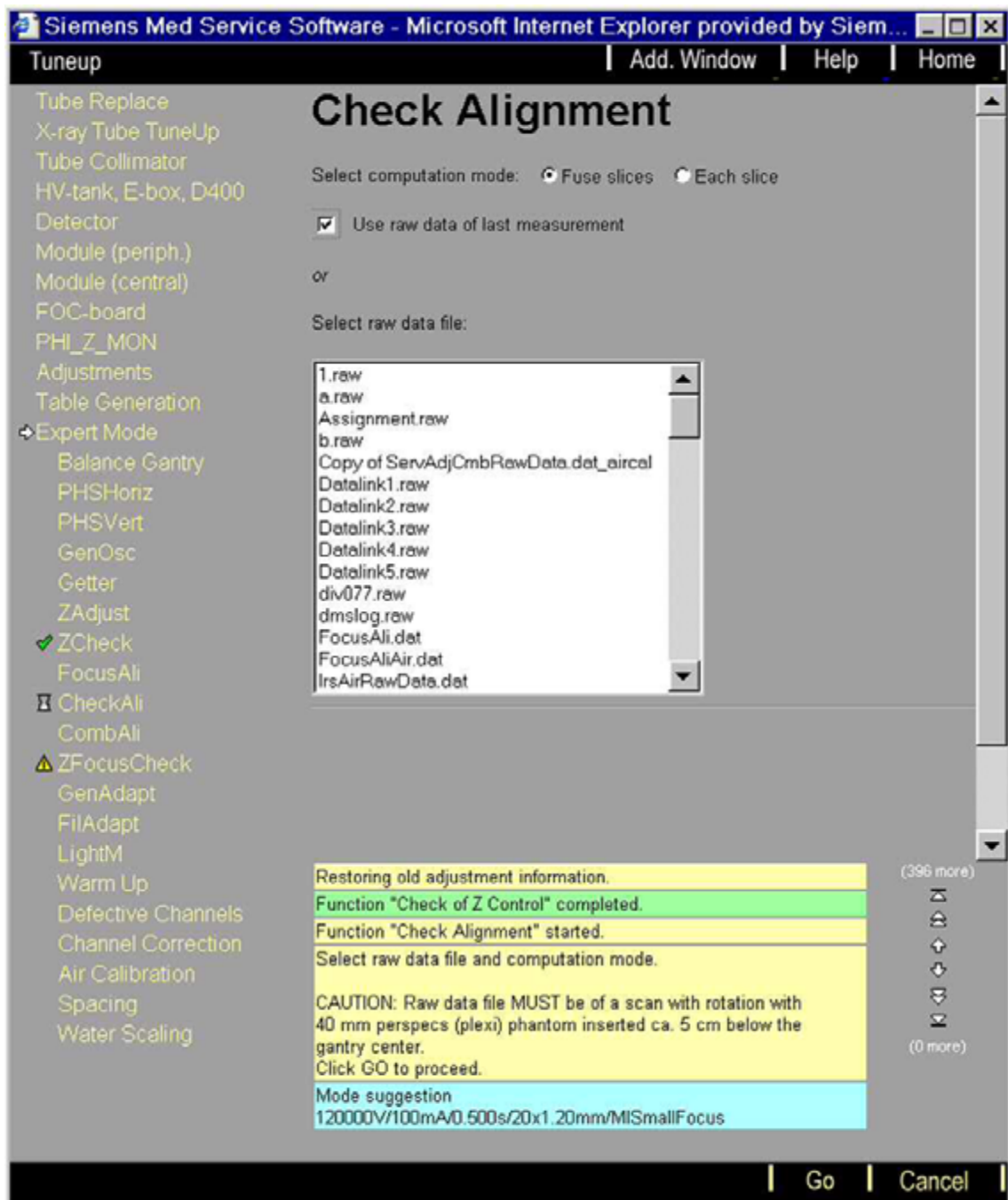


Fig. 114: Example of check alignment test

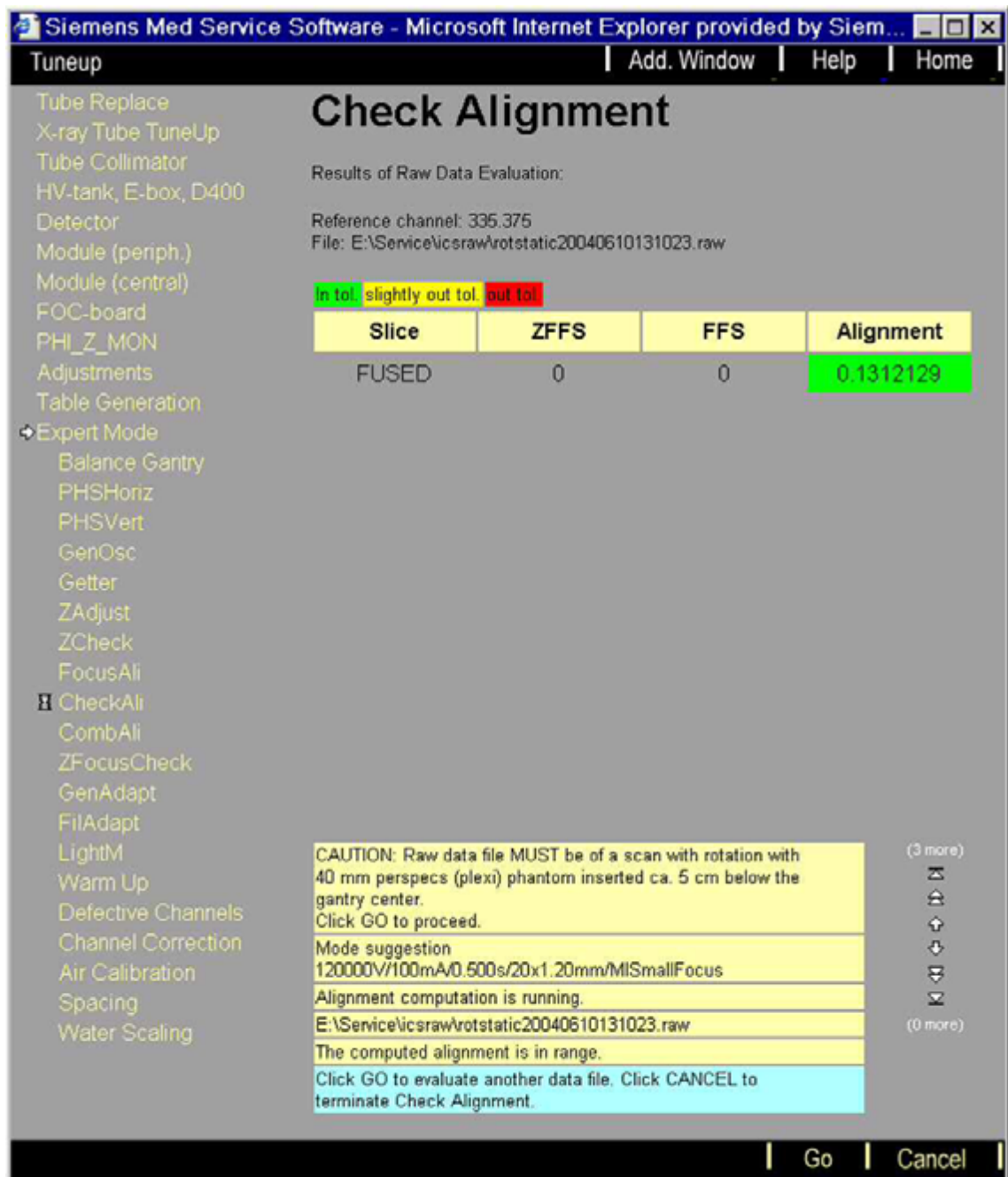


Fig. 115: Example of check alignment result

- ⇒ If the result is slightly out of tolerance
 - Perform a tune-up. Follow the guided tour for the X-Ray tube
- ⇒ If the result is not plausible
 - Check if FocAli was performed after the tube replacement
 - Troubleshoot tube collimator and FOC board

DMS mechanics

If there are problems with the UHR mechanics, follow the advice provided in the error message.

To prevent room-down because of a broken UHR, it is possible to deactivate the function for the UHR in the control system. ([How to deactivate the UHR comb / p. 222](#))

To understand the function of the UHR control (Implemented at FOC, D514), the current applied to the UHR motor is shown in the next figure.

UHR phi-z comb: Current measurement is not possible. Follow the mechanical checks.

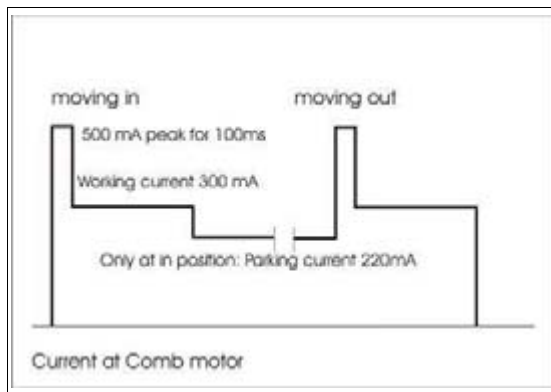


Fig. 116: Current of UHR motor

To identify problems of the UHR control, you may perform the following tests:

Mechanics

- Check whether the mechanics are operating smoothly by moving the UHR in and out by hand. (remove the UHR motor to be able to move the UHR without applying too much force)
 - ⇒ If the mechanics are not moving smoothly:
 - Check for parts blocking the UHR.
 - Grease the mechanics according to the maintenance instructions

Limit switches

- If the problem is related to the limit switches:
 - Check their functionality using the service switch S1 at D514 and move the UHR in and out by pressing S3 (UHR out) or S4 (UHR in) .
Ensure that the UHR LED at D514 is activated at both mechanical end stops.
 - ⇒ If the LED is not activated at the end stop on one side:
Position the UHR in the middle, so that no limit switch is activated.
Use a piece of metal, e.g., a screwdriver to check whether the limit switches are working properly.
 - ⇒ If the LED in the limit switch is not activating:
→Check the power supply for that switch at UHR control D584 X82 or X83
→Replace the corresponding limit switch
 - ⇒ If the LED's at the limit switches are working fine:
→Check cable W594
→Replace D514
→Replace limit switch (also if LED is working)
 - For checking the adjustment of limit switches, refer to the document Replacement of Parts: Gantry.

UHR motor

- If the problem is related to the UHR motor
 - Check their functionality using the service switch S1 at D514 and move the UHR in and out by pressing S3 (UHR out) or S4 (UHR in) .
Observe whether the UHR is moving.
 - ⇒ If the UHR is not moving:
Measure the voltage for the UHR motor at UHR control D584 X81 while activated via the service switches.
→If the motor is not moving while voltage is present, replace the motor.
→If there is no voltage, replace D514
 - ⇒ If the **UHR-plus comb** is not moving:
Measure the voltage for the UHR motor at UHR control D586 J1 (power) while activated via the service switches.
→If the motor is not moving while voltage is present, replace the motor.
→If there is no voltage, replace D586

Adjustment

- To ensure the UHR is adjusted correctly in the phi-direction, perform an MTF measurement.
If the results are in tolerance, the adjustment is OK.
If the results are out of tolerance, perform the COMB/ALI adjustment in the tune-up.
If you got a fully de-adjusted comb, you can identify the core position by evaluating a mean value plot of an UHR mode. ([Core position check of UHR comb and UHR-plus comb / p. 221](#))

Other sources of problems

To troubleshoot image quality related problems, especially weak rings, be sure to check the event log as well for environmental problems and error messages concerning the following parts.

Power supply

The power supply for the DMS is located in E607. To check the supply, perform a measurement at the front side of the DMS power supply.

In addition, the supply voltages can be checked as well on the left and right backplane.

Important: The tolerances given below are valid only for the test points at the backplanes.

If voltages are out of tolerance:

1. Adjust the voltage as described in "replacement of parts".
2. If not adjustable, exchange the DMS power supply. ([CT02-023.841.01 / DMS power supply](#))
3. To check the DMS power supply manually, evaluate the clearly marked test points at the **DMS power supply**:

⇒ If a voltage is missing, first check the correspondent fuse at the distribution board.

Test point	Ground	Description
+24V	AGND	24V / 6A controller / fan
pos_V_D	DGND	3.3V / 11A module digital
pos_V_A	AGND	4.5V / 21A module analog
neg_V_A	AGND	-3.7V / 18A module analog

- To check the voltages at the **left DMS backplane**, evaluate the following test points and LED's:

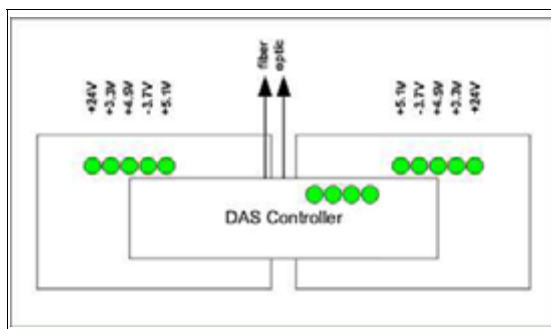


Fig. 117: LEDs of DMS backplane

LED	Test point	Ground	Description / Tolerance
V15 / +24V	TP59 +24V	TP60 / +24V_RET	24V for fan +/- 10% 100mV ripple
V14 / 3.3V	TP56 pos_V_D	TP58 / DGND	3.3V digital +/- 5% 30mV ripple
V12 / 4.5V	TP55 pos_V_A	TP61 / AGND	4.5V analog +/- 4% 20mV ripple
V13 / -3.7V	TP57 neg_V_A	TP61 / AGND	-3.7V analog +/- 4% 20mV ripple
V20 / 5.1V	TP70 +5.1V	TP60 / +24_RET	5.1V generated at backplane

- To check the voltages at the **right DMS backplane**, evaluate the following test points and LED's:

LED	Test point	Ground	Description / Tolerance
V11 / +24V	TP59 +24V	TP60 / +24V_RET	24V for fan +/- 10% 100mV ripple
V10 / 3.3V	TP56 pos_V_D	TP58 / DGND	3.3V digital +/- 5% 30mV ripple
V8 / 4.5V	TP55 pos_V_A	TP61 / AGND	4.5V analog +/- 4% 20mV ripple
V9 / -3.7V	TP57 neg_V_A	TP61 / AGND	-3.7V analog +/- 4% 20mV ripple
V14 / 5.1V	TP70 +5.1V	TP60 / +24_RET	5.1V generated at backplane

- The layout of the connectors X100 and X200

Pin	X100	X200
A1	pos_V_D module digital	pos_V_D module digital
A2	DGND digital ground	DGND digital ground
A3	pos_V_A module analog	pos_V_A module analog
A4	n.c.	n.c.
A5	AGND analog ground	AGND analog ground
A6	n.c.	n.c.
A7	neg_V_A module analog	neg_V_A module analog
1	+24V controller and fan	+24V controller and fan

Pin	X100	X200
2	+24V controller and fan	+24V controller and fan
3	24V return	24V return
4	24V return	24V return
5	n.c.	+ sense pos_V_D
6	n.c.	- sense pos_V_D
7	n.c.	+ sense pos_V_A
8	n.c.	- sense pos_V_A
9	n.c.	+ sense 24V
10	n.c.	- sense 24V
11	n.c.	+ sense neg_V_A
12	n.c.	- sense neg_V_A
13	+ inhibit pos_V_D	n.c.
14	- inhibit pos_V_D	n.c.
15	n.c.	+ inhibit neg_V_A and pos_V_A
16	n.c.	- inhibit neg_V_A and pos_V_A
17	n.c.	n.c.

- To track down a short circuit in the power supply, cables, backplanes, DAS Controller, or modules, perform the following:

Before following this advice, be sure the mains input for the DMS power supply is ok.

(e. g. check if DPU is powered on, which is wired in parallel)

Check if the fan is working

- ⇒ If the fan is not working
 - Replace the DMS power supply

Disconnect the following parts and check if the output of the power supply is present

- X30 at left DMS backplane

- ⇒ If power supply is present now
 - Problem is related to:
 - Left backplane
 - DAS controller
 - Reference Monitor board
 - All modules connected to the left backplane
- continue with disconnecting above parts to isolate the fault

- X31 at right DMS backplane
 - ⇒ If power supply is present now
 - Problem is related to:
 - Right backplane
 - DAS controller
 - All modules connected to right backplane
 - continue with disconnecting above parts to isolate fault
- If X30 and X31 is disconnected but no output is present
 - Disconnect X100 and X200 at the DMS power supply
 - ⇒ If power supply is present now
 - Cable W592 or W593 is defective
 - ⇒ If power supply is still missing:
 - Check fuses at the distribution board
 - If fuses are ok, replace the power supply

ADU

The **ADU** (D513) consists of two controllers:

- **DOM** which performs the dose modulation
- **ACU** which performs the control of the data acquisition

Checking the DOM section

1. **Options > Service > Local Service > Test Tools > Controller > DOM**
2. Select the radio button
 - Selftest** or
 - Dose signal test**
3. Press **Go**
 - ⇒ If the test was successful **Test passed** is output
 - ⇒ Otherwise, check the Somaris log for errors

Checking the ACU section

1. **Options > Service > Local Service > Test Tools > Controller > ACU**
2. Select the radio button **DAS Communication signals**
3. Press **Go**
 - ⇒ If the test was successful, **Test passed** is output
 - ⇒ Otherwise, check the Somaris log for errors

Environmental Conditions

Environmental conditions are a possible reason in case of weak ring artifacts.

Check the Somaris log for errors and warnings .

Gantry

Inside the gantry, the temperature and the humidity are regulated.

To read out the actual values for temperature and humidity, use the control window.

The output of the table may be selected in decimal or hexadecimal values.

The gantry has to be in standby mode when accessing the control system

- ⇒ To read out the actual **temperature** of the gantry, perform the following:
 1. **Options > Service > Local Service > Control > Table load/modify**
 2. Select radio button **CONTROL SYSTEM** under TABLE SOURCE
 3. In the list box TABLE NAMES select **T00**
 4. Press **Go**

The parameter set for the table T00 is shown.
 Use the slider to move to the desired information
 - gantry temperature [degrees C]
 The temperature has to be in the range of 26°C +/- 4°C
- ⇒ To read out the actual **humidity** of the gantry, perform the following:
 1. **Options > Service > Local Service > Control > Table load/modify**
 2. Select radio button **CONTROL SYSTEM** under TABLE SOURCE
 3. In the list box TABLE NAMES select **T00**
 4. Press **Go**

The parameter set for the table T00 is shown.
 Use the slider to move to the desired information
 - gantry humidity [%]

DMS

- Environmental conditions for the DMS are:

- Temperature :

- ⇒ Check the Somaris log for errors and warnings concerning temperature problems.
- ⇒ Ensure that the DMS is powered on over night.
- ⇒ If raw data is available for the images the customer is complaining about, use Analyze to read out the temperature from the header data.
 This indicates the temperature at the time of data acquisition.
- ⇒ To read out the actual temperatures for the DMS, perform the following:
 2. Select radio button **CONTROL SYSTEM** under TABLE SOURCE
 3. In the list box TABLE NAMES select **TCC**
 4. Press **Go**

The parameter set for the table TCC is shown.
 Use the slider to move to the desired information
 - DMS temp (channel 80) [grad]
 - DMS temp (channel 250) [grad]
 - DMS temp (channel 400) [grad]
 - DMS temp (channel 600) [grad]
 Hint: Ensure that the radio button "Dec" is activated for decimal display.
 A difference between left and right temperatures is normal.

- Humidity :

- ⇒ Check the Somaris log for errors and warnings concerning humidity problems.

Overview -Temperature switches

- F1 at right DMS backplane:
 - opens at 38°C +/- 3°C
 - closes at 27°C +/- 3°C
- Temperature switch in the gantry:
 - opens at 43°C +/- 1.7°C

Troubleshooting the data path

This document describes a detailed service procedure with respect to tracking and identifying problems via the data transmission paths.

General descriptions

1) Global architecture of the high speed data paths

The data transmission paths in the P30F scanner connect the DAS Controller - DCON board in DMS to the ADU board in the C-BOX and to G-REC receiver in the IRS over high-speed data slip rings. The global architecture of data paths is depicted in the figure below:

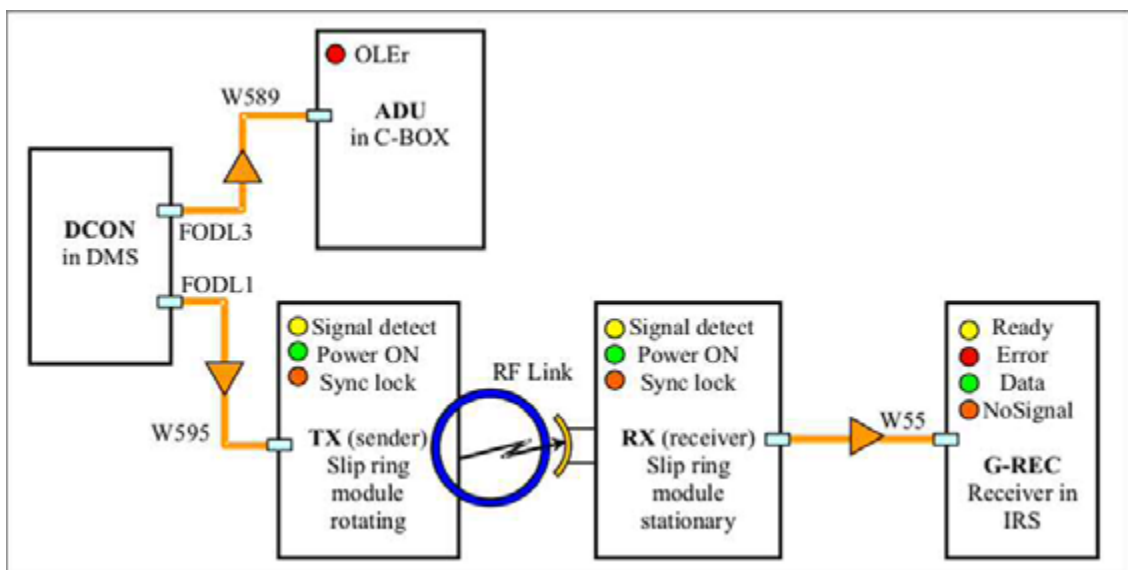


Fig. 118: Architecture of data links

There are two main segments of data paths:

A. DCON to ADU is a Simplex fiber optic link connection operating at 800Mbps. The DCON FODL3 (DOM) output of the fiber optic cable generates light at a wavelength of 850nm and has a fiber optic cable connector, type SC.

B. DCON to IRS is a Simplex connection over a high-speed data slip ring operating at 2.5Gbps. This link consists in a fiber optic cable connection between DCON and the slip ring TX module, an electrical transmission over the data ring and a fiber optic cable connection between the slip ring RX module and G-REC in IRS. The DCON FODL1 (IRS) optical output generates light at a wavelength of 850nm and has an optic fiber connector, type LC. The two DCON fiber optic cable outputs are only optically compatible, i.e. they operate at the same wavelength and at about the same optical power levels. However they are not exchangeable with each other for any functional purpose.

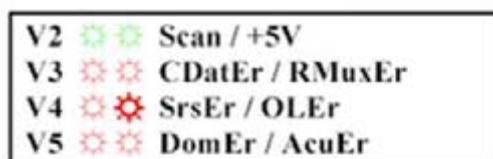


Fig. 119: ADU front panel diagnostic LED's

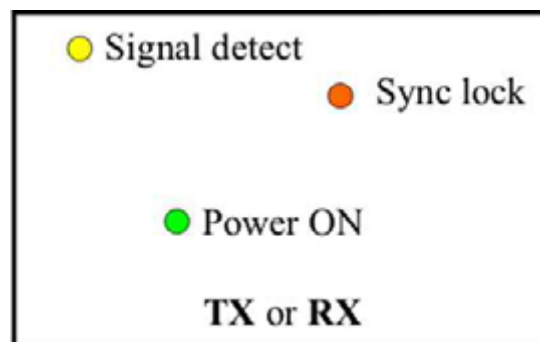


Fig. 120: LED definitions for slip ring modules

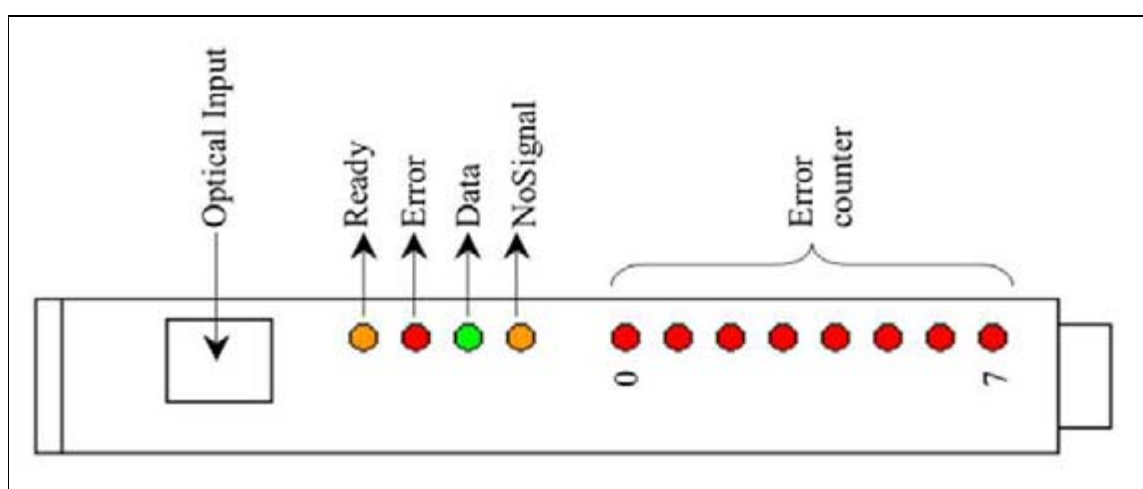


Fig. 121: D24 G-REC receiver in IRS

2) Theory of operation

Transmission DCON → IRS

The DCON board in the DMS collects measurement and bias (offset) data from all integrator modules and builds a serial bit stream containing data packets that are sent to IRS over the data slip ring. One data packet (reading) contains all measurement values for the selected rows of detectors. The detector values are corrected for offset and converted into logarithmic format (already in the DCON), and are used directly by the IRS for image reconstruction.

Transmission DCON → ADU

Parallel to data processing for the IRS, the DCON also generates a DOM data frame for use in dose modulation functions. A DOM frame is a compressed version of any IRS reading sent to ADU via the optical link DMS ADU. The ADU board builds an angular attenuation profile used to modulate on-line the radiation profile applied to the patient. The communication DCON IRS is more sensitive to transmission errors due to the slip-ring joint between the rotating and stationary part of gantry and due to jitter and noise effects in different segments chained in this link. There is no retransmission for defective packets. In both ADU and IRS the defective packets are simply interpolated. The communication

DCON ADU is less sensitive to errors and moreover ADU uses only a fraction of the entire data. The ADU board will never abort a scan in case of excessive defective data. However, it will abort the on-line dose modulation and switch to sinusoidal dose control. Conversely, the IRS accepts only a very limited number of defective packets in each rotation and will abort the scan in case there are too many errors.

Check the signal path integrity

1) Quick Check

The data path is able to transport information only when each segment in the path is working satisfactorily. A broken segment or defective component will break the data flow completely. In order to quickly detect the location where the data path is broken, each link is provided with status indicators. When the optical path is conducting a modulated signal with an adequate power level and signal quality (low jitter), the LED indicators are as follows:

Link DCON →ADU

- When the red LED OLError on the ADU is OFF, this means that the light receiver in the ADU detects enough modulated light at the input. Also the quality of the received signal is sufficient (the jitter is sufficiently low) so that the receiving PLL is locked into the incoming bit frequency. This means that the FODL3 output is working and that the fiber optic cable between DAS and ADU is OK. Otherwise, this LED is ON or it is blinking continuously.

Link DCON →slip ring

- Green Power OK LED on the slip ring TX module is ON, i.e. the module receives power from the power supply (C-BOX).
- Yellow Signal detect LED on the slip ring TX module is ON, i.e. the module receives modulated light from DCON. This means that the FODL1 output is working and that the fiber optic cable between DCON and TX is OK.
- Orange Sync lock LED on the slip ring TX module is ON, i.e. the module receives a clean signal from DCON and the internal PLL is locked into the incoming bit frequency.

This means that the jitter in this link is low enough and that the connection between DCON and TX is OK. Otherwise, this LED has reduced light intensity due to continuously blinking.

Link slip ring TX →slip ring RX

- Green Power OK LED on the slip ring RX module is ON, i.e. the module receives power from the power supply (MCU).
- Yellow Signal detect LED on the slip ring RX module is ON, i.e. the internal link over the slip ring between the TX and RX modules is working and the receive antenna detects an appropriate high signal.

- Orange Sync lock LED on the slip ring RX module is ON, i.e. the module receives a clean signal from TX over the slip ring and the internal PLL is locked into the incoming bit frequency. This means that the jitter in this link is low enough and that the connection between TX and RX is OK. Otherwise, this LED has reduced light intensity due to continuously blinking, possibly depending on the gantry's angular position.

Link slip ring →G-REC

- Red NoSignal LED on receiver is OFF, i.e. the G-REC in the IRS receives modulated light from the gantry. This means that the optical output in slip ring RX is working and that the fiber optic cable between the gantry and the IRS is OK.
- Red Error LED on G-REC is OFF. This means that the quality of the signal received from the gantry is good enough (the jitter is low enough), so that the receiving PLL is locked into the incoming bit frequency and there is no CRC error detected in the data received. Otherwise, this LED blinks continuously, possibly depending on the gantry's angular position. Note that this indicator is only active in the scan mode, i.e. immediately after the mode load when the green data LED is also ON!
- The red eight-bit error counter LEDs Err7...0 are all OFF. This means that there is no error has been detected so far in the data received. Otherwise, the counter increments each time an error is detected. The presence of errors may be depend on the gantry position. Note that this indicator is only active in the scan mode, i.e. immediately after the mode load when the green data LED is also ON!

Concluding remarks

It may be that different segments of the data path are defective due to e.g broken fiber optic cables, power missing at the components, excess jitter, defective parts, etc. In this case no signal, and as a consequence no data reach the destination, that is either the ADU or IRS. It is also possible to check each link / segment to localize these problems.

For a quick check of the data path, is it required to have a "known as good" or independently tested spare fiber optic cable (W55) of 30 m length. Use this fiber optic cable to patch a certain segment of data link in order to isolate the problem. For example, connect the DCON FODL1 output directly to the G-REC input to fully isolate the slip ring (DO NOT FORGET to disable gantry rotation in this case). It is also useful to have a fiber optic cable that is at least 1.5m long with a SC connector at one end and an LC connector at the other end (SC to LC Simplex fiber optic cable).

2) Debugging the signal path

A) Debug the signal path DCON →ADU

Check the redOLER LED on the ADU front plate

IF (red OLER LED on ADU is either ON or blinking) **THEN**

- ⇒ DCON's FODL3 output or DCON is defective, or it is missing power **OR**
- ⇒ The W589 fiber optic cable between the DCON and the ADU is broken / needs cleaning **OR**

⇒ The ADU is defective or the DOM FW is damaged or is not loaded

END IF.

Description: After configuration of the DOM FPGA by DOM FW during the INIT phase, the red OLEr LED on ADU must go OFF. Otherwise, there is a problem with the fiber optic link DCON →ADU.

To check the DCON's FODL3 output: The best way to check the fiber optic DCON outputs is to measure the fiber optic cable power at the DCON output by means of an optical power meter (see appendix). Other simpler / empirical tests are also possible:

1. First make sure that the yello signal detect LED on the slip ring TX module is working, i.e. the LED goes OFF when removing the W595 optic cable connecting the FODL3 DCON output to the TX module, and goes ON again by re-plugging this cable.
2. Use a special adapter for the fiber optic cable with a SC connector at one end and a LC connector at the other end to connect the FODL3 output to the input of the TX slip ring module.

IF (yellow signal detect LED on slip ring TX module turns ON) **THEN**

- ⇒ DCON"s FODL3 output is **OK**
- ⇒ The W589 fiber optic cable between DCON and ADU is defective or needs cleaning **OR**
- ⇒ ADU is defective or the DOM FW is damaged or is not loaded

END IF.

NOTE : The test above may be performed as well by connecting the DCON"s FODL3 to the G-REC input by a long enough special cable or SC-LC connector adapter and checking the red NoSignal LED on G-REC.

Check the W589 fiber optic cable between the DCON"s FODL3 (DOM) output and ADU:

The best way to check the fiber optic cable is to measure the power at the DCON output and at the cable output by means of an optical power meter (see appendix). The cable insertion loss should be a maximum of 0.5dB. Otherwise, the cable must be replaced or cleaned. Other simpler / empirical tests are also possible. Make sure that the DCON FODL3 output is working (see above). Use a spare "known as a satisfactory fiber optic cable" to replace the W589 cable.

IF red OLEr LED on ADU turns OFF)

- ⇒ **THEN** the W589 cable is defective →Replace or clean the W589 fiber optic cable.

END IF.

To check the ADU board:

Reload DOM FW or replace the board.

B) Debug the signal path DCON →slip ring TX module**Check the power at the RX module**

IF (green Power LED on TX module is OFF) **THEN**

- ⇒ the TX module is defective **OR**
- ⇒ The power-signal cable from C-BOX to TX is defective **OR**
- ⇒ DPU in the C-BOX or the C-BOX backplane is defective

END IF.

- Check the power-signal cable, check the supply voltage and the TX module, and replace whatever is necessary until the green Power LED indicator on TX goes ON.
- Perform the Data Link Test with the Service Platform and check the gantry error monitor for errors reported by the ACU: Power error.

Check the fiber optic signal at TX module

IF (yellow Signal detect LED on TX module is OFF) **THEN**

- ⇒ the DCON's FODL1 (IRS) fiber optic cable output is defective **OR**
- ⇒ the W595 fiber optic cable between DCON and slip ring is defective / needs cleaning **OR**
- ⇒ the TX module is defective or it receives no power from the C-BOX

END IF.

To check the DCON's FODL1 (IRS) output:

The best way to check the DCON fiber optic cable outputs is to measure the power at the DCON output by means of an optical power meter (see appendix). Other simpler / empirical tests are possible. Use the spare fiber optic cable to connect the DCON's FODL1 output directly to the G-REC input. If the red NoSignal LED on G-REC turns OFF, then the DCON output is good. Otherwise, the DCON FODL1 is probably defective or DCON is missing power.

To check the W595 fiber optic cable between the DCON's ACQ output and slip ring TX:

The best way to check the fiber optic cable is to measure the power at the DCON output and at the cable output by means of an optical power meter (see appendix). The cable insertion loss should be a maximum of 0.5dB. Otherwise, the cable must be replaced or cleaned. Other simpler / empirical tests are also possible. Make sure that the DCON output is working and the TX module has power. Use the spare fiber optic cable to replace / patch the W595 cable. If the yellow signal detect LED on TX turns ON, then the W595 cable is defective. →Replace or clean the W595 fiber optic cable.

To check the fiber optic cable input of the TX module:

- In case the DCON's FODL3 (DOM) output is OK (see **Debug the signal path DCON to ADU** above), use a "known as good" special SC-LC adapter cable to connect the DCON's FODL3 (DOM) output to the TX fiber optic cable input. If the yellow signal detect LED on TX turns ON, then the TX module and its signal detect indicator are OK. Otherwise, replace the TX module.
- Perform the Data Link Test with the Service Platform and check the gantry error monitor for errors reported by the ACU: Signal error.

Check the quality of the data signal (jitter) at the TX modules

Note: When plugging the fiber optic cable into the TX module at the same time the yellow signal detect LED is activated, there will be a noticeable increase in light intensity at the orange Sync Lock LED indicator on the TX module. Otherwise, if the orange Sync Lock LED on the TX module shows a permanently reduced intensity (i.e. due to blinking at a high rate), the PLL in the TX module does not succeed in locking into the incoming bit frequency most likely due to poor signal quality (high jitter).

If (orange Sync Lock LED on TX module shows reduced intensity due to fast blinking) **THEN**

- ⇒ DCON's output is defective **OR**
- ⇒ The W595 fiber optic cable between DCON and slip ring is defective / needs cleaning **OR**
- ⇒ The TX module is defective

END IF.

To check the DCON output:

Use the patch cable to connect the DCON FODL1 (IRS) output directly to the IRS and perform the Data Link Test with the Service Platform for static modes (rotation switch disabled). If this test fails, replace DCON.

Check the W595 fiber optic cable between the DCON's FODL1 output and slip ring TX:

Replace or clean the cable.

To check the TX module:

- Perform the Data Link Test with the Service Platform and check the gantry error monitor for errors reported by the ACU: Link error.
- Replace the module.

C) Debug the RF signal path between the slip ring TX →RX module

Note : Make sure that the data link segment between DCON and TX is working before proceeding with this step. Otherwise, first debug the segment DCON → TX as described above.

Check the power at the RX module

IF (green Power LED on the RX module is OFF) **THEN**

- ⇒ the RX module is defective **OR**
- ⇒ The power-signal cable from MCU to RX is defective **OR**

⇒ the MCU board is defective

END IF.

- Check the power-signal cable, check the supply voltage and the TX module, and replace whatever is necessary until the green Power LED indicator on TX goes ON.
- Perform the Data Link Test with the Service Platform and check the gantry error monitor for errors reported by the ACU: Power error.

Check the electrical RF signals at the RX module

The RX module receives an RF signal via the capacitive coupling between the transmitting rotating slip antenna and the stationary receiving segment-antenna. The RF coupling condition depends on the mechanical tolerances. The tests below have to be performed by manually rotating the gantry by 360 degrees.

IF (yellow signal detect LED on RX module is OFF) **THEN**

- ⇒ The TX module (the RF output) is defective **OR**
- ⇒ The slip ring antenna system is defective or needs realignment **OR**
- ⇒ The RX module (the RF input) is defective

END IF.

To check the RF output of the TX module:

Replace the unit with a new one.

To check the slip ring antenna system:

Refer to the instruction for checking the microstrip ring antenna structure ([Mechanical slip ring system / p. 104](#)), termination resistors ([How to check the transmitting antenna / p. 223](#)) and the alignment of the receiver antenna

To check the RF input of the RX module:

Replace the unit with a new one.

Check the quality of the digital signal (jitter) at the RX module

Note: Before proceeding with this step, ensure that the quality of the data signal at the TX module is OK (see above). When plugging the fiber optic cable into the TX module at the same time the yellow signal detect LED is activated, there will be a noticeable increase in light intensity at the orange Sync Lock LED indicator on both the TX and RX modules. Otherwise, if the orange Sync Lock LED on the RX module shows a permanently reduced intensity (i.e. due to blinking at a high rate), the PLL in the RX module does not succeed in locking into the incoming bit frequency, probably due to poor signal quality (high noise or high jitter). This condition can change at various gantry angular positions.

IF (orange Sync Lock LED on RX module has reduced intensity due to fast blinking) **THEN**

- ⇒ The TX module is defective **OR**
- ⇒ The slip ring antenna system is defective or needs realignment **OR**

⇒ The RX module is defective

END IF.

To check the TX module:

Replace the unit with a new one.

To check the slip ring antenna system:

Refer to the instructions for checking the microstrip ring antenna structure ([Mechanical slip ring system / p. 104](#)), terminations resistors ([How to check the transmitting antenna / p. 223](#)) and the alignment of the receiver antenna

To check the RX module:

- Perform the Data Link Test with the Service Platform and check the gantry error monitor for errors reported by the MCU: Link error.
- Replace the unit with a new one.

D) Debug the link between the RX module → G-REC in IRS

Note: Before proceeding with this step, ensure that the data signal and its quality at the RX module is OK (see above).

Check the power at G-REC (IRS)

If the ACQ/IRS is powered and the FPGA devices on the G-REC boards are configured, the yellow Ready LED is blinking. Otherwise, debug IRS/ACQ first.

Check the fiber optic signal at the G-REC input

IF (red No Signal LED on G-REC is ON) **THEN**

- ⇒ The RX module (fiber optic output) is defective **OR**
- ⇒ The W55 fiber optic cable between RX and G-REC is broken **OR**
- ⇒ G-REC fiber optic input is defective

END IF.

To check the RX fiber optic output:

The best way to check the RX fiber optic output is to measure the power at the output by means of an optical power meter (see appendix). Other simpler / empirical tests are possible. First perform the two steps below to make sure that the fiber optic cables between RX and G-REC and the G-REC fiber optic input are OK. If this is true, then the RX fiber optic output is probably defective. Replace the unit with a new one.

To check the fiber optic cable between RX and G-REC:

Replace the cable with a new one.

To check the G-REC fiber optic cable input:

Use the spare fiber optic cable to connect the DCON's FODL1 output directly to the G-REC input. If the red NoSignal LED on G-REC remains ON, the G-REC fiber optic input is probably defective. Replace the board with a new one.

Check the quality of data signal (jitter) at the G-REC input

- Perform the Data Link Test with the Service Platform and check the CRC error sum and Data error sum counters.
- Alternatively load a scan to activate the Error indicators at G-REC. The green Data LED should be ON. In case the Error LED blinks or the Err 8-bit counter increments, errors are detected by G-REC.
- In case of transmission errors, perform an error location analysis as described later in this document.

IRS permanent communication problems

Sometimes the IRS is not able to receive any valid scan data, although the signal path is OK. To trace the problem, it is recommended to isolate the slip ring:

- Disable gantry rotation!
- Use the spare fiber optic cable to connect the DCON FODL1 output directly to the G-REC input.
- Perform exactly the same (static) scan that fails consistently with a normal configuration.

If the scan fails again after isolating the slip ring, the problem is in the DCON or IRS.

Random errors in link DCON → IRS

Within the link DCON→IRS, transmission errors in the data path are often random. Communication mostly works, however, with transitory errors that may lead to a scan abort when accumulating over the tolerated threshold. The random transmission errors may occur even if the scan is not aborted. Therefore it is important to be able to know if random errors are occurring with a system.

1) Identify random problems

There are many ways to check if random transmission errors occur in the data path:

1. Check the log book for IRS warnings
2. Check the error LED counter on G-REC
3. Perform the data link test

A) Check the log book for IRS warnings

At the end of the scan, the IRS SW counts the data errors encountered in the scan and puts a warning message in the logbook. Check the logbook for IRS errors to find out how many data packets were detected to be unsatisfactory due to CRC errors and were therefore interpolated by IRS. Only the CRC errors are significant for data transmission. Other types of errors detected by IRS, e.g. HV_DROP, are not related to data transmission.

B) Check the error LED counter on G-REC

The error counter at the G-REC input electronics counts the CRC and synchronization errors for the duration of the scan, i.e. not only during radiation but also during the bias phase and during pauses between sequential scans. This errors counter is displayed in binary code with an 8-bit red LED bar on G-REC. This counter is only active during the scan with data transfer and after mode load until the scan ends. Check that the green DATA LED on G-REC is ON, when investigating for random data transmission errors. To use this counter as a random error detector, it is necessary to load a scan (load the mode) and to perform investigations before or during the scan.

C) Perform the Data Link Test with the Service Platform**Theory of operation**

The data link test is a sequence of scans performed automatically with the DMS configured to generate test data (assignment mode). The data received are automatically analyzed in the IRS/ICS for every incorrect bit. Error statistics are generated that allow for locating sources for error. If this test is successful, the transmission DMS to IRS is working well. This test can be started from the Service Platform. The results are presented as in the figure below:

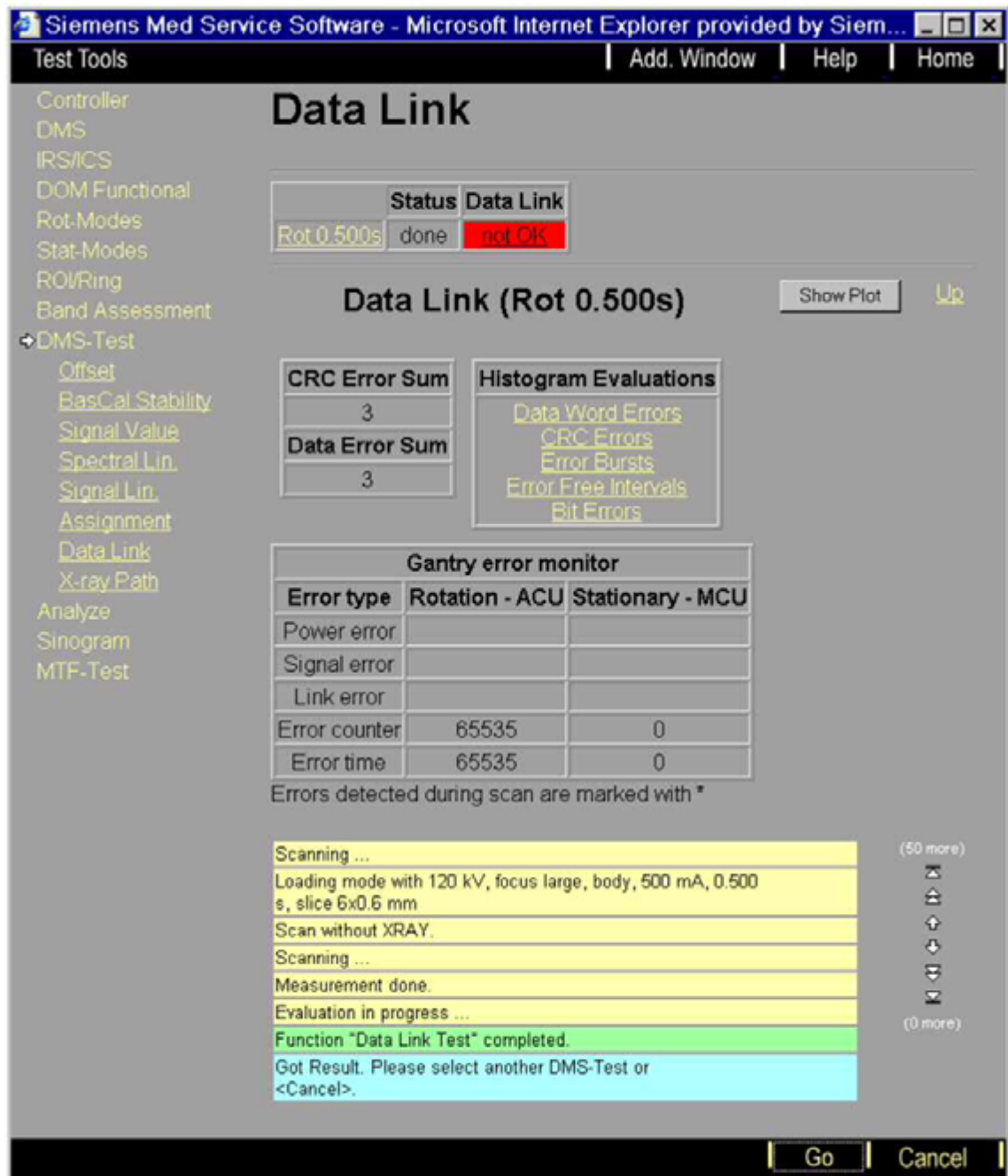


Fig. 122: The results of data link test without X-ray / ROT mode 0.5 sec / rot.

- The data link test may be performed at different rotation speeds or w/o rotation (ROT or STATIC modes) in the static mode with or w/o active braking of the gantry via the rotation driver, with radiation at different levels of x-ray power or w/o x-ray.
- It is advisable to perform the data link test after a careful analysis of IRS logbook errors. Search for those modes (rotation speed, x-ray power level, gantry tilt, etc...) that are most prone to transmission errors and test the data link with these particular modes.
- The analysis SW will increment an error counter for each CRC error. The total number of transmission errors is available and displayed as test result. The error counter runs over all readings into the scan that include at least five complete rotations in order to

find the relationship between tube angle and error occurrence. In the case of an error counter that is not null, its time evolution can be plotted for all scans together with the tube angle position as depicted below:

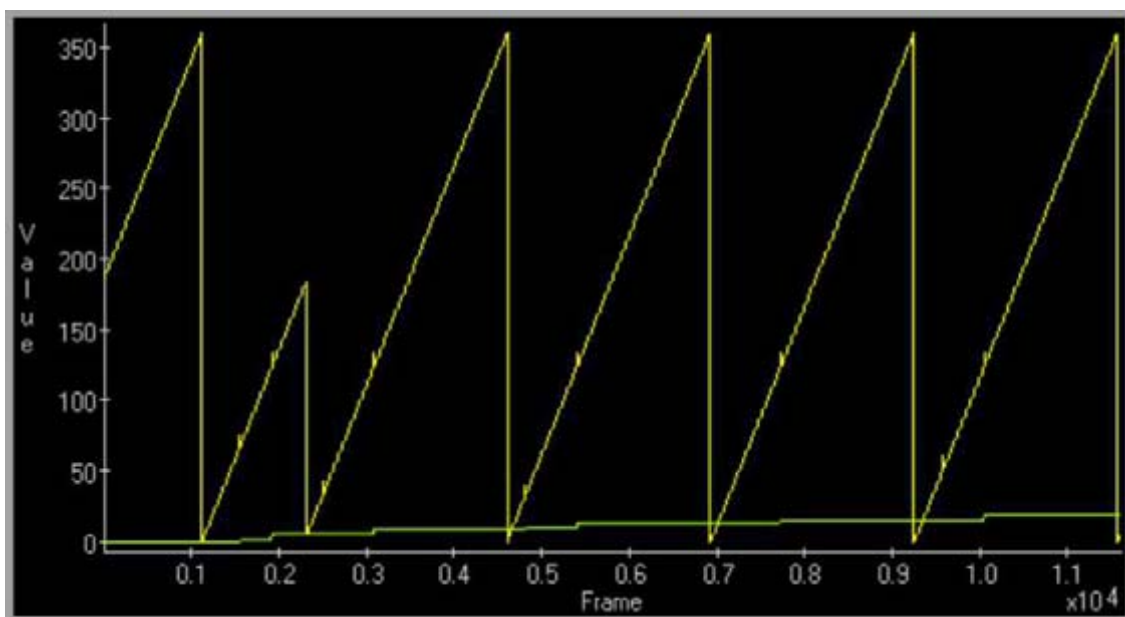


Fig. 123: Example of tube angle and error count versus frames

The X axis of the plot is the reading number. The vertical axis of Channel 2 (green plot) is the number of errors. The vertical axis of Channel 1 (yellow plot) is the angular tube position (0...360°C) as explained below.

Gantry angle (deg)	0	30	60	90	120	150	180	210	240	270
Tube position (x o'clock) from the front	3	4	5	6	7	8	9	10	11	12
Tube position from the front	East (right)			South (bottom)			West (left)			North (up)

Selectively check or uncheck channel 1,2 checkboxes to change the vertical scale.

The gantry error monitor

Additionally, the power, signal and link errors reported by the TX and RX electronic modules in the slip ring are monitored dynamically during the scan via the ACU on the rotating gantry and via the MCU on the stationary gantry. These errors are the same debug signals that are indicated by the green, red and orange LEDs available on both RX and TX modules. Both ACU and MCU HW count the number of times these errors were activated during the scan (error counter) as well as how long the errors were active during the scan (error time). This feature is useful for identifying transitory errors that occur due to loose contacts caused by mechanical vibrations during rotation that are otherwise not detectable.

2) Debug for random transmission errors

A) Isolate the slip ring and the fiber optic cables

Debugging such transitory problems is very difficult; it is therefore advisable to reduce the problems as much as possible.

STEP 1 A quick and useful test is to isolate the slip ring:

- Disable gantry rotation
- Use the spare fiber optic cable to connect the DCON outputs directly to the G-REC input.
- Perform exactly the same (static) scan that is subject to random errors with a normal configuration. If the random error continues to persist, the problem is within DCON or IRS. Otherwise, the debug procedure has to be refined.

STEP 2

Furthermore, the fiber optic cables between DCON →slip ring or between slip ring →IRS may be partially broken or have dirty ferrules in the connectors and therefore manifest larger insertion loss than normal. This is especially applicable to connectors in the gantry that may be contaminated by carbon dust from the slip ring brushes. In this case, the signal at the fiber optic cable output is lower than usual. However, the signal detect indicators may not trigger although the signal quality is poor. To isolate the problem, measure the fiber optic cable loss factors as described in the appendix and replace the fiber optic cable in case of higher losses. Alternatively, use the spare fiber optic cable to successively patch the fiber optic cable in question. If you get better results (reduced number of transitory errors or no error at all), clean the ferrules or replace the fiber.

B) Localize the source for error

The data link test automatically performs a statistical analysis regarding error distribution to locate the source for transmission errors.

Check for errors occurring at about the same rotation angle

Problems with the mechanical alignment of slip ring antennas generally produces supposed random errors. However, in reality these errors are concentrated at almost the same angular gantry position. The Histogram Evaluation option is useful for investigating such problems. Select both Data Words Error and CRC Errors to display a histogram plot of error distribution around the rotation. The Data Words Error analysis may be slower but it checks for every bit error. The CRC Errors analysis checks only for block errors and is faster.

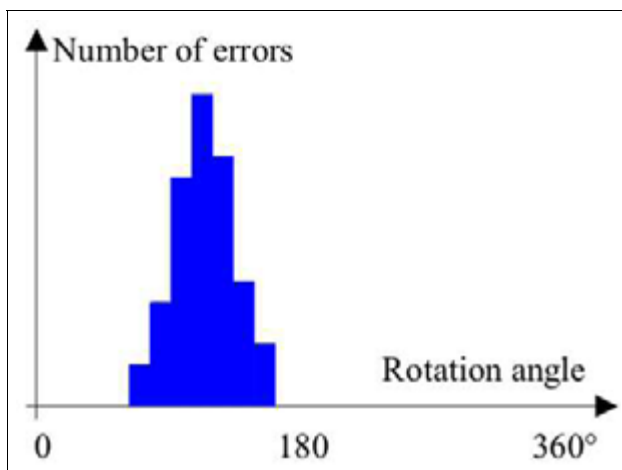


Fig. 124: Histogram of error versus rotation angle

A histogram distribution as depicted above suggests that the errors are concentrated at the same sector of the slip ring. This is indicative of mechanical problems with the micro-strip structure or mechanical inaccuracies in this sector. Rotate the gantry by hand in this angular position and check the structure for defects and mechanical alignments. Use the relationship between gantry angle and X-ray tube angle in the table above (0° is X-ray tube at 3 o'clock).

Check for errors occurring in the DMS or IRS

In case of defective electronic components in the DCON or the G-REC, the data errors are apparently random and apparently due to transmission. However, in reality the error occurs before transmission in the DCON and after the transmission in the G-REC. The Histogram Evaluation option is very useful for checking this type of scenario. Select the Bit Errors option to get a histogram plot of the bit error position within the 16-bit data word as depicted below.

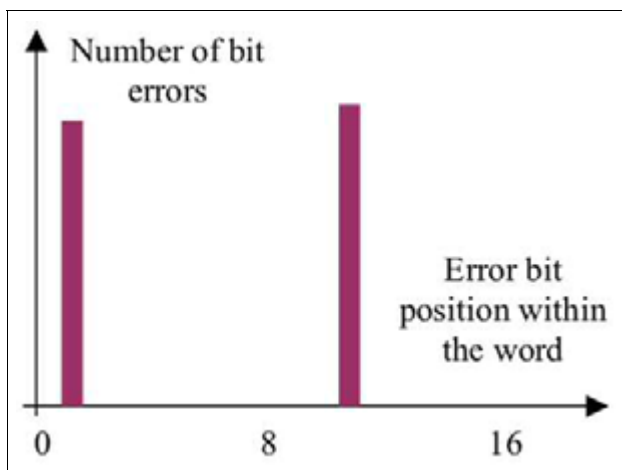


Fig. 125: Error bit position histogram

If the errors always occur at the same bit position(s) within the 16-bit word when repeating the data link test, this is indicative that the source for errors is either the DCON or the G-REC and not the transmission across the slip ring. Note that a test with a bypassed slip ring (cable connected directly between DCON and G-REC and no rotation) cannot detect intermittent contact problems in DCON that occur only during rotation.

C) Searching for external interference as the source for error

Random errors in the data path DCON → IRS may be caused by the following external factors:

1. The frequency inverter that drives the rotation motor generates an increased level of electrical noise between ROT and STAT GND and therefore injects noise in the RF link TX →RX electronic modules in the slip ring;
2. The HV generator or E-BOX on the rotating gantry generates HF equalization currents through the PE brush connection between ROT and STAT GND and therefore injects noise in the RF link between TX →RX electronic modules in the slip ring;
3. The slip ring is extremely sensitive to HF common mode noise between ROT and STAT GND because of extreme mechanical asymmetries in the antennae system.

The Histogram Evaluation option is very useful for checking this type of scenario. Select both the Error Bursts and Error Free Intervals options to get two histograms that plot the error distribution within the data stream (distribution of the error burst length - EBL and error free interval - EFI) and perform a correlation analysis as explained below.

Case 1

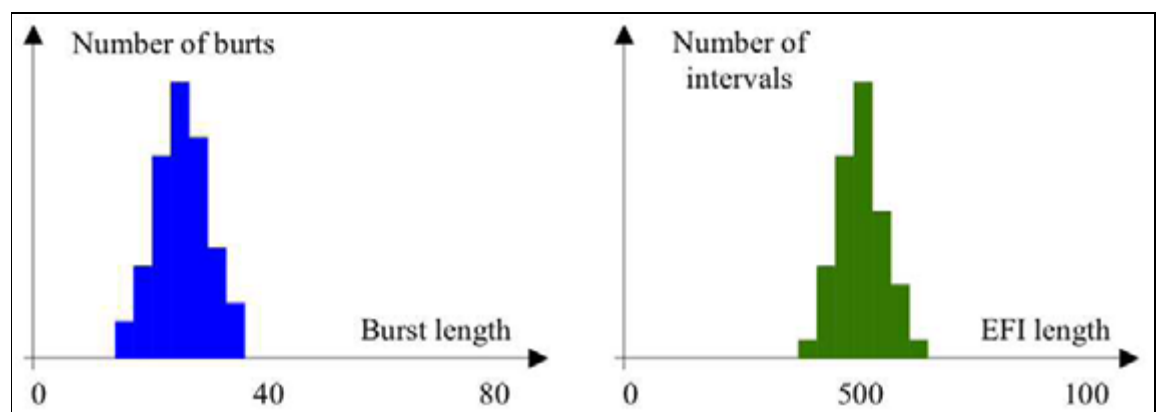


Fig. 126: Burst and EFI histogram case 1

In this case both histograms have a Gauss (bell) shape as depicted above. This suggests a periodic perturbation source such as the rotation driver inverter or other power supply noise. Check the high-power connections at the generator, the grounding connection and the high power carbon brushes. Perform tests with the rotation driver disabled or without radiation to better identify the perturbation source.

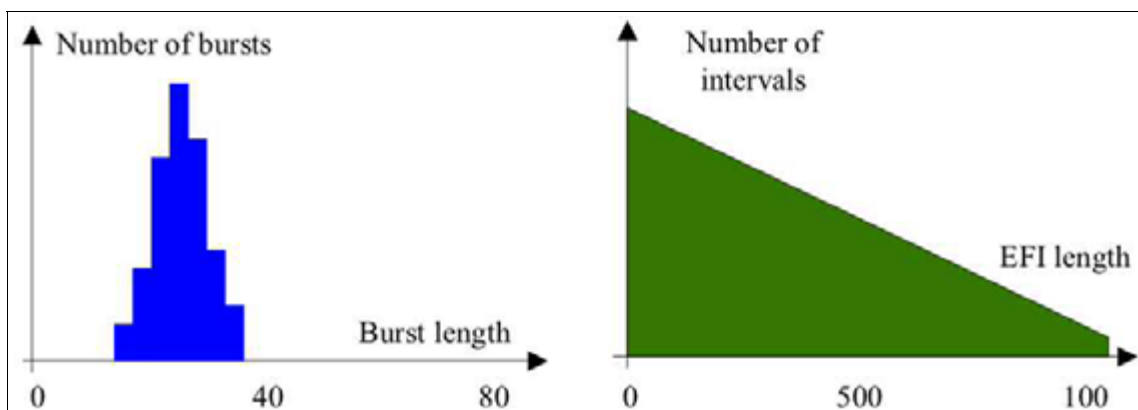
Case 2

Fig. 127: Burst and EFI histogram case 2

In this case only the burst histograms have a Gauss (bell) shape as shown above, whereas the EFI histogram decreases linearly. This suggests a non-periodic perturbation source such as external equipment operated in the vicinity of the CT examination room and generating power line interference (e.g. therapy equipment, heating ovens, etc...). Check the quality of the hospital AC power lines (line transient spike analyzer) and the grounding of the CT system as well as other hospital equipment operating in the vicinity.

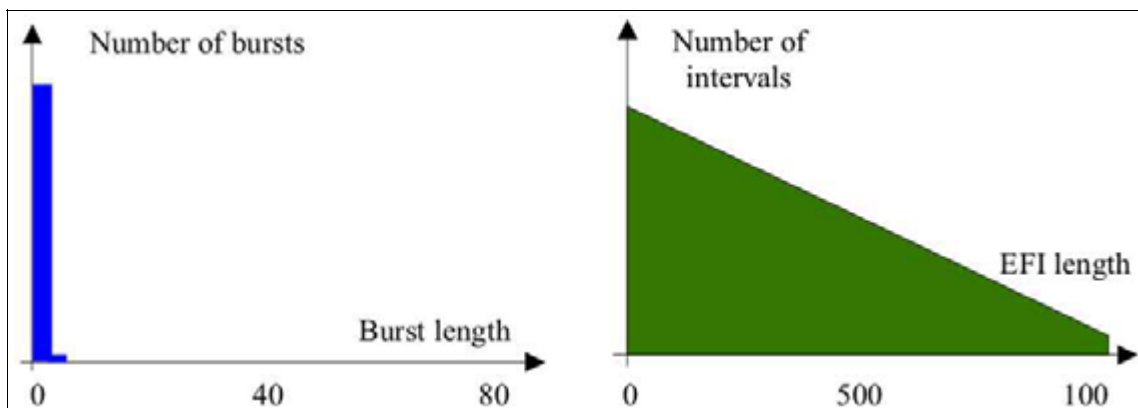
Case 3

Fig. 128: Burst and EFI histogram case 3

In case bursts histograms are concentrated and occur at very short burst lengths as shown above, whereas the EFI histogram decreases linearly, this is suggestive of a poor signal-to-noise ratio at some fiber optic input or at the RF input for the RX module. Check the power level of the fiber optic cables, check the fiber optic cables, clean the connectors, and align the RX receiving antenna.

D) Searching for dominant sources for error

In practical application, each error source may contribute to the error budget, but only one is usually dominant. The Data Link Test is a useful tool for locating the dominant source for errors. As mentioned earlier, the RF link across the data slip ring is subject to electromagnetic interference produced within the gantry. Major sources for this interference are the ring motor and its frequency inverter as well as the high-voltage generator and power supplies on the rotating gantry. In order to identify the dominant source for errors, the debug strategy eliminates one or more potential sources and investigates the way they influences

error statistics. Accordingly, the data link test provides predefined test modes that deal with this debug strategy. For example, to investigate the contribution of the X-ray generator to the error budget, perform the data link test using scans without radiation.

Investigate the initial error statistics

To get an idea about the initial error profile, perform the data link test with patient-like modes, i.e. modes that simulate a real scan. Use either topogram-like modes ("Static 0 deg, X-Ray" "Static 90 deg, X-Ray") or sequence-like modes ("Rot 0.500s, X-Ray"... "Rot 1.000s, X-Ray"). Try to identify, if the errors are related to particular parameters such as rotation speed, tube angle, etc...

Investigate the contribution of the rotation driver

To investigate the contribution of the rotation driver to the error budget, perform the data link test with the predefined modes "Static X-Ray1" and "Static X-Ray2". These modes fully disable rotation including active braking. The frequency inverter is disabled via SW and thus generates no interference at all. This is not equivalent to switching the rotation switch OFF! These tests may be performed only in the static mode. The gantry should be positioned either via the service platform or by hand at various angular tube positions or at positions where previous tests showed an increased error probability. If the error statistics change significantly as compared to previous tests, the dominant source for errors is the rotation driver and investigations should be pursued in this direction.

Investigate the contribution of the X-Ray generator

To investigate the contribution of the X-ray generator to the error budget, perform the data link test with the predefined modes "Rot 0.500s" ... "Rot 1.000s". These modes disable X-ray generation for various rotation speeds. Use the rotation speeds that previously showed increased error probability. If the error statistics change significantly as compared to previous tests, the dominant source for errors is the X-ray generator and investigations should be pursued in this direction.

Fine-tuned investigation

After having used the above steps, it may become obvious that other test mode(s) may be more suitable for checking out a particular situation. In this case, edit the user-defined test modes "Static20.dat" and "Rot20.dat" to fit the particular debug needs.

Appendix

Checking and cleaning an fiber optic cable

The fiber optic cables used for data paths are 62.5 / 125 μm multi-mode fiber optic cables operating at a 850 nm wavelength. The cable's performance may degrade because of dust or dirty covering the input core in the cable's receptacle or because of mechanical damage in the connectors, core and cladding due to extreme bending. In these situations, the overall insertion loss increases, degrading the amount of light coupled and released. The best way to check fiber optic cable is to measure its insertion loss by using appropriate measurement equipment such as e.g.:

- HP 8410A Portable Loss Test Meter fitted with modules HP 81400A and HP 81411A,

- HP 8153A Lightwave Multimeter with HP 81536A module.

A SC connector interface for 62.5 / 125 μ m MMF (multi-mode) fiber optic cable is required in both cases. The maximal insertion loss at 850 nm should be typically < 0.1dB and maximum 0.5 dB. In case of higher loss, try cleaning the ferrules at both sides according to manufacture's guidelines. If not successful, replace the cable.

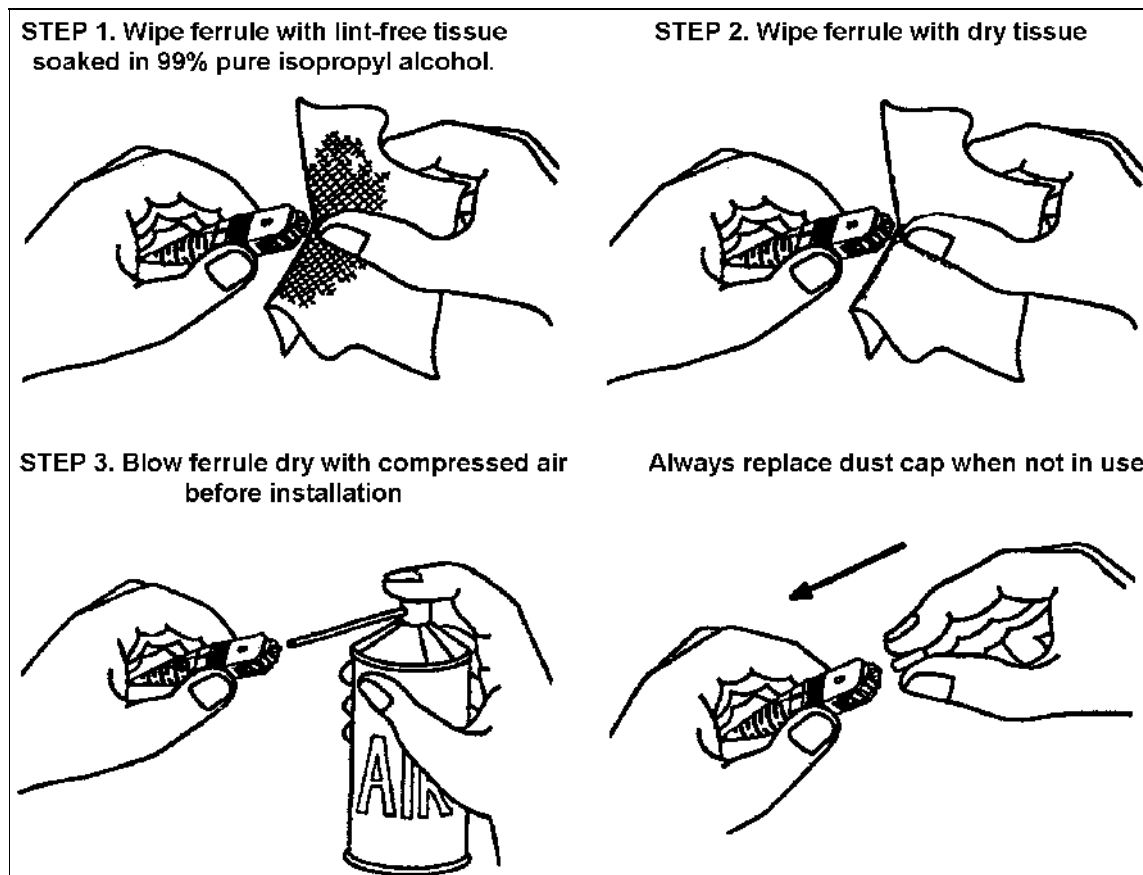


Fig. 129: Cleaning optical connectors

Note: A fiber optic cable mounted in the gantry is difficult to measure with the above instruments unless the cable is fully released from the gantry cable tree. To avoid this operation, you may use the spare fiber optic cable and an SC adapter to extend a cable end up to the measuring instrument. Consider additional loss in the spare cable to correct the final result.

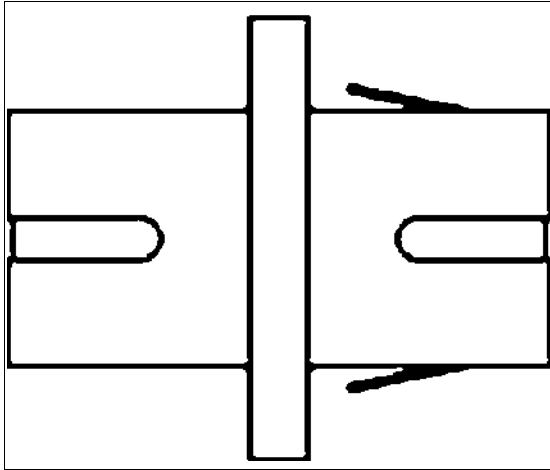


Fig. 130: SC or LC Simplex fiber optic adapter

SC or LC Simplex fiber optic cables adapters allow connecting two fiber optic cables end-to-end

Measuring the fiber optic signal along the link

The fiber optic cable power at both outputs of DCON and also at the fiber optic output of the slip ring RX module have to be high enough for reliable data transmission. The fiber optic receivers in the ADU, slip ring TX and G-REC have built-in signal level detectors that report an error when the signal falls below a safe minimum level. However, for error identification it is important to know if the problem is the incoming signal or the receiving part itself. For example, it may be possible that the power delivered via fiber optic cable to the input is high enough, however, the fiber optic receptacle in the receiver module is covered with dust or dirt, and the incoming light does not reach the destination. In case of doubt, the fiber optic signal can be measured at any point in the link using an optical power meter as mentioned above (e.g. the HP 8410A or HP 8153A). The power level of the fiber optic cable should be at least -9 dBm avg. at 850nm into a 62.5/125 MMF fiber optic cable. Measuring the power at the fiber optic cable termination connected to the ADU, slip ring TX module or G-REC should provide similar results and the fiber optic cable in-between is as tested as well.

The modulation signal (fiber optic cable extension ratio) can be optimally measured with an oscilloscope and a fiber optic probe, e.g. Tektronix 6701B. The signal amplitude is typically 400 μ W.

Handling of defective channels

Scope

The software automatically detects bad channels.

In order to find a bad channel, the Defective Channels procedure performs 4 scans without and with radiation.

From the result of these 4 measurements, the channels are determined to be either satisfactory or unsatisfactory.

Defective channels are handled in 3 different ways:

- During each CheckUp, the defective channel procedure is performed.
- Manual start of defective channel procedure from tune-up platform.
- Manual entry of a defective channel in the table.

Manual entry in the tune-up table is only necessary in case of a sporadic problem.

The center of the detector and the first and last slot (used for air slice normalization) have to be without defective channels: slot 1, slot 21, slot 22 and slot 42.

In all other slots, a certain amount of defects is allowed.

The only way to find out if a defect is allowed or not is to evaluate the entries in **Affected Modules** under Detector Statistics or Report.

General Information

Information for software version VB19x and VB20x

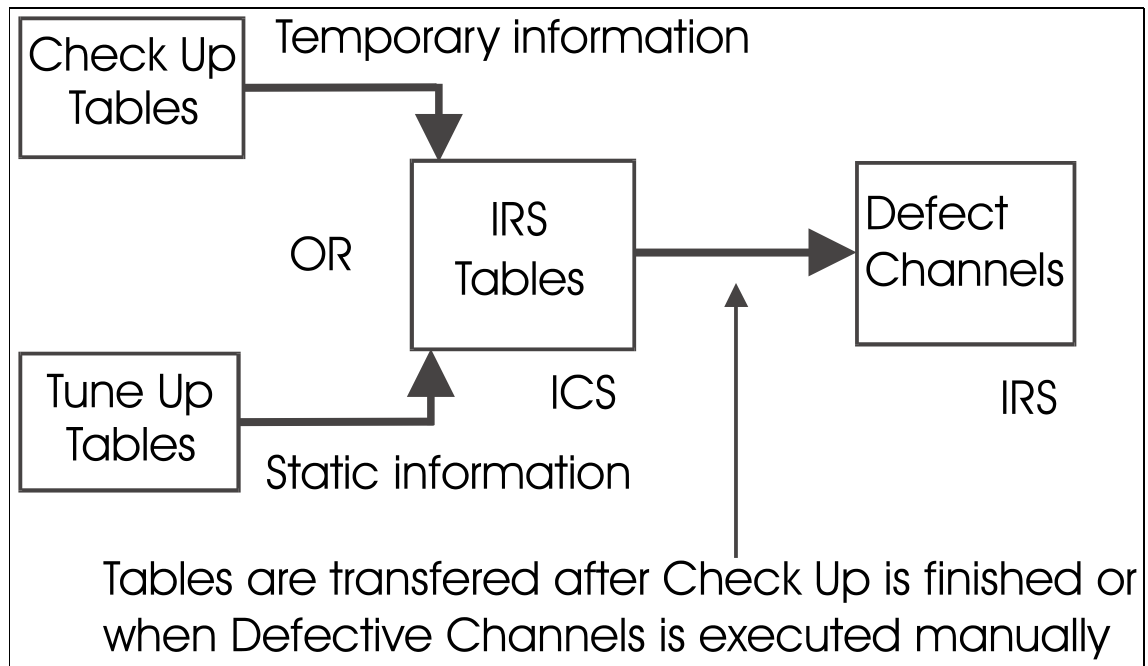


Fig. 131: Overview

- Entries found with the automatic detection under the service function will remain in the TuneUp table until the next detection is performed under the service function.
Manually entered defect entries will permanently remain in the table TuneUp.
Defects found during the checkup will be stored in the CheckUp table. The CheckUp table is emptied prior to the next checkup.
- If necessary, you may manually patch a channel as well as remove the manually patched channel again.
- Manually patched channels are not automatically deleted from the tune-up tables. These channels have to be deleted manually.
- After part replacement (e.g. module) or if you do not succeed removing unsatisfactory channels, delete the IRS tables in ICS as follows:
 1. Go to C:\Somaris\service\datastore\icstable
 2. Delete all files starting with badchannelsxxxxx
 3. Start an automatic detection under service to create an actual channel status.
 A partial tune-up as described in the guided tours follows.

- Several possibilities of checking the status of unsatisfactory channels:
 - Use the Defective Channels procedure and select Detector Statistics
 - Use the Report function
 - Options > Service > Local Service > Reports > Tune Up > Table Generations**
 - **Def Chan. Cust.**
 - ⇒ Shows status of last calibration
 - **Def. Channels**
 - ⇒ Shows status of last detection under tune-up
 - **Options > Service > Local Service > Reports > Def. Chann. History.**
 - ⇒ A history of all channels that have been patched or unpatched is shown

Information about the 3 types of defective channel tables

- **CheckUp** tables
 - Generated automatically at every check-up.
 - An empty table is available for every check-up (temporary table).
- **TuneUp** tables
 - Generated during automatic detection with the service function.
 - Only these tables may be edited manually.
 - The entries are updated with the next detection under service
- **IRS** tables
 - At checkup, the tables are derived by merging checkup and tune-up tables.
 - During detection under tune-up the IRS table is derived only from table tune-up
 - The tables are located on the ICS and are transferred at every check-up to the IRS.

Important:

- The information about the defects in the IRS table is derived as follows:
 - for **32 x 600** : all information is copied from the tune-up tables.
 - for **24 x 1200**: all information is copied from the tune-up tables.
 - for **all other slice combinations** the information is derived from these tables

How to find

Options > Service > Local Service > TuneUp > Expert Mode > Defective Channels

How to use

The direction of counting the slices is from head to foot.

After starting the program Defective Channels, a window pops up, enabling:

Editing the correction tables

1. Click **Correction Tables > TuneUp > 32x600**
2. Click the button for the desired detector slot (slots 1- 42)
Hint: A segment containing unsatisfactory elements is marked in red.
3. After a while, the user interface will show the corresponding channel / slice matrix.
 - ⇒ Now select the desired channel and slice. Select the element with the left mouse and click it. The color will change from green to red.
 - ⇒ Left-click to remove an unsatisfactory channel entry. The color will change from red to green.
4. Continue, if necessary, with Correction Tables > TuneUp > 24x1200

Starting automatic detection with software VB19x

Remove all absorbers !

1. Ensure that the checkbox **Automatic Detection after Editing** has been checked.
2. Click **Go**
 - ⇒ Measurements are performed automatically.
 - ⇒ To repeat an automatic detection in case of unstable / not plausible results, leave the upright checkbox checked and click Go.
3. Click **Go**
 - ⇒ You will be asked to accept the current calibration.
4. Click **Go**
 - ⇒ The IRS tables are transferred to the IRS
 - ⇒ If you click **Cancel**, no modifications to the tune-up and IRS tables will be stored.

Starting automatic detection with software VB20x

Remove all absorbers !

1. Click **Auto**
 - ⇒ Measurements are performed automatically.
 - ⇒ To repeat an automatic detection in case of unstable / not plausible results, leave the upright checkbox checked and click **Auto**.
2. You will be asked to accept the current calibration
3. Click on **Save**
 - ⇒ The IRS tables are transferred to the IRS
 - ⇒ If you click **Cancel**, no modifications to the tune-up and IRS tables will be stored.

Viewing the detector statistics

Select **Detector Statistics**

- ⇒ The number of defective elements are shown for the various collimations (tune-up and checkup).

Using Reports to view details of the Defective Channels Correction

1. Open an additional window and choose **Reports** .
2. From the menu bar TuneUp/QA/Tests choose:
TuneUp > Table Generation > Def. Channels
3. Double-click Report to view the results.
 - ⇒ **Detector Statistics** shows all currently defective channels.
 - Table Editing History** shows the history of the detection.
 - **Added by Automatic Detection** and
 - **Manually Added/Removed**
 - Table Conditions** shows all affected slice combinations within the IRS table.
 - Affected Modules** shows the slot as well as necessary replacements (change = has to be replaced).
 - Comments** shows additional information, in case a comment was added by the customer service engineer.

Adding a comment with software VB19x

1. Unmark the checkbox **Automatic Detection after Editing**.
2. Use the slider to move down to Comments
3. You can now enter a comment for a manually patched channel.
Generally used to document details in case of sporadic problems.

Adding a comment with software VB20x

1. Use the slider to move down to Comments
2. You can now enter a comment for a manually patched channel.
Generally used to document details in case of sporadic problems.

Appendix

Channel list

Note that all tests count the slots in clockwise direction.

For example, if DMS is on top, channel 1 is on the left side.

Slots 1 and 42 are used for air slice normalization and are not permitted to contain defective detector modules. Similarly, slots 21 and 22 are not allowed to have permanent defects.

Channels	Slot count	Channels	Slot count
1 - 16	1	337 - 352	22
17 - 32	2	353 - 368	23
33 - 48	3	369 - 384	24
49 - 64	4	385 - 400	25
65 - 80	5	401 - 416	26
81 - 96	6	417 - 432	27
97 - 112	7	433 - 448	28
113 - 128	8	449 - 464	29
129 - 144	9	465 - 480	30
145 - 160	10	481 - 496	31
161 - 176	11	497 - 512	32
177 - 192	12	513 - 528	33
193 - 208	13	529 - 544	34
209 - 224	14	545 - 560	35
225 - 240	15	561 - 576	36
241 - 256	16	577 - 592	37
257 - 272	17	593 - 608	38
273 - 288	18	609 - 624	39
289 - 304	19	625 - 640	40
305 - 320	20	641 - 656	41
321 - 336	21	657 - 672	42

How to switch off balancing

Using ScanEdit to switch off balancing.

1. Load the recon which you intend to evaluate without balancing.
2. Open the Windows Explorer and navigate to:
c:\Somaris\Tools\bin
and start **ScanEdit.exe** with a double click.
3. To get the mode data into the scan editor:
select **Protocol > GetProtocol**
Search for :
 - **Service** and change from 0 to 1
 - **OrganCharacteriticOverwrite** and change from Mloff to Mlon
 - **Balancing** and change from 1 to 0
4. Send this modified Settings to reconstruction:
Select **Porotocol > SendRecon**
5. Perform reconstruction in the recon tab card without balancing

How to interpret the physicist's line

To find the decoding for the physicist's line, go to:

- Somaris > Help > Contents and Index and
- select the tabcard Search and
- type in physicist's line
 - ⇒ The explanation is shown

Details of the static mode

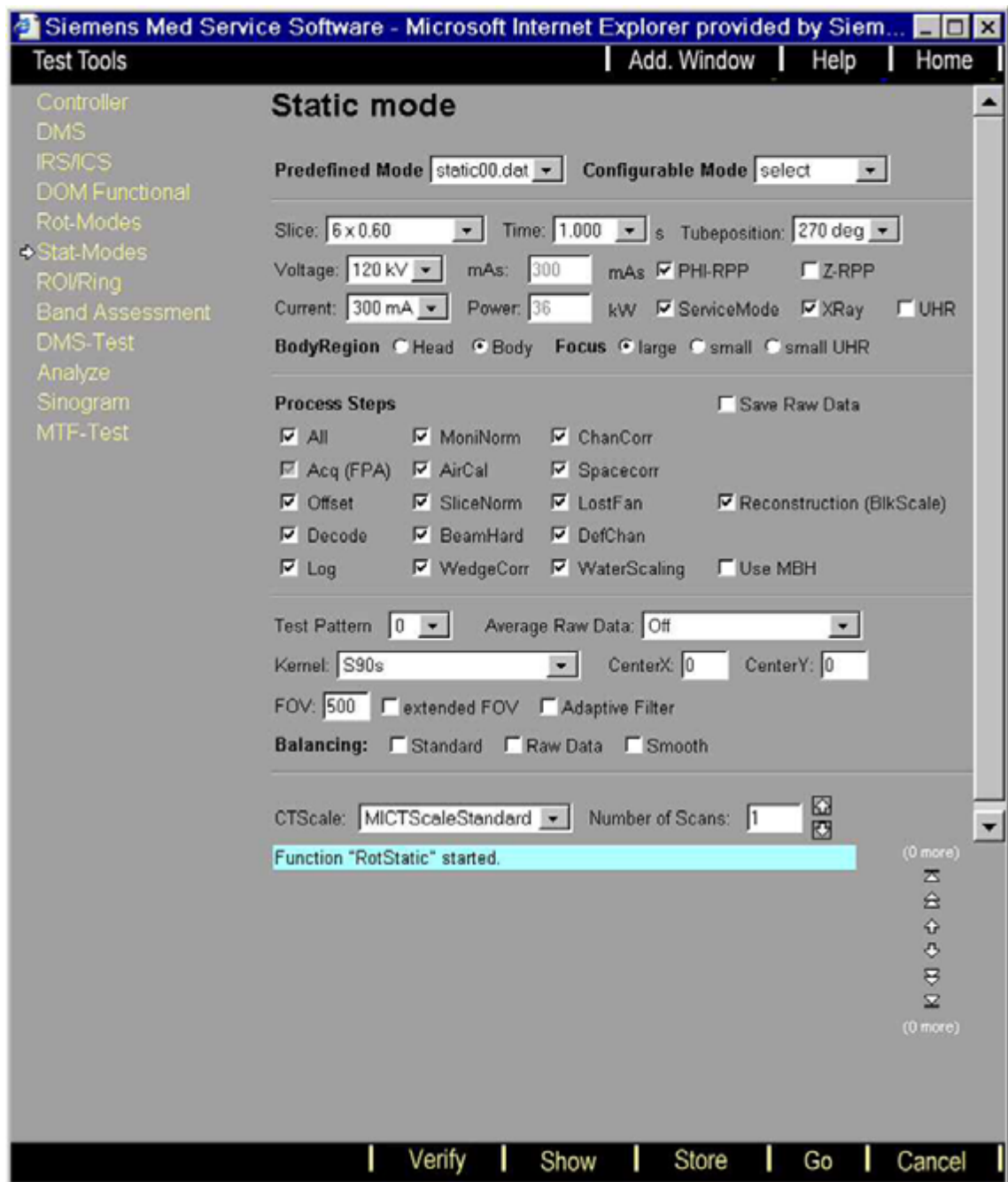


Fig. 132: Static Mode

Hints for using the static mode

- **Service Mode:**
When you deselect the checkbox, the mode will be checked by modelib such as a customer mode.
Only with the checkbox selected, all degrees of freedom necessary for service are available.
- **XRay:**
X-ray can be switched on or off using this checkbox

- PHI-RPP
- Z-RPP
- Test Pattern:
 - 1 = the data set for the disk test is selected
 - 0 = no test data selected (normal operation)
- Average Raw Data:

Readings will be averaged. While the smaller size of raw data is advantageous, today only statistical evaluations are possible. No image reconstruction is possible.

Additionally, deselect Reconstruction(BlkScale'), when using Average RawData.
- Balancing Raw Data is not used at the moment
- Using modes 32 x 0.6 mm
 - Always select Z-RPP
 - At 1 sec (all slices) rotational time, select PHI-RPP as well
- Using modes 20 x 1.2 mm or X x 1.2 mm
 - always deselect Z-RPP
- Using UHR modes
 - always select "Focus small UHR"

How to manually perform a focus difference

The result of the X-ray path test is filtered. As a result, a low frequency difference is not visible at all.

Contamination inside the tube or tube collimator may be not visible.

To get an output in this case, the difference has to be performed manually, using analyze A BaseCal with PHI FFS is used for evaluation in this example

- Go to:
 - Options > Service > Local Service > Test Tools > Analyze**
 - Select ICS Tables and press Go
 - Select a BaseCal table of the type ...**rpp2x1**...and press Go
 - ⇒ The plot of the BaseCal appears
 - Select
 - Arithmetics
 - Subtract
 - ICS-Tables
 - and select the same BaseCal as above
 - (e.g.30x600_120_500_small_standard_body_rpp2x1_orig)

- Under loaded Rawdata/Table (left side) select:
 - Frame 1 to 1
 - Channel 1 to 672
 - Slice 1 to 30
 - FFS: PHI:2, Z:1
 - Select to the right the second Rawdata/Table:
 - Frame 2 to 2
 - Channel 1 to 672
 - Slice 1 to 30
 - FFS: PHI:1, Z:1
- and press GO
- ➡ The result is shown
For further evaluation, contact HSC 221

Core position check of UHR comb and UHR-plus comb

If the UHR comb was largely maladjusted or the initial position was lost, you can try and find the core position by using analyze.

- Start Analyze
 - Options > Service > Local Service > Test Tools > Analyze**
- Select the radio button **ICS-Tables**
- Mark an UHR mode BaseCal and press **Go**

- The plot of the BaseCal appears

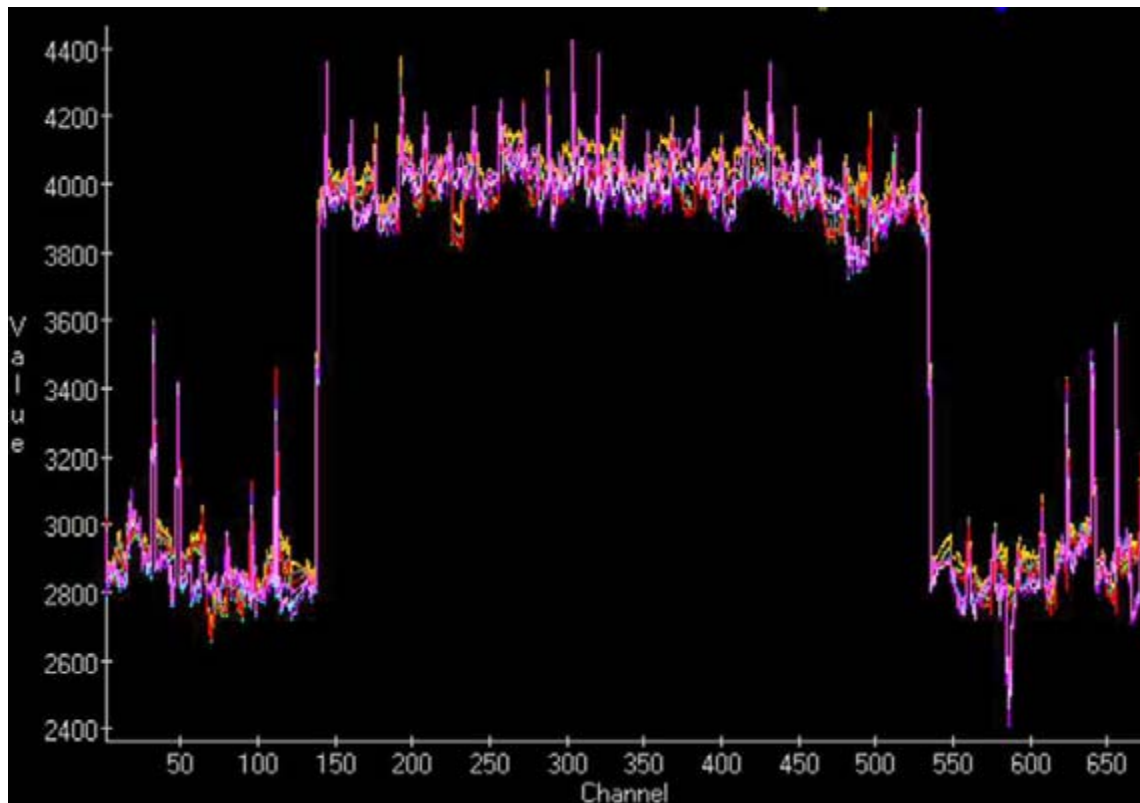


Fig. 133: Example of an UHR BaseCal plot

- Zoom both edges to see the channel where the UHR comb starts and ends.

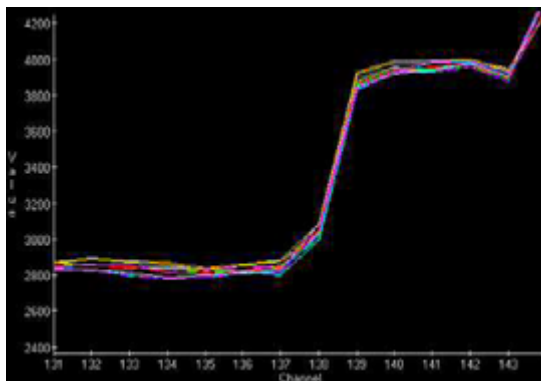


Fig. 134: Zoom of lower channel side

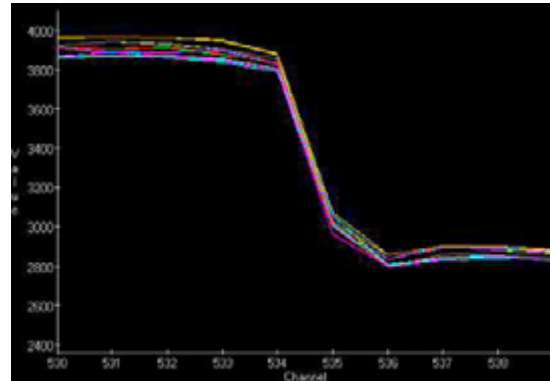


Fig. 135: Zoom of higher channel side

- ⇒ Compare your results with this example showing a well-adjusted UHR
- The first channel with a higher value is channel 139
- The last channel with a higher value is channel 534

How to deactivate the UHR comb

The UHR control can be disabled using the control function

- Go to:
Options > Service > Local Service > Control > Table load/modify
- Select:
 - **PERMANENT FILESYSTEM** under TABLE SOURCE and
 - **TAA** under TABLE NAMES
 and press **Go**
- Select the checkbox for "UHR not installed"
- Press the softkey **Write** to save the change

After the next init, the system works without UHR control

How to check the transmitting antenna

The transmitting antenna consist of:

- Antenna itself
- Termination

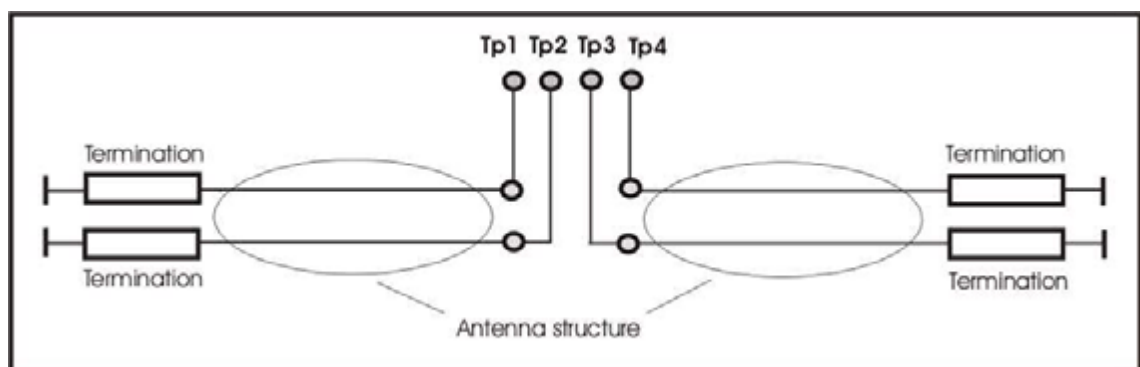


Fig. 136: Principle of transmitting antenna

To measure the antenna, perform the following steps:

1. Remove the transmitter
2. Use a digital multimeter to measure at the following test points:

Measure between	Nominal value
TP1 and TP2	79 Ohm +/- 1 Ohm
TP3 and TP4	79 Ohm +/- 1 Ohm
TP1 and Ground	39.5 Ohm +/- 0.5 Ohm
TP2 and Ground	39.5 Ohm +/- 0.5 Ohm
TP3 and Ground	39.5 Ohm +/- 0.5 Ohm
TP4 and Ground	39.5 Ohm +/- 0.5 Ohm

The following figures show the structure of the transmitting antenna and the position of the test points.



Fig. 137: Structure of transmitting antenna

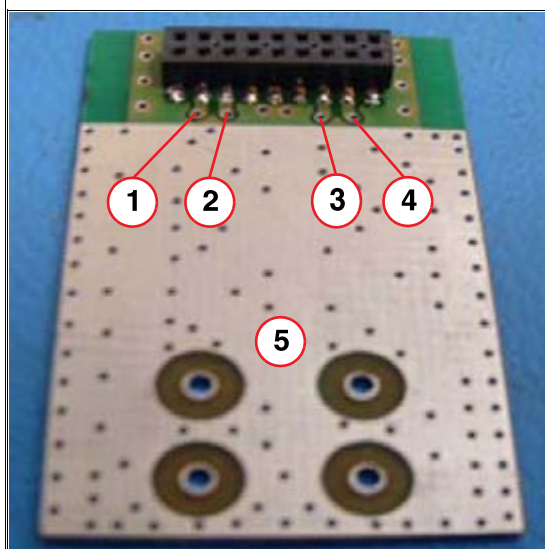


Fig. 138: Position of test points

- | | |
|--------|--------------|
| Pos. 1 | Test point 1 |
| Pos. 2 | Test point 2 |
| Pos. 3 | Test point 3 |
| Pos. 4 | Test point 4 |
| Pos. 5 | Ground |

Gantry door switches

Ensure that the system is in standby mode.

1. Log into the MedCom Service SW (**OPTIONS->LOCAL SERVICE**).
2. Browse to Test Tools/RotMode and select a 1.5 sec mode.
3. Press the Go button to start rotation (do not push the Start button).
4. Loosen the left cover screw.
 - ⇒ Verify that the gantry panels go off and the rotation stops.
5. An auto reset is performed. The gantry rotation starts again.
6. Secure the screw and loosen the right cover screw.
 - ⇒ Check that the gantry panels go off and the rotation stops.
7. Log out from the MedCom Service SW by closing the service window.

STOP push buttons

Test each stop switch listed with the following procedure.

1. Push the STOP button.
 - ⇒ Check that the gantry panels go off.
2. Push Continue on the navigator monitor to reset the gantry to standby mode.

Tab. 18 List of STOP switches

STOP switch	Notes
S302	Inside Gantry
S303	Inside Gantry
Front panel Right	
Front panel Left	
Rear panel right	
Rear panel left	
Control box	

Start button (MCU) tests

This test is used to check the basic function of a Start button. The system has to be in standby mode.

1. Select **Local Service -> Test Tools -> Controller -> MCU**.
2. **Select component and follow the instructions in the dialog.**
 - ⇒ **Start button test**
The user is prompted to press the start button twice. The test terminates automatically.
 - ⇒ **Footswitch test**
The user is asked to press the foot switch twice. The test terminates automatically.
 - ⇒ **External trigger test**
The user is asked to activate the external trigger twice. The test terminates automatically.

Stop Report Loop

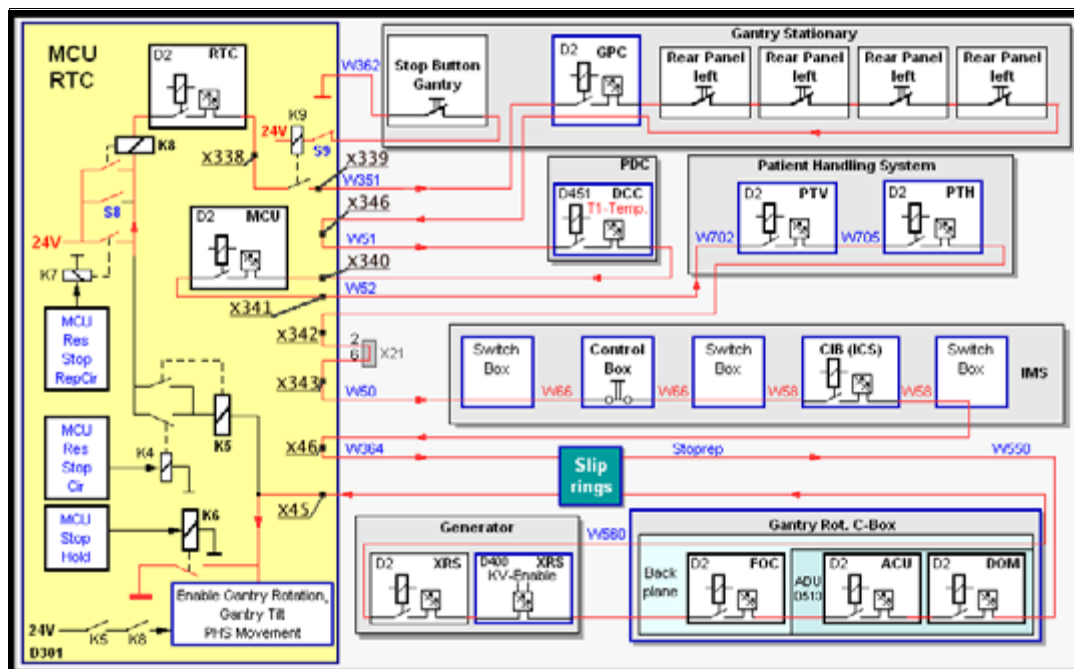


Fig. 139: Block diagram of STOP circuit

The stop report loop is made up of a number of contacts connected in series and fed with 24v via Relay K8.

Relay K6 is normally energized as long as power is available at the MCU. The stop loop is then set by the firmware closing K7 for 100ms. When all stop contacts are closed, K8 will energize and be held on by its self-holding contacts. Opening one of the stop contacts will cause K8 to release.

Controller watchdog test

This test checks the basic function of the watchdog trigger + MCB relays. The system has to be in standby mode. As a result, the test is useless in the case of a stop loop problem (stop loop not closed -> no standby!) The same test is performed during INIT. In case of a stop loop problem, test with [\(Using S8 / p. 229\)](#) or [\(Measuring controller diode / p. 231\)](#).

1. **Select Local Service -> Test Tools -> Controller -> RTC.**
2. Select controller and follow the instructions in the dialog.
 - ⇒ Test is OK, if **"Test passed"** is displayed.

Stop loop test

This test is used to check the basic function of the stop report loop. The system has to be in standby mode. As a result, the test is useless in the case of a stop loop problem (stop loop not closed -> no standby!) In case of a stop loop problem, use [\(Using S8 / p. 229\)](#) or [\(Measuring controller diode / p. 231\)](#).

1. Select **Local Service -> Test Tools -> Controller -> Stop.**
2. Select controller and follow the instructions in the dialog.
 - ⇒ Test is OK, if "**Test passed**" is displayed.

Using S8

If the stop report loop does not close, you may use the **S8** switch (Service STOP) on the MCU/RTC for troubleshooting. By activating **S8**, the K8 contact is bridged and 24 V (30 mA) is supplied to the stop report loop. Go down the test points. The first one **NOT** measuring 24 V indicates the device that is opening the loop.

For example, if X342 is the first test point at 0V then the PHS has an open stop contact.

- Measure **24 V** on the MCU board at test points provided in table, using X345 as ground. All test points are located slightly to the right of the stop service switches and in the center of the MCU/RTC.

Test point	Description
X338	RTC
X339	BUT (buttons inside gantry)
X346	GPC (and all 4 control panels)
X340	PDC (DCC)
X341	MCU
X342	PHS
X343	APESW (X21with link)
X45	ROT (from slip rings)
X46	SWBOX (Control box &ICS)
X345	GROUND

In addition, the stop loop is broken down by the GPC board by offering test points to isolate STOP loop problems caused by GPC and the four control panels. X200 on the GPC is used in the same manner; go down the test points until one does **NOT** have 24 V.

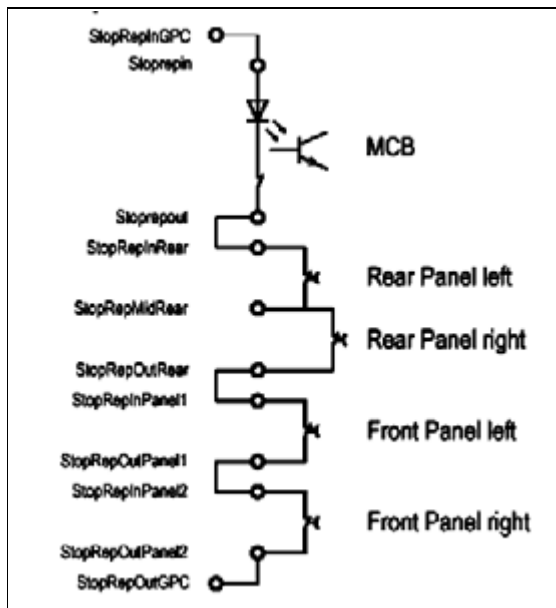


Fig. 140: STOP circuit test points on GPC board

The C-box also breaks down the stop loop additionally by offering test points to isolate STOP loop problems and is used in the same manner. Go down the test points until one does **NOT** have 24V.

The switch box routes the stop loop to the control box and the ICS.

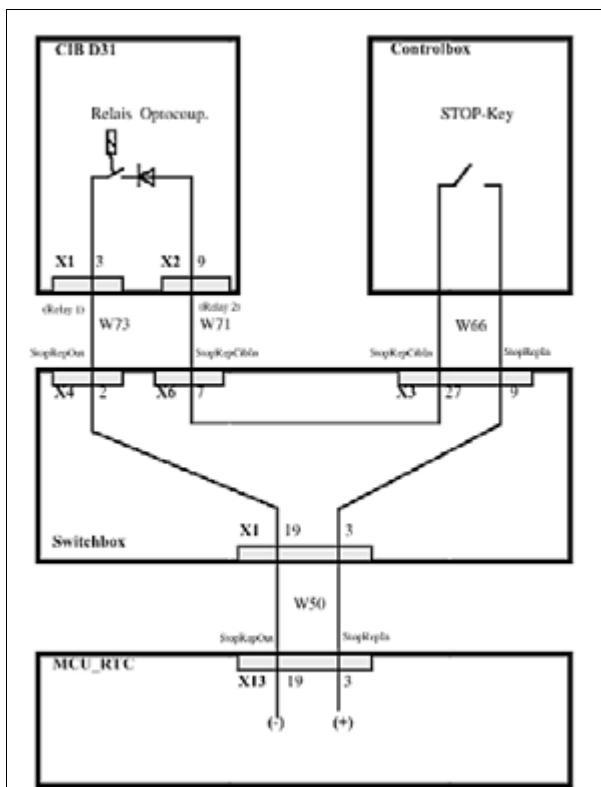


Fig. 141: Stop circuit detail of switch box

Measuring controller diode

To find a defective optocoupler on the MCB board, which may also be causing an open stop loop, check the voltage of each controller diode in the circuit. Disconnect the relevant cable on the MCU board (left gantry stand) and measure at the pins listed for the cable. Use a meter with selected diode measuring function to perform the measurement.

NOTE

Switch the system to COMP/ON while plugging in or unplugging connectors!

Switch the system to SYS/ON during measurement!

Tab. 19 Connectors for stop report loop

Cable/boards	Plug	Pin	Pin	Controller	Voltage
w50	x13	3	19	ICS	1V
w351	x3	3	10	GPC	1V
w51	x14	3	7	PDC	1V
w52	x16	3	7	PHS	2V
D2 (MCB) on MCU	X3	a4	b4	MCU	1V
D2 (MCB) on MCU	X3	a4	b4	RTC	1V

Switch off power and use an ohm meter to measure X21 and X9.

Tab. 20 Connectors for stop report loop

cable	plug	pin	pin	controller	voltage
n.a	x21	2	6	Jumper	0 Ohm = OK
w362	x9	8	15	stop button gantry	measure without power 0 Ohm = OK

- If all values measured are OK, the problem is in the rotating part where a diode measurement is not possible (all diodes are connected in series). In this case, follow the steps below for troubleshooting:
 - Change the MCB's stationary / rotation part.
 - Check the slip ring part (stop loop path).
 - Change the C-box.

CAN bus

The CAN bus is the main communication path for all controllers to communicate. Power for the CAN bus is provided by the MCU.

Tests

The CAN bus test checks the basic function of the CAN bus (communication between the MCU and one or all FW controllers). The system has to be in the standby mode. For troubleshooting purposes, the test can be performed with or without rotation (to isolate slip ring transmission problems). If you select rotation, you can choose predefined rotation times (0.5, 0.75, 1.0, or 1.5 secs.)

1. Select **Local Service -> Test Tools -> Controller -> CAN** and click **Go**.
2. Select controller and follow instructions in the dialog bar.
 - ⇒ Test is OK if "**Test passed**" is displayed. If an error message appears, troubleshoot according to the information in the error message.

Troubleshooting

The controllers on the main CAN bus are connected in parallel. Should one not respond, the software will indicate the controller having problems (unless the problem is with the MCU or CIB). Difficulties occur when one controller or the wiring to it has a short circuit. In this case, NO controllers can report because the CAN bus is down. Try using **Local service->Control->Table_Load/Modify** in the service platform to read table T00 from the control system. If you cannot read it, the CAN bus is not working between the CIB and MCU. Follow the instructions below to isolate the problem.

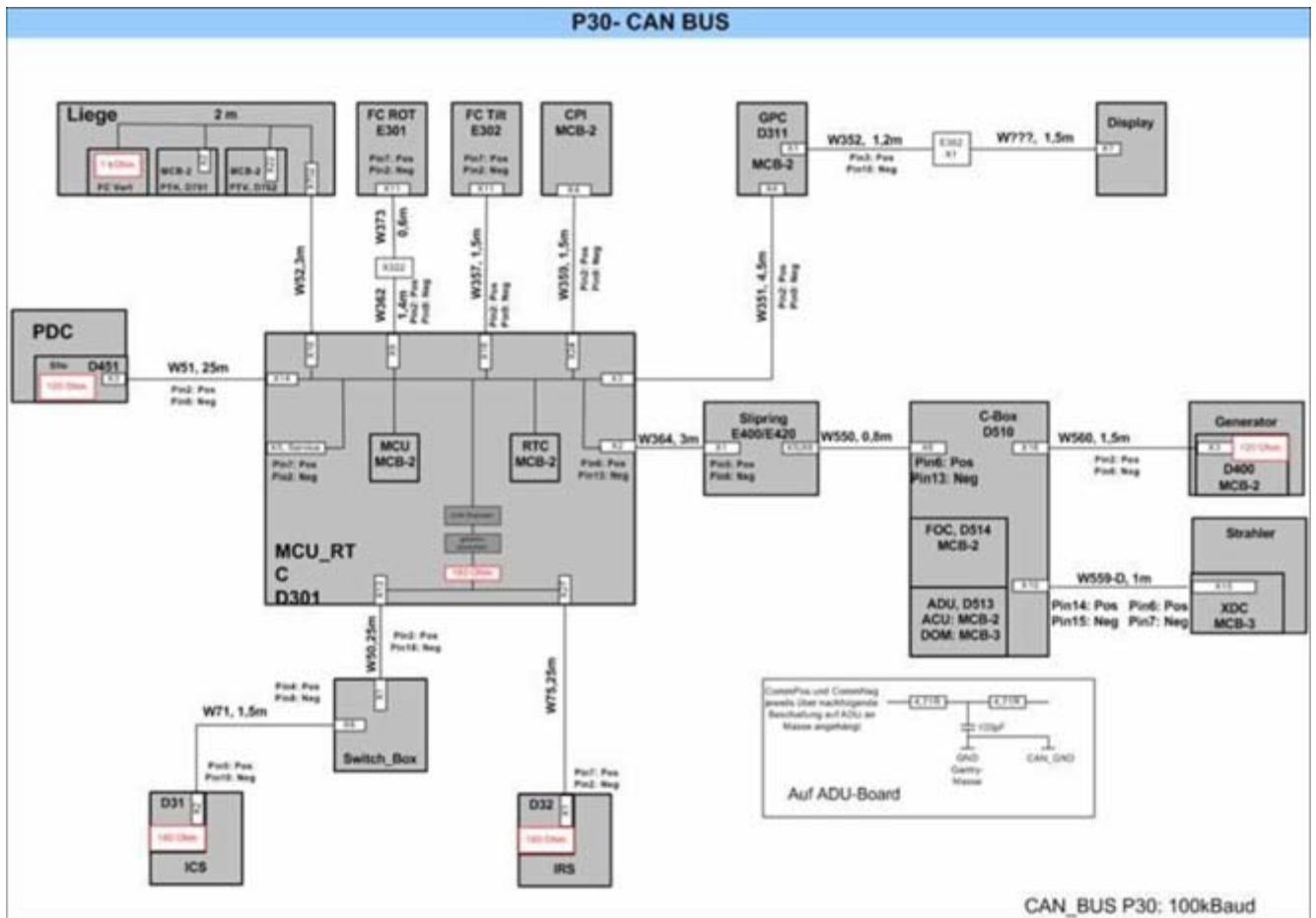


Fig. 142: CAN BUS Sensation: 100kBaud

The CAN Bus Sensation consists of 2 segments divided by a CAN repeater on the MCU board. The 2 segments are terminated separately, segment 1 on the D451 in the PDC and the D400 in the generator electronics box. (The 1 KOhm resistor in the PHS tests only the PHS independently of the rest of the system during fabrication).

Segment 2 is terminated on D31 in the ICS and on D32 in the IRS. CAN bus troubleshooting, especially sporadic errors, is quite difficult, since the bus is used by many components throughout the system. Sporadic messages 3003 from any controller mean that there are disturbances on the CAN bus. This error is not detected by firmware, but directly by the CAN ICs (integrated circuits). There are mainly 2 things to check in such case, the CAN bus termination and the signal quality.

Basic Troubleshooting

- Check the power supplies to the MCU/RTC and use the hardware tests for the CIB in the ICS.
- Turn system to Computer ON at the control box and then remove X2 at the MCU board.
- Turn system to system ON and use service program to read table T00.

- If you can now read T00, then the short is in the rotating part of the gantry or the slip rings. Continue troubleshooting by re-connecting X2 and disconnecting each board or controller in the rotating part until you can read T00. Also, try disconnecting the slip ring connections.
- If you cannot read T00, then turn system to Computer ON at the control box and then remove X16 at the MCU board.
- Turn system to system ON and use the service program to read table T00.
- If you can read T00, the short circuit is in the PHS. Reconnect X16 and isolate the problem.
- Continue in this way for X3, X14, X27 and X24. Then try X9 and X18 in case the frequency converters are causing the problem.
- Replace the MCU/RTC board D301

NOTICE

Before you replace the D301 check:

If following error message appears: “CAN OVERRUN”, check the switch position of S5 on the board D301,

⇒ **The switch S5 must be set to “RTC”**

- Replace the ICS

Extended Troubleshooting

1. Check correct termination

Segment 1

- System must be switched off.
- Measure the resistance between Commpos (X345) and Commneg (X346) on the slip ring.
 - ⇒ The result should be ~ 60 Ohms.
- Measure Commpos (X345) and Commneg (X346) against Gnd.
 - ⇒ The result should be in the MOhms range.
 - ⇒ If the result is not correct, measure the terminator resistor on D 541 PDC and on D400 GEL.
- The terminator resistor on D451 PDC
- Remove D451 PDC X3 and measure on board X3 between Commpos (pin 2) and Commneg (pin 6)
 - ⇒ The result should be ~ 120 Ohms.
- The terminator resistor on D400.
- Remove D400 X3 and measure on board X3 between Commpos (pin 2) and Commneg (pin6)
 - ⇒ The result should be ~ 120 Ohms.

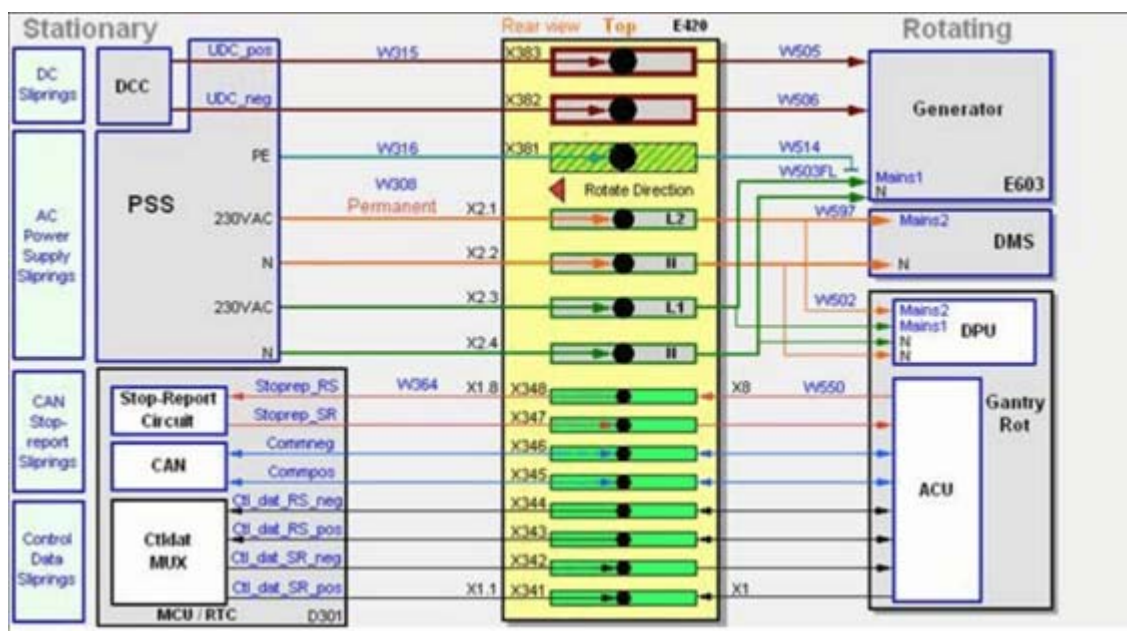


Fig. 143: Overview

Segment 2

- System must be switched off
- Use 9 pin adapter part. Nr. 46 83 314 on MCU X13. Measure the resistance between Commpos (pin 2) and Commneg (pin 18) on X13.
 - ⇒ The result should be ~ 60 Ohms.
- Measure Commpos (pin 2) and Commneg (pin 18) against Gnd.
 - ⇒ The result should be in the MOhms range.
 - ⇒ If the termination is not correct, measure the terminator resistor on D31 ICS, D32 IRS and on the MCU
- The terminator resistor on D31 ICS
- Remove MCU X13 and measure on cable X13 between Commpos (pin 2) and Commneg (pin18)
 - ⇒ The result should be ~ 180 Ohms.
- The terminator resistor on D32 IRS
- Remove MCU X27 and measure on cable X27 between Commpos (pin 7) and Commneg (pin2)
 - ⇒ The result should be ~ 180 Ohms.
- The terminator resistor on MCU

- Remove MCU X13 and measure on board X13 between Commpos (pin 2) and Commneg (pin18)
 - ⇒ The result should be ~ 180 Ohms.

2. Check signal quality on CAN bus

Measuring points: use an old cable with a 9 pin Cannon connector, cut 15 cm and strip wires to pin 7 (Commpos) and pin 6 (Commneg) or use 9 pin adapter, part. Nr. 46 83 314.

- Use GND on MCU

NOTE

To measure the CAN signals correctly, a differential probe is usually required, but disturbances can be seen better, if Commpos or Commneg is measured individually against GND.

Connect an oscilloscope channel one to Commpos, channel two to Commneg signal, GND on MCU board and use the index pulse signal on MCU board as external trigger. Use different rotating speeds and set time base and level to display the CAN bus signals. (see examples below)

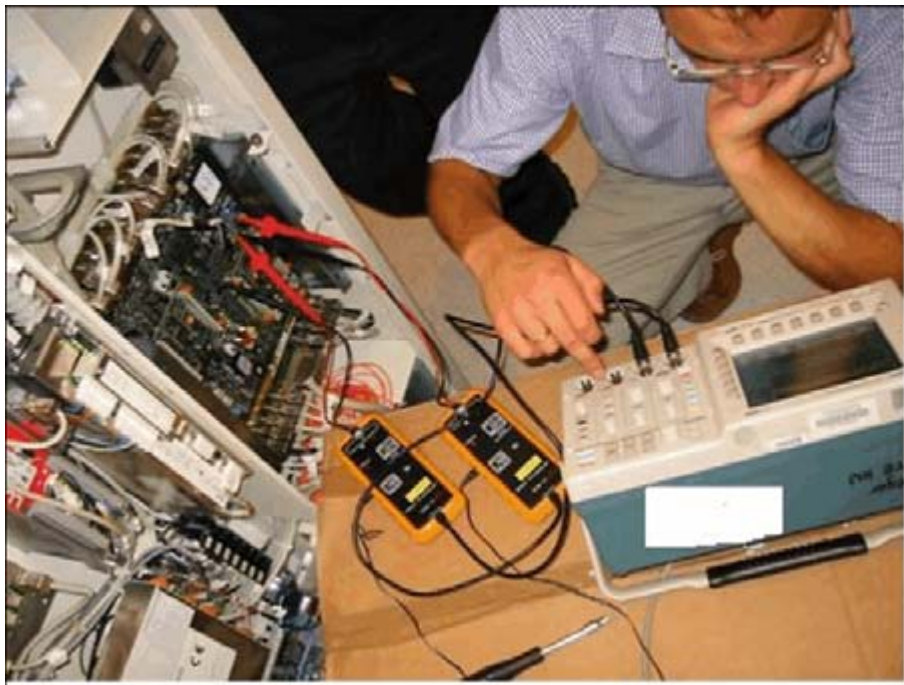


Fig. 144: Example of setup

- Check signal quality with different rotating speeds, some problems might only show up with fast rotation speeds

- Check signal quality with X-ray on or off to determine influence of X-ray generation on CAN bus signal quality.

NOTE

If signal disturbance appears on Commpos or Commneg, especially with fastest rotation, check GND brushes on Pwr slip ring.

Contact problems on the contact plates of GND brushes can cause CAN bus signal disturbance.



Fig. 145: Example of corroded contact plates

NOTE

If signal disturbance appears only with X-ray, troubleshoot X-ray components, check for all copper GND straps connected to H1, GEL etc.

NOTE

For isolating the gantry from the rest of the system by disconnecting the CAN bus connectors on the MCU, a CANpro is needed to rotate the gantry.

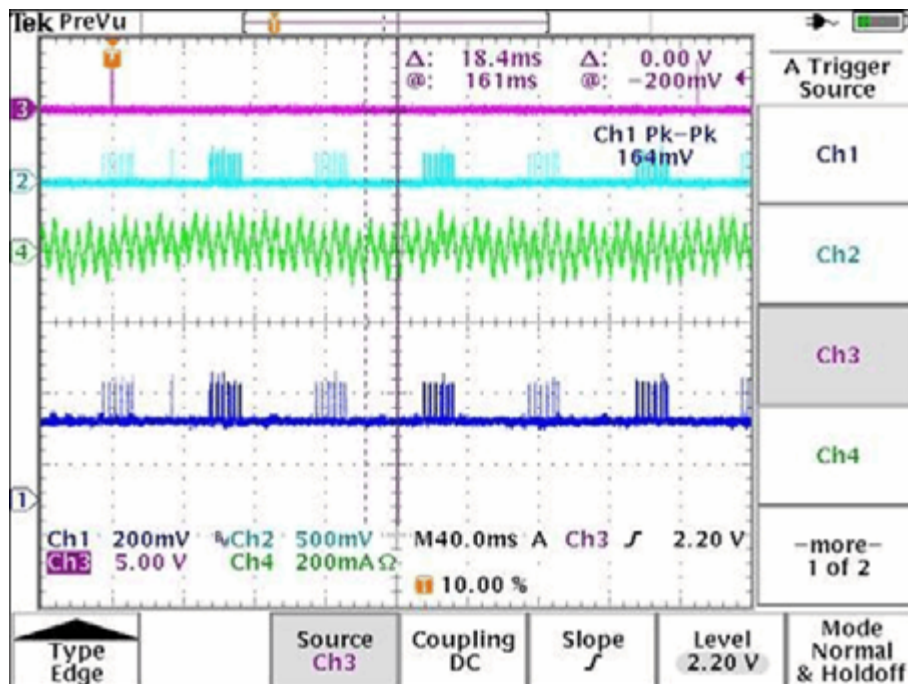


Fig. 146: Example of correct CAN bus signals

Channel **one** here Commpos or Commneg is measured against GND

Channel **two** . Commpos is measured differentially against Commneg with a differential probe.

Channel **three** is the index pulse signal from MCU used as the trigger.

Channel **four** is the current through the Main GNF connection on the power slip ring, usually not needed.

⇒ If the CAN bus signal packets can be seen, the shape of the signals is correct.

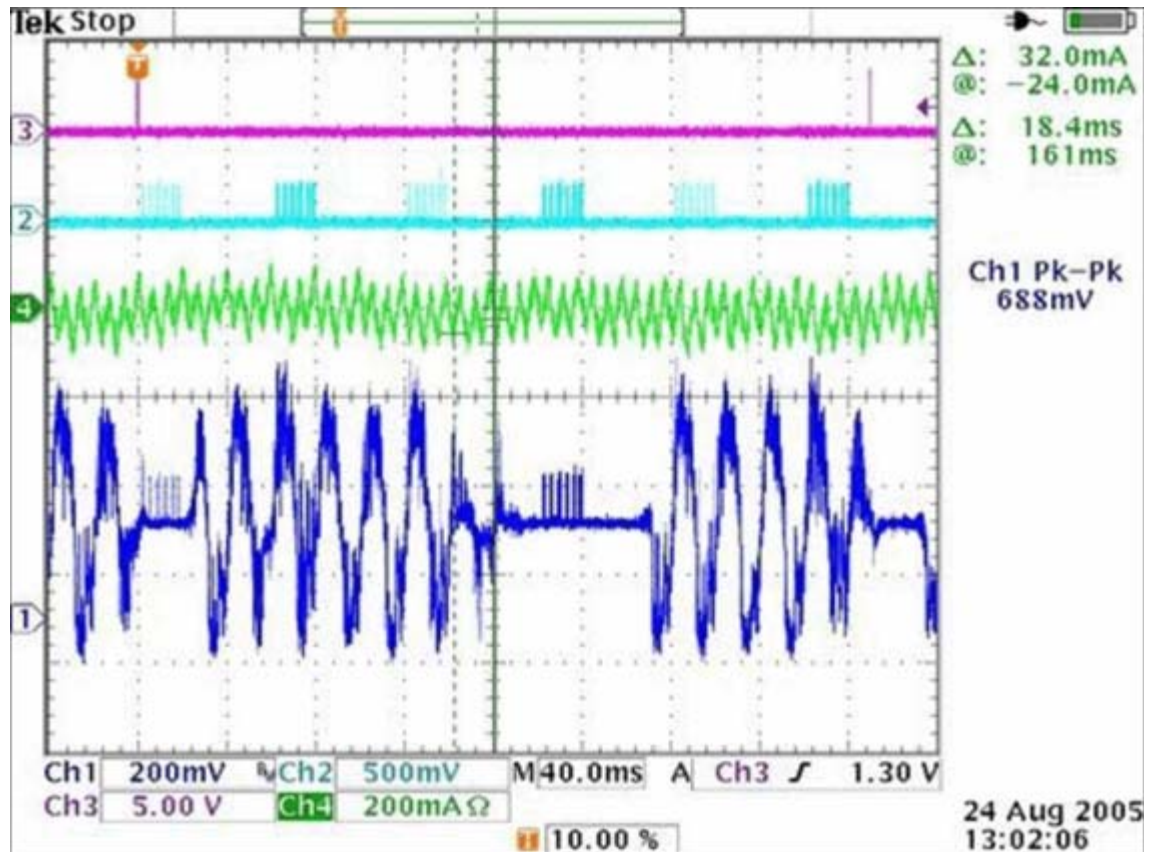


Fig. 147: Example of bad signal quality

The channel assignment is same as in the previous example. (example of correct CAN bus signals). The CAN signal measured differentially does not show a problem, but if Commpos or Commneg is measured differentially, massive disturbances are visible.

RTC

Rotation and tilt test

The following test is used to check the basic rotation and tilt functions. The system has to be in standby mode. As a result, the test is useless in case of a rotation or tilt problem (-> no standby!). In case of a rotation or tilt problem, use item **1** for troubleshooting.

1. Select **Local Service -> Test Tools -> Controller -> RTC**.
2. Select the controller and follow the instructions in the dialog.
 - ⇒ Test is OK, if "**Test passed**" is displayed.

Motor/Inverter overload protection

Ixt (current * time) monitoring provides electronic motor and inverter protection against high thermal loads.

General rule

Within 10 minute periods and with a continuous current of $I=1.5 \times I_n$ (150% overload), an overload of 1 minute is acceptable.

Max. speed limit test

A rotation with a speed higher than the absolute rotational maximum speed is performed to test the 5% monitoring of the rotational speed. The monitoring is always active. The test terminates automatically.

Tilt test

Tests the gantry tilt function by tilting the gantry to different positive and negative positions. The test terminates automatically.

FC interface tilt

Tests the interface to the frequency converter (FC) for the gantry tilt. RTC sends a telegram and waits for an acknowledgement from FC. The test is terminated by pressing the <DONE> button.

FC interface rotation

Tests the interface to the LUST frequency converter (FC) for the gantry rotation. RTC_MCB sends a telegram and waits for an acknowledgement from the FC. The test is terminated by pressing the <DONE> button.

AP test

Evaluates the time interval between angle pulses. After one rotation, the minimum and maximum time interval with the appropriate angle positions are displayed. In addition, 24 time intervals are indicated. These values are used for balancing. The test terminates automatically.

Performing the test

The following test checks the function of the AP pulses. The system has to be in standby mode.

1. Select **Local Service -> Test Tools -> Controller -> RTC**.
2. Select controller and follow the instructions in the dialog
 - ⇒ Test is OK, if "**Test passed**" is displayed.
3. If test does not pass, proceed as follows:
 - Clean slot ring.
 - Check slot ring for mechanical damage
 - Check adjustment of LPS board D522

Refer to **CT02-023.841.01.xx.02** , LPS board.

Switches, LED's & test points

Tab. 21 Service Switches

Switch	Function	LED	Status
service RTC S6	service RTC	yellow	on during service
S1	rot CW		
S2	rot CCW		
S3	tilt pos		
S4	tilt neg		
S7	Brake switch		
S8	Service Stop Report		
S10	service STOP on	yellow	on during service
S11	service STOP off		

Tab. 22 MCU board test points

Test point	Function	Test point	Function
X329	AP90	X333	Reserved
X328	AP0	X334	Ground
X327	Index pulse	X335	Vcc
		X330	Service in
		X331	Service out
		X348	X-ray

ADU

AP test

Evaluates the time interval between angle pulses. After one rotation, the minimum and maximum time interval with the appropriate angle positions are displayed. In addition, 24 time intervals are indicated. These values are used for gantry balancing. The test terminates automatically.

Performing the test

The following test checks the function of the AP pulses. The system has to be in standby mode.

1. Select **Local Service -> Test Tools -> Controller --> ACU**
2. Select the controller, set Rot-time to 1sec. and follow the instructions in the dialog
⇒ Test is OK, if "**Test passed**" is displayed.
3. If test does not pass, proceed as follows:
 - Clean slot ring.
 - Check slot ring for mechanical damage
 - Check adjustment of LPS board D522Refer to **CT02-023.841.01.xx.02** , LPS board.

Additional information

Position: The following picture shows the relation between the **slot ring** and **AP position** . Viewed from the back of the gantry .

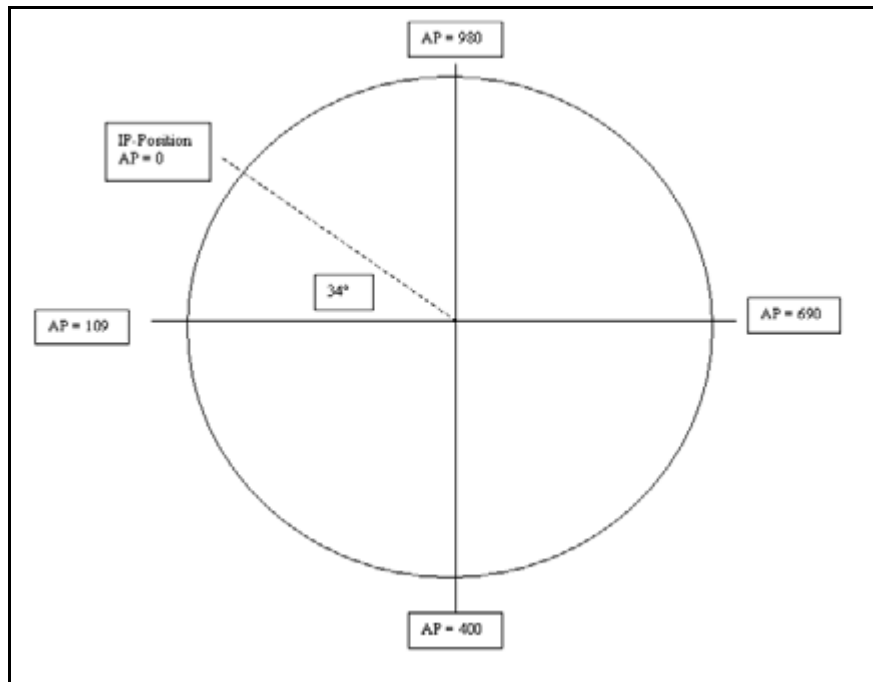


Fig. 148: Counting the AP pulses

You may rotate the gantry to the desired position using a static mode within the service software. Just calculate

Tube position = AP position x (360/1160)

Additionally, you may read out the gantry position via table T20.

Tolerances :

AP0-Pulse:1160 +/- 0

AP0 time 0.42sec: 362usec +/-7%

AP0 time 0.5sec: 431usec +/-7%

AP0 time 0.75sec: 646usec +/-7%

AP0 time 1.0sec: 862usec +/-7%

AP0 time 1.5sec: 1293usec +/-7%

GPC

The functions listed are terminated via the CAN bus (ABORT) or STOP/STOP REPORT.

The CAN bus is used as the communications medium for controlling or executing GPC functions. All commands (tilt, table movement, etc.) initiated from the operating panels or on the GPC/D311 are transmitted to the corresponding components and sub-assemblies via the same path.

Three different LED's on the board are used to monitor 5 V, 12 V, and 24 V power.

Service switches

Tab. 23 Service switches GPC

Switch	Function
S1	Tilts gantry positive/forward (top of gantry)
S2	Tilts gantry negative/backward (top of gantry)
S3	Moves table horizontally, positive/away from gantry
S10	Moves table horizontally, negative/toward gantry
S41	Moves table horizontally, negative/toward gantry at accelerated speed
S42	Moves table horizontally, positive/away from gantry at accelerated speed
S12	Moves table vertically, positive/downward
S13	Moves table vertically, negative/upward
S20	Moves patient from outer light marker to scan slice position
S21	Moves table away from gantry for patient loading or unloading
S23	Resets horizontal table position to zero
S30	Switches light marker on/off

Tab. 24 Test points

X200 STOP Test point	X230 Power Test point
INGPC	Test Trig
INMCB	Ground_24
OUTMCB	24V
INREAR	Ground_12ext
MIDREAR	12V external
OUTREAR	Ground_12int
IN1	12V internal
OUT1	Ground_5
In2	5V
Out2	Ground_3.3
OutGPC	3.3V
-	Ground_2.5
-	2.5V

The stop test points are used to diagnose problems in the STOP loop. Refer to [\(Using S8 / p. 229\)](#) for details on the tests.

Intercom

D346 Noise Filter

LED's

The four green LED's on D346 indicate the following.

Tab. 25 LED display on D346

LED	Meaning
LED 1	On when power supply OK
LED 2	Blinking when signal processor is running
LED 3 LED 4	Actual state of microphone input selection. Blinking in test modes without microphone input.

Tab. 26 Coding for LED 3 and LED 4 on D346

LED 3	LED 4	Input selection
off	off	Mic 1 back left
off	on	Mic 2 back right
on	off	Mic 3 front left
on	on	Mic 4 front right

Chapter	Change	Reason
1	A more detailed Troubleshooting Description for the CAN BUS	Change No.: 516608

